MA4M3 Local Fields

Lecture Notes

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Contents

Course Outline

The real numbers $\mathbf{R}$ are defined as the completion of the rational numbers $\mathbf{Q}$ in the usual metric. However, this metric is not that well-suited to arithmetic study; for example, the integers are discrete in $\mathbf{R}$.

In number theory, one is often more interested in p-adic numbers $\mathbf{Q}_p$, the completion of $\mathbf{Q}$ in the *p-adic* metric. In the p-adic metric, a number is very close to zero if it is highly divisible by a prime p (for example, whilst 1,000,000,000 is 'large' in the usual metric, it is highly divisible by 2 and 5, so it is very small in the 2-adic and 5-adic metrics). The integers are not discrete in the p-adic metric (as e.g. one can arbitrarily approximate 0 by integers p-adically), so p-adic numbers are much better suited to arithmetic, and have accordingly become fundamental in number theory and arithmetic geometry.

The real and p-adic numbers are examples of *local fields.* This module will give an introduction to local fields, with an emphasis on the p-adic numbers/non-archimedean local fields, and describe some of their beautiful properties, including: the classification of local fields, Hensel's lemma and applications to solubility of polynomials, and extensions and Galois theory of local fields.

The course will also treat some notable applications in number theory and arithmetic geometry, in particular the Hasse–Minkowski theorem on solubility of quadratic forms and the Kronecker—Weber theorem on abelian extensions of $\mathbf{Q}$.

Administrative matters

(∗) **Remark.** *A number of remarks and proofs in these notes are included for background/wider interest only, and are* **not examinable**. *These remarks are marked with a* (∗)*, like this comment. They will typically go beyond what is covered in the lectures themselves. Abridged lecture notes, containing the bare minimum of examinable content, will be made available as the course proceeds.*

Recommended reading. A primary resource for this course is Cassels' book *Local Fields* [Cas86]. Serre's book *A Course in Arithmetic* [Ser73] contains a very nice introduction to p-adic fields, as well as detailed proofs for all the results we include on quadratic forms. Serre also has a book titled *Local Fields* [Ser79]; this is considerably more advanced than most of the material treated here, but it is a good resource for students who want to follow up the course with deeper material.

Assessment details. The course will be assessed via written exam in Summer 2022, comprising 85% of the module score.

The remaining 15% will be assessed through exercise sheets. There will be four assignments, with deadlines in weeks 3, 6, 8 and 10 of term 2. Your grade will comprise the scores from your best three of the four sheets, with each worth 5%.

Contact details. Any questions/comments/corrections about the course or these notes will be gratefully received, and may be directed to me at: `christopher.d.williams@warwick.ac.uk`.

Acknowledgement. I would like to heartily acknowledge the very strong influence of Tom Fisher in preparing these notes; many of the topics are borrowed from his own excellent course on Local Fields, which I attended in my own fresh-faced student days. He recently advised me that his own course was in turn borrowed from elsewhere, in a mathematical twist on the circle of life. His presentation was sufficiently inspiring that, ten years later, I'm yet to ever leave this field of mathematics, which deserves acknowledgement in itself!

Thanks also to AJ Fong, Kate Harrison, Muyang Li, Hanwen Liu, Ruth Pugh, Thomas Read, Stephen Richardson, and Kenji Terao, all of whom took this course and noticed errors/gaps in the course materials or suggested nicer proofs than some given in lectures. (Apologies, also, to anybody who gave comments and corrections that I missed off this list).

Some waffly motivation: solving quadratic forms

Rational solutions to quadratic equations

In number theory, one is often interested in solving polynomial equations. Many of these problems are comprehensible to school children, but can lead to wonderful and deep mathematics.

One class of equations that you definitely meet at school is *quadratic equations.* Given a quadratic equation $x^2 + aX + b = 0$, the quadratic formula tells you there are either 0 or 2 real solutions, and tells you exactly what they are.

However, number theorists are much more interested in *rational* solutions. This is already more subtle; for example:

(1a) *Does $X^2 = 3$ have any solutions in $\mathbf{Q}$?*

The answer is of course *no*, and there is a good chance this will have been the first real mathematical proof you ever saw. Suppose there is a solution $a/b \in \mathbf{Q}$; then we may take a, b coprime by dividing out common factors. Then $a^2/b^2 = 3$, so $a^2 = 3b^2$ and hence 3 divides a. Writing $a = 3c$ we see that $3c^2 = b^2$, so 3 divides b. But this is a contradiction.

Consider the related question:

(1b) *Does $X^2 = -3$ have any solutions in $\mathbf{Q}$?*

In this case, you know that there are no solutions in $\mathbf{R}$, so there certainly aren't in $\mathbf{Q}$!

If we pass to two variables, things get a little more complicated. For example:

(2a) *Does $X^2 + Y^2 = 3$ have any solutions in $\mathbf{Q}$?*

After some thought, you can deduce that the answer is again *no.* Clearing denominators and dividing by common factors, we can reduce this question to whether the three-variable equation $x^2 + y^2 = 3z^2$ has any *integer* solutions, with x, y, z having no joint common factor. Now the only quadratic residues modulo 3 are 0 and 1; so if $x^2 + y^2 \equiv 0 \pmod 3$, then we must have both x and y divisible by 3. But writing $x = 3u, y = 3v$ we see $3(u^2 + v^2) = z^2$, so z is also divisible by 3, and we get a contradiction.

This was more complicated. However, considering the related question:

(2b) *Does $X^2 + Y^2 = -3$ have any solutions in $\mathbf{Q}$?*

There are still has no real solutions, hence no rational solutions.

Let's ramp up to three variables:

(3a) *Does $X^2 + Y^2 + 5Z^2 = 3$ have any solutions in $\mathbf{Q}$?*

We reduce again to finding integer solutions to $x^2 + y^2 + 5z^2 = 3w^2$. This time, we can't use the same modulo 3 argument. We instead cast around for a bit and alight upon working modulo 8, where the quadratic residues are 0,1,4. By examining a stack of cases, we see that w must be even, as $x^2 + y^2 + 5z^2$ can never be 3 modulo 8. Hence $3w^2 \equiv 0$ or $4 \pmod 8$. Now one of x, y, z must be odd; but then, again by exhausting all the cases, one can see that there is no way for at least one of x, y, z to be odd and $x^2 + y^2 + 5z^2 \equiv 0$ or $4 \pmod 8$. So there are no solutions.

...this was even *more* involved, and required enumerating all the possible cases one by one, which is quite nasty. However, again, considering:

(3b) *Does $X^2 + Y^2 + 5Z^2 = -3$ have any solutions in $\mathbf{Q}$?*

There are still no solutions in $\mathbf{R}$, hence no solutions in $\mathbf{Q}$.

An abstract principle

Each time we increase the difficulty, the argument for the type (a) questions gets more involved; but the type (b) questions are trivial, because we know there are no real solutions to any of them. This is a very simple principle: since $\mathbf{Q} \subset \mathbf{R}$, if there are no solutions in $\mathbf{R}$ there cannot be any in $\mathbf{Q}$.

Now, what if we liberated ourselves from $\mathbf{R}$? We could consider an abstract principle: if K is a field with $\mathbf{Q} \subset K$, and there are no solutions in K, then there are no solutions in $\mathbf{Q}$. In the first three years of an undergraduate degree, the chances are that the only such fields K you will have seen are $\mathbf{R}$ and $\mathbf{C}$. The field $\mathbf{C}$ is hopeless for applying this principle, since there are *always* complex solutions to polynomial equations (in any number of variables).

Towards p-adic fields

One might take inspiration from the solutions we gave to the type (a) questions. In (1a) and (2a), our arguments were heavily based around the prime 3. In (3a), we used the prime 2. This suggests that the prime numbers are heavily involved. Indeed, one might posit the existence of a field $\mathbf{Q}_3 \supset \mathbf{Q}$ in which the equations in (1a) and (2a) have no solution, and a field $\mathbf{Q}_2 \supset \mathbf{Q}$ in which (3a) has no solution.

In this course, we'll see that such 'p-adic' fields do exist – and, indeed, that there is a field $\mathbf{Q}_p \supset \mathbf{Q}$ for *every* prime p. As over $\mathbf{R}$, it will be very easy to check whether or not an equation of the form

$$a_1X_1^2 + \cdots + a_nX_n^2 = b \tag{0.1}$$

has a solution in $\mathbf{Q}_p$, and if there are no such solutions we can immediately rule out there being a rational solution.

However, it turns out these fields are even more powerful. Something we'll see in the course is that there is a converse to our principle! During the course, we'll see the following 'local–global principle:

Theorem (Hasse–Minkowski)**.** *Suppose there are solutions to* (0.1) *in* $\mathbf{R}$ *and in* $\mathbf{Q}_2$, $\mathbf{Q}_3$, $\mathbf{Q}_5$, $\mathbf{Q}_7$, $\mathbf{Q}_{11}, \ldots$ *(i.e. in* $\mathbf{Q}_p$ *for all* p*). Then* (0.1) *has a solution in* $\mathbf{Q}$*.*

Example. The following was an audience-generated example during the first lecture: does

$$71X^2 + 7Y^2 + 61Z^2 = 11$$

have any rational solutions? The answer is *yes.* It clearly has real solutions, and during the course we'll see it's easy to read off that it has solutions in $\mathbf{Q}_p$ for all primes p. Thus Hasse–Minkowski tells us that it has a solution in $\mathbf{Q}$. (A computer search shows that $(X, Y, Z) = (\frac{5}{42}, 0, \frac{17}{42})$ or $(\frac{1}{3}, \frac{2}{3}, 0)$ are both solutions).

(Aside: in the very close equation $73X^2 + 7Y^2 + 61Z^2 = 11$, there exist rational solutions, e.g. $(\frac{5}{38}, -\frac{41}{76}, \frac{27}{76})$. Moreover for all solutions, one can show that $XYZ \neq 0$).

Once you have been led to define $\mathbf{Q}_p$, you realise that the p-adic number systems are very natural places to do arithmetic: indeed, p-adic methods have shaped the modern number theory landscape to an extraordinary degree. During this course, we'll introduce the p-adic numbers and some of their beautiful properties.

Part I

Introduction to Local Fields

In Part I, we introduce the main concepts in the study of local fields. We start in Chapter 1 by defining the field $\mathbf{Q}_p$ of *p-adic numbers*, for p a prime. In particular, we describe its basic properties: the p-adic absolute value, the p-adic valuation, p-adic expansions, and the p-adic integers $\mathbf{Z}_p$. This is all fairly down-to-earth, largely involving explicit computations with rational numbers and working modulo powers of p.

Many of the concepts introduced in Chapter 1 generalise widely, and in Chapters 2 and 3, we describe these more general analogues. In particular, in Chapter 2, we describe the general theory of *absolute values* on a field, and discuss notions of when absolute values are equivalent, the dichotomy between archimedean (e.g. the usual absolute value on $\mathbf{R}$) and non-archimedean (e.g. the p-adic) absolute values. We show that these are essentially the only absolute values on $\mathbf{Q}$, and then classify the complete archimedean fields: they are exactly $\mathbf{R}$ and $\mathbf{C}$.

Most of the rest of the course is taken up with the (more *arithmetically* interesting) case of non-archimedean fields, such as $\mathbf{Q}_p$. A non-archimedean absolute value may be described equivalently in terms of a *valuation* on the field, which we discuss in detail in Chapter 3. We describe connections between valuation theory and commutative algebra, particular in terms of local rings (rings with a unique maximal ideal). We introduce the *discrete valuations*, a particularly nice class of valuations, and describe properties of *discrete valuation rings*, more general analogues of the ring $\mathbf{Z}_p$ of p-adic integers. We give another important example of non-archimedean fields: the field of Laurent series over a finite field, which can be viewed as a characteristic p analogue of $\mathbf{Q}_p$.

In Chapter 4, we give an alternative description of $\mathbf{Z}_p$ in terms of *inverse limits*, a general mathematical notion. This allows us to study more topological questions about non-archimedean fields. In particular, we introduce the notion of being *locally compact.* Finally we bring together all of these basics into a definition of the primary object of study in the entire course, namely, the definition of a *local field.* By definition, this is a field that is locally compact with respect to an absolute value. We state the classification of local fields: any local field is either $\mathbf{R}$ or $\mathbf{C}$, or a finite extension of $\mathbf{Q}_p$, or a field of Laurent series over a finite field.

In Parts II and III, we will apply the concepts introduced here to attack more arithmetic questions: the study of polynomial equations over the rationals, and of Galois extensions of $\mathbf{Q}$ and its completions.

Chapter 1

Introduction to p-adic numbers

To define p-adic numbers, it is first a good idea to remember how we first defined $\mathbf{R}$.

Definition 1.1. (i) Let $(a_n)_{n\in\mathbf{N}} \subset \mathbf{Q}$ be a sequence of rationals. This sequence is *Cauchy* if for all $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that if $m, n > N$, we have

$$|a_m - a_n| < \epsilon. \tag{1.1}$$

(ii) The real numbers are

$$\mathbf{R} = \{\text{Cauchy sequences in } \mathbf{Q}\}/(\text{equivalence}).$$

Example. The real number π is the limit of the sequence 3, 3.1, 3.14, 3.141, 3.1415, etc. Any real number can be written as an explicit sequence in this way by successive truncations of its decimal expansion.

This definition depends crucially on the absolute value/norm function $|\cdot|$ in (1.1). Throughout your undergraduate life to date, the notation $|x|$ will have meant 'the distance between 0 and x'. If we want to generalise this, then we must ask:

Question: *What does 'distance' mean?*

1.1. Absolute values

Definition 1.2. Let K be a field. An *absolute value* on K is a function

$$|\cdot| : K \longrightarrow \mathbf{R}$$

such that for all $x, y \in K$, we have:

(i) $|x| \geqslant 0$, with equality if and only if $x = 0$;

(ii) $|xy| = |x| \cdot |y|$;

(iii) $|x + y| \leqslant |x| + |y|$.

Examples. – If $K = \mathbf{Q}, \mathbf{R}, \mathbf{C}$ and $x = a + ib \in K$ with $a, b \in \mathbf{R}$, then the function

$$|a + ib|_\infty := |a + ib| = \sqrt{a^2 + b^2} \geqslant 0$$

is an absolute value.

– For any field K, the function $|\cdot| : K \to \mathbf{R}$ given by

$$|x| = \begin{cases} 1 : x \neq 0 \\ 0 : x = 0 \end{cases}$$

is an absolute value, called the *trivial absolute value.*

Lemma 1.3. *Let $|\cdot|$ be an absolute value on a field K and let $x \in K$.*

(1) We have $|1| = 1$.

(2) If $x^n = 1$ for some n, then $|x| = 1$.

(3) We have $|x| = |-x|$.

Proof. (1) Let $y \in K$ be non-zero; then $|1| \cdot |y| = |1 \cdot y| = |y|$ by (ii), and $|y| \neq 0$ by (i), so cancelling we see $|1| = 1$.

(2) We have $|x|^n = |x^n| = |1| = 1$, so $|x| \in \mathbf{R}_{\geqslant 0}$ is a root of unity. But the only such root of unity is 1.

(3) We know $|-1| = 1$ by (2); and then $|x| = |-1||x| = |-x|$. □

Corollary 1.4. *If K is a finite field, then the only absolute value is the trivial absolute value.*

Proof. Let $n := \#K$. Then $K^\times$ is cyclic of order $n-1$, so if $x \neq 0$, then $x^{n-1} = 1$, and hence $|x| = 1$ by Lemma 1.3. □

Definition 1.5. A *valued field* is a field K together with an absolute value $|\cdot|$ on K.

A valued field is a metric space with the distance

$$d(x, y) = |x - y|.$$

Hence it is a topological space, with open sets given by unions of balls of the form

$$B(x, r) := \{y \in K : d(x, y) < r\}.$$

Definition 1.6. Let $(K, |\cdot|)$ be a valued field. The *completion* of K, denoted $\widehat{K}$, is

$$\widehat{K} := \{\text{Cauchy sequences in } K \text{ for } |\cdot|\}/(\text{equivalence}).$$

Lemma 1.7. *There are unique continuous extensions of $\times, +$ and $|\cdot|$ making $\widehat{K}$ into a valued field.*

Proof. If (a_n) and (b_n) are Cauchy sequences, then we define $(a_n)+(b_n) = (a_n+b_n)$ and $(a_n)\times(b_n) = (a_n b_n)$. It is an exercise to see these are well-defined extensions and give $\widehat{K}$ a ring structure.

To see $\widehat{K}$ is a field, we must show that if $0 \neq a \in \widehat{K}$, then a is invertible. Let $a_n \to a$ be a Cauchy sequence. Since $a \neq 0$ and a_n is Cauchy, there exists $C > 0$ and N such that for $n \geqslant N$ we have

$$|a_n| \geqslant C.$$

After throwing out the first N terms, we consider $(1/a_n)$. This is Cauchy, since

$$\left|\tfrac{1}{a_n} - \tfrac{1}{a_m}\right| = \left|\tfrac{a_m - a_n}{a_m a_n}\right| = \tfrac{|a_m - a_n|}{|a_m| \cdot |a_n|} \leqslant C^{-2}|a_m - a_n|,$$

which tends to 0 as $m, n \to \infty$. Then $(1/a_n) \to 1/a \in \widehat{K}$, and $a \in \widehat{K}^\times$.

Finally, we extend $|\cdot|$ to $\widehat{K}$ by defining $|a| := \lim_{n\to\infty} |a_n|$. Again, it is an exercise to show this is well-defined: that is, if (a'_n) is a Cauchy sequence equivalent to (a_n), then $\lim_{n\to\infty}|a_n| = \lim_{n\to\infty}|a'_n|$. □

($*$) **Remark.** If K is a field and d a general metric on K, then we can still define the completion of K with respect to d. But this is not a field in general! (You *need* multiplicativity of the absolute value).

We're yet to see the p-adic metric, but when we have, you might like to come back to this remark and consider the following exercise/example. We can define a 6-adic metric on $\mathbf{Q}$, via the 6-adic valuation: if $\alpha \in \mathbf{Q}$, write $\alpha = 6^n \frac{a}{b}$, where $6 \nmid a$ and $(6, b) = 1$, and define $v_6(\alpha) = n$. Show that the function $|\cdot|_6 : \mathbf{Q} \to \mathbf{R}, \alpha \mapsto 6^{-v_6(\alpha)}$ is not an absolute value, but does define a metric on $\mathbf{Q}$ via $d(\alpha, \beta) = |\alpha - \beta|_6$. By the Chinese Remainder Theorem, there exist integers $\alpha_n \in \mathbf{Z}$ such that $\alpha_n \equiv 1 \pmod{3^n}$ and $\alpha_n \equiv 0 \pmod{2^n}$. Show that α_n is Cauchy for $|\cdot|_6$, hence tends to a (non-zero) limit α in the completion $\mathbf{Q}_6$ of $\mathbf{Q}$ for $|\cdot|_6$. Show also that α is not invertible in $\mathbf{Q}_6$, i.e. $\mathbf{Q}_6$ is not a field.

Examples.
- The completion of $\mathbf{Q}$ in $|\cdot|_\infty$ is $\mathbf{R}$. The completion of $\mathbf{Q}(i)$ in $|\cdot|_\infty$ is $\mathbf{C}$ (where $|\cdot|_\infty$ is defined after fixing an embedding of $\mathbf{Q}(i)$ into $\mathbf{C}$).
- The completion of any field K with respect to the trivial absolute value is K itself.
- If $(K, |\cdot|)$ is a valued field, then the completion of $(\widehat{K}, |\cdot|)$ is $\widehat{K}$ (that is, 'completions are complete').

1.2. The p-adic valuation and p-adic absolute value

In this course, we are most interested in the *p-adic absolute value* on $\mathbf{Q}$, which we now define.

First, we introduce the notion of 'valuation at a prime p', which tracks how much p divides a rational number. Let p be a prime. If $0 \neq \alpha \in \mathbf{Q}$, then there exists a unique n such that

$$\alpha = p^n \frac{a}{b}, \qquad p \nmid a, \qquad p \nmid b.$$

Definition 1.8. Define a function

$$\begin{aligned} v_p : \mathbf{Q}^\times &\longrightarrow \mathbf{Z}, \\ \alpha = p^n \tfrac{a}{b} &\longmapsto n. \end{aligned}$$

Example. Let $\alpha = 12/5$. Then we have:

$$\begin{array}{lll} \text{for } p = 2: & \alpha = 2^2 \cdot \frac{3}{5} & \Rightarrow v_2(\alpha) = 2, \\ \text{for } p = 3: & \alpha = 3^1 \cdot \frac{4}{5} & \Rightarrow v_2(\alpha) = 1, \\ \text{for } p = 5: & \alpha = 5^{-1} \cdot \frac{12}{1} & \Rightarrow v_5(\alpha) = -1, \\ \text{for } p > 5: & \alpha = p^0 \cdot \frac{12}{5} & \Rightarrow v_p(\alpha) = 0. \end{array}$$

Lemma 1.9. *Let $\alpha, \beta \in \mathbf{Q}^\times$.*

(i) We have $v_p(\alpha\beta) = v_p(\alpha) + v_p(\beta)$.

(ii) We have $\alpha \in \mathbf{Z}$ if and only if $v_p(\alpha) \geqslant 0$ for all p.

(iii) We have $v_p(\alpha + \beta) \geqslant \min(v_p(\alpha), v_p(\beta))$.

Proof. Let $\alpha = p^n \frac{a}{b}$ and $\beta = p^m \frac{c}{d}$.

(i) We have

$$\alpha\beta = p^{m+n} \tfrac{ac}{bd},$$

so $v_p(\alpha\beta) = m + n = v_p(\alpha) + v_p(\beta)$.

(ii) For each prime, by definition $v_p(\alpha) < 0$ if and only if p divides the denominator of α. If $\alpha \in \mathbf{Z}$, the denominator is 1, and this has no prime divisors. Conversely if the denominator is not 1, then it must have a prime divisor p, and $v_p(\alpha) < 0$.

(iii) Suppose without loss of generality that $m \leqslant n$, i.e. $\min(v_p(\alpha), v_p(\beta)) = v_p(\beta) = m$. Then we have

$$\alpha + \beta = p^n \tfrac{a}{b} + p^m \tfrac{c}{d} = p^m \left(p^{n-m} \tfrac{a}{b} + \tfrac{c}{d}\right) = p^m \frac{p^{n-m}ad + bc}{bd}.$$

By (i), we know

$$v_p(\alpha + \beta) = v_p\left(p^m \frac{p^{n-m}ad + bc}{bd}\right) = v_p(p^m) + v_p(p^{n-m}ad + bc) - v_p(bd).$$

Note $v_p(p^m) = m$. As bd is coprime to p, we have $v_p(bd) = 0$; and $v_p(p^{n-m}ad + bc) \geqslant 0$ by (ii). So

$$v_p(\alpha + \beta) = m + v_p(p^{n-m}ad + bc) \geqslant m = \min(v_p(\alpha), v_p(\beta)),$$

as required. □

Definition 1.10. Define a function

$$\begin{aligned} |\cdot|_p : \mathbf{Q} &\longrightarrow \mathbf{R}, \\ \alpha &\longmapsto \begin{cases} p^{-v_p(\alpha)} & : \alpha \neq 0 \\ 0 & : \alpha = 0. \end{cases} \end{aligned}$$

Example. For $\alpha = 12/5$ as above, we have

$$|\alpha|_2 = \tfrac{1}{4}, \qquad |\alpha|_3 = \tfrac{1}{3}, \qquad |\alpha|_5 = 5, \qquad |\alpha|_p = 1 \text{ for } p > 5.$$

Note that the more divisible α is by the prime p, then the *smaller* $|\alpha|_p$ is.

Proposition 1.11. *$|\cdot|_p$ defines an absolute value, called the p-*adic absolute value*.*

Proof. We must check the axioms of Definition 1.2:

(i) For any $n \in \mathbf{Z}$, we have $p^n > 0$, so (i) holds.

(ii) By Lemma 1.9(i), we have

$$|\alpha\beta|_p = p^{-v_p(\alpha\beta)} = p^{-[v_p(\alpha)+v_p(\beta)]} = p^{-v_p(\alpha)} \cdot p^{-v_p(\beta)} = |\alpha|_p \cdot |\beta|_p.$$

(iii) By Lemma 1.9(iii), we have

$$\begin{aligned} |\alpha + \beta|_p = p^{-v_p(\alpha+\beta)} &\leqslant p^{-\min(v_p(\alpha), v_p(\beta))} \\ &\leqslant \max(p^{-v_p(\alpha)}, p^{-v_p(\beta)}) \\ &\leqslant \max(|\alpha|_p, |\beta|_p). \end{aligned}$$

But $\max(|\alpha|_p, |\beta|_p) \leqslant |\alpha|_p + |\beta|_p$, giving (iii). □

We've actually proved a stronger version of the inequality in (iii)!

Definition 1.12. Let $|\cdot|$ be an absolute value on $\mathbf{Q}$. We say $|\cdot|$ is a *non-archimedean absolute value* if we have

$$|\alpha + \beta| \leqslant \max(|\alpha|, |\beta|) \tag{1.2}$$

for all $\alpha, \beta \in \mathbf{Q}$. We say it is an *archimedean absolute value* if (1.2) is not true.

In the process of the proof of Proposition 1.11, we showed:

Corollary 1.13. *For a prime p, $|\cdot|_p$ is a non-archimedean absolute value.*

Notation. The following is convenient notation. If $\alpha, \beta \in \mathbf{Q}^\times$, then we write

$$\alpha \equiv \beta \pmod{p^n}$$

to mean $v_p(\alpha - \beta) \geqslant n$. (Note this agrees with the usual notion of modular arithmetic when $\alpha, \beta \in \mathbf{Z}$. More generally, it means that $\alpha \in \beta + p^r\mathbf{Z}$.)

1.3. The p-adic numbers

Recall Definition 1.1: the real numbers are the completion of $\mathbf{Q}$ with respect to the usual absolute value (which we now denote $|\cdot|_\infty$), that is, the collection of all limits of sequences in $\mathbf{Q}$ that are Cauchy for $|\cdot|_\infty$. The *p-adic numbers* are the completion of $\mathbf{Q}$ with respect to $|\cdot|_p$. For $|\cdot|_p$, we have an alternative criterion for being Cauchy, based on v_p:

Lemma 1.14. *Let $(a_n)_n$ be a sequence in $\mathbf{Q}$. This sequence is Cauchy for $|\cdot|_p$ if and only if:*

$$\text{for all } N \in \mathbf{Z}, \text{ there exists } M \in \mathbf{Z} \text{ such that for all } m, n \geqslant M, \text{ we have } v_p(a_m - a_n) > N, \tag{1.3}$$

that is, if and only if

$$a_m \equiv a_n \pmod{p^N} \qquad \forall m, n \geqslant M.$$

Proof. Suppose the sequence is Cauchy for $|\cdot|_p$, and let $N \in \mathbf{Z}$. There exists $\epsilon > 0$ such that $\epsilon < p^{-N}$. Since the sequence is Cauchy, there exists $M \in \mathbf{Z}$ such that for all $m, n > M$ we have

$$|a_m - a_n|_p = p^{-v_p(a_m - a_n)} < \epsilon < p^{-N}.$$

But this means $v_p(a_m - a_n) > N$, so (1.3) holds.

Conversely, suppose (1.3) holds. For any $\epsilon > 0$ there exists N such that $p^{-N} < \epsilon$. By assumption there exists M such that for all $m, n \in \mathbf{Z}$ we have $v_p(a_m - a_n) > N$, so

$$|a_m - a_n|_p = p^{-v_p(a_m - a_n)} < p^{-N} < \epsilon,$$

and the sequence is Cauchy. $\square$

Examples. (1) For any prime p, the sequence $1, p, p^2, p^3, p^4, ...$ is Cauchy for $|\cdot|_p$. For any $\epsilon > 0$, choose M such that $p^{-M} < \epsilon$; then for all $m, n \geqslant M$, we have $|p^m - p^n|_p \leqslant p^{-M} < \epsilon$.

The limit of this sequence is 0.

(2) Consider the sequence 3,33,333,3333,..., i.e. the sequence with

$$a_n = \frac{10^n - 1}{3} = 3 \cdot \sum_{j=0}^{n-1} \cdot 10^j.$$

If without loss of generality $m > n$, then

$$a_m - a_n = 3 \cdot \sum_{j=m}^{n-1} 10^j = 10^m \cdot 3 \cdot \sum_{j=0}^{n-m-1} 10^j$$

is divisible by 10^m. Thus

$$v_2(a_m - a_n) = m, \qquad v_5(a_m - a_n) = m,$$

so the sequence is Cauchy for the 2-adic and 5-adic absolute values. We see also that

$$a_n + \tfrac{1}{3} = 10^n,$$

so that

$$\left|a_n + \tfrac{1}{3}\right|_2 = 2^{-n}, \qquad \left|a_n + \tfrac{1}{3}\right|_5 = 5^{-n},$$

and it follows that $a_n \to -1/3$ in both the 2-adic and 3-adic absolute values.

In both the above examples, the limits are rational numbers. But this does not always have to be the case.

Definition 1.15. For a prime p, the *p-adic numbers* are the completion of $\mathbf{Q}$ with respect to $|\cdot|_p$, that is,

$$\mathbf{Q}_p := \{|\cdot|_p\text{-Cauchy sequences in } \mathbf{Q}\}/(\text{equivalence}).$$

We always have $\mathbf{Q} \subset \mathbf{Q}_p$, since if $\alpha \in \mathbf{Q}$, we may take the Cauchy sequence $\alpha, \alpha, \alpha, \ldots$. But this inclusion is *always* strict: we shall see $\mathbf{Q}_p$ is much bigger than $\mathbf{Q}$. For now, we just show this in a special case.

Example 1.16. Let $p = 5$. We construct a $|\cdot|_5$-Cauchy sequence (a_n) of rationals such that for all $n \geqslant 1$, we have

(1) $a_n^2 + 1 \equiv 0 \pmod{5^n}$, and

(2) $a_{n+1} \equiv a_n \pmod{5^n}$.

This sequence is Cauchy by (2), but we'll see this has no limit in $\mathbf{Q}$.

To construct such a sequence, take $a_1 = 2$. For $n \geqslant 1$, suppose a_n is already chosen, and write $a_n^2 + 1 = 5^n c$, for some $c \in \mathbf{Z}$, using (1). For any $b \in \mathbf{Z}$, we have

$$\begin{aligned}(a_n + 5^n b)^2 + 1 &= 1 + a_n^2 + 5^n \cdot 2a_n b + b^2 5^{2n} \\ &\equiv 5^n(c + 2a_n b) \pmod{5^{n+1}}.\end{aligned}$$

Choose b such that $c + 2a_n b \equiv 0 \pmod 5$, which is always possible since $2a_n$ is coprime to 5. Then define

$$a_{n+1} = a_n + 5^n b.$$

By construction, this satisfies (1) and (2).

The sequence is Cauchy for $|\cdot|_5$ by (2). Thus by definition it tends to a limit $C \in \mathbf{Q}_5$. Then

$$\begin{aligned}|C^2 + 1|_5 &= |C^2 - a_n^2 + a_n^2 + 1|_5 \\ &\leqslant |C^2 - a_n^2|_5 + |a_n^2 + 1|_5 \\ &\longrightarrow 0 \quad \text{as } n \to \infty.\end{aligned}$$

Thus $C^2 = -1$, which implies $C \notin \mathbf{Q}$.

From this example, we also see that the 5-adic numbers $\mathbf{Q}_5$ and real numbers $\mathbf{R}$ are genuinely different: the equation $X^2 + 1 = 0$ is solvable in $\mathbf{Q}_5$, but not in $\mathbf{R}$.

By Lemma 1.7, there are unique continuous extensions of $+$, $\times$ and $|\cdot|_p$ to $\mathbf{Q}_p$. We also have:

Lemma 1.17. *There is a unique continuous extension of v_p from $\mathbf{Q}^\times$ to $\mathbf{Q}_p^\times$.*

Proof. Note that $|\mathbf{Q}|_p = \{p^r : r \in \mathbf{Z}\} \cup \{0\}$. Also $\{p^r : r \in \mathbf{Z}\}$ is *discrete* in $\mathbf{R}$: for every $r \in \mathbf{Z}$, there exists an open ball $U_m \subset \mathbf{R}$ around p^r for which $U_m \cap |\mathbf{Q}_p^\times|_p = \{p^r\}$. In particular, any convergent sequence in $|\mathbf{Q}|_p$ either:

- must converge to 0,
- or is eventually constant.

Let (a_n) be a $|\cdot|_p$-Cauchy sequence in $\mathbf{Q}$, tending to $\alpha \in \mathbf{Q}_p$, and assume $\alpha \neq 0$. By convergence, for all sufficiently large n, we must have $a_n \neq 0$. Since $|\cdot|_p$ is continuous, we know that $|\alpha_n|_p$ must converge in $|\mathbf{Q}^\times|_p \subset \mathbf{R}$. By above, $|\alpha_n|_p$ must be eventually constant. In particular, there exists some $M, r \in \mathbf{Z}$ such that $|\alpha_m|_p = |\alpha_n|_p = p^r$ for all $m, n > M$, and then $|\alpha|_p = \lim_{n\to\infty}|\alpha_n|_p = p^r$. Then we take $v_p(\alpha) = -r$. By construction, this must be the unique extension of v_p to $\mathbf{Q}_p^\times$. $\square$

Notation. The following is convenient (if possibly misleading) notation. If $\alpha, \beta \in \mathbf{Q}_p$, and $n \in \mathbf{Z}$, then we often write

$$\alpha \equiv \beta \pmod{p^n}$$

if $v_p(\alpha - \beta) \geqslant n$.

If $\alpha, \beta \in \mathbf{Z}$, then this recovers exactly the notion of reduction modulo p^n that you are already familiar with. (Note that two integers are congruent modulo p^n if and only if their difference is divisible by p^n!). Another way of thinking about this is that $\alpha \equiv \beta \pmod{p^n}$ if and only if α and β represent the same class in the (finite) quotient group $\mathbf{Z}/p^n\mathbf{Z}$.

This generalises directly to elements of $\mathbf{Z}_p$: we have $v_p(\alpha - \beta) \geqslant n$ if and only if $\alpha - \beta \in p^n\mathbf{Z}_p$ if and only if α and β represent the same class in the quotient group $\mathbf{Z}_p/p^n\mathbf{Z}_p$.

For more general elements $\alpha, \beta \in \mathbf{Q}_p$, and $n \in \mathbf{Z}$, note that we still have $v_p(\alpha - \beta) \geqslant n$ if and only if $\alpha - \beta \in p^n\mathbf{Z}_p$, that is, there exists $\gamma \in \mathbf{Z}_p$ with $\alpha - \beta = p^n\gamma$. When we write $\alpha \equiv \beta \pmod{p^n}$, it really means that $\alpha - \beta \in p^n\mathbf{Z}_p$; that is, α and β represent the same class in the (now infinite!) quotient $\mathbf{Z}_p$-module $\mathbf{Q}_p/p^n\mathbf{Z}_p$.

It is worth clarifying: if $\alpha \in \mathbf{Z}_p$, then α is congruent modulo p to a unique element of the finite set $\{0, ..., p-1\}$. For $\alpha \in \mathbf{Q}_p \backslash \mathbf{Z}_p$, this is not true! But the *relative* notion of comparing elements in $\mathbf{Q}_p/p\mathbf{Z}_p$ still makes sense.

1.4. The p-adic integers

In $\mathbf{R}$, every Cauchy sequence of integers is eventually constant. But in $\mathbf{Q}_p$, there are many interesting Cauchy sequences of integers! It's hard to overstate how important this is in number theory. It lets us do (p-adic) analysis on the integers.

This also means it's interesting to consider the completion of $\mathbf{Z}$ in the p-adic metric. The resulting completion, the p-adic integers, has a simple definition as a subring of $\mathbf{Q}_p$, as follows:

Definition 1.18. Define the *p-adic integers* $\mathbf{Z}_p$ to be

$$\mathbf{Z}_p = \{\alpha \in \mathbf{Q}_p : |\alpha|_p \leqslant 1\} = \{\alpha \in \mathbf{Q}_p : v_p(\alpha) \geqslant 0\}.$$

Recall that in a number field K, we have the ring of integers $\mathcal{O}_K$, that is, the subring of elements that are roots of monic irreducible integer polynomials over $\mathbf{Q}$; and the ring of integers in $\mathbf{Q}$ is just $\mathbf{Z}$. The ring $\mathbf{Z}_p$ plays a similar role inside $\mathbf{Q}_p$.

Proposition 1.19. *(1) We have $\mathbf{Z} \subset \mathbf{Z}_p$.*

(2) When $\mathbf{Z}_p$ is equipped with the subspace topology from $\mathbf{Q}_p$, the integers $\mathbf{Z}$ are dense in $\mathbf{Z}_p$ (that is, for every $\alpha \in \mathbf{Z}_p$, there exists a $|\cdot|_p$-Cauchy sequence (α_n), with $\alpha_n \in \mathbf{Z}$, such that $\alpha_n \to \alpha$).

Proof. (1) If $\alpha \in \mathbf{Z}$, then $v_p(\alpha) \geqslant 0$ by Lemma 1.9(ii). Thus $|\alpha|_p \leqslant 1$ and $\alpha \in \mathbf{Z}_p$.

(2) Note $\mathbf{Q}$ is dense in $\mathbf{Q}_p$ by construction. Also $\mathbf{Z}_p = B_{1.1}(0)$ is an open set in $\mathbf{Q}_p$. Thus $\mathbf{Q} \cap \mathbf{Z}_p$ is dense in $\mathbf{Z}_p$, since we take the subspace topology. Observe that

$$\begin{aligned} \mathbf{Q} \cap \mathbf{Z}_p &= \{\alpha \in \mathbf{Q} : |\alpha|_p \leqslant 1\} \\ &= \{\alpha \in \mathbf{Q} : v_p(\alpha) \geqslant 0\} = \left\{\tfrac{a}{b} \in \mathbf{Q} : p \nmid b\right\} = \mathbf{Z}_{(p)}, \end{aligned}$$

the localisation of $\mathbf{Z}$ at the prime ideal (p).

Let $a/b \in \mathbf{Z}_{(p)}$. Since b is prime to p, for each $n \geqslant 1$ we may choose $y_n \in \mathbf{Z}$ such that $by_n \equiv 1 \pmod{p^n}$. Hence (y_n) is a $|\cdot|_p$-Cauchy sequence of integers tending to $1/b$, and ay_n tends to a/b. In particular, $\mathbf{Z}$ is dense in $\mathbf{Z}_{(p)}$, which is dense in $\mathbf{Z}_p$; so $\mathbf{Z}$ is dense in $\mathbf{Z}_p$. ☐

Exercise. Determine all primes p such that $37/84 \in \mathbf{Z}_p$. Find a sequence of integers tending to $37/84$ in the 5-adic metric.

Lemma. *The open balls B in $\mathbf{Q}_p$ in the p-adic topology are all of the form*

$$B = a + p^r\mathbf{Z}_p, \qquad a \in \mathbf{Q}_p, r \in \mathbf{Z}.$$

Proof. An open ball in the p-adic topology has the form $B(a,\epsilon)$, where $a \in \mathbf{Q}_p$ and $\epsilon > 0$. It is convenient to take $\epsilon = p^{1-r}$, with $r \in \mathbf{Z}$. Then compute:

$$\begin{aligned} B(a,\epsilon) = B(a,p^{1-r}) &= \{x \in \mathbf{Q}_p : |x-a|_p < p^{1-r}\} \\ &= \{x \in \mathbf{Q}_p : v_p(x-a) > r-1\} \\ &= \{x \in \mathbf{Q}_p : v_p(x-a) \geqslant r\} \qquad \text{(since } v_p \text{ takes values in } \mathbf{Z}\text{)} \\ &= \{x \in \mathbf{Q}_p : x - a \in p^r\mathbf{Z}_p\} \qquad \text{(since } v_p(\alpha) \geqslant r \iff \alpha \in p^r\mathbf{Z}_p\text{)} \\ &= \{x \in \mathbf{Q}_p : x \in a + p^r\mathbf{Z}_p\} = a + p^r\mathbf{Z}_p. \end{aligned}$$

So the open balls in $\mathbf{Z}_p$ are exactly the sets $a + p^r\mathbf{Z}_p$, for $a \in \mathbf{Z}_p$, $r \geqslant 0$. ☐

1.5. p-adic expansions

In practice, we never really use the definition of $\mathbf{R}$ as the set of limits of Cauchy sequences: we have more concrete ways of describing real numbers, for example via decimal expansions:

Example. The real number $\pi = 3.1415\cdots$ can be written in the form

$$\begin{aligned} \pi &= 3 \cdot 10^0 + 1 \cdot 10^{-1} + 4 \cdot 10^{-2} + 1 \cdot 10^{-3} + 5 \cdot 10^{-4} + \cdots \\ &= \sum_{r=0}^{\infty} a_n 10^{-n}, \end{aligned}$$

where $a_n \in \{0, \ldots, 9\}$.

In fact, the elements of $\mathbf{Q}_p$ have similar, p-adic, expansions.

Definition 1.20. Let $r \in \mathbf{Z}$, and let $a_r, a_{r+1}, a_{r+2}, ...$ be a sequence of elements of $\{0, \dots, p-1\}$. Define a sequence

$$\beta_N = a_r p^r + a_{r+1} p^{r+1} + \cdots + a_{r+N} p^{r+N}.$$

Note that for all $m, n \geqslant N$, we have $\beta_m \equiv \beta_n \pmod{p^{N+1}}$, that is, (β_N) is Cauchy for $|\cdot|_p$. It thus converges to a limit in $\mathbf{Q}_p$; we write this limit as

$$\beta = a_r p^r + a_{r+1} p^{r+1} + a_{r+2} p^{r+2} + \cdots = \sum_{n=r}^{\infty} a_n p^n \in \mathbf{Q}_p.$$

We call such an expression a *p-adic expansion of β*.

Note r is allowed to be negative in the above construction.

The main result of this section is Proposition 1.22, which says that *every* element of $\mathbf{Q}_p$ can be written in this form.

In the proof, we'll need a lemma:

Lemma 1.21. *Let $0 \neq \alpha \in \mathbf{Z}$. Then for every $M \in \mathbf{Z}_{\geqslant 0}$, there are unique integers $a_0, \dots, a_M \in \{0, \dots, p-1\}$ such that*

$$\alpha \equiv a_0 + a_1 p + \cdots + a_n p^n \pmod{p^{M+1}}.$$

Proof. We induct on M. For $M = 0$, this is the statement that every integer is congruent to a unique element of $\{0, \dots, p-1\}$ modulo p.

Suppose the statement is true for $M-1$. Fix a_0 such that $\alpha \equiv a_0 \pmod{p}$. Then $(\alpha - a_0)/p$ is an integer, and by the induction step we may write

$$\frac{\alpha - a_0}{p} = a_1 + a_2 p + \cdots + a_M p^{M-1} \pmod{p^M},$$

for some integers $a_1, \dots, a_{M-1} \in \{0, \dots, p-1\}$. But then, multiplying by p and adding a_0, we find

$$\alpha = a_0 + a_1 p + a_2 p^2 + \cdots + a_M p^M \pmod{p^{M+1}}.$$

This shows existence.

For uniqueness, suppose

$$\alpha = a_0 + a_1 p + \cdots + a_M p^M = a_0' + a_1' + \cdots + a_M' p^M \pmod{p^{M+1}}$$

are different expansions. Let i be the minimal integer such that $a_i \neq a_i'$. Then

$$0 = \alpha - \alpha \equiv (a_i - a_i') p^i \pmod{p^{i+1}},$$

that is, $a_i - a_i' \equiv 0 \pmod{p}$. But since both a_i and a_i' are in $\{0, \dots, p-1\}$, this forces $a_i = a_i'$. $\square$

Proposition 1.22. *Let $0 \neq \alpha \in \mathbf{Q}_p$. Then α has a unique p-adic expansion; that is, α can be written uniquely in the form*

$$\alpha = a_r p^r + a_{r+1} p^{r+1} + a_{r+2} p^{r+2} + \cdots = \sum_{n=r}^{\infty} a_n p^n,$$

with $a_n \in \{0, \dots, p-1\}$ and $a_r \neq 0$.

Proof. Without loss of generality, we may scale α by a (unique) power of p and assume $v_p(\alpha) = 0$, that is, $\alpha \in \mathbf{Z}_p$.

Write $\alpha = \lim_{n\to\infty}\alpha_n$, with (α_n) a $|\cdot|_p$-Cauchy sequence in $\mathbf{Q}$. By Proposition 1.19(2), we may assume $\alpha_n \in \mathbf{Z}$ for each n.

First, we construct a better Cauchy sequence tending to α. For each $N \in \mathbf{Z}$, there exists $M_N \in \mathbf{Z}$ such that $\alpha_m \equiv \alpha_n \pmod{p^{N+1}}$ for all $m, n > M_N$. But then by Lemma 1.21 α_m and α_n have the same expansion

$$\alpha_m \equiv \alpha_n \equiv a_0 + a_1 p + \cdots + a_N p^N \pmod{p^{N+1}},$$

and this is thus true of *all* $n > M_N$. Define $\beta_N = a_0 + a_1 p + \cdots + a_N p^N$. This gives a new sequence $\beta_1, \beta_2, \beta_3, \ldots$ in $\mathbf{Z}$. By construction, we have

$$\beta_N \equiv \beta_m \equiv \beta_n \pmod{p^{N+1}}$$

for all $m, n \geqslant N$, so (β_N) is Cauchy; and by construction

$$\beta_N \to \sum_{n=0}^{\infty} a_n p^n \qquad \text{as } N \to \infty.$$

Note that by construction, letting $R = \max(N, M_N)$ we have

$$\beta_N = \beta_m = \alpha_n \pmod{p^{N+1}}$$

for all $m, n > R$; hence (β_N) is equivalent to (α_n). Thus $\beta_N \to \alpha$. Since limits of Cauchy sequences are unique, we finally deduce

$$\alpha = \sum_{n=0}^{\infty} a_n p^n,$$

which shows *existence* of p-adic expansions.

For uniqueness, we argue as in Lemma 1.21. Suppose

$$\alpha = \sum_{n=r}^{\infty} a_n p^n = \sum_{n=r'}^{\infty} a'_n p^n.$$

If the expansion is not unique, then there exists a minimal i such that $a_i \neq a'_i$. Then

$$0 = (a_i - a'_i)p^i \pmod{p^{i+1}},$$

so $a_i \equiv a'_i \pmod{p}$, a contradiction. Thus the expansion is unique. □

Lemma 1.23. *Let $\alpha = \sum_{n=r}^{\infty} a_n p^n$, with $a_r \neq 0$. Then*

$$r = v_p(\alpha).$$

Proof. For $N \geqslant r$, write $\alpha_N = a_r p^r + \cdots + a_N p^N$. Then $\alpha_N \to \alpha$. We conclude since $v_p(\alpha_N) = r$ for all N. □

Corollary 1.24. *Let $\alpha = \sum_{n=r}^{\infty} a_n p^n \in \mathbf{Q}_p$. Then $\alpha \in \mathbf{Z}_p$ if and only if $a_n = 0$ for all $n < 0$, i.e. there are no negative powers of p in its p-adic expansion.*

Proof. We have $\alpha \in \mathbf{Z}_p$ if and only if $v_p(\alpha) \geqslant 0$ if and only if $a_n = 0$ for all $n < r$. □

Examples: how to compute p-adic expansions. In this example, we explain two methods for computing p-adic expansions: a brute force way, and a slick way. We apply to the examples:

(1) the 5-adic expansion of $1/2$;

(2) the 7-adic expansion of $3/4$.

Brute force method:
We compute the 5-adic expansion of $1/2$. Write

$$1/2 = \sum_{n \geqslant r} a_n 5^n. \tag{1.4}$$

We compute:

- Note $v_5(\alpha) = 0$, so $a_n = 0$ for $n < 0$.
- We compute a_0 by reducing both sides of (1.4) modulo 5. Doing this, we find

$$\tfrac{1}{2} = a_0 \pmod 5.$$

  To compute $1/2 \pmod 5$, note that $2 \times 3 = 6 \equiv 1 \pmod 5$, so $3 \equiv 1/2 \pmod 5$; so $a_0 = 3$.
- We now reduce (1.4) modulo 25. This gives

$$\tfrac{1}{2} \equiv a_0 + a_1 5 = 3 + a_1 5 \pmod{25}.$$

  Thus

$$\begin{aligned} \tfrac{1}{2} - 3 \equiv a_1 5 \pmod{25} \qquad &\Rightarrow \qquad -\tfrac{5}{2} \equiv 5a_1 \pmod{25} \\ &\Rightarrow \qquad -\tfrac{1}{2} \equiv a_1 \pmod 5. \end{aligned}$$

  Now $2 \times 2 \equiv 4 \equiv -1 \pmod 5$, so $2 \equiv -1/2 \pmod 5$, and we deduce $a_1 = 2$.
- Similarly

$$\begin{aligned} a_2 \equiv \frac{\alpha - a_0 - a_1 5}{25} = \frac{\frac{1}{2} - 3 - 2 \cdot 5}{25} & \\ & \equiv \frac{-25/2}{25} = -\frac{1}{2} \equiv 2 \pmod 5. \end{aligned}$$

- Continuing in this manner, we may inductively compute

$$a_r = \frac{\alpha - (a_0 + a_1 5 + \cdots + a_{r-1} 5^{r-1})}{5^{r-1}} \pmod 5.$$

If you continue in this fashion, you find $a_3 = a_4 = a_5 = \cdots = 2$, i.e. the coefficients are 3,2,2,2,2,2....

We apply this method to the 7-adic expansion of $\beta = 3/4 = \sum_{n \geqslant 0} b_n 7^n$.

- Note $1/4 \equiv 2 \pmod 7$ (since $4 \times 2 \equiv 1 \pmod 7$). So $3/4 \equiv 6 \pmod 7$, and $b_0 = 6$.
- We compute

$$b_1 \equiv \frac{\beta - b_0}{7} = \frac{3/4 - 6}{7} = -\frac{21/4}{7} = -3/4 \equiv 1 \pmod 7.$$

  So $b_1 = 1$.

- We compute
$$b_2 \equiv \frac{\beta - b_0 - b_1 7}{7^2} = \frac{3/4 - 6 - 1\cdot 7}{7^2} = -\frac{49/4}{49} = -1/4 \equiv 5 \pmod 7,$$
using that $4 \times 5 = 20 \equiv -1 \pmod 7$. We deduce $b_2 = 5$.

If we continued computing, we would get that the coefficients are 6,1,5,1,5,1,5,....

Slick way: We can see all of this in a more slick way. Recall the notion of geometric series: expanding formally, we have

$$\begin{aligned}(1-X)(1+X+X^2+\cdots) &= (1+X+X^2+\cdots) - (X+X^2+X^3+\cdots)\\ &= 1,\end{aligned}$$

so (where things converge!) we have

$$1 + X + X^2 + \cdots = \frac{1}{1-X}.$$

Putting, for example, X to be a power of p, then the series converges, and we deduce that $1/(1-p)$ has the p-adic expansion
$$\frac{1}{1-p} = 1 + p + p^2 + p^3 + \cdots .$$
In our first (5-adic) example, we have
$$\begin{aligned}\alpha = \frac{1}{2} = 1 - \frac{1}{2} = 1 - \frac{2}{4} &= 1 + 2\cdot\frac{1}{1-5}\\ &= 1 + 2(1 + 5 + 5^2 + 5^3 + \cdots)\\ &= 3 + 2\cdot 5 + 2\cdot 5^2 + 2\cdot 5^3 + \cdots .\end{aligned}$$

Recall our other example: the 7-adic expansion $\alpha = 3/4$. We want to expand in terms of powers of 7, so we want to exploit a geometric series of the form

$$\frac{1}{1-7^r} = 1 + 7^r + 7^{2r} + \cdots ,$$

for some integer r. Now observe that $1 - 7^2 = -48 = -12 * 4$, so we have

$$\begin{aligned}3/4 = 1 + \frac{1}{-4} = 1 + \frac{12}{-48} &= 1 + 12\cdot\frac{1}{1-7^2}\\ &= 1 + 12(1 + 7^2 + 7^4 + 7^6 + 7^8 + \cdots)\\ &= 1 + (5 + 1\cdot 7)(1 + 7^2 + 7^4 + \cdots)\\ &= 6 + 1\cdot 7 + 5\cdot 7^2 + 1\cdot 7^3 + 5\cdot 7^4 + 1\cdot 7^5 + \cdots .\end{aligned}$$

That is, we recover the brute force result from above.

for some $r \geqslant 0$, $a \in \mathbf{Z}$, and $b \in \{0, ..., p^r - 1\}$; then

We can use this trick to compute the p-adic expansion of *any* rational prime to p!

Lemma. *Let $m/n \in \mathbf{Q}$ with $p \nmid n$. Then we may write m/n in the form*

$$m/n = a + b\cdot\frac{1}{1-p^r},$$

where:

- $r \geqslant 0$,
- $b \in \{0, ..., p^{r-1} - 1\}$,
- $a \in \mathbf{Z}$.

Proof. Given such m/n, let r be such that $p^r \equiv 1 \pmod n$ (which is possible, since n is prime to p). Then $p^r - 1 = An$, for some A. We write

$$\frac{m}{n} = \frac{Am}{An} = \frac{Am}{p^r - 1} = \frac{-Am}{1 - p^r}.$$

Now $-Am$ is congruent modulo $p^r - 1$ to a unique integer $b \in \{0, ..., p^{r-1} - 1\}$, whence $-Am = -a(p^r - 1) + b$ for some integer $a \in \mathbf{Z}$. But now

$$\frac{m}{n} = \frac{-Am}{1 - p^r} = \frac{-a(p^r - 1) + b}{1 - p^r} = a + b \cdot \frac{1}{1 - p^r},$$

as required. □

Now the p-adic expansion of $a + b \cdot \frac{1}{1-p^r}$ is the sums of the p-adic expansion of a (which should be easy to compute) and of

$$b \cdot \frac{1}{1 - p^r} = b + bp^r + bp^{2r} + \cdots,$$

and writing $b = b_0 + b_1 p + \cdots + b_{r-1}p^{r-1}$, we get the (periodic) p-adic expansion with coefficients

$$b_0, b_1, \ldots, b_{r-1}, b_0, b_1, \ldots, b_{r-1}, b_1, \ldots.$$

This only leaves the rationals m/n where p divides n. Suppose we are given $M/N \in \mathbf{Q}$. We can factor out the unique factor of p and write $M/N = p^r \frac{m}{n}$, with $p \nmid m, n$. If the p-adic expansion of m/n is

$$m/n = a_0 + a_1 p + a_2 p^2 + \cdots,$$

then the p-adic expansion of M/N is

$$M/N = a_0 p^r + a_1 p^{r+1} + \cdots,$$

that is,

$$M/N = b_r p^r + b_{r+1} p^{r+1} + \cdots, \qquad b_i = a_{i-r}.$$

Exercise. What is the p-adic expansion of -1?

Exercise. Prove that $\mathbf{Q}_p$ is uncountable. Deduce that $\mathbf{Q} \neq \mathbf{Q}_p$ for all p.

Remark. Let $\alpha, \beta \in \mathbf{Q}_p$, and write

$$\alpha = \sum_{n=r}^{\infty} a_n p^n, \qquad \beta = \sum_{n=s}^{\infty} b_n p^n.$$

Note that for $N \in \mathbf{Z}$, we have $\alpha \equiv \beta \pmod{p^N}$ if and only if $a_n = b_n$ for all $n < N$.

Chapter 2

Absolute Values

We already defined absolute values on a general field K in Definition 1.2. In this chapter, we explore this definition, classify the absolute values on the field $\mathbf{Q}$, and study extensions of absolute values in field extensions.

Recall: a *valued field* is a field together with an absolute value, that is, a multiplicative function $|\cdot| : K \to \mathbf{R}_{\geqslant 0}$ that vanishes only at 0 and satisfies the triangle inequality $|x+y| \leqslant |x| + |y|$. An absolute value is *non-archimedean* if it additionally satisfies the ultrametric law (or 'strong triangle inequality') $|x+y| \leqslant \max(|x|,|y|)$, and *archimedean* if it does not satisfy this.

Recall also that if $|\cdot|$ is an absolute value on K, then we get an associated metric $d(x,y) = |x-y|$ on K, and this turns K into a topological space.

2.1. Non-archimedean absolute values

We now describe some basic properties of non-archimedean absolute values, and some odd features that arise in their topologies. For example, striking examples are that series converge if and only if their terms tend to 0, and any element of an open ball is the centre.

Lemma 2.1. *Let $(K, |\cdot|)$ be non-archimedean. Then:*

(i) If $|x| < |y|$, then $|x \pm y| = |y|$.

(ii) $|x_1 + \cdots + x_n| \leqslant \max |x_i|$, with equality if (without loss of generality) $|x_1| > |x_i|$ for $i \geqslant 2$.

(iii) If $(K, |\cdot|)$ is complete, then any series $\sum_{i=1}^{\infty} a_i$ converges if and only if $a_i \to 0$.

Proof. (i) We know
$$|x+y| \leqslant \max(|x|,|y|) = |y|.$$
But also
$$|y| = |x+y-x| \leqslant \max(|x+y|,|x|) = |x+y|,$$
since $|y| > |x|$. Combining gives the result.

(ii) Proof by induction. For the equality, apply (i) with $y = x_1, x = x_2 + \cdots + x_n$.

(iii) Let $S_n = \sum_{i=1}^{n} a_i$. Then S_n converges as $n \to \infty$ implies $S_{n+1} - S_n = a_{n+1} \to 0$. To see 'if', note that for $n \geqslant m$, we have $|S_n - S_m| = |a_{m+1} + \cdots a_n| \leqslant \max |a_i| \to 0$ as $m \to \infty$, so (S_n) is Cauchy, and thus converges as K is complete. $\square$

Recall that if K is a valued field, $x \in K$, and $r \geqslant 0$, then we define the 'open' ball
$$B(x,r) = \{y \in K : |x-y| < r\},$$

and similarly we have the 'closed' ball

$$\overline{B}(x,r) = \{y \in K : |x-y| \leqslant r\}.$$

In a non-archimedean field, 'every point of a ball is the centre', 'open' balls are also closed, and 'closed' balls are also open.

Lemma 2.2. *Let $(K, |\cdot|)$ be non-archimedean. Then:*

(i) If $y \in B(x,r)$, then $B(x,r) = B(y,r)$.

(ii) If $y \notin B(x,r)$, then $B(x,r) \cap B(y,r) = \varnothing$.

(iii) If $y \in \overline{B}(x,r)$, then $\overline{B}(x,r) = \overline{B}(y,r)$.

(iv) If $y \notin \overline{B}(x,r)$, then $\overline{B}(x,r) \cap \overline{B}(y,r) = \varnothing$.

(v) $B(x,r)$ is both open and closed.

(vi) $\overline{B}(x,r)$ is both open and closed.

Proof. (i) If $z \in B(x,r)$, then $|y-z| = |(y-x)+(x-z)| \leqslant \max(|x-y|, |x-z| < r$, so $B(x,r) \subset B(y,r)$. But switching x, y gives $B(y,r) \subset B(x,r)$.

(ii) If $z \in B(x,r) \cap B(y,r)$, then $B(x,r) = B(z,r) = B(y,r)$ by (i), contradiction.

(iii, iv) Identical to (i) and (ii).

(v) $B(x,r)$ is open by definition of the topology. Now if $y \notin B(x,r)$, then $B(x,r) \cap B(y,r) = \varnothing$, so

$$K \backslash B(x,r) = \bigcup_{y \notin B(x,r)} B(y,r)$$

is open, and $B(x,r)$ is hence closed.

(vi) $\overline{B}(x,r) = |\cdot|^{-1}([0,r])$, and hence is closed because $|\cdot|$ is continuous. If $y \in \overline{B}(x,r)$, then $B(y,r/2) \subset \overline{B}(y,r) = \overline{B}(x,r)$, so it is open. $\square$

Recall that a topological space X is *totally disconnected* if for all $x, y \in X$ with $x \neq y$, there exist open sets $A, B \subset X$ with $x \in A$, $y \in B$ and $X = A \sqcup B$ a disjoint union (i.e. $X = A \cup B$, $A \cap B = \varnothing$). (This is like being 'maximally not connected', or like being Hausdorff on steroids).

Corollary 2.3. *If $(K, |\cdot|)$ is non-archimedean, then K is totally disconnected.*

Proof. Suppose $x \neq y$, and let $r > 0$ with $2r < |x-y|$. Then $K = B(x,r) \cup [K \backslash B(x,r)]$, both are open, and $y \in K \backslash B(x,r)$. $\square$

Remark. Note the huge difference between this and the archimedean case: $\mathbf{R}$ is connected, but $\mathbf{Q}_p$ is as far from connected as possible!

Recall that for any field K, there is a canonical map $\varphi : \mathbf{Z} \to K$ sending 1 to 1. We usually drop the φ and just write n for its image in K.

Lemma 2.4. *An absolute value $|\cdot|$ is non-archimedean if and only if $|n|$ is bounded for $n \in \mathbf{Z}$.*

Proof. Suppose $|\cdot|$ is non-archimedean. As $|n| = |-n|$, we may assume $n \geqslant 1$. Then

$$|n| = |1 + \cdots + 1| \leqslant \max(|1|, |1|, \ldots, |1|) = 1.$$

Conversely, suppose $|n|$ is bounded for $n \in \mathbf{Z}$, that is, there exists $B \in \mathbf{R}$ such that $|n| \leqslant B$ for all n. Let $x, y \in K$. Then

$$\begin{aligned} |x+y|^m = \left|\sum_{r=0}^m \binom{m}{r} x^r y^{m-r}\right| &\leqslant \sum_{r=0}^m \left|\binom{m}{r} x^r y^{m-r}\right| \\ &\leqslant B \sum_{r=0}^m \left|x^r y^{m-r}\right| \\ &\leqslant B(m+1)\max(|x|^m, |y|^m) = B(m+1)\max(|x|,|y|)^m. \end{aligned}$$

In particular, we see that for all m,

$$|x+y| \leqslant [B(m+1)]^{1/m} \cdot \max(|x|,|y|).$$

But $[B(m+1)]^{1/m} \to 1$ as $m \to \infty$, and hence $|x+y| \leqslant \max(|x|,|y|)$. $\square$

Corollary 2.5. *If* $\mathrm{char}(K) > 0$*, then* every *absolute value is non-archimedean.*

Proof. If $\mathrm{char}(K) = p$, then $\mathbf{Z} \to \mathbf{R}, n \mapsto |n|$ is periodic with period (at most) p, hence bounded. $\square$

2.2. Equivalence of absolute values

The following encodes when two absolute values are 'the same'.

Lemma 2.6. *Let* $|\cdot|_1, |\cdot|_2$ *be non-trivial absolute values on a field* K*. Then the following are equivalent:*

(1) $|\cdot|_1$ *and* $|\cdot|_2$ *induce the same topology on* K*;*

(2) if $x \in K$*, then* $|x|_1 < 1$ *if and only if* $|x|_2 < 1$*;*

(3) There exists $c > 0$ *such that* $|x|_1 = |x|_2^c$*.*

Proof. (1)⇒(2): Note that for any absolute value $|\cdot|$, we have $|x| < 1$ if and only if $x^n \to 0$ as $n \to \infty$ in the topology induced by $|\cdot|$. Now if $|\cdot|_1$ and $|\cdot|_2$ induce the same topology, then

$$\begin{aligned} |x|_1 < 1 &\iff x^n \to 0 \text{ with respect to } |\cdot|_1 \\ &\iff x^n \to 0 \text{ with respect to } |\cdot|_2 \\ &\iff |x|_2 < 1. \end{aligned}$$

(2)⇒(3): Let $a \in K^\times$ with $|a|_1 < 1$. Let $x \in K^\times$ and $m, n \in \mathbf{Z}$ with $n > 0$. Note that for all rationals m/n, we have

$$\begin{aligned} m/n < \frac{\log|x|_1}{\log|a|_1} &\iff m\log|a|_1 < n\log|x_1| \\ &\iff \log|a^m|_1 - \log|x^n|_1 < 0 \\ &\iff \log\left|\frac{a^m}{x^n}\right|_1 < 1 \\ &\iff \log\left|\frac{a^m}{x^n}\right|_2 < 1 \\ &\iff m/n < \frac{\log|x|_2}{\log|a|_2}. \end{aligned} \tag{2.1}$$

Thus we see that
$$\frac{\log|x|_1}{\log|a|_1} = \frac{\log|x|_2}{\log|a|_2}$$
for all x; for if this was not true, then there would be a rational number between them, and this cannot happen by (2.1). In particular, we have $\log|x|_1 = c\log|x|_2$ for $c = \log|a|_1/\log|a|_2$, and taking exponentials gives the result.

(3)⇒(1): Given (3), the basis of open balls in the respective topologies is the same. □

Definition 2.7. We say that $|\cdot|_1$ and $|\cdot|_2$ are *equivalent*, denoted $|\cdot|_1 \sim |\cdot|_2$, if the three equivalent conditions of Lemma 2.6 are satisfied. (This is easily seen to give an equivalence relation on to set of absolute values on K).

Definition 2.8. An equivalence class of absolute values on K is called a *place*. We typically denote a place by v (or w).

Exercise. Prove that if p and q are distinct primes, then $|\cdot|_p$ and $|\cdot|_q$ are not equivalent absolute values on $\mathbf{Q}$. Prove that $|\cdot|_p$ and $|\cdot|_\infty$ are not equivalent.

If absolute values are not equivalent, then they are independent in a very strong sense. The following is an analogue of the Chinese remainder theorem for general valued fields.

Theorem 2.9 (Weak approximation)**.** *Let $|\cdot|_1, \dots, |\cdot|_r$ be pairwise inequivalent non-trivial absolute values on a field K. Let $\beta_1, \dots, \beta_r \in K$. Then for all $\epsilon < 0$, there exists an element $\alpha \in K$ such that*
$$|\alpha - \beta_i|_i < \epsilon \qquad \text{for all } i = 1, \dots, r.$$

In the proof, we use the following lemma:

Lemma 2.10. *If $|\cdot|_1, \dots, |\cdot|_r$ are pairwise inequivalent non-trivial absolute values, there exists an element $c \in K$ such that*
$$|c|_1 > 1, \qquad |c|_i < 1 \text{ for } i = 2, \dots, r.$$

Proof. We argue by induction on r. First consider $r = 2$. Without loss of generality, by Lemma 2.6 there exists $d \in K$ with $|d|_1 \geqslant 1$ and $|d|_2 < 1$. Since $|\cdot|_1$ is non-trivial, there exists some $z \in K$ with $|z|_1 > 1$. Then we take
$$c = d^m z,$$
which for sufficiently large m satisfies $|c|_1 > 1$ and $|c_2| < 1$.

Now consider general $r \geqslant 3$. By the induction step, there exists d such that $|d|_1 > 1$, and $|d|_i < 1$ for $i = 2, \dots, r$; and there also exists $z \in K$ such that $|z|_1 > 1$ and $|z|_r < 1$. Then we set
$$c = \begin{cases} d & : |d|_r < 1 \\ d^m z & : |d|_r = 1 \\ \frac{d^m}{1+d^m} z & : |d|_r > 1, \end{cases}$$
for some sufficiently large m. □

Proof of Theorem 2.9. For $i = 1, \dots, r$, let $c_i \in K$ such that $|c_i|_i > 1$, and $|c_j|_i < 1$ for $j \neq i$. Then let
$$\alpha_n = \sum_{i=1}^{r} \frac{c_i^n \beta_i}{c_i^n + 1}.$$

Note that

$$\begin{aligned}|\alpha_n - \beta_i|_i &= \left|\frac{-\beta_i}{c_i^n+1} + \sum_{j\neq i}\frac{c_j^n\beta_i}{c_j^n+1}\right|_i \\ &\leqslant |\beta_i|\left(\left|\frac{1}{c_i^n+1}\right|_i + \sum_{i\neq j}\left|\frac{c_j^n}{c_j^n+1}\right|_i\right).\end{aligned}$$

By definition of the c_j, each term tends to 0 as $n \to \infty$; so for any $\epsilon > 0$, there always exists n such that α_n fits the conditions of the theorem. □

2.3. Classification of absolute values on Q

In this section, we prove Ostrowski's theorem, which states that the usual real absolute value, and the p-adic absolute values for p prime, are the 'only' absolute values on $\mathbf{Q}$ (up to equivalence).

Theorem 2.11 (Ostrowski)**.** *Any non-trivial absolute value $|\cdot|$ on $\mathbf{Q}$ is equivalent to either $|\cdot|_\infty$ or $|\cdot|_p$ for a prime p.*

Proof. First suppose $|\cdot|$ is archimedean. We are trying to prove that:

$|\cdot| \sim |\cdot|_\infty$, *i.e. there exists $c > 0$ with $|a| = |a|_\infty^c$ for all $a \in \mathbf{R}$.*

If $x \in \mathbf{Q}^\times$ write $x = \pm a/b$, with $a, b \in \mathbf{Z}_{\geqslant 1}$, so $|x| = |a|/|b|$. Also $|1| = 1$. We deduce:

$|\cdot| \sim |\cdot|_\infty \iff$ *there exists $c > 0$ with $|a| = |a|_\infty^c = a^c$ for all $a \in \mathbf{Z}_{>1}$.*

After taking logs, an equivalent way of writing this is

$|\cdot| \sim |\cdot|_\infty \iff$ *there exists $c > 0$ with $c = \frac{\log|a|}{\log a}$ for all $a \in \mathbf{Z}_{>1}$,*

and further we can rewrite this as

$|\cdot| \sim |\cdot|_\infty \iff$ *for all $a, b \in \mathbf{Z}_{>1}$, we have*

$$\frac{\log|a|}{\log a} = \frac{\log|b|}{\log b} \qquad (= c, \text{ for some } c > 0).$$

We prove this latter statement.

Let $a, b \in \mathbf{Z}_{>1}$. Write b^n in base a, we have

$$b^n = c_m a^m + \cdots + c_0,$$

with $0 \leqslant c_i < a$. We have $m \leqslant n\log_a(b) = \frac{\log(b)}{\log(a)}$.

Note

$$|c_i| \leqslant B := \max(|0|, |1|, \cdots, |a-2|, |a-1|),$$

so that

$$\begin{aligned}|b|^n &\leqslant (m+1)B \cdot \max(|a|^m, 1) \\ &\leqslant (n\log_a(b) + 1)B \cdot \max(|a|^{n\log_a(b)}, 1).\end{aligned}$$

Taking nth roots, we obtain

$$|b| \leqslant [n\log_a(b)+1)B]^{1/n} \cdot \max(|a|^{\log_a(b)}, 1)$$

for all n. As $[n\log_a(b)B]^{1/n} \to 1$ as $n \to \infty$, we deduce

$$|b| \leqslant \max(|a|^{\log_a(b)}, 1). \tag{2.2}$$

Since $|\cdot|$ is archimedean, $|\mathbf{Z}|$ is not bounded by Lemma 2.4, so there exists $b > a$ such that $|b| > 1$. Then

$$|b| \leqslant |a|^{\log_a(b)} = |a|^{\frac{\log(b)}{\log(a)}}. \tag{2.3}$$

Taking logs,

$$\frac{\log|b|}{\log(b)} \leqslant \frac{\log|a|}{\log(a)}. \tag{2.4}$$

As $b > a$, we have $\log_a(b) > 1$; so by (2.3), we see $|a| > 1$ as well. Since a was arbitrary, we see $|a| > 1$ for all $a \in \mathbf{Z}_{>1}$; and the choice of b could have been made arbitrarily in $\mathbf{Z}_{>1}$! This allows us to swap a and b, and running the same argument, we deduce that

$$\frac{\log|a|}{\log(a)} \leqslant \frac{\log|b|}{\log(b)}, \tag{2.5}$$

that is

$$\frac{\log|a|}{\log(a)} = \frac{\log|b|}{\log(b)},$$

as we wanted. Thus $|\cdot| \sim |\cdot|_\infty$, by the chain of equivalences above.

Now suppose $|\cdot|$ is non-archimedean. The ultrametric law implies that $|n| \leqslant 1$ for all $n \in \mathbf{Z}$. Since $|\cdot|$ is non-trivial, there must exist an integer $n > 1$ with $|n| < 1$. Factorising, we may write $n = p_1^{e_1} \cdots p_r^{e_r}$, for $e_i \geqslant 1$. We deduce there exists a prime $p = p_i$ with $|p| < 1$.

Suppose $|p| < 1$ and $|q| < 1$ for two distinct primes p, q. Then $1 = ap + bq$, for some integers a, b, and $1 = |ap + bq| \leqslant \max(|ap|, |bq|) \leqslant \max(|p|, |q|) < 1$; contradiction. Thus there is a unique p with $|p| = \alpha < 1$, and $|q| = 1$ for all other primes q. Then $|\cdot| \sim |\cdot|_p$. □

2.4. Archimedean absolute values

The primary focus of this course is on non-archimedean absolute values. However, our new viewpoint – of finding absolute values on a field, and considering their completion with respect to the absolute value – throws up an interesting question:

> *Are there other* archimedean *fields we should be studying? Are there more general analogues of* $\mathbf{R}$ *and* $\mathbf{C}$*, complete with respect to an archimedean absolute value?*

Another theorem of Ostrowski tells us that the answer is *no*. In this section, we state his theorem, and give a very loose sketch of its proof.

Theorem 2.12 (Ostrowski)**.** *Let K be a field complete with respect to an archimedean absolute value. Then $K \cong \mathbf{R}$ or $\mathbf{C}$ as topological fields.*

Proof. *This proof is only a sketch. It is non-examinable. A complete proof is given in [Cas86, §3].*

By Corollary 2.5, K necessarily has characteristic 0; thus it contains $\mathbf{Q}$. The restriction of $|\cdot|$ to $\mathbf{Q}$ is equivalent to $|\cdot|_\infty$ by Theorem 2.11, so since K is complete, K must contain $\mathbf{R}$. We may as well adjoin a root of X^2+1 and assume $\mathbf{C} \subset K$.

Suppose $\mathbf{C} \neq K$, so there exists $\alpha \in K\backslash\mathbf{C}$. Let $c \in \mathbf{C}$ such that $|\alpha - c|$ is minimised: note the minimum is attained, since $\mathbf{C}$ is complete, and positive, since $\alpha \notin \mathbf{C}$. Replacing α with $\alpha - c$, we have $|\alpha - z| \geqslant |\alpha|$ for all $z \in \mathbf{C}$. Then

$$|\alpha^n - z^n| = |\alpha - z| \cdot |\alpha - \zeta z| \cdots |\alpha - \zeta^{n-1} z| \geqslant |\alpha - z| \cdot |\alpha|^{n-1},$$

where ζ is a primitive nth root of unity. If $|z| < 1$, then as $n \to \infty$, this shows that $|\alpha| \geqslant |\alpha - z|$. Hence, by minimality, $|\alpha| = |\alpha - z|$. There is a way to repeat this argument with $\alpha - z$ in place of α (though this takes some work, since one may not renormalise α again) to show $|\alpha - z| = |\alpha - 2z|$, and so on; so $|\alpha - nz| = |\alpha|$. In particular,

$$|nz| = |-nz| = |\alpha - nz - \alpha| \leqslant |\alpha - nz| + |\alpha| = 2|\alpha|,$$

so $|n| \leqslant 2|\alpha/z|$ is bounded. Thus $|\cdot|$ is non-archimedean, contradicting Lemma 2.4. □

Corollary 2.13. *Let K be a number field, let $|\cdot|$ be an archimedean absolute value, and let $\widehat{K}$ be the completion of K with respect to $|\cdot|$. Then $\widehat{K} \cong \mathbf{R}$ or $\mathbf{C}$ as topological fields.*

Proof. This corollary is an obvious consequence of Theorem 2.12.

This result is important, however, and it has a simpler direct proof. In particular, Ostrowski's (first) theorem (Theorem 2.11) implies that the restriction of $|\cdot|$ to $\mathbf{Q}$ is equivalent to $|\cdot|_\infty$, so we deduce $\mathbf{R} \subset \widehat{K}$.

Write $K = \mathbf{Q}(\alpha)$ for some α. Then $K \subset \mathbf{R}(\alpha)$ is dense with respect to $|\cdot|$, and $\mathbf{R}(\alpha) = \mathbf{R}$ or $\mathbf{C}$ (as $\mathbf{C}$ is algebraically closed). Since $\mathbf{R}$ and $\mathbf{C}$ are complete, we must have $\widehat{K} = \mathbf{R}$ or $\mathbf{C}$. □

Remark 2.14. From this corollary, we may deduce that the archimedean places of K correspond to the real embeddings $K \hookrightarrow \mathbf{R}$ and the pairs of complex embeddings $K \hookrightarrow \mathbf{C}$. The correspondence is as follows: to an archimedean place v of K, we attach the embedding $K \hookrightarrow \widehat{K} = \mathbf{R}$ or the pair of embeddings $\iota, c \circ \iota : K \hookrightarrow \widehat{K} = \mathbf{C}$, where c is complex conjugation. Conversely any embedding ι induces a place via $|\alpha| = |\iota(\alpha)|_\infty$, for $\alpha \in K$, and two embeddings induce the same place if and only if they are complex conjugates.

2.5. The $\mathfrak{p}$-adic absolute value

We've classified all possible archimedean absolute values on a number field K. But what about non-archimedean absolute values on K? Ostrowski's theorem says the only non-archimedean absolute values on $\mathbf{Q}$ (up to equivalence) are the p-adic ones $|\cdot|_p$. We now generalise this to number fields.

The p-adic absolute value relied on the fact that any integer has a unique factorisation into products of primes. As explored in a course on Algebraic Number Theory[1], in a general number field this is no longer true at the level of *elements*, but rather only at the level of *ideals*.

Let K be a number field with ring of integers $\mathcal{O}_K$. The following is a summary of important facts from algebraic number theory:

[1]E.g. the third year course MA3A6 Algebraic Number Theory at Warwick.

Theorem 2.15. *(a) Any fractional ideal $I \subset K$ has a unique factorisation*

$$I = \mathfrak{q}_1^{r_1} \cdots \mathfrak{q}_m^{r_m},$$

where each $\mathfrak{q}_i \subset \mathcal{O}_K$ is a prime ideal and $r_m \in \mathbf{Z}$.

(b) If $\mathfrak{p} \subset \mathcal{O}_K$ is a prime ideal, then there is a unique rational prime p with $p \in \mathfrak{p}$.

(c) If p is prime, the principal ideal $p\mathcal{O}_K$ factorises as

$$p\mathcal{O}_K = \prod_{\mathfrak{p}} \mathfrak{p}^{e(\mathfrak{p}/p)},$$

where the product is over all prime ideals of $\mathcal{O}_K$ with $p \in \mathfrak{p}$ and $e(\mathfrak{p}/p) \in \mathbf{Z}_{\geqslant 1}$ is the ramification index *of $\mathfrak{p}$ over p.*

Definition 2.16. Let $\mathfrak{p} \subset \mathcal{O}_K$ prime, lying above a rational prime p. If $\alpha \in K^\times$, write

$$\alpha\mathcal{O}_K = \mathfrak{p}^r \cdot \mathfrak{q}_1^{r_1} \cdots \mathfrak{q}_m^{r_m}$$

be the factorisation of the principal ideal $\alpha\mathcal{O}_K$ (specifically, r is the power of $\mathfrak{p}$ appearing). Let

$$v_{\mathfrak{p}}(\alpha) := r.$$

Define the $\mathfrak{p}$-*adic absolute value* to be the function

$$\begin{aligned} |\cdot|_{\mathfrak{p}} : K^\times &\longrightarrow \mathbf{R}_{\geqslant 0} \\ \alpha &\longmapsto p^{-\frac{v_p(\alpha)}{e(\mathfrak{p}/p)}} \end{aligned}$$

(and extend to K by setting $|0|_{\mathfrak{p}} = 0$).

(Implicitly here if $-v_p(\alpha)/e(\mathfrak{p}/p) = m/n$ is not an integer, we define $p^{-v_p(\alpha)} = (p^{1/n})^m$, where $p^{1/n}$ is the unique positive real solution to $X^n - p = 0$.)

Exercise. Prove that:

(i) $|\cdot|_{\mathfrak{p}}$ is a non-archimedean absolute value;

(ii) the restriction of $|\cdot|_{\mathfrak{p}}$ to $\mathbf{Q}$ is equal to $|\cdot|_p$.

(Part (ii) is why we scale $|\cdot|_{\mathfrak{p}}$ by $e(\mathfrak{p}/p)$).

Definition 2.17. Let $K_{\mathfrak{p}}$ denote the completion of K with respect to $|\cdot|_{\mathfrak{p}}$.

Exercise. Prove that $K_{\mathfrak{p}}$ is a field extension of $\mathbf{Q}_p$.

Later in the course, we'll prove:

Theorem 2.18. *$K_{\mathfrak{p}}/\mathbf{Q}_p$ is a* finite *field extension of degree at most $[K : \mathbf{Q}]$, and that every non-archimedean absolute value on K is equivalent to $|\cdot|_{\mathfrak{p}}$ for some prime ideal $\mathfrak{p} \subset \mathcal{O}_K$.*

Chapter 3

Valuations

We have seen, in the p-adic and $\mathfrak{p}$-adic absolute values, that one source of (non-archimedean) absolute values is through *valuations*. We now describe a general version of this theory, and use it to describe generalisations of $\mathbf{Q}_p$ and $\mathbf{Z}_p$.

Throughout, K will be a field.

3.1. Recap from commutative algebra: local rings

We briefly recap some relevant definitions from commutative algebra. Throughout R will be a commutative integral domain with 1, and $R^\times$ will denote its (multiplicative) unit group, that is, the group of elements that have a multiplicative inverse.

Definition 3.1. An ideal $I \subset R$ is *proper* if $I \neq R$. It is a *maximal ideal* if it is proper and it is not contained in any larger proper ideal of R. We say R is a *local ring* if it has a unique maximal ideal $\mathfrak{m}$.

Lemma 3.2. *(i) If R is a local ring, then its maximal ideal is $\mathfrak{m} = R \backslash R^\times$.*

(ii) Conversely, if $R \backslash R^\times$ is an ideal, it is maximal and R is local.

Proof. First note: if $I \subset R$ is any proper ideal, we must have $I \subset R \backslash R^\times$, since if $x \in R^\times \cap I$, then $xR = R \subset I$ and $R = I$, contradicting properness.

(i) Suppose R is local. We know $\mathfrak{m} \subset R \backslash R^\times$. If $\mathfrak{m} \neq R \backslash R^\times$, then there exists $y \in R \backslash R^\times$ with $y \notin \mathfrak{m}$. But then y must be contained in a maximal ideal $\mathfrak{m}'$ not equal to $\mathfrak{m}$, contradiction.

(ii) By the above arguments $R \backslash R^\times$ must be maximal. Moreover it also contains any proper ideal of R, so it must be the unique maximal ideal. □

Definition 3.3. Let $\mathfrak{p} \subset R$ be a prime ideal (i.e. if $xy \in \mathfrak{p}$, then either $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$). Let $S_\mathfrak{p} := R \backslash \mathfrak{p}$. The *localisation of R at $\mathfrak{p}$* is the ring

$$R_\mathfrak{p} := S_\mathfrak{p}^{-1} R := \left\{ \tfrac{r}{s} \in \mathrm{Frac}(R) : r \in R, s \in S_\mathfrak{p} \right\}$$

with addition and multiplication operations inherited from $\mathrm{Frac}(R)$.

($*$) **Remark.** We have crucially used that R is a domain in this definition (to define the fraction field). More generally, one can use this as inspiration to find a precise definition as follows: define an equivalence relation on $R \times S_\mathfrak{p}$ by

$$(r, s) \sim (r', s') \iff \exists u \in S_\mathfrak{p} \text{ such that } (rs' - r's)u = 0.$$

Then $R_\mathfrak{p}$ is the ring of equivalence classes $[(r,s)]$ in $R \times S_\mathfrak{p}$, with $\frac{r}{s}$ denoting the equivalence class of (r,s).

Lemma 3.4. *Let $\mathfrak{p} \subset R$ be a prime ideal. The localisation $R_\mathfrak{p}$ is a local ring.*

Proof. It is easy to see that $\mathfrak{p}R_\mathfrak{p} = \{\frac{r}{s} \in R_\mathfrak{p} : r \in \mathfrak{p}\} \subset R_\mathfrak{p}$ is an ideal. Moreover every element of $R_\mathfrak{p} \backslash \mathfrak{p}R_\mathfrak{p}$ is invertible, hence $R_\mathfrak{p} \backslash \mathfrak{p}R_\mathfrak{p} = R_\mathfrak{p}^\times$. Thus $\mathfrak{p}R_\mathfrak{p} = R_\mathfrak{p} \backslash R_\mathfrak{p}^\times$ and we conclude by Lemma 3.2(ii). □

Example. Take $R = \mathbf{Z}$ and $\mathfrak{p} = (p) \subset \mathbf{Z}$. Then $R_\mathfrak{p} = \mathbf{Z}_{(p)} = \{m/n \in \mathbf{Q} : (m,n) = 1, p \nmid n\}$ is a local ring, with maximal ideal $p\mathbf{Z}_{(p)}$. If $x \in \mathbf{Z}_{(p)}$ is not in $p\mathbf{Z}_{(p)}$, then $x = m/n$ with $p \nmid m, n$; so $x^{-1} = n/m \in \mathbf{Z}_{(p)}$. We deduce $\mathbf{Z}_{(p)}^\times = \mathbf{Z}_{(p)} \backslash p\mathbf{Z}_{(p)}$, as in Lemma 3.2.

3.2. Basic properties and definitions of valuations

Definition 3.5. A *valuation* on a field K is a function

$$v : K^\times \longrightarrow \mathbf{R}$$

such that for all $x, y \in K$, we have:

- $v(xy) = v(x) + v(y)$;
- $v(x+y) \geqslant \min(v(x), v(y))$.

Examples.

- We have the trivial valuation $v(x) = 0$. (Henceforth: all valuations will be non-trivial without comment).
- The p-adic valuation v_p is a valuation on $\mathbf{Q}$. More generally, if K is a number field and $\mathfrak{p}$ a prime ideal of K, then $v_\mathfrak{p}$, as defined in Definition 2.16, is a valuation on K.
- Let k be any field, and let $K = k(T) = \mathrm{Frac}(k[T])$ be the field of rational functions over k. Any $f(T) \in k(T)$ can be written uniquely in the form

  $$f(T) = T^r \cdot \frac{g(T)}{h(T)}, \qquad g, h \in k[T], \qquad g(0) \neq 0 \neq h(0)$$

  (i.e. both g and h have non-zero constant term). Then the function

  $$\begin{aligned} v_0 : k(T)^\times &\longrightarrow \mathbf{Z}, \\ T^r \cdot \tfrac{g(T)}{h(T)} &\longmapsto r \end{aligned}$$

  is a valuation on $k(T)$.
- More generally, if $P(T) \in k[T]$ is any irreducible polynomial, then any $f(T) \in k(T)$ can be written uniquely in the form

  $$f(T) = P(T)^r \frac{g(T)}{h(T)}, \qquad g, h \in k[T] \backslash P(T)k[T].$$

  Then the function

  $$\begin{aligned} v_P : k(T)^\times &\longrightarrow \mathbf{Z}, \\ P(T)^r \cdot \tfrac{g(T)}{h(T)} &\longmapsto r \end{aligned}$$

  is a valuation on $k(T)$. Note $v_0 = v_T$ is a special case.

Examples. *(Continued).* Let us explain how to compute v_T in more detail. For an element $G(T) \in k[T]$, write

$$G(T) = a_0 + a_1T + \cdots + a_nT^n, \qquad a_i \in k.$$

Then

$$v_T(G) = \min\{i : a_i \neq 0\},$$

that is, the index of the lowest non-zero term. For example,

$$v_T(T^4 + 3T^3 + 2T^2) = 2, \qquad v_T(T^{17} + T^{15}) = 15.$$

What's really going on is that we're factorising G into irreducibles, and counting how many times the irreducible polynomial 'T' occurs: e.g. in the above examples

$$T^4 + 3T^3 + 2T^2 = T^2 \cdot (T+1) \cdot (T+2), \qquad T^{17} + T^{15} = T^{15} \cdot (T^2+1).$$

This is well-defined since $k[T]$ is a Euclidean Domain, hence a Unique Factorisation Domain.

Now, a general element f of $k(T) = \mathrm{Frac}(T)$ has form $G(T)/H(T)$, where $G, H \in k[T]$ are polynomials. We can always divide out by common irreducible factors so that G and H are coprime. Then write $G(T) = T^r \cdot g(T)$ and $H(T) = T^s h(T)$, where $g(T) \neq 0 \neq h(T)$ (that is, both g and h have non-zero constant term). Then $r = v_T(G)$, $s = v_T(H)$, and

$$f(T) = \tfrac{G(T)}{H(T)} = T^{r-s} \cdot \tfrac{g(T)}{h(T)} \xrightarrow{v_T} r - s = v_T(G) - v_T(H).$$

has valuation $r - s$.

Some concrete examples: let $k = \mathbf{Q}$, let $G(T) = 2T^2 + 3T^3 + T^4$ and $H(T) = 1 + T^2$. Then $H(T)$ is irreducible and $G(T)$ factorises as $G(T) = T^2(T+1)(T+2)$. Then

$$\begin{aligned} f(T) &:= \tfrac{G(T)}{H(T)} \\ &= \tfrac{T^2(T+1)(T+1)}{T^2+1} \xrightarrow{v_T} 2 - 0 = 2. \end{aligned}$$

Further concrete examples are:

- $v_T\left(\frac{T^3+1}{T^4+1}\right) = 0,$
- $v_T\left(\frac{T^3+T}{T^4+1}\right) = 1,$
- $v_T\left(\frac{T^3+1}{T^4+3T^3}\right) = -3,$
- $v_T\left(\frac{T^3+2T^2}{T^2+5T}\right) = v_T\left(\frac{T^2+T}{T+5}\right) = 1$ (after dividing out the common factor of T).

The more general valuations v_P, for $P(T)$ irreducible, can be computed similarly. For example, take

$$f(T) = \frac{G(T)}{H(T)} = \frac{T^4 + 3T^3 = 2T^2}{T^2+1} = \frac{T^2(T+1)(T+2)}{T^2+1}$$

as above. Then

$$\begin{aligned} v_T(f) &= 2 && \text{(as above)}, \\ v_{T+1}(f) &= 1 && \text{(since } T+1 \text{ divides } G(T) \text{ exactly once and does not divide } H(T)), \\ v_{T+2}(f) &= 1 && \text{(since } T+1 \text{ divides } G(T) \text{ exactly once and does not divide } H(T)), \\ v_{T^2+1}(f) &= -1 && \text{(since } T^2+1 \text{ does not divide } G(T) \text{ and divides } H(T) \text{ exactly once)}, \\ v_P(f) &= 0 && \text{for all other irreducibles } P \text{ (since no other irreducible divides } G \text{ or } H). \end{aligned}$$

Note the *strong* similarity to v_p as p ranges over the prime numbers!

Henceforth we will always assume all valuations are non-trivial without further comment.

Lemma 3.6. *Let v be a valuation on a field K. Then:*

(1) $v(1) = 0$.

(2) $v(x) = -v(x^{-1})$ *for all* $x \in K^\times$.

Proof. (1) We have
$$v(1) = v(1 \cdot 1) = v(1) + v(1) = 2v(1),$$
so $v(1) = 0$.

(2) Note
$$0 = v(1) = v(x \cdot x^{-1}) = v(x) + v(x^{-1}),$$
so $v(x) = -v(x^{-1})$. □

This compares directly to Lemma 1.3. We could now go through and prove lots of basic properties about valuations; but it turns out that we can just transfer most of the theory of (non-archimedean) absolute values directly into the theory of valuations, since the two are closely related by the following lemma. In particular, the slogan would be:

"every valuation is the log of a non-archimedean absolute value;
...and every non-archimedean absolute value is the exponential of a valuation."

Lemma 3.7. *Let* $0 < \alpha < 1$.

(1) If v is a valuation on a field K, then the function
$$\begin{aligned} |\cdot| : K &\longrightarrow \mathbf{R}_{\geqslant 0} \\ x &\longmapsto \alpha^{v(x)} \end{aligned}$$
is a non-archimedean absolute value on K.

(2) Conversely, if $|\cdot|$ is a non-archimedean absolute value on K, then
$$\begin{aligned} v : K^\times &\longrightarrow \mathbf{R}, \\ x &\longmapsto \log_\alpha |x| \end{aligned}$$
is a valuation on K.

Proof. Exercise (this is proved in the same way we studied the special case v_p and $|\cdot|_p$ on $\mathbf{Q}$). □

Remark. If $v : K^\times \to \mathbf{R}$ is a valuation, it is customary to extend v to
$$v : K \to \mathbf{R} \cup \{+\infty\}$$
by setting $v(0) = +\infty$. (In particular, note $v(0) \geqslant v(x)$ for all $x \in K^\times$). This is compatible with Lemma 3.7, since "$|0| = \alpha^{v(0)} = \alpha^\infty = 0$" (as $\alpha < 1$).

Definition 3.8. Two valuations v_1, v_2 on K are *equivalent* if there exists $c > 0$ such that
$$v_1(x) = cv_2(x).$$

Definition 3.9. If v is a valuation on a field K, with associated absolute value $|\cdot|$, we define:

- $\mathcal{O} := \{x \in K : |x| \leqslant 1\} = \{x \in K : v(x) \geqslant 0\}$, the *valuation ring.*
- $\mathfrak{m} = \{x \in K : |x| < 1\} = \{x \in K : v(x) > 0\}$, the *maximal ideal.*
- $k = \mathcal{O}/\mathfrak{m}$, the *residue field.*

Examples. (1) Suppose $K = \mathbf{Q}$ and $v = v_p$. Then:

- The valuation ring is

$$\mathcal{O} = \mathbf{Z}_{(p)} = \left\{ \tfrac{m}{n} \in \mathbf{Q} : (m,n) = 1, \quad p \nmid n \right\},$$

that is, the localisation of $\mathbf{Z}$ at the prime ideal (p).

- The unit group is

$$\mathcal{O} = \mathbf{Z}_{(p)}^{\times} = \left\{ \tfrac{m}{n} \in \mathbf{Z}_{(p)} : p \nmid m \right\},$$

the multiplicative group of invertible elements in $\mathbf{Z}_{(p)}$.

- The maximal ideal is

$$\mathfrak{m} = p\mathbf{Z}_{(p)} = \left\{ \tfrac{m}{n} \in \mathbf{Z}_{(p)} : p|m \right\}.$$

- The residue field is

$$k = \mathbf{Z}_{(p)}/p\mathbf{Z}_{(p)} \cong \mathbf{Z}/p\mathbf{Z} \cong \mathbf{F}_p.$$

(2) Let $K = \mathbf{Q}_p$ with $v = v_p$. Then $\mathcal{O} = \mathbf{Z}_p$, $\mathfrak{m} = p\mathbf{Z}_p$, and $k = \mathbf{Z}_p/p\mathbf{Z}_p \cong \mathbf{F}_p$. (This last isomorphism is proved on the first example sheet).

Exercise. Let $K = k(T)$ with absolute value v_0. If $f \in k(T)$, give criteria in terms of $f(0)$ for f to be in the valuation ring, unit group, and maximal ideal. What is the residue field?

Lemma 3.10. *Let v be a valuation on a field K, with valuation ring $\mathcal{O}$.*

(1) We have

$$\mathcal{O}^{\times} = \{x \in K : v(x) = 0\}.$$

(2) $\mathcal{O}$ is local, with maximal ideal $\mathfrak{m}$.

Proof. (1) If $v(x) = 0$, then $v(x^{-1}) = 0$ by Lemma 3.6, so $x^{-1} \in \mathcal{O}$ and $x \in \mathcal{O}^{\times}$.

(2) $\mathfrak{m}$ is an ideal, since:

- If $x, y \in \mathfrak{m}$, then $v(x+y) \geqslant \min(v(x), v(y)) > 0$, so $x + y \in \mathfrak{m}$;
- If $r \in \mathcal{O}$ and $x \in \mathfrak{m}$, then $v(rx) = v(r) + v(x) \geqslant v(x) > 0$, so $rx \in \mathfrak{m}$.

Note $\mathfrak{m} = \mathcal{O} \backslash \mathcal{O}^{\times}$ by (1); so $\mathcal{O}$ is local, with maximal ideal $\mathfrak{m}$, by Lemma 3.2. □

Lemma 3.11. *(1) If $x, y \in K^{\times}$, then $x\mathcal{O} \subset y\mathcal{O}$ if and only if $v(x) \geqslant v(y)$.*

(2) If $0 \neq x \in \mathfrak{m}$, then $K = \mathcal{O}\left[\frac{1}{x}\right]$. In particular, $K = \mathrm{Frac}(\mathcal{O})$.

Proof. (1) Note $y\mathcal{O}$ is an ideal of $\mathcal{O}$, so $x\mathcal{O} \subset y\mathcal{O}$ if and only if $x \in y\mathcal{O}$.
Suppose $x \in y\mathcal{O}$. Then $x = yr$, some $r \in \mathcal{O}$; i.e. $v(r) \geqslant 0$. So

$$v(x) = v(y) + v(r) \geqslant v(y).$$

Conversely, if $v(x) \geqslant v(y)$, then $v(x/y) \geqslant 0$, so

$$x/y \in \mathcal{O} \;\Rightarrow\; x \in y\mathcal{O}.$$

(2) Let $z \in K^\times$. Then $v(zx^n) = v(z) + nv(x) \geqslant 0$ for $n \gg 0$. So $zx^n \in \mathcal{O}$, whence

$$z \in x^{-n}\mathcal{O} \subset \mathcal{O}\left[\tfrac{1}{x}\right].$$

Since $0 \in \mathcal{O}$, It follows that $K \subset \mathcal{O}\left[\frac{1}{x}\right] \subset K$, from which we conclude. $\square$

Example. Consider the familiar special case $K = \mathbf{Q}_p$, $\mathcal{O} = \mathbf{Z}_p$. Then this lemma recovers the fact that $p^n\mathbf{Z}_p \subset p^m\mathbf{Z}_p$ if and only if $m \leqslant n$, and that $\mathbf{Q}_p = \mathbf{Z}_p[\frac{1}{p}] = \mathrm{Frac}(\mathbf{Z}_p)$.

Lemma 3.12. *The ring $\mathcal{O}$ is integrally closed in K.*

(Recall: this means that if

$$f(X) = X^m + a_{m-1}X^{m-1} + \cdots + a_1X + a_0 \in \mathcal{O}[X]$$

is monic irreducible, and $x \in K$ with $f(x) = 0$, then $x \in \mathcal{O}$).

Proof. Suppose $x \in K$ is an algebraic integer over $\mathcal{O}$, that is, there exist $a_i \in \mathcal{O}$ such that

$$x^m + a_{m-1}x^{m-1} + \cdots + a_1x + a_0 = 0.$$

Then

$$\begin{aligned} |x^m| = |a_{m-1}x^{m-1} + \cdots + a_0| &\leqslant \max_{0\leqslant i\leqslant m-1} |a_ix^i| \\ &\leqslant \max_{0\leqslant i\leqslant m-1} |x^i| \qquad (\text{since } |a_i| \leqslant 1) \\ &= \max(|x|^{m-1}, 1). \end{aligned}$$

If $|x| > 1$, this inequality cannot hold, so $|x| \leqslant 1$ and $x \in \mathcal{O}$. $\square$

Lemma 3.13. *K is complete if and only if $\mathcal{O}$ is complete.*

Proof. $\Rightarrow$: If K is complete, then $\mathcal{O}$ is complete since it is closed in K.
$\Leftarrow$: Conversely: note that

$\mathcal{O}$ is complete $\Rightarrow$ $x + \mathcal{O}$ is complete for all $x \in K$

(as if you translate a Cauchy sequence by x, then it remains Cauchy, and the limit is scaled by x).

If (α_n) is any Cauchy sequence in K, then (setting $\epsilon = 1$) there exists N such that $|\alpha_N - \alpha_n| < 1$ for all $n \geqslant N$. In particular

$$\alpha_n \in \alpha_N + \mathcal{O}$$

for all $n \geqslant N$. Then α_n converges to a limit in $\alpha_N + \mathcal{O}$, hence in K. $\square$

3.3. Discrete valuation rings

Definition 3.14.

- If (K, v) is a valuation, then the *value group* is $v(K^\times) \subset \mathbf{R}$. This is a subgroup of $\mathbf{R}$.
- We say v is a *discrete valuation* if $v(K^\times)$ is discrete. It is a *normalised discrete valuation* if $v(K^\times) = \mathbf{Z}$.
- A *discretely valued field* is a field K with a (non-trivial) discrete valuation v.

Example. The valuations v_p on $\mathbf{Q}$, $v_{\mathfrak{p}}$ on a number field K, and v_0 and v_P on $k(T)$ are all normalised discrete valuations.

($*$ Non-examinable:) The following is a non-discrete valuation on a field. Let $K_n = \mathbf{Q}_p[X]/(X^n - p) \cong \mathbf{Q}_p(p^{1/n})$, a finite extension of $\mathbf{Q}_p$, with unique valuation extending v_p on $\mathbf{Q}_p$ (so that $v_n(p^{1/n}) = 1/nv_p(p) = 1/n$. It is not at all obvious that such a valuation exists; we will prove this later in the course). Then $K_n \subset K_{n+1}$, and $v_{n+1}|_{K_n} = v_n$ Let $K = \bigcup_n K_n$. The valuations v_n extend to a unique valuation v on K; but $v(K^\times) = \mathbf{Q}$ is not discrete.

Lemma 3.15. *The following are equivalent:*

(1) v is discrete;

(2) $\mathcal{O}$ is a principal ideal domain (PID);

(3) $\mathcal{O}$ is Noetherian (i.e. every ideal is finitely generated);

(4) $\mathfrak{m}$ is principal.

Proof. (1) $\Rightarrow$ (2): Let $I \subset \mathcal{O}$ be a non-zero ideal. Let $a \in I$ such that

$$v(a) = \min\{v(x) : x \in I\} \subset \mathbf{R}_{\geqslant 0}.$$

Since v is discrete, this minimum is always attained. Note $a\mathcal{O} \subset I$.

Now if $x \in I$, then $v(x) \geqslant v(a)$ by assumption, so $x\mathcal{O} \subset a\mathcal{O}$ by Lemma 3.11; so $I \subset a\mathcal{O}$. Thus

$$a\mathcal{O} = I,$$

that is, I is principal.

(2) $\Rightarrow$ (3): If I is principal, it is certainly finitely generated, so $\mathcal{O}$ is Noetherian.

(3) $\Rightarrow$ (4): Let $\mathfrak{m} = x_1\mathcal{O} + \cdots + x_m\mathcal{O}$ (which is possible since $\mathfrak{m}$ must be finitely generated). Without loss of generality, assume

$$v(x_1) \leqslant v(x_2) \leqslant \cdots \leqslant v(x_m).$$

But then, by Lemma 3.11, we must have

$$x_1\mathcal{O} \supset x_2\mathcal{O} \supset \cdots \supset x_m\mathcal{O}.$$

So $x_2, ..., x_m \in x_1\mathcal{O}$, hence $\mathfrak{m} = x_1\mathcal{O}$ is principal.

(4) $\Rightarrow$ (1): Write $\mathfrak{m} = \pi\mathcal{O}$ for $\pi \in \mathcal{O}$. Let $c \coloneqq v(\pi) > 0$. If $x \in K^\times$, then $v(x) > 0$ if and only if $x \in \mathfrak{m}$, hence $v(x) \geqslant c$. Thus

$$c = v(\pi) = \min\{v(x) : v(x) > 0\}.$$

We deduce $v(K^\times)$ is thus discrete.[1] □

[1] Here we have used that if G is additive subgroup of $\mathbf{R}$ with a minimal positive element $c > 0$, then $G = c\mathbf{Z}$ is discrete. For suppose not; then there exists $d \in G \backslash c\mathbf{Z}$. Suppose $ac < d < (a+1)c$, with strict inequality since $d \notin c\mathbf{Z}$. Then $0 < d - ac < c$, but $d - ac \in G$, contradicting minimality of c

Example. Consider the familiar example of $\mathbf{Z}_p$, which satisfies all of these conditions. The value group is $\mathbf{Z}$, hence is discrete. Every ideal of $\mathbf{Z}_p$ is of the form $p^n\mathbf{Z}_p$, for some $n \geqslant 0$, and all are principal. To see this: we know every ideal is principal by the lemma. We must show that if $x \in \mathbf{Z}_p$ is e a general element, then $x\mathbf{Z}_p = p^n\mathbf{Z}_p$ for some n. Let $n = v(x)$. Then $n \geqslant v(x)$, so $p^n\mathbf{Z}_p \subset x\mathbf{Z}_p$; and $v(x) \geqslant n$, so $x\mathbf{Z}_p \subset p^n\mathbf{Z}_p$ (both using Lemma 3.11). Combining, $x\mathbf{Z}_p = p^n\mathbf{Z}_p$.

Definition 3.16. A ring R is a *discrete valuation ring*, or *DVR*, if:

(1) R is a local ring, and

(2) R is a PID, and

(3) R is *not* a field.

The following picture describes the position DVRs sit in the hierarchy of rings:

$$\begin{tikzcd}
 & \boxed{\text{UFDs}} \arrow[dl] & & \boxed{\text{PIDs}} \arrow[ll] & & & \boxed{\text{Fields}} \\
\boxed{\text{RINGS}} & & & & \boxed{\text{Local PIDs}} \arrow[ul] \arrow[dl] & = & \sqcup \\
 & & & \boxed{\text{Local rings}} \arrow[ulll] & & & \boxed{\textbf{DVRs}}
\end{tikzcd}$$

($*$) **Remark.** For those also taking MA4 Algebraic Curves, or who are interested in geometry or pursuing a PhD in number theory, the following illustrates a wider context to think about DVRs. If X is an algebraic curve, and $x \in X$ is a point, then

$$x \text{ is smooth in } X \iff \text{the local ring at } x \text{ is a DVR.}$$

Here recall that the local ring $\mathcal{O}_x$ at a point x is the set of all algebraic functions on X that are regular (i.e. do not have a pole) at x. The associated normalised discrete valuation on $\mathcal{O}_x$ is the map sending a function to its order of vanishing at x. Note in particular that this implies that if X is an elliptic curve, then *all* the local rings on X are DVRs (since elliptic curves are everywhere smooth).

Definition 3.17. If R is a DVR, then its unique maximal ideal $\mathfrak{m}$ is principal. A *uniformiser* of R is any generator π of $\mathfrak{m}$.

Examples.

- $\mathbf{Z}_p$ is a DVR. To see this, note it is the valuation ring of $(\mathbf{Q}_p, v_p)$. Thus:
  - it is local by Lemma 3.10;
  - it is a PID by Lemma 3.15, as v_p is discrete;
  - it is not a field.

  The maximal ideal is $p\mathbf{Z}_p$, so the element $p \in \mathbf{Z} \subset \mathbf{Z}_p$ is a uniformiser.
- $\mathbf{Z}_{(p)}$ is also a DVR, and again p is a uniformiser.
- A non-example: $\mathbf{Z}$ is a PID with discrete valuations, but it is not a DVR, as it has infinitely many maximal ideals (hence is not local).

Remark. Uniformisers are not unique!! For example, 6, 12, 15, 21, 24, 30, or even non-integers such as $3/2$, $3/5$, $6/5$, etc. are all uniformisers in $\mathbf{Z}_3$. All of these are of the form $3 \times (\text{elt of } \mathbf{Z}_3^\times)$. In particular, note

$$x \in \mathbf{Z}_p \text{ is a uniformiser} \iff v_p(x) = 1.$$

Proposition 3.18. *Let R be a DVR, with uniformiser π. Let $K = \mathrm{Frac}(R)$. Every element $x \in K^\times$ can be written uniquely in the form $x = u\pi^r$, for $u \in R^\times$ and $r \in \mathbf{Z}$.*

Proof. A DVR is a PID, hence a UFD. Thus if such an expression exists, it is unique.

We now show *existence* of such an expression. It suffices to prove this for $0 \neq x \in R$, for if $x/y \in K = \mathrm{Frac}(R)$, with $x = u\pi^r$ and $y = v\pi^s$, then $x/y = (uv^{-1}) \cdot \pi^{r-s}$.

Let $0 \neq x \in R$. Recall that since R is local, we have (Lemma 3.2)

$$R = R^\times \sqcup \mathfrak{m} = R^\times \sqcup \pi R.$$

– If $x \in R^\times$, then

$$x = x \cdot \pi^0$$

has the correct form (i.e. $u = x$, $r = 0$).

– If $x \notin R^\times$, then $x \in \mathfrak{m} = \pi R$, so $x/\pi \in R$.

Now we repeat this process with x/π:

– If $x/\pi \in R^\times$, then set $u = x/\pi$ and $r = 1$, and get

$$x = (x/\pi) \cdot \pi = u \cdot \pi^1.$$

– If $x/\pi \notin R^\times$, then $x/\pi \in \mathfrak{m} = \pi R$, and we consider $x/\pi^2 \in R$.

We continue, considering x/π^r as $r \to \infty$:

$$\begin{tikzcd}[column sep=small]
 & & & x \in R \arrow[dl] \arrow[dr] & & & \\
 & & x \in R^\times \arrow[dl, Rightarrow] & \text{or} & x \in \pi R \arrow[d, Rightarrow] & & \\
 & x = x \cdot \pi^0 & & & x/\pi \in R \arrow[dl] \arrow[dr] & & \\
 & & & x/\pi \in R^\times \arrow[dl, Rightarrow] & \text{or} & x/\pi \in \pi R \arrow[d, Rightarrow] & \\
 & & x = x/\pi \cdot \pi^1 & & & x/\pi^2 \in R \arrow[dl] \arrow[dr] & \\
 & & & & x/\pi^2 \in R^\times \arrow[dl, Rightarrow] & \text{or} & x/\pi^2 \in \pi R \arrow[d, Rightarrow] \\
 & & & x = (x/\pi^2) \cdot \pi^2 & & & \ddots
\end{tikzcd}$$

Then one of two things happens:

(1) This process terminates, i.e. there exists r with $x/\pi^r \in R^\times$. Then we may write

$$x = (x/\pi^r) \cdot \pi^r = u \cdot \pi^r.$$

(2) This process does not terminate. Then $x/\pi^r \in \mathfrak{m}$ for all $r \geqslant 0$. In particular,

$$x \in \pi^r \mathfrak{m} = \mathfrak{m}^{r+1} \ \forall r \geqslant 1 \qquad \Rightarrow \qquad x \in \bigcap_{r \geqslant 0} \mathfrak{m}^r.$$

To complete the proof, it suffices to prove (2) never happens. But the *Krull intersection theorem*[2] implies that $\bigcap_{r\geqslant 0}\mathfrak{m}^r = 0$; so if $x \in \mathfrak{m}^r$ for all r, then $x = 0$. But we assumed $x \neq 0$, so this is a contradiction. $\square$

The following lemma justifies the terminology 'discrete valuation ring'.

Lemma 3.19. *(i) If K is a field with a non-trivial discrete valuation, then its valuation ring $\mathcal{O}$ is a DVR.*

(ii) If R is a DVR, then there exists a unique normalised discrete valuation v on $K = \mathrm{Frac}(R)$ such that

$$R = \mathcal{O} = \{x \in K : v(x) \geqslant 0\}$$

is the valuation ring in K and $\mathfrak{m} = \{x \in K : v(x) > 0\}$ its maximal ideal.

Proof. (i) Note:

- $\mathcal{O}$ is local by Lemma 3.10.
- Since v is discrete, $\mathcal{O}$ is a PID by Lemma 3.15.
- Since v is non-trivial, there exists $x \in \mathcal{O}$ with $v(x) \neq 0$; then $x^{-1} \notin \mathcal{O}$, so $\mathcal{O}$ is not a field.

Hence $\mathcal{O}$ is a DVR.

(ii) Let π be a uniformiser in K. Any $x \in K^\times$ can be written uniquely as $x = u\cdot\pi^r$, for $u \in R^\times$. We define

$$\begin{aligned} v : K^\times &\longrightarrow \mathbf{Z} \\ x = u\pi^r &\longmapsto r. \end{aligned}$$

Note $v(x) = \max\{n \in \mathbf{Z} : x \in \mathfrak{m}^n\}$, so it is independent of π.

$$v(x) \geqslant 0 \iff r \geqslant 0 \iff x \in R,$$

so $R = \mathcal{O}$ is the valuation ring (and the maximal ideals then match). $\square$

Remark 3.20. Lemma 3.19 shows we have equivalences

$$\begin{tikzcd}
\boxed{\text{DVRs}} \arrow[rr, bend left, "\text{Frac}"] & & \boxed{\text{Discretely valued fields}} \arrow[ll, bend left, "\text{valuation ring}"] \\
\cup & & \cup \\
\boxed{\text{Complete DVRs}} \arrow[rr, leftrightarrow, "\text{Lemma 3.13}"'] & & \boxed{\text{Complete discretely valued fields}}
\end{tikzcd}$$

It means the study of discretely valued fields is equivalent to the study of DVRs. Moreover, Lemma 3.13 shows that under this correspondence, the complete DVRs and complete discretely valued fields are identified.

[2] The Krull intersection theorem states that: *if R is a Noetherian local ring, with maximal ideal $\mathfrak{m}$, then*

$$\bigcap_{r\geqslant 0}\mathfrak{m}^r = \{0\}.$$

To prove it: note that $I := \bigcap_{r\geqslant 0}\mathfrak{m}^r$ is an ideal in R, and that $\mathfrak{m}I = \mathfrak{m}$. Then Nakayama's lemma implies $I = 0$.

($*$) **Remark.** (Non-examinable) Not every local ring is a DVR. Let $R = K[X,Y]$ be the polynomial ring in two variables. The ideal $(X,Y) \subset R$ is prime (even maximal), so we may consider the localisation $R_{(X,Y)}$, a local ring; but its maximal ideal is not principal.

This is bound up tightly with the notion of *dimension* of a ring, and with related notions in algebraic geometry. There is a notion of *regular local ring*, which captures the notion of smoothness of an algebraic variety; and a DVR is a regular local ring of dimension 1. In particular, a curve X is smooth at a point $x \in X$ if and only if its local ring is regular, which happens if and only if this local ring is a DVR. (The example $R_{(X,Y)}$ above is a regular local ring of dimension 2; it is the local ring of the smooth 2-dimensional affine plane $\mathbf{A}^2_K$ at the point $(0,0)$).

The following 'π-adic expansion' is a more general analogue of the p-adic expansions of Proposition 1.22.

Proposition 3.21. *Let K be a discretely valued field, let $\mathcal{O} \subset K$ its valuation ring, and $\mathfrak{m} = \pi\mathcal{O}$ its maximal ideal. Let $A \subset \mathcal{O}$ be a set of coset representatives for the quotient $k = \mathcal{O}/\pi\mathcal{O}$.*

(i) Every $x \in K$ has a unique π-adic expansion

$$x = \sum_{i=v(x)}^{\infty} a_i \pi^i \qquad \text{for } a_i \in A \text{ and } M = v(x) \in \mathbf{Z}.$$

(ii) K is complete if and only if every $\sum_{i=M}^{\infty} a_i\pi^i$, with $a_i \in A$, converges in K.

Example. Put another way: this proposition says that

(i) Passing to π-adic expansions gives an injective map

$$K \hookrightarrow \left\{ \sum_{i=m}^{\infty} a_i\pi^i : m \in \mathbf{Z}, a_i \in A \right\}.$$

(ii) K is complete if and only if this map is bijective.

For example, consider the discretely valued field $(\mathbf{Q}, v_p)$. On the first exercise sheet, you saw that every rational has a unique p-adic expansion (part (i)); and and that the above map is not surjective (since rational numbers have *periodic* p-adic expansions, and not all p-adic expansions are periodic!). However we do have a bijection between elements of $\mathbf{Q}_p$ and p-adic expansions, and $\mathbf{Q}_p$ is complete, giving an example of (ii) in this case.

Proof. (i) We leave this as an exercise: the proof closely follows that of Proposition 1.22, just with more abstract notation.

(ii) Suppose K is complete. Note that if $a_i \in A$, then $|a_i| \leqslant 1$. Since $|\pi| < 1$, we see $|a_i\pi^i| = |a_i| \cdot |\pi|^i \to 0$ as $i \to \infty$, so this series converges by Lemma 2.1.

Conversely, suppose every sum converges. Let (x_n) be a Cauchy sequence in K, and write

$$x_n = \sum a_i(x_n) \cdot \pi^i, \qquad a_i(x_n) \in A.$$

As the sequence is Cauchy, for any $r \in \mathbf{Z}$, there exists $N \in \mathbf{N}$ such that $v(x_n - x_m) \geqslant r+1$ for all $m, n > N$. This implies that the expansions of x_n and x_m agree up to r, that is

$$a_i(x_n) = a_i(x_m) \qquad \forall i \leqslant r,\ \forall m,n > N.$$

In particular, for each $i \in \mathbf{Z}$, eventually $a_i(x_n)$ is constant, say equal to a_i; and $a_i = 0$ for all $i < M \in \mathbf{Z}$, where $M = \lim_{n\to\infty} v(x_n)$. Define

$$x = \sum_{i=M}^{\infty} a_i \pi^i.$$

This converges in K by assumption, and $x_n \to x$. □

3.4. Power series rings and fields

The following is a very important example of a DVR.

Definition 3.22. For a field k, let $R = k[[T]]$ be the ring of formal power series over k, i.e.

$$k[[T]] = \left\{ \sum_{i=0}^{\infty} a_i T^i : a_i \in k \right\}.$$

Let $K = k((T))$ be the field of Laurent series

$$k((T)) = \left\{ \sum_{i=m}^{\infty} a_i T^i : m \in \mathbf{Z}, a_i \in k \right\}$$

(where each element is allowed finitely many negative powers of T).

Lemma 3.23. *$k((T))$ is a field.*

Proof. Let

$$0 \neq f(T) = a_r T^r + a_{r+1} T^{r+1} + \cdots, \qquad a_r \neq 0, r \in \mathbf{Z}.$$

We must show $f^{-1} \in k((T))$.

Note $T^{-r} \in k((T))$, so $T^r \in k((T))^\times$, and we may rescale, assuming without loss of generality that

$$f(T) = a_0 + a_1 T + a_2 T^2 + \cdots, \qquad a_0 \neq 0.$$

If an inverse $g(T) = \sum_{j=0}^{\infty} b_j T^j$ exists, then

$$1 = \left(\sum a_i T^i\right) \cdot \left(\sum b_j T^j\right) = a_0 b_0 + (a_1 b_0 + a_0 b_1) T + \cdots.$$

Comparing coefficients, we get the infinite collection of equations

$$\begin{aligned} 1 &= a_0 b_0 \\ 0 &= a_0 b_1 + a_1 b_0 \\ 0 &= \cdots. \end{aligned}$$

Since $a_0 \neq 0$, we inductively define

$$\begin{aligned} b_0 &= a_0^{-1} \\ b_1 &= -\frac{a_1 b_0}{a_0} \\ &\vdots \\ b_n &= -\frac{a_n b_0 + \cdots + a_1 b_{n-1}}{a_0} \\ &\vdots \end{aligned},$$

and this gives a solution to the equations, hence an inverse to f. Thus $f \in k((T))$ and $k((T))$ is a field. $\square$

The field $k((T))$ shares many features with $\mathbf{Q}_p$, which we now summarise.

Definition 3.24. Define a function

$$\begin{aligned} v_T : k((T))^\times &\longrightarrow \mathbf{Z} \\ \sum a_i T^i &\longmapsto \min\big\{i : a_i \neq 0\big\}. \end{aligned}$$

Lemma 3.25. *(1) v_T defines a normalised discrete valuation on $k((T))$.*

(2) $k[[T]]$ is a DVR, with maximal ideal $T \cdot k[[T]]$, and residue field k.

Proof. (1) Exercise.

(2) Note $k[[T]] = \{f \in k((T)) : v_T(f) \geqslant 0\}$ is the valuation ring in $k((T))$. It is thus a DVR by Lemma 3.19. $\square$

Recall the field $k(T)$ from Examples 3.2, equipped with the valuation sending $T^r \frac{f(T)}{g(T)} \mapsto r$.

Lemma 3.26. *(i) We have*

$$k((T)) = \mathrm{Frac}\big(k[[T]]\big).$$

(ii) $k((T))$ contains $k(T)$, the two valuations v_T agree.

(iii) $k[[T]]$ is the completion of $k[T]$ with respect to v_T.

(iv) $k((T))$ is the completion of $k(T)$ with respect to v_T.

In particular, we have the following picture summarising the analogues between $\mathbf{Q}_p$ and $k((T))$:

Underlying ring R	Fraction field K	Valuation	Finite expansion
$\mathbf{Z}$	$\mathbf{Q}$	v_p	$a_0 + a_1 p + \cdots + a_m p^m$
$k[T]$	$k(T)$	v_T (or v_P)	$a_0 + a_1 T + \cdots + a_m T^m$

Completion of R	Uniformiser	Infinite expansion	Completion of K
$\mathbf{Z}_p$	p	$a_0 + a_1 p + a_2 p^2 + \cdots$	$\mathbf{Q}_p$
$k[[T]]$	T	$a_0 + a_1 T + a_2 T^2 + \cdots$	$k((T))$

Proof. (i) From above, $k[[T]]$ is the valuation ring in $k((T))$ for v_T. Thus $k((T)) = \mathrm{Frac}(k[[T]])$ by Lemma 3.11.

(ii) Since $k[T] \subset k[[T]]$, we have

$$k(T) = \mathrm{Frac}(k[T]) \subset \mathrm{Frac}(k[[T]]) = k((T)).$$

Write v_0 for the valuation v_T on $k(T)$, and v_1 for the valuation on $k((T))$. If $f \in K(T)$, we must show $v_0(f) = v_1(f)$, so that v_T is the same on both fields. Write $f(T) = T^r \frac{g(T)}{h(T)}$, with $g(0) = h(0) \neq 0$. Then $v_0(f) = r$. Write

$$g(T) = a_0 + a_1 T + \cdots + a_m T^m, \qquad h(T) = b_0 + b_1 T + \cdots + b_n T^n.$$

Then we have $a_0, b_0 \neq 0$. Since $b_0 \neq 0$, we can construct an explicit inverse $h^{-1} = c_0 + c_1 T + c_2 T^2 + \cdots \in k[[T]]$ as a power series, with $c_0 \neq 0$, by the process in the proof of Lemma 3.23. Then

$$\begin{aligned} f(t) &= T^r \cdot (a_0 + a_1 T + \cdots + a_m T^m) \cdot (c_0 + c_1 T + c_2 T^2 + \cdots) \\ &= a_0 c_0 T^r + O(T^{r+1}) \end{aligned}$$

In particular, the minimal non-zero coefficient in the power series of $f(T)$ is for T^r, so $v_1(f) = r = v_0(f)$. Thus the valuations v_T on $k(T)$ and $k((T))$ agree.

(iii) We know $k[T]$ is dense in $k[[T]]$: if

$$f(T) = \sum_{i=0}^{\infty} a_i T^i \in k[[T]],$$

then the sequence

$$f_n := \sum_{i=0}^{n} a_i T^i \to f \qquad \text{as } n \to \infty,$$

and $f_n \in k[T]$. Hence it suffices to prove that $k[[T]]$ is complete with respect to v. This follows from Proposition 3.21(ii).

(iv) Similar to (iii). $\square$

Chapter 4

Local Fields

Given this course is called 'Local Fields', it is perhaps strange that we've not yet given a precise definition of what a local field is! In this chapter we rectify this. First we give a very ad-hoc definition of local fields, based on properties we've seen already, and (rather unnaturally) separating the archimedean and non-archimedean cases. Then we examine these definitions more closely, and give a uniform definition: namely, a local field is a locally compact valued field.

Importantly, we've already seen *all* the examples of local fields! We've not seen enough to prove this yet: instead we just state the classification in this chapter, and prove it later.

4.1. An ad-hoc definition of local fields

Definition 4.1. Let $(K, |\cdot|)$ be archimedean. We say K is an *(archimedean) local field* if $K \cong \mathbf{R}$ or $\mathbf{C}$ as topological fields.

Definition 4.2. Let $(K, |\cdot|)$ be non-archimedean, with associated valuation v and residue field $k = \mathcal{O}/\mathfrak{m}$. We say K is a *(non-archimedean) local field* if:

- K is complete,
- v is discrete,
- and k is finite.

Theorem 4.3. *The non-archimedean local fields are precisely:*

- *finite extensions of* $\mathbf{Q}_p$*, for some prime* p*;*
- $\mathbf{F}_q((T))$*, for* $\mathbf{F}_q$ *a finite field.*

Proof. We will prove this later in the course. The characteristic 0 case is Theorem 8.13, and positive characteristic case is Theorem 5.10. □

The whole purpose of this chapter is to give a cleaner, and uniform, equivalent definition of local fields in terms via their topological properties. In particular, the local fields are exactly the *locally compact valued fields* (whether archimedean or non-archimedean).

($*$) **Remark.** For completeness, we also include the following (non-examinable) definition. We say a field K is a *global field* if it is either:

- a number field (i.e. finite extension of $\mathbf{Q}$),
- or a finite extension of $\mathbf{F}_q(T)$ for some finite field $\mathbf{F}_q$.

Every example of the second kind is the function field of some algebraic curve over a finite field.

Every local field is then the completion of a global field with respect to some absolute value.

4.2. Inverse limits

We now describe $\mathbf{Z}_p$ another way: via *inverse limits.*

Definition 4.4. (i) An *inverse system* (A_n) is a collection $\{A_n : n \in \mathbf{N}\}$ of sets together with *transition maps*

$$\cdots \xrightarrow{f_{n+1}} A_{n+1} \xrightarrow{f_n} A_n \xrightarrow{f_{n-1}} \cdots \xrightarrow{f_3} A_3 \xrightarrow{f_2} A_2 \xrightarrow{f_1} A_1.$$

(ii) The *inverse limit* of an inverse system (A_n) is the set

$$\varprojlim_n A_n = \Big\{(a_n) = (a_1, a_2, a_3, a_4, ...) : a_n \in A_n, f_n(a_{n+1}) = a_n\Big\},$$

i.e. the set of all sequences (a_n) such that

$$\begin{tikzcd} \cdots \arrow[r] & A_{n+1} \arrow[r, "f_n"] & A_n \arrow[r, "f_{n-1}"] & \cdots \arrow[r, "f_2"] & A_2 \arrow[r, "f_1"] & A_1 \\ & \Psi & \Psi & & \Psi & \Psi \\ \cdots \arrow[r, mapsto] & a_{n+1} \arrow[r, mapsto] & a_n \arrow[r, mapsto] & \cdots \arrow[r, mapsto] & a_2 \arrow[r, mapsto] & a_1. \end{tikzcd}$$

Example. – Let $A_n = \mathbf{Z}/p^n\mathbf{Z}$, with $f_n : \mathbf{Z}/p^{n+1} \to \mathbf{Z}/p^n$ the map $x \mapsto x \pmod{p^n}$. We get an inverse system

$$\cdots \to \mathbf{Z}/p^n\mathbf{Z} \xrightarrow{(\mathrm{mod}\, p^{n-1})} \mathbf{Z}/p^{n-1}\mathbf{Z} \to \cdots \to \mathbf{Z}/p^2\mathbf{Z} \xrightarrow{(\mathrm{mod}\, p)} \mathbf{Z}/p.$$

– Here is an example element of the inverse limit $\varprojlim_n \mathbf{Z}/p^n\mathbf{Z}$:

$$\begin{tikzcd} \cdots \arrow[r] & \mathbf{Z}/p^4\mathbf{Z} \arrow[r, "(\mathrm{mod}\, p^3)"] & \mathbf{Z}/p^3\mathbf{Z} \arrow[r, "(\mathrm{mod}\, p^2)"] & \mathbf{Z}/p^2\mathbf{Z} \arrow[r, "(\mathrm{mod}\, p)"] & \mathbf{Z}/p\mathbf{Z} \\ & \Psi & \Psi & \Psi & \Psi \\ \cdots \arrow[r, mapsto] & 1+p+p^2+p^3 \arrow[r, mapsto] & 1+p+p^2 \arrow[r, mapsto] & 1+p \arrow[r, mapsto] & 1. \end{tikzcd}$$

It would be reasonable to think of this element as being, somehow, the infinite sum

$$1 + p + p^2 + p^3 + \cdots ...$$

...which should bring to mind a p-adic expansion!

Remark 4.5. If the A_n are rings, and the transition maps are ring homomorphisms, then the inverse limit is also a ring, with pointwise operations:

$$(a_i) + (b_i) = (a_i + b_i), \qquad (a_i) \cdot (b_i) = (a_i \cdot b_i).$$

So $\varprojlim \mathbf{Z}/p^n\mathbf{Z}$ is a ring.

Lemma 4.6. *The map*

$$\begin{aligned} \phi : \mathbf{Z}_p &\longrightarrow \varprojlim_n \mathbf{Z}/p^n\mathbf{Z} \\ a &\longmapsto (a \pmod{p}, a \pmod{p^2}, a \pmod{p^3}, \dots) \end{aligned}$$

defines an isomorphism of rings.

Proof. Since reduction modulo p^n is a ring homomorphism, ϕ is also a ring homomorphism (since the ring operations on $\varprojlim \mathbf{Z}/p^n\mathbf{Z}$ are computed pointwise).

This map is injective: suppose $a \in \ker(\phi) \subset \mathbf{Z}_p$. Then

$$a \mapsto (0, 0, 0, \dots) = (a \bmod p,\ a \bmod p^2,\ a \bmod p^3,\ \dots),$$

i.e. $a \equiv 0 \pmod{p^n}$ for all n. But then $a = 0$.

To see ϕ is surjective, suppose

$$(b_0, b_1, b_2, b_3, \dots) \in \varprojlim \mathbf{Z}/p^n\mathbf{Z},$$

with $b_{i-1} \in \mathbf{Z}/p^i\mathbf{Z}$. We may uniquely represent b_0 by an element $a_0 \in \{0, ..., p-1\}$. As (b_i) is an element of the inverse limit, we know

$$f_1(b_1) = b_0 \Rightarrow b_1 \equiv b_0 \pmod{p}.$$

In particular,

$$b_1 = a_0 + a_1 p \pmod{p^2},$$

for a unique $a_1 \in \{0, ..., p-1\}$. Similarly

$$b_2 \equiv b_1 \pmod{p^2},$$

so

$$b_2 = b_1 + a_2 p^2 = a_0 + a_1 p + a_2 p^2 \pmod{p^2}.$$

for a unique $a_2 \in \{0, ..., p-1\}$. Continuing, we end up with an element

$$a = a_0 + a_1 p + a_2 p^2 + \cdots \in \mathbf{Z}_p,$$

uniquely determined by its p-adic expansion; and by construction, $\phi(a) = (b_0, b_1, b_2, ...)$. $\square$

Example. If K is a field, then the same proof shows

$$K[[T]] \cong \varprojlim K[T]/(T^n),$$

where the transition maps $K[[T]]/(T^{n+1}) \to K[T]/(T^n)$ are reduction modulo n.

4.3. Topological properties of inverse limits

Inverse limits allow us to give a neat description of the topology on $\mathbf{Z}_p$. Recall that to define a topology on a set A, it suffices to give a collection of 'open balls'; and then the open sets are all possible unions of open balls. In the p-adic topology on $\mathbf{Z}_p$, the open balls are of the form

$$B(a, p^{-r}) = a + p^{r+1}\mathbf{Z}_p.$$

On an inverse limit, we define a collection of open balls as follows:

Definition 4.7. Let (A_n) be an inverse system and $A \coloneqq \varprojlim_n A_n$.

- For each m, we have a natural projection map

$$\begin{aligned} \mathrm{pr}_n : A &\longrightarrow A_n \\ (a_1, a_2, a_3, ...) &\longmapsto a_n. \end{aligned}$$

- In the *inverse limit topology* on $A := \varprojlim_n A_n$, the open balls are the sets of the form

$$\mathrm{pr}_m^{-1}(a_m) \subset A, \qquad \forall a_m \in A_m,\ \forall m \in \mathbf{N}.$$

($*$) **Remark.** This is a natural definition: we have equipped each A_n with the discrete topology, and then the inverse limit topology is the 'weakest topology on A such that the projection maps are all continuous'. If we equip the A_n with more general topologies we can make a similar, more general, definition of the topology on A by taking the basic open sets to be $\mathrm{pr}_m^{-1}(X_m)$, as m ranges over $\mathbf{N}$ and X_m over the open sets in A.

Lemma 4.8. *On $\mathbf{Z}_p = \varprojlim_n \mathbf{Z}/p^n\mathbf{Z}$, the inverse limit topology is the same as the p-adic topology.*

Proof. To prove two topologies are equal, it suffices to show that the sets of open balls are the same in both topologies. We already saw that the open balls in the p-adic topology are of the form $a + p^r\mathbf{Z}_p$, where $a \in \mathbf{Z}_p$, $r \geqslant 0$.

Open balls in the inverse limit topology. The projection maps are given by

$$\begin{aligned} \mathrm{pr}_r : \mathbf{Z}_p \cong \varprojlim \mathbf{Z}/p^n\mathbf{Z} &\longrightarrow \mathbf{Z}/p^r\mathbf{Z}, \\ x &\longmapsto x \pmod{p^r}. \end{aligned}$$

Let $\overline{a} \in \mathbf{Z}/p^r\mathbf{Z})$, and lift to $a \in \mathbf{Z}$. Then

$$\begin{aligned} \mathrm{pr}_r^{-1}(\overline{a}) &= \{x \in \mathbf{Z}_p : x \equiv a \pmod{p^r}\} \\ &= \{x \in \mathbf{Z}_p : x - a \equiv 0 \pmod{p^r}\} \\ &= \{x \in \mathbf{Z}_p : x - a \in p^r\mathbf{Z}_p\} \\ &= a + p^r\mathbf{Z}_p. \end{aligned}$$

This is the same as above, so

$$\{\text{open balls in } \mathbf{Z}_p \text{ for } p\text{-adic topology}\} = \{\text{open balls in } \mathbf{Z}_p \text{ for inverse limit topology}\},$$

that is,

$$\text{the } p\text{-adic topology on } \mathbf{Z}_p = \text{the inverse limit topology on } \mathbf{Z}_p,$$

as required. □

Definition 4.9. A topological space A is *profinite* if it is an inverse limit of finite sets.

For example, $\mathbf{Z}_p$ is profinite.

Recall from a course on topology: let X be a metric space; then

$$\begin{aligned} X \text{ is } \textit{compact} &\iff \text{every open cover of } X \text{ has a finite subcover} \\ &\iff X \text{ is sequentially compact (every sequence has a convergent subsequence).} \end{aligned}$$

Lemma 4.10. *Let $A = \varprojlim_n A_n$ be a profinite topological space (i.e. each A_n is finite). Then A is compact.*

(On the second exercise sheet, you are asked to prove this for the special case $A = \mathbf{Z}_p$).

Proof. We only sketch the proof, so in this generality, it is non-examinable. The case where $A = \mathbf{Z}_p$, as it appears on the second example sheet, remains examinable.

We make use of the following:

Fact: *Let $A = \varprojlim_n A_n$. There is a metric on A such that the induced topology is the inverse limit topology. In particular, A is a metric space.*

Note that in the special case of $A = \mathbf{Z}_p$, we already proved this in Lemma 4.8. In general, the fact is proved in a non-examinable section at the end of this chapter. As a consequence of the fact, it suffices to show that if A is profinite, then A is *sequentially* compact.

Let $x_1, x_2, x_3, ...$ be a sequence in A. Then write

$$x_r = (x_{r,1},\ x_{r,2},\ x_{r,3},\ ...) \in A_1 \times A_2 \times A_3 \times \cdots .$$

Now we construct a convergent subsequence iteratively using the pigeonhole principle:

(1) Since A_1 is finite, there exists $a_1 \in A_1$ such that $x_{r,1} = a_1$ for infinitely many r. Pass to the subsequence where $x_{r,1} = a_1$. Let y_1 be any element of this subsequence; so $y_1 = (a_1, -, -, ...)$ starts with a_1.

(2) Now since A_2 is finite, there exists $a_2 \in A_2$ such that $x_{r,2} = a_2$ for infinitely many r. Pass to the subsequence where $x_{r,2} = a_2$, and let y_2 be any element of the resulting subsequence. Then
$$y_2 = (a_1, a_2, -, -, ...).$$

...

(n) Continuing inductively, we eventually end up with $a_n \in A_n$ and an element y_n with
$$y_n(a_1, a_2, \dots, a_n, -, -, \dots).$$

Then the subsequence y_n tends to $a := (a_1, a_2, a_3, \dots) \in A$. □

($*$) **Remark.** When A is additionally a topological *group*, you can upgrade this statement with a converse. Indeed, a topological group A is profinite if and only if A is:

- Hausdorff,
- compact,
- and totally disconnected.

The proof that profinite implies Hausdorff and totally disconnected is very similar to Corollary 2.3. To prove that the three topological properties imply profinite, you show that

$$G \cong \varprojlim_N G/N,$$

where the inverse limit is over all open normal subgroups. Every such subgroup must have finite index, or the set of cosets $\{gN : g \in G\}$ would be an infinite cover of G with no finite subcover.

Let now $(K, |\cdot|)$ be non-archimedean. Let $\mathcal{O}$ be the valuation ring and $\pi \in \mathfrak{m}$ (note it does *not* need to be a uniformiser). The following generalises Lemma 4.6.

Proposition 4.11. *The following are equivalent:*

(i) K is complete with respect to $|\cdot|$,

(ii) The map

$$\begin{aligned} \phi : \mathcal{O} &\longrightarrow \varprojlim \mathcal{O}/\pi^n \\ a &\longmapsto (a \pmod{\pi^n})_n \end{aligned} \tag{4.1}$$

is an isomorphism.

Proof. (i) $\Rightarrow$ (ii) follows exactly as in the proof of Lemma 4.6, where it is proved for $K = \mathbf{Q}_p$.

(ii) $\Rightarrow$ (i): Suppose (4.1) is an isomorphism. By Lemma 3.13, to show K is complete it suffices to prove $\mathcal{O}$ is complete. We can prove this directly in a very similar way to the proof of Proposition 3.21(2) (see footnote[1]).

Alternatively, the $\mathcal{O}$-analogue of Proposition 3.21(2) shows that:

$\mathcal{O}$ *is complete* $\iff$ *every series* $a_0 + a_1\pi + a_2\pi^2 + a_3\pi^3 + \cdots$ *converges in* $\mathcal{O}$.

For such a sequence, consider the element

$$(a_0, a_0 + a_1\pi,\ a_0 + a_1\pi + a_2\pi^2, ...) \in \varprojlim_n \mathcal{O}/\pi^n.$$

Because (4.1) is an isomorphism, there exists $a \in \mathcal{O}$ with

$$a \pmod{\pi^n} = a_0 + a_1\pi + \cdots a_{n-1}\pi^{n-1}.$$

But then $a_0 + a_1\pi + a_2\pi + \cdots$ converges to a. $\square$

Corollary 4.12. *If $(K, |\cdot|)$ is a non-archimedean local field, then:*

(1) $\mathcal{O}$ is compact;

(2) $x + \mathcal{O}$ and $x\mathcal{O}$ are compact for every $x \in K$.

Proof. (1) Recall K is local if:

– K is complete, hence $\mathcal{O} \cong \varprojlim \mathcal{O}/\pi^n$ by Proposition 4.11;

– K is discrete, so has a uniformiser π, with $\mathfrak{m} = \pi\mathcal{O}$;

– its residue field $\mathcal{O}/\mathfrak{m} = \mathcal{O}/\pi\mathcal{O}$ is finite, hence

$$\mathcal{O}/\pi^n\mathcal{O} = \{a_0 + a_1\pi + \cdots + a_{n-1}\pi^{n-1} : a_i \in \mathcal{O}/\pi\mathcal{O}\}$$

is finite for all n.

Thus if K is a local field, we deduce $\mathcal{O}$ is profinite. Part (1) then follows from Lemma 4.10.

(2) We show $x + \mathcal{O}$ is sequentially compact. If (a_n) is a sequence in $x + \mathcal{O}$, then $(a_n - x)$ is a sequence in $\mathcal{O}$. As $\mathcal{O}$ is compact, it has a convergent subsequence $(b_n - x)$. But then (b_n) is a convergent subsequence of (a_n).

Similarly, any sequence (a_n) in $x\mathcal{O}$ has a convergent subsequence (by considering $(a_n/x) \subset \mathcal{O}$; note if $x = 0$, then $x\mathcal{O} = \{0\}$ is compact, so we may assume $x \neq 0$ here). $\square$

[1]Take a Cauchy sequence (x_n) in $\mathcal{O}$. The Cauchy property means that for every $n \in \mathbf{N}$, for $r \gg 0$ the value $x_r \pmod{\pi^n}$ is constant, say equal to $a_n \pmod{\pi^n}$. Moreover, we have $x_r \pmod{\pi^{n-1}}$ is also constant, equal to $a_n \pmod{\pi^{n-1}}$ and (inductively) to $a_{n-1} \pmod{\pi^{n-1}}$, so $a_n \equiv a_{n-1} \pmod{\pi^{n-1}}$. Thus

$$(a_n)_n \in \varprojlim \mathcal{O}/\pi^n \cong \mathcal{O}$$

yields an element $a \in \mathcal{O}$, and $x_n \to a$ by construction, so $\mathcal{O}$ is complete, as required.

4.4. Locally compact fields

We've shown that for every $x \in \mathbf{Q}_p$, the set $x + \mathbf{Z}_p$ is open and compact: that is,

> "$\mathbf{Q}_p$ contains a open compact subset around every point x",

that is,

> "locally around any point, $\mathbf{Q}_p$ looks compact."

We generalise this in the following definition.

Definition 4.13. A metric space X is *locally compact* if for all $x \in X$, there exists an open ball U and a compact subset C with $x \in U \subset C \subset X$. We call C a *compact neighbourhood of* x.

Remark. You might ask: why is the definition not 'for all $x \in X$, there exists a compact subset containing x'? This would be a bad definition, because $\{x\}$ is always a compact set containing x, so *every* topological space satisfies this property. The addition of the open set U ensures that the compact set C is 'large enough' to make this definition interesting.

Examples.

– $\mathbf{R}$ is locally compact: if $x \in \mathbf{R}$, and $\epsilon > 0$, then

$$x \in (x - \epsilon, x + \epsilon) \subset [x - \epsilon, x + \epsilon].$$

Note the open interval $(x - \epsilon, x + \epsilon)$ is open, and the closed interval $C = [x - \epsilon, x + \epsilon]$ is compact. So C is a compact neighbourhood of x.

– $\mathbf{Q}_p$ is locally compact: any $x \in \mathbf{Q}_p$ is contained in $x + \mathbf{Z}_p$, and this is compact (since $\mathbf{Z}_p$ is compact).

– $\mathbf{Q}$ with the topology from $\mathbf{R}$ is not locally compact. Why? Recall that if X is a metric space, then

$$X \text{ is compact} \iff X \text{ is complete and totally bounded.} \tag{4.2}$$

Let X be a subset of $\mathbf{Q}$. If X contains an open ball in $\mathbf{Q}$, i.e. if it contains an interval $(x - \epsilon, x + \epsilon) \cap \mathbf{Q}$, then it cannot be complete. In particular, no compact subset of $\mathbf{Q}$ can ever contain an open ball, so no point can have a compact neighbourhood.

Explicitly, to see the closed interval $[0, 1]$ is not compact: fix a convergent sequence of irrationals $0 < c_1 < c_2 < c_3 < \cdots < 1$, with $c_i \to c < 1$ irrational. Then

$$[0, 1] = [0, c_1] \sqcup (c_1, c_2) \sqcup (c_2, c_3) \sqcup \cdots \sqcup (c, 1]$$

is an infinite disjoint open cover, so has no finite subcover.

In this last example, we exploited the fact that $\mathbf{Q}$ has 'holes' at every irrational, i.e. it is not complete. In fact, we have the following result, related to (4.2).

Lemma 4.14. *Any locally compact field K is complete.*

Proof. Let C be a compact neighbourhood of 0, containing an open ball $B(0, \varepsilon)$ for some $\varepsilon > 0$. For any $x \in K$, we have

$$x \in B(x, \varepsilon) \in x + C,$$

so $x+C$ is a compact neighbourhood of x.

Let (x_n) be a Cauchy sequence. Then there exists N such that $|x_N - x_n| < \varepsilon$ for all $n \geqslant N$. In particular, for $n \geqslant N$, (x_n) is contained in

$$B(x_N, \varepsilon) \subset x_N + C.$$

As $x_N + C$ is compact, it is complete; so x_n converges in $x_N + C$, hence in K. $\square$

Proposition 4.15. *Let $(K, |\cdot|)$ be non-archimedean, with $|\cdot|$ non-trivial. The following are equivalent:*

(i) K is locally compact;

(ii) the valuation ring $\mathcal{O}$ is compact;

(iii) K is complete, the associated valuation v is discrete and the residue field $\mathcal{O}/\mathfrak{m}$ is finite.

Recall part (iii) was exactly the definition of non-archimedean local field.

Proof. (i) $\Rightarrow$ (ii): Let C be a compact neighbourhood of 0. By definition C contains an open ball $B(0,\varepsilon)$. Since $|\cdot|$ is non-trivial, there exists $0 \neq x \in K$ with $|x| < 1$, so $|x^n| < \varepsilon$ for $n \gg 0$. Then

$$x^n\mathcal{O} \subset B(0,\varepsilon) \subset C.$$

Since $x^n\mathcal{O}$ is closed, and a subset of a compact set, it is thus compact. But $x^n\mathcal{O}$ is compact if and only if $\mathcal{O}$ is compact (as we can rescale every open set in an open cover by x^n), so $\mathcal{O}$ is compact.

(ii) $\Rightarrow$ (i): If $\mathcal{O}$ is compact, and $x \in K$, then $x + \mathcal{O} \subset K$ is a compact neighbourhood of x.

(ii) $\Rightarrow$ (iii): Suppose $\mathcal{O}$ is a compact metric space. We must check:

(K complete) Since $\mathcal{O}$ is compact, it is complete. Thus K is complete by Lemma 3.13.

(k is finite) Let $0 \neq x \in \mathfrak{m}$. We have an open cover

$$\mathcal{O} = \bigcup_{y \in \mathcal{O}} y + x\mathcal{O}.$$

Since $\mathcal{O}$ is compact, there is a finite subcover

$$\mathcal{O} = \Big(y_1 + x\mathcal{O}\Big) \cup \cdots \cup \Big(y_r + x\mathcal{O}\Big),$$

and hence

$$\mathcal{O}/x\mathcal{O} \subset \{y_1 \pmod{x\mathcal{O}}, \dots, y_r \pmod{x\mathcal{O}}\}$$

is finite. In particular, $k = \mathcal{O}/\mathfrak{m}$ is a quotient of the finite space $\mathcal{O}/x\mathcal{O}$, hence k is finite.

(v is discrete) Suppose the valuation is not discrete. Then valuations are allowed to be arbitrarily small and positive: for all $x \in K$ with $v(x) > 0$, there exists $y \in K$ with

$$v(x) > v(y) > 0.$$

(If not, then there exists an element of minimal positive valuation, say $v(x) = c > 0$. Then $v(K^\times) = c\mathbf{Z}$, so v is discrete). Thus there exists a sequence $x_1, x_2, x_3, \dots \in \mathfrak{m}$ with

$$v(x_1) > v(x_2) > v(x_3) > \cdots > 0.$$

Then by Lemma 3.11 we have

$$x_1\mathcal{O} \subsetneq x_2\mathcal{O} \subsetneq x_3\mathcal{O} \subsetneq \cdots.$$

We claim that $x_1, x_2, x_3, \dots$ all have distinct images in $\mathcal{O}/x_1\mathcal{O}$. Suppose not; then there exist $i < j$ with

$$x_i \equiv x_j \pmod{x_1\mathcal{O}},$$

i.e. $x_i = x_j + rx_1$ for some $r \in \mathcal{O}$. But then $x_j = x_i - rx_1 \in x_i\mathcal{O}$, so $x_j\mathcal{O} \subset x_i\mathcal{O}$. As $i < j$, we know $x_i\mathcal{O} \subset x_j\mathcal{O}$; so $x_i\mathcal{O} = x_j\mathcal{O}$. As $v(x_i) > v(x_j)$, this contradicts Lemma 3.11, and we conclude that all the x_i have distinct images in $\mathcal{O}/x_1\mathcal{O}$.

We conclude since $\mathcal{O}/x_1\mathcal{O}$ is finite (but the argument to prove k finite above, as $x_1 \in \mathfrak{m}$).

(iii) Since these conditions define a non-archimedean local field, this is exactly Corollary 4.12. □

This shows that any non-archimedean field $(K, |\cdot|)$ is a local field if and only if it is locally compact. We arrive at the following cleaner version of our ad-hoc definition:

Corollary 4.16. *Let $(K, |\cdot|)$ be a valued field with $|\cdot|$ non-trivial. Then*

$$K \text{ is a local field} \iff K \text{ is locally compact.}$$

Proof. The non-archimedean case is exactly Proposition 4.15.

If K is an archimedean locally compact field, then K is complete by Lemma 4.14, hence is isomorphic to $\mathbf{R}$ or $\mathbf{C}$ by Ostrowski's (second) theorem (Theorem 2.12). Conversely, $\mathbf{R}$ and $\mathbf{C}$ are locally compact, since the closure of any open ball $B(x, \varepsilon)$ is compact in either, hence a compact neighbourhood of x. □

4.5. (∗) Non-examinable extras

More general inverse limits. *The rest of Chapter 4 is non-examinable.* The inverse limits above are a special case of a more general phenomenon.

Definition. Let I be a partially ordered set such that for all $i, j \in I$, there exists $k \in I$ such that $k \leqslant i, k \leqslant j$. An *inverse system* (over I) is a collection $(A_i)_{i\in I}$ of sets with transition maps

$$f_{ij} : A_j \to A_i \qquad \text{for all } i \leqslant j,$$

with $f_{ii} = \mathrm{id}$, such that

$$i \leqslant j \leqslant k \Rightarrow f_{ik} = f_{ij} \circ f_{jk}.$$

Then the inverse limit of (A_i) is

$$\varprojlim_i A_i = \{(a_i)_{i\in I} : a_i \in A_i, f_{ij}(a_j) = a_i \forall i \leqslant j\}.$$

Examples.

- Let $I = \mathbf{N}$, ordered by divisibility (so $i \leqslant j$ if and only if $i|j$). Let $A_i = \mathbf{Z}/i\mathbf{Z}$, with transition maps $f_{ij} : \mathbf{Z}/j\mathbf{Z} \to \mathbf{Z}/i\mathbf{Z}$ given by reduction $\pmod i$. Then
$$\widehat{\mathbf{Z}} = \varprojlim_i \mathbf{Z}/i\mathbf{Z}.$$
- Let K be a perfect field with algebraic closure $\overline{K}$. Let
$$I = \{L/K \text{ finite Galois}\},$$
ordered by inclusion. Let $A_L := \mathrm{Gal}(L/K)$, with transition maps $f_{L,M}$ given by restriction of Galois automorphisms from M to L. Then
$$\mathrm{Gal}(\overline{K}/K) = \varprojlim_L \mathrm{Gal}(L/K).$$

Metrics on inverse limits. We saw that the inverse limit topology on $\mathbf{Z}_p$ is the same as the topology induced by the p-adic metric; so as a profinite space, $\mathbf{Z}_p$ is a metric space.

This is true more generally: we can define a metric on any profinite space such that the inverse limit topology is the one induced by that metric. To define this, let $A := \varprojlim_n A_n$. For each $x, y \in A$, define aa 'valuation'

$$v(x,y) = \min\{r : x_r \neq y_r \in A_r\} \in \mathbf{Z}_{\geqslant 0},$$

with the understanding that if $x = y$, then $v(x,y) = +\infty$. Then let $0 < \alpha < 1$, and define

$$d(x,y) = \alpha^{-v(x,y)}.$$

This defines a metric that satisfies the ultrametric/non-archimedean triangle inequality: if $x, y, z \in A$, then write $r = v(x,z)$. If $v(x,y) \leqslant r$, then

$$v(x,y) \geqslant v(x,z) \qquad \Rightarrow \qquad d(x,z) \leqslant d(x,y).$$

If $v(x,y) > r$, then $y_r = x_r \neq z_r$, so $v(y,z) \leqslant r$, and then

$$v(x,y) \geqslant v(y,z) \qquad \Rightarrow \qquad d(x,y) \leqslant d(y,z).$$

So we conclude

$$d(x,z) \leqslant \max\{d(x,y), d(x,z)\}.$$

I leave it as an exercise to show that the open balls induced from this metric are exactly the open balls defining the inverse limit topology.

Completions of rings. Via Proposition 4.11, we are led to the following algebraic definition of completion:

Let R be a DVR with maximal ideal $\mathfrak{m}$. The completion *of R, denoted $\widehat{R}$, is*

$$\widehat{R} := \varprojlim_n R/\mathfrak{m}^n.$$

It is an exercise to show that $\widehat{R}$ is a complete DVR, and that $\mathrm{Frac}(\widehat{R})$ is the completion of $\mathrm{Frac}(R)$ with respect to the valuation on R.

Summary so far

We briefly take stock of what we've covered so far. Fluency in this language will be useful for everything that follows.

If $(K, |\cdot|)$ is a valued field, we saw there is a dichotomy between the cases:

- $|\cdot|$ is archimedean (looks like the usual distance function you know and love),
- $|\cdot|$ is non-archimedean (i.e. $|\cdot|$ satisfies the strong triangle inequality, and $|n|$ is bounded for $n \in \mathbf{Z}$).

We studied the archimedean case in Chapter 2, and showed that there aren't many interesting examples: the only complete archimedean fields are $\mathbf{R}$ and $\mathbf{C}$. As such, the focus of this course is on the (much more *arithmetically* interesting) case of non-archimedean fields.

In Chapter 3, we introduced the notion of *valuation ring* $\mathcal{O} \subset K$, and showed (in Lemma 3.7) that a non-archimedean absolute value is equivalent to a valuation v on K. Properties of the field are then reflected in the valuation ring: for example,

$$K \xmapsto{\text{valuation ring}} \mathcal{O} := \{x \in K : |x| \leqslant 1\}$$

$$\begin{array}{ccccc}
\boxed{\text{non-archimedean fields}} & \iff & \boxed{\text{valuation rings}} & \subset & \boxed{\text{local integral domains}} \\
\cup & & \cup & & \\
\boxed{\text{discretely valued fields}} & \iff & \boxed{\text{DVRs}} & & \\
\cup & & \cup & & \\
\boxed{\text{complete discretely valued fields}} & \iff & \boxed{\text{complete DVRs}} & & \\
\cup & & \cup & & \\
\boxed{\text{non-archimedean local fields}} & \iff & \boxed{\text{complete DVRs w/ finite res field}} & &
\end{array}$$

$$\mathrm{Frac}(\mathcal{O}) \longleftarrow\!\shortmid \mathcal{O}$$

In Chapter 4, we then showed that complete DVRs $\mathcal{O}$ are isomorphic to an inverse limits, deduced that if $\mathcal{O}$ has finite residue field, then $\mathcal{O}$ is compact. We used this to deduce that if K is a non-archimedean local field, then K is locally compact, like $\mathbf{R}$ and $\mathbf{C}$, giving a conceptual and uniform classification of local fields.

Part II

Solving Equations over Local Fields

In Part II, we turn to applications of local fields – and, in particular, of p-adic fields – to classical arithmetic questions. We show that it is often rather easy to check when polynomial equations have roots in $\mathbf{Q}_p$, rather analogous to the case over $\mathbf{R}$ (where one simply has to check that a polynomial takes both positive and negative values to ensure it has a root).

Largely this is due to *Hensel's lemma*, which reduces the study of polynomials over $\mathbf{Z}_p$ to the study of polynomials over the residue field $\mathbf{F}_p$. In particular, if $f(X) \in \mathbf{Z}_p[X]$, then one knows that if $f(X) \pmod{p}$ has no roots, then f can have no roots in $\mathbf{Z}_p$. Hensel's lemma gives a converse for many f; if one has a root modulo p, then one can often lift this to a root in $\mathbf{Z}_p$ itself. In Chapter 5, we state and prove Hensel's lemma in the setting of general complete discretely valued fields, and describe some immediate applications. We classify all the quadratic extensions of $\mathbf{Q}_p$, and then describe the *Teichmüller map*, which is (for any non-archimedean local field K) an embedding of the residue field k into K.

In Chapter 6, we describe an application of this theory to the study of *quadratic forms*. We state (but do not prove) the Hasse–Minkowski theorem, one of the first examples of 'local–global' phenomena: a quadratic form over $\mathbf{Q}$ is soluble (i.e. has a zero) if and only if it is soluble in $\mathbf{R}$ and $\mathbf{Q}_p$ for all p. We focus on the applications of this to explicit multivariable polynomials of degree 2, proving simple criteria for many classes of quadratic forms to be soluble over $\mathbf{Q}_p$. We apply these results to prove classical results about the representation of integers as sums of squares: namely we show that any odd prime is a sum of two squares if and only if it is congruent to $1 \pmod{4}$, we classify the numbers that are sums of three squares, and we prove that *every* positive integer is a sum of four squares.

Chapter 5

Hensel's lemma

We now come to an extremely important result in the theory of valuations/non-archimedean absolute values: Hensel's lemma, which provides a simple criterion for a polynomial to have a root in a complete non-archimedean field K.

5.1. Statement and examples of Hensel's lemma

In both of the following theorems, we take:

- K complete with respect to a discrete valuation v, $\mathcal{O} \subset K$ its valuation ring, $\mathfrak{m} = \pi\mathcal{O}$ its maximal ideal, with v normalised so $v(\pi) = 1$;
- $f(X) \in \mathcal{O}[X]$ a polynomial;
- and $\overline{f} = f \pmod{\pi} \in k[X]$.

Theorem 5.1 (Hensel's lemma, version 1)**.** *Suppose $\overline{f}$ has a simple root $\overline{a} \in k$. Then there exists a* unique *$b \in \mathcal{O}$ such that*

$$f(b) = 0, \qquad b \equiv \overline{a} \pmod{\pi}.$$

Here a root is *simple* if $f(a) \equiv 0 \pmod{\mathfrak{m}}$, but $\frac{df}{dX}(a) = f'(a) \not\equiv 0 \pmod{\mathfrak{m}}$, where a is any lift of $\overline{a}$ to $\mathcal{O}$.

Examples. (a) Let $K = \mathbf{Q}_5$, $\mathcal{O} = \mathbf{Z}_5$. Let

$$f(X) = X^2 + 1.$$

Then 2 is a simple root of $\overline{f}(X)$, since

$$f(2) = 5 \equiv 0 \pmod{5}, \quad \text{and} \quad f'(X) = 4 \neq 0 \pmod{5}.$$

By Hensel's lemma, there exists an element $b \in \mathbf{Q}_5$ with $b^2 + 1 = 0$, that is, $X^2 + 1$ has a root in $\mathbf{Q}_5$. (Moreover, one can take $b \equiv 2 \pmod{5}$).

Note this is an example we've already seen: we explicitly constructed such an b in Example 1.16. The proof of Hensel's lemma follows a similar strategy.

(b) Let $p = 13$. The quadratic residues modulo 13 are 1, 3, 4, 9, 10 and 12. Let $N \in \mathbf{Z}$, and consider the polynomial $X^2 - N \in \mathbf{Q}_{13}[X]$. Then:

- If $N \equiv 1, 3, 4, 9, 10, 12 \pmod{13}$, then Hensel's lemma shows that $\sqrt{N} \in \mathbf{Q}_{13}$.

– If $N \equiv 2, 5, 6, 7, 8, 11 \pmod{13}$, then $\sqrt{N} \notin \mathbf{Q}_{13}$. (Suppose it was; then $X^2 - N$ has a root in $\mathbf{Q}_{13}$. Since $\mathbf{Z}_{13}$ is integrally closed by Lemma 3.12, it thus has a root in $\mathbf{Z}_{13}$. Reducing this root modulo 13 would give a root of $X^2 - N \pmod{13}$. But no such root can exist, as N is not a quadratic residue for 13).

As a special case, we see that as $-1 \equiv 12 \pmod{13}$, the polynomial $X^2 + 1$ has a root in $\mathbf{Q}_{13}$, as proved on the first homework sheet.

(c) What about multi-variable equations? Consider the equation $E : Y^2 = X^3 - X$. (Those who took MA4 Elliptic Curves will recognise this equation: it is the congruent number curve). If K is a field, define

$$E(K) = \{(x, y) \in K : y^2 = x^3 - x\}$$

for the set of K-solutions to this equation. It is a fact, that you can deduce from theorems from Elliptic Curves, that

$$E(\mathbf{Q}) = \{(0, 0), (1, 0), (-1, 0)\},$$

that is, the equation has only three rational solutions. However:

Lemma. *The set $E(\mathbf{Q}_5)$ is uncountable.*

Proof. Let $x \in 2 + 5\mathbf{Z}_5$, that is, $x \in \mathbf{Z}_5$ with $x \equiv 2 \pmod 5$. Consider the polynomial

$$f_x(Y) := Y^2 - x^3 + x \in \mathbf{Z}_5[Y],$$

so that $f_x'(Y) = 2Y$. Note that

$$f_x(1) = 1 - x^3 + x \equiv 1 - 2^3 + 2 = 5 \equiv 0 \pmod 5,$$

whilst

$$f_x'(1) = 2 \not\equiv 0 \pmod 5.$$

Hensel's lemma then implies there exists $y \in \mathbf{Z}_5$ with

$$f_x(y) = y^2 - x^3 + x = 0, \qquad y \equiv 1 \pmod 5.$$

In particular, we have

$$(x, y) \in E(\mathbf{Q}_5).$$

There are uncountably many different choices for x, since $2 + 5\mathbf{Z}_5$ is uncountable (see the first exercise sheet); so there are uncountably many elements of $E(\mathbf{Q}_5)$. $\square$

(c′) (non-examinable!) More generally, if $p > 7$ and $E : Y^2 = f(X)$ is an elliptic curve, with $f(X)$ a monic cubic, can you show that $E(\mathbf{Q}_p)$ is always infinite? Hint: use the Hasse bound (from the theory of elliptic curves) to count $E(\mathbf{F}_p)$. (This result is also true for smaller p, but you'd need something more involved from the theory of elliptic curves, e.g. formal groups).

Examples. (A non-example!)

(d) What about the equation $f(X) = X^2 + 7$ in $\mathbf{Q}_2$? Now $\overline{f}'(X) = 2X \equiv 0 \pmod 2$, so there are *no* simple roots, and we cannot apply Theorem 5.1!

In light of the final example, where version 1 of Hensel's lemma didn't apply, we'll actually prove a more general analogue – modulo higher powers of p – which *does* apply in this case.

Note that in version 1, we have

$$\begin{aligned} f(a) \equiv 0 \pmod{\mathfrak{m}} &\iff |f(a)| < 1, \\ f'(a) \not\equiv 0 \pmod{\mathfrak{m}} &\iff |f'(a)| = 1. \end{aligned}$$

We want to also be able to handle the case $f'(a) \equiv 0 \pmod{\mathfrak{m}}$, that is, $|f'(a)| < 1$. This is still possible, provided '$|f(a)|$ is sufficiently smaller than $|f'(a)|$'. More precisely:

Theorem 5.2 (Hensel's lemma, version 2)**.** *Suppose there exists $a \in \mathcal{O}$ with $|f(a)| < |f'(a)|^2$. Then there exists a* unique *$b \in \mathcal{O}$ such that*

$$f(b) = 0, \qquad |b - a| < |f'(a)|. \tag{5.1}$$

Example. (d′) Consider $f(X) = X^2 + 7$ in $\mathbf{Q}_2$ again. Note

$$f(1) = 8,$$

i.e. f has a root modulo 8, and

$$f'(1) = 2 \cdot 1 = 2.$$

Note

$$|f(1)|_2 = 2^{-3} < 2^{-2} = |f'(1)|_2^2,$$

so version 2 of Hensel's lemma applies, and shows there exists $b \in \mathbf{Z}_2$, wiith $b \equiv 1 \pmod 4$, such that $f(b) = b^2 + 7 = 0$.

5.2. Proof of Hensel's lemma

First we show that:

Lemma 5.3. *Theorem 5.2 implies Theorem 5.1.*

Proof. Suppose

$$\overline{f}(\overline{a}) = 0, \qquad \overline{f}'(\overline{a}) \neq 0. \tag{5.2}$$

Let a be any lift of $\overline{a}$ to $\mathcal{O}$. Then (5.2) means

$$f(a) \equiv 0 \pmod{\mathfrak{m}}, \qquad f'(a) \not\equiv 0 \pmod{\mathfrak{m}}.$$

Thus $|f(a)| < 1$ and $|f'(a)|^2 = 1$; so $|f(a)| < |f'(a)|$, and Theorem 5.2 applies. Thus there exists $b \in \mathcal{O}$ with $f(b) = 0$ and $|b - a| < |f'(a)| = 1$. But note $|b - a| < 1$ implies

$$b \equiv a = \overline{a} \pmod{\mathfrak{m}}. \qquad \square$$

It remains to prove Theorem 5.2. We closely follow the strategy used in Example 1.16.

Proof: Existence of b. **First translation.** Note

$$\begin{aligned} |f(a)| < |f'(a)|^2 &\iff v(f(a)) > 2v(f'(a)) \\ &\iff v(f(a)) \geqslant 2v(f'(a)) + 1, \end{aligned}$$

as v is normalised and discrete. Let $r = v(f'(a))$, so $v(f(a)) \geqslant 2r + 1$, that is,

$$f(a) \equiv 0 \pmod{\pi^{2r+1}}.$$

Strategy. Set $b_1 = a$, and construct a sequence $b_n \in \mathcal{O}$ such that:

(i) $f(b_n) \equiv 0 \pmod{\pi^{2r+n}}$,

(ii) $b_{n+1} \equiv b_n \pmod{\pi^{r+n}}$.

Such a sequence is Cauchy by (ii), so tends to a limit $b \in \mathcal{O}$ (as K, hence $\mathcal{O}$, is complete). Note

$$f(b_n) \to 0 \text{ as } n \to \infty$$

by (i). But f is polynomial, hence continuous, so

$$f(b_n) \to f(b).$$

We deduce $f(b) = 0$, as limits are unique.

Also,

$$b_n \equiv b_{n-1} \equiv \cdots \equiv b_1 = a \pmod{\pi^{r+1}} \forall n \qquad \Rightarrow \qquad b \equiv a \pmod{\pi^{r+1}}.$$

Thus $v(b - a) \geqslant r + 1 > r = v(f'(a))$; hence $|b - a| < |f'(a)|$.

Inductive step, set-up. Recall we put $b_1 = a$. We proceed to construct b_n by induction. Suppose b_n has been constructed satisfying (i) and (ii). Then

$$\begin{aligned} \text{(i)} &\Rightarrow f(b_n) \equiv 0 \pmod{\pi^{2r+n}} \\ &\Rightarrow f(b_n) = c_n \pi^{2r+n}, \text{ some } c \in \mathcal{O}. \end{aligned} \tag{5.3}$$

Then let

$$b_{n+1} = b_n + c_n \pi^{r+n}, \qquad \text{some } d_n \in \mathcal{O};$$

by definition this satisfies (ii), so the trick is to choose a good d_n to satisfy (i).

Inductive step: Taylor expansions. We want

$$f(b_{n+1}) = f(b_n + c_n \pi^{r+n}) \equiv 0 \pmod{\pi^{2r+n+1}}.$$

It is natural to consider $f(X + Y)$. For this, we take a Taylor expansion of f.

Claim. *We have*

$$f(X + Y) = f(X) + f'(X)Y + g(X, Y)Y^2, \qquad g(X, Y) \in \mathcal{O}[X, Y].$$

To see this, one can expand $f(X + Y)$ term by term using the binomial theorem. Alternatively, we may collect together the powers of Y, and write

$$\begin{aligned} f(X + Y) &= f_0(X) + f_1(X)Y + f_2(X)Y^2 + \cdots \\ &= f_0(X) + f_1(X)Y + g(X, Y)Y^2, \end{aligned}$$

with $f_i(X), g(X, Y) \in \mathcal{O}[X]$. Then:

- setting $Y = 0$ shows that $f_0(X) = f(X)$,
- and we have
$$\frac{f(X+Y) - f(X)}{Y} = f_1(X) + f_2(X)Y + \cdots ;$$
as $Y \to 0$, the left-hand side tends to $f'(X)$ whilst the right-hand side tends to $f_1(X)$, so
$$f_1(X) = f'(X).$$

This proves the claim.

Now

$$\begin{aligned} f(b_{n+1}) = f(b_n + c_n\pi^{r+n}) &= f(b_n) + f'(b_n)\pi^{r+n} + g(b_n)\pi^{2r+2n} \\ &\equiv f(b_n) + f'(b_n)c_n\pi^{r+n} \pmod{\pi^{2r+n+1}} \qquad \text{(since } n \geqslant 1). \end{aligned}$$

Thus we need to find d_n such that

$$f(b_n) + f'(b_n)c_n\pi^{r+n} \equiv 0 \pmod{\pi^{2r+n+1}}.$$

Inductive step: choosing d_n. By considering (ii), we have

$$b_n \equiv b_{n-1} \equiv \cdots \equiv b_1 \equiv a \pmod{\pi^{r+1}},$$

so we have
$$f'(b_n) \equiv f'(a) \equiv 0 \pmod{\pi^r}, \qquad f'(b_n) \equiv f'(a) \not\equiv 0 \pmod{\pi^{r+1}}.$$

Thus
$$v(f'(b_n)) = v(f'(a)) = r,$$

and we can write $f'(b_n) = u\pi^r$ with $u \in \mathcal{O}^\times$. In particular, substituting (5.3), we deduce

$$f(b_{n+1}) = (c_n + ud_n)\pi^{2r+n} \pmod{\pi^{2r+n+1}}.$$

If we take $d_n = -u^{-1}c_n$, then
$$f(b_{n+1}) \equiv 0 \pmod{\pi^{2r+n+1}},$$

and b_{n+1} satisfies (i).

Thus we've constructed the sequence b_n, and $b_n \to b$ with $f(b) = 0$ and $|b - a| < |f'(a)|$ as above, and the proof of *existence* of b is complete.

All that remains is *uniqueness* of b.

Lemma 5.4. *In Theorem 5.2, the element $b \in \mathcal{O}$ with $f(b) = 0$ and $|b - a| < |f'(a)|$ is unique.*

Proof. Suppose also that $b' \in \mathcal{O}$ with $f(b') = 0$ and $|b' - a| < |f'(a)|$.

$$c \coloneqq b - b'.$$

Suppose $c \neq 0$ (i.e. b and b' are different solutions).

As above, we have

$$b \equiv a \equiv b' \pmod{\pi^{r+1}} \qquad \Rightarrow |f'(a)| = |f'(b)|.$$

So $|b - a|, |b' - a| < |f'(b)|$. Then

$$|c| = |b - b'| = |b - a + a - b'| \leqslant \max(|b - a|, |b' - a|) < |f'(b)|. \tag{5.4}$$

But note that

$$0 = f(b') = f(b-c) = f(b) - f'(b)c + g(b)c^2,$$

with $g(X)$ as above. Note $f(b) = 0$ and $|g(b)| \leqslant 1$, so

$$|f'(b)c| = |g(b)c^2| \leqslant |c|^2.$$

As we assumed $c \neq 0$, we may divide by $|c|$ to get

$$|f'(b)| \leqslant |c|.$$

But this contradicts (5.4), so $c = 0$ and $b = b'$. $\square$

This then completes the proof of Hensel's lemma.

($*$) **Remark.** This is a non-archimedean analogue of the classical Newton–Raphson method: we use an iteration $b_{n+1} = b_n - f(b_n)/f'(b_n)$.

5.3. Quadratic extensions of $\mathbf{Q}_p$

A vital way of studying the arithmetic of a field K is to study its Galois theory, namely, its algebraic extensions. For the field $\mathbf{R}$, this is very straightforward: $\mathbf{R}$ has exactly one algebraic extension, namely the field $\mathbf{C}$, of degree 2. The Galois theory of $\mathbf{Q}$ is, by contrast, very complicated; the inverse Galois problem asks if every finite group is the Galois group of some extension of $\mathbf{Q}$, and this remains an open problem.

We will see that the Galois theory of $\mathbf{Q}_p$ lies somewhere in between. It is much richer than that of $\mathbf{R}$; there are infinitely many algebraic extensions of $\mathbf{Q}_p$. However, we shall later see that it is much simpler than that of $\mathbf{Q}$ (for example, the inverse Galois problem fails over $\mathbf{Q}_p$: we'll prove there is no Galois extension of $\mathbf{Q}_p$ with Galois group Sym_5).

We now make our first steps towards the Galois theory of $\mathbf{Q}_p$ by using Hensel's lemma to completely classify its possible quadratic extensions.

Notation. If R is a ring, and $n \in \mathbf{N}$, let

$$(R^\times)^n := \{r^n : r \in R^\times\} \subset R^\times$$

be the subset of all nth powers in $R^\times$.

Example.

- $(\mathbf{R}^\times)^2 = \{x \in \mathbf{R} : x > 0\}$, so

$$\mathbf{R}^\times/(\mathbf{R}^\times)^2 \cong \{\pm 1\}$$

(and the only non-trivial quadratic extension is $\mathbf{R}(\sqrt{-1})$).

- $\mathbf{Q}^\times/(\mathbf{Q}^\times)^2$ is infinite, since 2,3,5,7,11,... all have distinct images: if $p \neq q$ prime, then p/q is not square in $\mathbf{Q}$!

Proposition 5.5. *Let p be odd.*

(1) We have

$$\mathbf{Z}_p^\times/(\mathbf{Z}_p^\times)^2 \xrightarrow{\sim} \mathbf{F}_p^\times/(\mathbf{F}_p^\times)^2 \cong \mathbf{Z}/2\mathbf{Z}.$$

One may take representatives $\{1, u\}$, where $u \in \mathbf{Z}_p^\times$ is a non-quadratic-residue modulo p.

(2) *We have*

$$\mathbf{Q}_p^\times/(\mathbf{Q}_p^\times)^2 \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

One may take representatives $\{1, p, u, up\}$.

(3) *There are precisely three quadratic extensions of* $\mathbf{Q}_p$*, namely*

$$\mathbf{Q}_p(\sqrt{u}), \mathbf{Q}_p(\sqrt{p}), \text{ and } \mathbf{Q}_p(\sqrt{up}).$$

Proof. (1) Consider the natural map

$$\phi : \mathbf{Z}_p^\times \xrightarrow{(\mathrm{mod}\, p)} \mathbf{F}_p^\times \xrightarrow{(\mathrm{mod}\, (\mathbf{F}_p^\times)^2)} \mathbf{F}_p^\times/(\mathbf{F}_p^\times)^2.$$

Both maps, hence the composition, are surjective, so

$$\mathbf{Z}_p^\times/\ker(\phi) \cong \mathbf{F}_p^\times/(\mathbf{F}_p^\times)^2.$$

Note also that if $b \in \mathbf{Z}_p^\times$, then

$$\begin{aligned} b \in \ker(\phi) &\iff b \pmod p \in (\mathbf{F}_p^\times)^2 \\ &\iff X^2 - b \text{ has a root } \pmod p \\ &\iff X^2 - b \text{ has a root } a \text{ in } \mathbf{Z}_p \text{ with } a \pmod p \in \mathbf{F}_p^\times \\ &\iff X^2 - b \text{ has a root in } \mathbf{Z}_p^\times \\ &\iff b \in (\mathbf{Z}_p^\times)^2. \end{aligned}$$

Here we have used Hensel's lemma version 1, noting that the roots of $X^2 - b \pmod p$ are simple because p is odd. In particular,

$$\ker(\phi) = (\mathbf{Z}_p^\times)^2,$$

and hence

$$\mathbf{Z}_p^\times/(\mathbf{Z}_p^\times)^2 \cong (\mathbf{F}_p^\times)/(\mathbf{F}_p^\times)^2.$$

Note $\mathbf{F}_p^\times$ is cyclic of even order, so $\mathbf{F}_p^\times/(\mathbf{F}_p^\times)^2 \cong \mathbf{Z}/2\mathbf{Z}$; we get an isomorphism

$$\mathbf{Z}_p^\times/(\mathbf{Z}_p^\times)^2 \xrightarrow{\sim} \mathbf{F}_p^\times/(\mathbf{F}_p^\times)^2 \cong \mathbf{Z}/2\mathbf{Z}.$$

Any non-quadratic-residue $u \in \mathbf{Z}_p^\times$ is a representative of the non-trivial element in $\mathbf{Z}/2\mathbf{Z}$.

(2) Note that

$$\mathbf{Q}_p^\times \cong \mathbf{Z}_p^\times \times \mathbf{Z}, \qquad up^r \mapsto (u, r),$$

and that

$$(\mathbf{Q}_p^\times)^2 = (\mathbf{Z}_p^\times)^2 \times 2\mathbf{Z} \subset \mathbf{Z}_p^\times \times \mathbf{Z}.$$

We see that

$$\mathbf{Q}_p^\times/(\mathbf{Q}_p^\times)^2 \cong \left[\mathbf{Z}_p^\times/(\mathbf{Z}_p^\times)^2\right] \times \mathbf{Z}/2\mathbf{Z} \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

The representatives for the middle term are are $(1,0), (1,1), (u,0), (u,1)$, which translate to the elements $1, p, u, up$ respectively. This proves (2).

Any quadratic extension of $\mathbf{Q}_p$ is of the form $\mathbf{Q}_p(\sqrt{\alpha})$, where $\alpha \in \mathbf{Q}_p^\times \backslash (\mathbf{Q}_p^\times)^2$. But if

$$\alpha \equiv \alpha' \pmod{(\mathbf{Q}_p^\times)^2},$$

then $\alpha = \alpha'\beta^2$ for some $\beta \in \mathbf{Q}_p^\times$, and then

$$\mathbf{Q}_p(\sqrt{\alpha}) = \mathbf{Q}_p(\sqrt{\alpha'\beta^2}) = \mathbf{Q}_p(\sqrt{\alpha'}).$$

So the only quadratic extensions are $\mathbf{Q}_p(\sqrt{u}), \mathbf{Q}_p(\sqrt{p}), \mathbf{Q}_p(\sqrt{up})$. $\square$

Example. When $p = 3$, we have 2 is not a quadratic residue mod 3, so the quadratic extensions of $\mathbf{Q}_3$ are

$$\mathbf{Q}_3(\sqrt{2}), \qquad \mathbf{Q}_3(\sqrt{3}), \qquad \mathbf{Q}_3(\sqrt{6}).$$

What if we adjoin other square roots? Consider for example $\mathbf{Q}(\sqrt{17})$. Then as $17 \equiv 2 \pmod 3$ we have $\mathbf{Q}_3(\sqrt{17}) = \mathbf{Q}_3(\sqrt{2})$. Similarly consider adjoining $\sqrt{21}$; write this in the form $21 = vp = 7 \cdot 3$. As $7 \equiv 1 \pmod 3$, it lies in the same class as 1 in $(\mathbf{Z}_3^\times/\mathbf{Z}_3^\times)^2$, i.e. $\mathbf{Q}_3(\sqrt{7}) = \mathbf{Q}_3$, and $\mathbf{Q}_3(\sqrt{21}) = \mathbf{Q}_3(\sqrt{3})$.

Proposition 5.6. *Let $p = 2$.*

(1) We have

$$\mathbf{Z}_2^\times/(\mathbf{Z}_2^\times)^2 \xrightarrow{\sim} (\mathbf{Z}/8\mathbf{Z})^\times \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

(2) We have

$$\mathbf{Q}_p^\times/(\mathbf{Q}_p^\times)^2 \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

One may take representatives $2^r(-1)^s5^t$ for $r, s, t \in \{0, 1\}$.

(3) There are exactly 7 quadratic extensions of $\mathbf{Q}_2$, namely

$$\mathbf{Q}_2(\sqrt{2}), \mathbf{Q}_2(\sqrt{-1}), \mathbf{Q}_2(\sqrt{5}), \mathbf{Q}_2(\sqrt{-2}), \mathbf{Q}_2(\sqrt{-5}), \mathbf{Q}_2(\sqrt{10}), \text{ and } \mathbf{Q}_2(\sqrt{-10}).$$

Proof. (1) Reduction modulo 8 induces a map

$$\phi : \mathbf{Z}_2^\times \longrightarrow (\mathbf{Z}/8\mathbf{Z})^\times.$$

We claim $\ker(\phi) = (\mathbf{Z}_2^\times)^2$.

Let $b \in \mathbf{Z}_2^\times$. Since the quadratic residues modulo 8 are $0, 1, 4$, and b is odd, we have

$$b \in (\mathbf{Z}_2^\times)^2 \qquad \Rightarrow \qquad b \equiv 1 \pmod 8.$$

Conversely, suppose $b \equiv 1 \pmod 8$, and let $f(X) = X^2 - b$. Then

$$|f(1)| \leqslant 2^{-3} < 2^{-2} = |f'(1)|^2.$$

By version 2 of Hensel's lemma, f has a root in $\mathbf{Z}_2$, so $b \in (\mathbf{Z}_2^\times)^2$. Thus

$$\mathbf{Z}_2^\times/(\mathbf{Z}_2^\times)^2 \cong (\mathbf{Z}/8\mathbf{Z})^\times \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

(2) We conclude $\mathbf{Q}_2^\times/(\mathbf{Q}_2^\times)^3 \cong (\mathbf{Z}/2\mathbf{Z})^3$ exactly as in Proposition 5.5(2).

(3) As in Proposition 5.5(3), the quadratic extensions of $\mathbf{Q}_2$ correspond to the 7 non-trivial elements of $\mathbf{Q}_2^\times/(\mathbf{Q}_2^\times)^2$. ☐

We'll return to this theme in later sections, when we have introduced the notion of *ramification*. We can treat more general cases in this manner. The following is another example, which we didn't cover in lectures.

Proposition. *We have $\mathbf{Q}_3^\times/(\mathbf{Q}_3^\times)^3 \cong (\mathbf{Z}/3\mathbf{Z})^2$.*

Proof. Following the strategy above, since $\mathbf{Q}_3 \cong \mathbf{Z}_3^\times \times \mathbf{Z}$, it suffices to prove $\mathbf{Z}_3^\times/(\mathbf{Z}_3^\times)^3 \cong \mathbf{Z}/3\mathbf{Z}$.

We have a composition

$$\mathbf{Z}_3^\times \longrightarrow (\mathbf{Z}/9\mathbf{Z})^\times \longrightarrow (\mathbf{Z}/9\mathbf{Z})^\times/\{\pm 1\} \cong \mathbf{Z}/3\mathbf{Z}.$$

(Note here that the cubes in $(\mathbf{Z}/9\mathbf{Z})^\times$ are ± 1, so this is $(\mathbf{Z}/9\mathbf{Z})^\times/(\mathbf{Z}/9\mathbf{Z})^{\times,3}$). Let $b \in \mathbf{Z}_3^\times$ with $b \equiv 1 \pmod{9}$; then writing $b = 1 + 9c$, we have

$$b^3 \equiv (1+3c)^3 \pmod{27}.$$

Consider $f(X) = X^3 - b$, and note

$$|f(1+3c)| \leqslant 3^{-3} < 3^{-2} = |f'(1+3c)|^2.$$

Thus version 2 of Hensel's lemma shows that f has a root, that is,

$$b \in (\mathbf{Z}_3^\times)^3.$$

If $b' \equiv -1 \pmod{9}$, then $-b' \equiv 1 \pmod{9}$, so $-b' = d^3$ for some d; then $b' = (-d)^3 \in (\mathbf{Z}_3^\times)^3$. In particular, reduction modulo 9 yields an isomorphism

$$\mathbf{Z}_3^\times/(\mathbf{Z}_3^\times)^3 \cong (\mathbf{Z}/9\mathbf{Z})^\times/\{\pm 1\} \cong \mathbf{Z}/3\mathbf{Z},$$

as required. □

5.4. The Teichmüller map

Let K be a non-archimedean local field, recalling that this means

- K is complete with respect to a discrete valuation v, and
- the residue field k of K is finite, say with $\#k = q$.

Let $\mathcal{O}$ be the valuation ring and $\mathfrak{m}$ the maximal ideal in K. Let

$$f(X) = X^q - X \in \mathcal{O}[X].$$

Since $k^\times$ is cyclic of order $q-1$, for each $b \in k$ we have $f(b) = 0$, so

$$f(X) \equiv \prod_{b\in k}(X - b) \pmod{\mathfrak{m}},$$

that is, each $b \in k$ is a simple root of $f \pmod{\mathfrak{m}} \in k[X]$. By Theorem 5.1, for each $b \in k$ there exists a unique $[b] \in \mathcal{O}$ with

$$f([b]) = 0, \qquad [b] \equiv b \pmod{\mathfrak{m}}.$$

Definition 5.7. We call $[b]$ the *Teichmüller representative* of $b \in k$. It is the unique element of $\mathcal{O}$ such that $[b]^q = [b]$ and $[b] \equiv b \pmod{\mathfrak{m}}$.

This yields a map

$$[-] : k \hookrightarrow \mathcal{O},$$

which is injective since $[b] = [b']$ implies $b \equiv [b] = [b'] \equiv b' \pmod{\mathfrak{m}}$ implies $b = b'$ in k.

Lemma 5.8. *The map* $[-] : k \hookrightarrow \mathcal{O}$ *is multiplicative. Hence* $\mathcal{O}^\times$ *contains* $k^\times$ *as a multiplicative subgroup.*

Proof. If $b, c \in k$, we compute that

$$([b][c])^q = [b]^q[c]^q = [b][c], \qquad \text{and} \qquad [b][c] \equiv bc \pmod{\mathfrak{m}}.$$

As $[bc]$ is the *unique* element of $\mathcal{O}$ satisfying both of these properties, we deduce $[b][c] = [bc]$, and $[-]$ is multiplicative.

The second statement is immediate from the first, which says that $k^\times \hookrightarrow \mathcal{O}^\times$ is a homomorphism of groups. □

Let $x \in \mathcal{O}$, and let $\overline{x} \in k$ denote $x \pmod{\mathfrak{m}}$. We have

$$x \equiv \overline{x} \equiv [\overline{x}] \pmod{\mathfrak{m}}.$$

Note $x \in \mathcal{O}^\times$ if and only if $\overline{x} \in k^\times$, so $[\overline{x}] \notin \mathfrak{m}$; and then by Lemma 3.10 we see $[\overline{x}] \in \mathcal{O}^\times$. Then

$$x/[\overline{x}] \equiv 1 \pmod{\mathfrak{m}} \qquad \Rightarrow \qquad x/[\overline{x}] \in 1 + \pi\mathcal{O}.$$

Now, we have:

Corollary 5.9. *The map*

$$\begin{aligned} \phi : \mathcal{O}^\times &\xrightarrow{\sim} k^\times \times \big(1 + \pi\mathcal{O}\big) \\ x &\longmapsto (\overline{x},\ x/[\overline{x}]) \end{aligned}$$

is an isomorphism of groups. (Here $\overline{x} = x \pmod{\pi}$*).*

Proof. Note that ϕ is a homomorphism by Lemma 5.8, since for $x, y \in \mathcal{O}^\times$, we have

$$\begin{aligned} \phi(xy) &= \left(\overline{xy}, \frac{xy}{[xy]}\right) \\ &= \left(\overline{x} \cdot \overline{y}, \frac{xy}{[x][y]}\right) \qquad \text{by Lemma 5.8} \\ &= \left(\overline{x}, \frac{x}{[\overline{x}]}\right) \cdot \left(\overline{y}, \frac{y}{[\overline{y}]}\right) = \phi(x)\phi(y), \end{aligned}$$

since multiplication in $k^\times \times (1 + \pi\mathcal{O})$ is computed pointwise. Also, ϕ is bijective, because the map

$$\begin{aligned} \varphi : k^\times \times (1 + \pi\mathcal{O}) &\longrightarrow \mathcal{O} \\ (y, z) &\longmapsto [y]z \end{aligned}$$

is an inverse to ϕ. □

We are in a position to classify complete discretely valued fields with finite residue fields. The following proves that any local field of characteristic p is of the form $k((T))$ for k finite, showing another part of Theorem 4.3.

Theorem 5.10. *Let* K *be a local field, and suppose* $\mathrm{char}(K) = p > 0$*. Let* k *be the residue field of* K*.*

(1) The map $[-] : k \hookrightarrow K$ *is a field embedding.*

(2) We have $K = k((T))$.

Proof. (1) As $\mathrm{char}(K) > 0$, K is non-archimedean by Corollary 2.5. Thus, by definition, K is complete with respect to a discrete valuation v, and k is finite.

Claim. *We also have* $\mathrm{char}(k) = p$.

Proof of claim: K has characteristic p, then $1 + \cdots + 1 = p = 0$ in K. Thus $p = 0 \pmod{\mathfrak{m}}$, so

$$p \equiv 1 \pmod{m} + \cdots 1 \pmod{m} = 0 \qquad \text{in k},$$

proving the claim. (All the sums here have p terms!)

Since k is finite, we thus have $\#k = q = p^n$ for some n.
Let $b, c \in k$. Then by binomial expansion, we have

$$\begin{aligned}([b] + [c])^q &= [b]^q + p(\cdots) + [c]^q \qquad &&\text{(for some expression } (\cdots)\text{)}\\ &= [b]^q + [c]^q &&\text{(since we are in characteristic } p\text{)}\\ &= [b] + [c],\end{aligned}$$

where the first equality follows from the standard fact that

$$p \text{ divides } \binom{q}{i} \text{ for all } 0 < i < q.$$

Also $[b] + [c] \equiv b + c \pmod{\mathfrak{m}}$. In particular, since $[b + c]$ is the *unique* element of $\mathcal{O}$ satisfying both these properties, we deduce that

$$[b + c] = [b] + [c],$$

i.e. $[-]$ is also *additive*. In particular, the map

$$[-] : k \hookrightarrow K$$

is a field embedding, as claimed.

(2) Recall Proposition 3.21(i): if A is a set of representatives for $\mathcal{O}/\mathfrak{m}$, and π is a uniformiser, then every element $x \in K$ has a unique expression of form

$$x = \sum_{i=m}^{\infty} a_i \pi^i.$$

We may take $A = \mathrm{Im}([-])$, which we identify with k by (1). Then every element of K has the form $\sum_{i=m}^{\infty} a_i \pi^i$ with $a_i \in k$.

Now define a map

$$\begin{aligned} K &\longrightarrow k((T)),\\ \sum a_i \pi^i &\longmapsto \sum a_i T^i.\end{aligned}$$

This is well-defined by uniqueness of π-adic expansion. It is a field homomorphism, which crucially uses the identification of A with k, and which I leave as an exercise. It is injective from the definition. Since K is complete, it is surjective by Proposition 3.21(ii). Thus it is an isomorphism, as required. □

Chapter 6

The Hasse–Minkowski Theorem

We come to a fundamental application of local fields: so-called 'local–global phenomena'. The prototypical example of this is the *Hasse–Minkowski theorem*, which describes when quadratic forms over $\mathbf{Q}$ have rational solutions in terms of simply checked 'local' criteria: namely, a quadratic form has a solution over $\mathbf{Q}$ if and only if it has solutions over $\mathbf{R}$ and $\mathbf{Q}_p$ for all primes p.

We will not prove this theorem, since all existing proofs are rather long and require some deep background material. Instead, we will show its utility in showing that rational quadratic forms do or do not have a rational solution.

Many of the results of this section are proved in full in Serre's book [Ser73].

6.1. Quadratic forms

Let K be a field of characteristic not equal to 2. Let $\mathbf{X} = (X_1, ..., X_n)$.

Definition 6.1. (1) A *quadratic form* over K is a homogeneous polynomial of degree 2 over K, that is, a polynomial of the form

$$Q(\mathbf{X}) = Q(X_1, ..., X_n) = \sum_{i \leqslant j} a_{ij} X_i X_j \in K[X_1, ..., X_n], \qquad a_{ij} \in K.$$

(2) A *diagonal* quadratic form has form

$$Q(\mathbf{X}) = b_1 X_1^2 + \cdots + b_n X_n^2.$$

(3) Two quadratic forms Q, Q' are *equivalent*, denoted $Q \sim Q'$, if there exists a linear change of variables taking Q to Q', i.e. if there exists $P \in \mathrm{GL}_n(K)$ with

$$Q'(\mathbf{X}) = Q(\mathbf{X}P).$$

Example. – Consider $Q(X, Y) = X^2 + 4XY + 7Y^2$ and $Q'(X, Y) = X^2 + 3Y^2$. Completing the square, we get

$$\begin{aligned} Q(X, Y) &= (X + 2Y)^2 + 3Y^2 \\ &= Q'(X + 2Y, Y) \\ &= Q'\left[(X, Y)\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\right], \end{aligned}$$

so Q and Q' are equivalent, with $P = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$. Note Q' is diagonal.

– Consider $Q(X,Y) = XY$ and $Q'(X,Y) = X^2 - Y^2$. Then note

$$Q\left[(X,Y)\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\right] = Q(X-Y, X+Y) = X^2 - Y^2 = Q'(X,Y).$$

Thus Q and Q' are equivalent, with $P = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$. Note Q' is diagonal.

Remark. Note that in this example we saw why we took K not to have characteristic 2: firstly we implicitly divide by 2 whenever we complete the square, and in the second example, the matrix $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has determinant 2, so is invertible only when $\mathrm{Char}(K) \neq 2$!

We'll see that every quadratic form is equivalent to a diagonal one. This needs:

Connection to symmetric matrices. If

$$Q(X_1, ..., X_n) = \sum_{i \leqslant j} a_{ij} X_i X_j$$

is a quadratic form, then we define a symmetric matrix

$$A = \begin{pmatrix} a_{11} & \frac{a_{12}}{2} & \cdots & \frac{a_{1n}}{2} \\ \frac{a_{12}}{2} & a_{22} & & \frac{a_{2n}}{2} \\ \vdots & & \ddots & \vdots \\ \frac{a_{1n}}{2} & \frac{a_{2n}}{2} & \cdots & a_{nn} \end{pmatrix}.$$

Then we have

$$Q(\mathbf{X}) = \mathbf{X}A\mathbf{X}^T = (X_1, ..., X_n)A\begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}.$$

This defines a bijective correspondence

$$\boxed{\text{Symmetric } n \times n \text{ matrices}/K} \longleftrightarrow \boxed{\text{Quadratic forms over } K \text{ in } n \text{ variables}}.$$

Example. Let $Q(X,Y) = X^2 + 5XY + 2Y^2$ over $\mathbf{Q}$. It is represented by the symmetric matrix

$$A = \begin{pmatrix} 1 & \frac{5}{2} \\ \frac{5}{2} & 2 \end{pmatrix},$$

as

$$(X,Y)\begin{pmatrix} 1 & \frac{5}{2} \\ \frac{5}{2} & 2 \end{pmatrix}\begin{pmatrix} X \\ Y \end{pmatrix} = (X,Y)\begin{pmatrix} X + \frac{5}{2}Y \\ \frac{5}{2}X + 2Y \end{pmatrix} = X^2 + 5XY + 2Y^2.$$

Lemma 6.2. *Let Q, Q' be quadratic forms with associated symmetric matrices A, A'. Then $Q \sim Q'$ if and only if there exists $P \in \mathrm{GL}_n(K)$ with*

$$A' = PAP^T.$$

Proof. We have $Q \sim Q'$ if and only if there exists P such that

$$\begin{aligned} \mathbf{X}A'\mathbf{X}^T &= Q'(\mathbf{X}) \\ &= Q(\mathbf{X}P) \\ &= \mathbf{X}PA(\mathbf{X}P)^T \\ &= \mathbf{X}PAP^T\mathbf{X}, \end{aligned}$$

i.e. if and only if $A' = PAP^T$. □

Lemma 6.3. *Every quadratic form is equivalent to a* diagonal *form, i.e. to a form of the shape*

$$b_1X_1^2 + \cdots + b_nX_n^2,$$

where each b_i is in a set of representatives for $K^\times/(K^\times)^2$.

Proof. Let Q be a quadratic form corresponding to a symmetric matrix A.

Recall from linear algebra: if A is a symmetric matrix, then A can be diagonalised: there exists a diagonal matrix

$$D = \begin{pmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{pmatrix}$$

such that

$$A = PDP^T, \qquad P \in \mathrm{GL}_n(K).$$

Then Q is equivalent to

$$b_1X_1^2 + \cdots + b_nX_n^2.$$

Now, replacing X_i with αX_i rescales b_i to $b\alpha^{-2}$ in K, so we can rescale so that all the b_i are in a set of representatives for $K^\times/(K^\times)^2$. $\square$

Example. Consider

$$Q(X,Y,Z) = X^2 + 6Y^2 + 3Z^2 + 4XY + 4YZ + 2XZ.$$

Completing the square to remove the XY: let $W = X + 2Y$. Then eliminating X,

$$\begin{aligned} &= (X+2Y)^2 + 2Y^2 + 3Z^2 + 4YZ + 2XZ \\ &= W^2 + 2Y^2 + 3Z^2 + 4YZ + 2(W-2Y)Z \\ &= W^2 + 2Y^2 + 3Z^2 + 2WZ. \end{aligned}$$

Now complete the square to remove the WZ term, by setting $V = W + Z$. Then

$$\begin{aligned} &= (W+Z)^2 + 2Y^2 + 2Z^2 + 2YZ \\ &= V^2 + 2Y^2 + 2Z^2, \end{aligned}$$

which is now diagonal.

Corollary 6.4. *(i) Any quadratic form over $\mathbf{C}$ is equivalent to one of form $X_1^2 + \cdots + X_n^2$.*

(ii) Any quadratic form over $\mathbf{R}$ is equivalent to one of the form

$$\big(X_1^2 + \cdots X_r^2\big) - \big(X_{r+1}^2 + \cdots + X_n^2\big).$$

(iii) For any positive integer n and any prime p, there are finitely many equivalence classes of quadratic forms in n variables over $\mathbf{Q}_p$.

(iv) For any positive integer n, there are infinitely many equivalence classes of quadratic forms in n variables over $\mathbf{Q}$.

Proof. (1) $\mathbf{C}^\times/(\mathbf{C}^\times)^2 = \{1\}$.

(2) $\mathbf{R}^\times/(\mathbf{R}^\times)^2 = \{\pm 1\}$.

(3) $\mathbf{Q}_p^\times/(\mathbf{Q}_p^\times)^2$ is finite (by Section 5.3).

(4) $\mathbf{Q}^\times/(\mathbf{Q}^\times)^2$ is infinite (e.g. all the primes have distinct images). $\square$

6.2. The Hasse–Minkowski theorem

Definition 6.5. If $c \in K$, we say that a quadratic form Q *represents* c if there exist $x_1, ..., x_n \in K$, not all zero, with

$$Q(x_1, ..., x_n) = c.$$

We say Q is *soluble* if Q represents 0.

Lemma 6.6. *If $Q \sim Q'$, then Q represents c if and only if Q' represents c. In particular, solubility is respected by equivalence.*

Proof. We know $Q'(\mathbf{x}) = Q(\mathbf{x}P)$ for some $P \in \mathrm{GL}_n(K)$. Then

$$Q'(\mathbf{x}) = c \iff Q(\mathbf{x}P) = c. \qquad \square$$

If Q is a quadratic form over K, and L is a field containing K, then we can consider Q as a quadratic form over L in the obvious way.

Theorem 6.7 (Hasse–Minkowski)**.** *Any quadratic form Q over $\mathbf{Q}$ is soluble if and only if it is soluble over $\mathbf{R}$ and $\mathbf{Q}_p$ for all primes p.*

Remark. Clearly if Q is soluble over $\mathbf{Q}$, it is soluble over $\mathbf{R}$ and $\mathbf{Q}_p$ for all p; so the content of the theorem is the converse.

The case of two variables, i.e. $n = 2$, is simple. By Lemmas 6.3 and 6.6 we need only consider the case $Q(X, Y) = aX^2 + bY^2$. For $x, y \neq 0$ we have

$$ax^2 + by^2 = 0 \iff (x/y)^2 = -b/a,$$

so Q is soluble in a field $K/\mathbf{Q}$ if and only if $-b/a \in (K^\times)^2$. We see

$$\begin{aligned} -b/a \in (\mathbf{R}^\times)^2, (\mathbf{Q}_p^\times)^2 &\Rightarrow -b/a > 0 \text{ and } v_p(-b/a) \text{ even for all } p \\ &\Rightarrow -b/a \in (\mathbf{Q}^\times)^2. \end{aligned}$$

The cases of 3 and 4 variables are proved separately, and then the case of $\geqslant 5$ variables is proved by induction. However all of these proofs are beyond the scope of this course, using things such as local class field theory, or are too long to include here. (See e.g. [Cas86], Chapter 11.2, for a proof for 3 variables, using Minkowski's convex body theorem. The full theorem is proved in Serre's *A course in arithmetic*, Chapter IV.).

Hasse–Minkowski then reduces the study of quadratic forms over $\mathbf{Q}$ – which is, as we've already seen, very complicated, with infinitely many equivalence classes for any fixed number of variables – to the study of quadratic forms over $\mathbf{R}$ and $\mathbf{Q}_p$, which is *much* simpler.

6.3. Quadratic forms over R

We've already seen that any quadratic form Q over $\mathbf{R}$ is equivalent to one of form

$$(X_1^2 + \cdots + X_r^2) - (X_{r+1}^2 + \cdots + X_n^2).$$

Definition 6.8.

- The *signature* of Q is the pair $(r, n - r)$.
- We say Q is *definite* if it has signature $(n, 0)$ or $(0, n)$ (so all the coefficients have the same sign) and *indefinite* otherwise (when both positive and negative terms appear).

The following is immediate.

Lemma 6.9. *Q is soluble over $\mathbf{R}$ if and only if it is indefinite.*

6.4. Quadratic forms over complete discretely valued fields

We now show that if Q is a quadratic form over a complete discretely valued field, then there are often similarly simple criteria for Q to be soluble. The results of this section are largely exercises in Hensel's lemma, which we use repeatedly.

Let K be any complete discretely valued field, with valuation v, valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{m} = \pi\mathcal{O}$ and residue field $k = \mathcal{O}/\mathfrak{m}$. If $Q(X_1, ..., X_n) = a_1X_1^2 + \cdots + a_nX_n^2$, then rescaling the X_i by powers of π, we may assume without loss of generality that $v(a_i) \in \{0, 1\}$.

For odd p, the following proposition completely determines when a quadratic form in 3 variables is soluble over $\mathbf{Q}_p$.

Proposition 6.10. *Suppose* $\mathrm{char}(k) \neq 2$. *Let*

$$Q(X, Y, Z) = aX^2 + bY^2 + cZ^2 \in K[X, Y, Z]$$

with $v(a), v(b), v(c) \in \{0, 1\}$. *Then:*

(i) If $v(a) = v(b) = v(c) = 0$ *(i.e.* $a, b, c, \in \mathcal{O}^\times$*), then* Q *is soluble over* K.

(ii) If $v(a) = 1$ *and* $v(b) = v(c) = 0$, *then*

$$Q \text{ is soluble over } K \iff -b/c \text{ is square modulo } \pi.$$

(iii) If $v(a) = v(b) = 1$ *and* $v(c) = 0$, *then*

$$Q \text{ is soluble over } K \iff -a/b \text{ is square modulo } \pi.$$

(iv) If $v(a) = v(b) = v(c) = 1$, *then* Q *is soluble over* K.

Proof. (i) Note

$$aX^2 + bY^2 + cZ^2 \text{ is soluble} \iff -\tfrac{a}{c}X^2 - \tfrac{b}{c}Y^2 - Z^2 \text{ is soluble.}$$

Dividing through by $-c$, we may without loss of generality assume

$$Q(X, Y, Z) = AX^2 + BY^2 - Z^2 \qquad \text{with } A = -\tfrac{a}{c}, B = -\tfrac{b}{c} \in \mathcal{O}^\times.$$

The functions

$$\begin{aligned} \phi_1 : \mathcal{O} \longrightarrow k, \qquad & x \mapsto Ax^2 \pmod{\pi}, \\ \phi_2 : \mathcal{O} \longrightarrow k, \qquad & y \mapsto 1 - By^2 \pmod{\pi} \end{aligned}$$

each have image of size $\frac{q+1}{2}$, where $q = \#k$. By the pigeonhole principle, their images cannot be disjoint (as then $\mathrm{Im}(\phi_1) \cup \mathrm{Im}(\phi_2)$ would have size $q + 1 > q = \#k$), so there exist $x, y \in \mathcal{O}$ such that

$$\phi_1(x) = \phi_2(y) \Rightarrow Ax^2 + By^2 = 1 \pmod{\pi}.$$

Then $Ax^2 + By^2 \in (\mathcal{O}^\times)^2$, i.e. there exists $z \in \mathcal{O}^\times$ such that

$$Ax^2 + By^2 - z^2 = 0,$$

so Q is soluble. (To see that $Ax^2 + By^2 \in (\mathcal{O}^\times)^2$, apply Hensel's lemma to the polynomial $f(T) = T^2 - (Ax^2 + By^2)$. Since $\mathrm{char}(k) \neq 2$, Hensel's lemma implies $f(T)$ has a root z in $\mathcal{O}^\times$. This is the same as the arguments of §5.3, and fluency in this sort of argument is important for the course!)

(ii) Suppose a solution (x, y, z) exists. Then:

Claim. *A solution (x, y, z) exists with $x \in \mathcal{O}$ and $y, z \in \mathcal{O}^\times$.*

To see the claim: note

$$(x, y, z) \text{ is a solution} \iff (\pi x, \pi y, \pi z) \text{ is a solution.}$$

Thus without loss of generality we can scale so that $x, y, z \in \mathcal{O}$, with one of $x, y, z \in \mathcal{O}^\times$.

Suppose $y \in \pi\mathcal{O}$; then

$$0 = ax^2 + by^2 + cz^2 \equiv cz^2 \pmod{\pi},$$

and $v(c) = 0$ implies $z \in \pi\mathcal{O}$. Then

$$0 = ax^2 + by^2 + cz^2 \equiv ax^2 \pmod{\pi^2},$$

from which we deduce $x \in \pi\mathcal{O}$, as $v(a) = 1$. Thus if one of $y, z \in \pi\mathcal{O}$, then all of $x, y, z \in \pi\mathcal{O}$. But we assumed at least one of $x, y, z \in \mathcal{O}^\times$, so we get a contradiction, and $y \in \mathcal{O}^\times$. The same argument shows $z \in \mathcal{O}^\times$, so we deduce the claim.

Now reducing the equation modulo π, we see that

$$(z/y)^2 = -b/c \pmod{\pi},$$

so $-b/c$ is square modulo π.

Conversely, suppose there exists $\overline{w} \in k$ with $\overline{w}^2 \equiv -b/c \pmod{\pi}$. Then Hensel's lemma provides an element $w \in \mathcal{O}$ with $w^2 = -b/c$, and then $Q(0, 1, w) = 0$, so Q is soluble.

(iii) Replacing Z with πW, and dividing by π, we get $Q \sim \frac{a}{\pi}X^2 + \frac{b}{\pi}Y^2 + c\pi W^2$. Reordering X, Y and Z, this puts us in the setting of (ii), from which the result follows.

(iv) If $v(a) = v(b) = v(c) = 1$, then $\pi^{-1}Q$ is soluble by (i), so Q is soluble. $\square$

Corollary 6.11. *Let Q be a quadratic form over $\mathbf{Q}$ in $n \geqslant 3$ variables. Then $\mathbf{Q}$ is soluble over $\mathbf{Q}_p$ for all but finitely many p.*

Proof. Without loss of generality, write $Q(X_1, ..., X_n) = a_1X_1^2 + \cdots + a_nX_n^2$ with $a_i \in \mathbf{Q}^\times$. If $v_p(a_1) = v_p(a_2) = v_p(a_3) = 0$, then Q is soluble over $\mathbf{Q}_p$ by Proposition 6.10(i). This is true of all but finitely many p. $\square$

Corollary 6.12. *Suppose* $\mathrm{char}(k) \neq 2$. *Any quadratic form Q over K in $n \geqslant 5$ variables is soluble.*

Proof. Write $Q(X_1, ..., X_n) = a_1X_1^2 + \cdots + a_nX_n^2$. Rescaling each variable by powers of the uniformiser, without loss of generality we may assume $v(a_i) \in \{0, 1\}$. Since $n \geqslant 5$, there exist three a_i with the same valuation. Thus,without loss of generality, either:

(1) $v(a_1) = v(a_2) = v(a_3) = 0$, in which case by (i) there exist $x_1, x_2, x_3 \in K$ not all zero with $a_1x_1^2 + a_2x_2^2 + a_3x_3^2 = 0$. Then $Q(x_1, x_2, x_3, 0, \dots, 0) = 0$.

(2) $v(a_1) = v(a_2) = v(a_3) = 1$. Then $\pi^{-1}Q$, hence Q, is soluble by (1). $\square$

The condition on the residue characteristic in Corollary 6.12 rules out the cases where K is a finite extension of $\mathbf{Q}_2$, or $K = \mathbf{F}_{2^n}((T))$. However, analogous results are still true for finite extensions of $\mathbf{Q}_2$:

Proposition 6.13. *Let $Q(X, Y, Z) = aX^2 + bY^2 + cZ^2 \in \mathbf{Z}_2[X, Y, Z]$. Suppose $2 \nmid abc$. Then Q is soluble over $\mathbf{Q}_2$ if and only if one of $a + b$, $b + c$ or $c + a$ is divisible by 4.*

Proof. This is a guided exercise on the exercise sheets. □

Proposition 6.14. *Let K be a finite extension of $\mathbf{Q}_2$, and let Q be a quadratic form over K in n variables. If $n \geqslant 5$, then Q is soluble.*

Proof. Omitted for space reasons. A proof of this, using the Hilbert symbol, is contained in Serre's *A course in arithmetic*, Theorem 6 on page 37. □

Corollary 6.15. *Let Q be a quadratic form over $\mathbf{Q}$ in n variables. If $n \geqslant 5$, then Q is soluble if and only if it is indefinite over $\mathbf{R}$.*

Proof. By Corollary 6.12 and Proposition 6.14, Q is soluble over $\mathbf{Q}_p$ for all p. By Lemma 6.9 it is soluble over $\mathbf{R}$ if and only if it is indefinite. The result follows from Hasse–Minkowski. □

6.5. Examples

Example 6.16. Recall the audience-generated example from lecture 1:

$$Q(X,Y,Z,W) = 71X^2 + 7Y^2 + 61Z^2 - 11W^2.$$

- This clearly has a solution over $\mathbf{R}$.
- If $p \neq 2, 71, 7, 61, 11$ then Q has a solution in $\mathbf{Q}_p$ by Proposition 6.10.
- If $p = 71$, then the equation $7Y^2 + 61Z^2 - 11W^2$ has a solution (y,z,w) in $\mathbf{Q}_{71}$ by Proposition 6.10. So $Q(0,y,z,w) = 0$ and Q is soluble over $\mathbf{Q}_{71}$.
- Similarly if $p = 7$, then $71X^2 + 61Z^2 - 11W^2$ is soluble over $\mathbf{Q}_7$; and $71X^2 + Y^2 - 11W^2$ is soluble over $\mathbf{Q}_{61}$; and $71X^2 + 7Y^2 + 61Z^2$ is soluble over $\mathbf{Q}_{11}$.
- When $p = 2$, note that $7 + 61 \equiv 0 \pmod 4$. So $71X^2 + 7Y^2 + 61^2$ is soluble over $\mathbf{Q}_2$ by Proposition 6.13.

Thus Q is soluble over $\mathbf{R}$ and $\mathbf{Q}_p$ for all p, hence over $\mathbf{Q}$!

Example 6.17. – The quadratic form $Q(X,Y,Z) = 3X^2 + 4Y^2 - 5Z^2 = 0$ has no rational solution. Indeed, note $4/5 = 1 \cdot 2^{-1} = 2$ is not square in $\mathbf{F}_3$, so Q is not soluble over $\mathbf{Q}_3$ by Proposition 6.10(ii).

– The quadratic form $Q(X,Y,Z) = 17X^2 + 351Y^2 - 1531Z^2$ is not soluble over $\mathbf{Q}_{17}$ by Proposition 6.10(ii), since $1531/351 \equiv 11 \pmod{17}$ is not a quadratic residue. In particular, it is not soluble over $\mathbf{Q}$.

Example 6.18. Let p be a prime. The quadratic form $Q(X,Y,Z) = X^2 + Y^2 - pZ^2$ is indefinite, so it is soluble over $\mathbf{R}$. It is soluble over $\mathbf{Q}_q$ for all primes $q \neq 2, p$ by Proposition 6.10(i). Now, by Proposition 6.10(2), is is soluble over $\mathbf{Q}_p$ if and only if

$$-\tfrac{1}{1} = -1 \text{ is square } \pmod p \iff \left(\frac{-1}{p}\right) = 1 \iff p \equiv 1 \pmod 4.$$

Finally, if $p \equiv 1 \pmod 4$, then $1 - p \equiv 0 \pmod 4$, so Q is soluble over $\mathbf{Q}_2$ by Proposition 6.13. Thus

$$Q \text{ is soluble over } \mathbf{Q} \iff p \equiv 1 \pmod 4.$$

In particular, if $p \equiv 1 \pmod 4$, then there exist $x, y, z \in \mathbf{Q}$ with $x^2 + y^2 = pz^2$. Moreover $z \neq 0$ (as otherwise $x = y = z = 0$), so rescaling, we deduce $p = (x/z)^2 + (y/z)^2$ is a sum of two rational squares.

6.6. (∗) Sums of squares

This section was not covered in the lectures, and is non-examinable.

The last example of §6.5 shows any prime $p \equiv 1 \pmod 4$ can be written as the sum of two *rational* squares. In fact, the Hasse–Minkowski theorem has applications to more general classical questions about which integers can be written as sums of *integer* squares. For this, we require the following result:

Lemma 6.19 (Davenport–Cassels). *Let $Q(X_1, \dots, X_n) = a_1X_1^2 + \cdots a_nX_n^2 \in \mathbf{Z}[X_1, \dots, X_n]$ be an integral quadratic form. Suppose that:*

$$\text{for every } \mathbf{x} \in \mathbf{Q}^n, \text{ there exists } \mathbf{y} \in \mathbf{Z}^n \text{ such that } Q(\mathbf{x} - \mathbf{y}) < 1. \tag{6.1}$$

Let N be an integer. If Q represents N over $\mathbf{Q}$, then Q represents N over $\mathbf{Z}$.

Explicitly this means:

$$\text{if there exists } \mathbf{q} \in \mathbf{Q}^n \text{ with } Q(\mathbf{q}) = N, \text{ then there exists } \mathbf{z} \in \mathbf{Z}^n \text{ with } Q(\mathbf{z}) = N.$$

Proof. This proof is non-examinable.

Let A be the matrix of Q, so $Q(\mathbf{x}) = \mathbf{x}A\mathbf{x}$.

Let S denote the set of $s \in \mathbf{Z}_{\geqslant 1}$ such that Q represents Ns^2 over $\mathbf{Z}$. This set is non-empty; taking any $\mathbf{q} \in \mathbf{Q}^n$ such that $Q(\mathbf{q}) = N$, there exists $s \in \mathbf{Z}_{\geqslant 1}$ such that $s\mathbf{q} \in \mathbf{Z}^n$, and then $Q(s\mathbf{q}) = s^2N$, whence $s \in S$. Fix t to be the smallest element of S, and fix some $\mathbf{z} \in \mathbf{Z}^n$ such that $Q(\mathbf{z}) = t^2N$. We must show that $t = 1$.

Applying hypothesis (6.1) to $\mathbf{z}/t \in \mathbf{Q}^n$, there exists $\mathbf{y} \in \mathbf{Z}^n$ with

$$\tfrac{\mathbf{z}}{t} - \mathbf{y} = \mathbf{w} \in \mathbf{Q}^n, \qquad Q(\mathbf{w}) < 1.$$

Note that:

$$\text{if } Q(\mathbf{w}) = 0, \text{ then } \mathbf{w} = 0, \text{ hence } \mathbf{z}/t = \mathbf{y} \in \mathbf{Z}^n \text{ and } t = 1, \tag{6.2}$$

where the last statement follows by minimality of t.

Assume then that $0 < Q(\mathbf{w}) < 1$. We prove this case cannot happen by contradicting minimality of t. Define:

$$\begin{aligned} a &:= Q(\mathbf{y}) - N \in \mathbf{Z}, \\ b &:= 2(Nt - \mathbf{z}A\mathbf{y}) \in \mathbf{Z}, \\ t' &:= at + b \in \mathbf{Z}^n, \\ \mathbf{z}' &:= a\mathbf{z} + b\mathbf{y} \in \mathbf{Z}^n. \end{aligned}$$

We claim:

Claim 6.20. *We have $Q(\mathbf{z}') = (t')^2N$, with $t' < t$.*

The lemma follows from this claim; it shows that if $0 < Q(\mathbf{w}) < 1$, then $t' \in S$, which contradicts minimality of t. Thus $Q(\mathbf{w}) = 0$, and $t = 1$ as in (6.2).

Proof of claim: We compute that

$$\begin{aligned} Q(\mathbf{z}') &= (a\mathbf{z} + b\mathbf{y})A(a\mathbf{z} + b\mathbf{y}) \\ &= a^2Q(\mathbf{z}) + 2ab(\mathbf{z}A\mathbf{y}) + b^2Q(\mathbf{y}) \\ &= a^2t^2N + ab(2Nt - b) + b^2(N + a) \\ &= (a^2t^2 + 2abt + b^2)N \\ &= (t')^2N. \end{aligned}$$

Moreover,

$$\begin{aligned} tt' &= at^2 + bt = t^2Q(\mathbf{y}) - Nt^2 + 2Nt^2 - 2t(\mathbf{z}A\mathbf{y}) \\ &= t^2Q(\mathbf{y}) - 2t(\mathbf{z}A\mathbf{y}) + Q(\mathbf{z}) \\ &= (t\mathbf{y} - \mathbf{z})A(t\mathbf{y} - \mathbf{z}) \\ &= t^2Q(\mathbf{w}) < t^2, \end{aligned}$$

since $0 < Q(\mathbf{w}) < 1$. Thus $t' < t$, proving the claim, and hence the lemma. □

Theorem 6.21. *An odd prime p is a sum of two integer squares if and only if $p \equiv 1 \pmod 4$.*

Proof. If $p \equiv 3 \pmod 4$, then p is clearly not a sum of two squares, since the quadratic residues (mod 4) are 0 and 1. If $p \equiv 1 \pmod 4$, then by the examples of §6.5, we deduce:

($*$) there exist $x, y \in \mathbf{Q}$ such that $x^2 + y^2 = p$.

Let
$$R(X, Y) = X^2 + Y^2.$$

This satisfies hypothesis (6.1); if $(x, y) \in \mathbf{Q}^2$, then let (x', y') be the nearest integer point. Then $x - x', y - y' \leqslant 1/2$, so
$$R(x - x', y - y') \leqslant (1/2)^2 + (1/2)^2 < 1.$$

By ($*$), we know R represents p over $\mathbf{Q}$. But then by Lemma 6.19 it represents p over $\mathbf{Z}$, whence there exist $a, b \in \mathbf{Z}$ with $a^2 + b^2 = p$. □

Theorem 6.22. *Let $N \in \mathbf{Z}_{\geqslant 1}$. Then N is a sum of three integer squares* except *if $N = 4^r(8s+7)$, with $r, s \in \mathbf{Z}_{\geqslant 0}$.*

Proof. We can write N uniquely in the form $4^r(8s + t)$, with $t \in \{1, 2, 3, 5, 6, 7\}$. Suppose there exist $a, b, c \in \mathbf{Z}$ with
$$a^2 + b^2 + c^2 = N.$$

We first reduce to the case $r = 0$; for if $r > 0$, then considering quadratic residues mod 8, we see a, b and c must all be even, so we may replace N with $N/4$ and a, b, c with $a/2, b/2, c/2$. Thus N is a sum of three integers squares if and only if $4N$ is, so without loss of generality we suppose $N = 8s + t$.

If $t = 7$, then by considering quadratic residues modulo 8, we see immediately that N cannot be a sum of three integer squares.

Now suppose $t \in \{1, 2, 3, 5, 6\}$,, and consider the quadratic form

$$Q(X, Y, Z, W) = X^2 + Y^2 + Z^2 - NW^2.$$

Claim 6.23. *Q is soluble over $\mathbf{Q}$.*

Proof of claim: Note:

- Q is indefinite, hence soluble over $\mathbf{R}$.
- Q is soluble over $\mathbf{Q}_p$ for all odd p, since $X^2 + Y^2 + Z^2$ is soluble over $\mathbf{Q}_p$ by Proposition 6.10.

To prove the claim, by Hasse–Minkowski it then suffices to prove Q is soluble over $\mathbf{Q}_2$. We treat different cases:

(i) If $t = 1, 5$, then $1 - N \equiv 1 - t \equiv 0 \pmod 4$, and Q is soluble over $\mathbf{Q}_2$ by Proposition 6.13.

(ii) If $t = 2$, let $z, w \in \mathbf{Z}_2^\times$ be arbitrary, and let $f(X) = X^2 + z^2 - Nw^2$. If $x \in \mathbf{Z}_2^\times$, then

$$f(x) = x^2 + z^2 - Nw^2 \equiv 1 + 1 - 2 = 0 \pmod 8,$$

so

$$|f(x)| = 2^{-3} < 2^{-2} = |f'(x)|^2,$$

and Hensel's lemma provides $x' \in \mathbf{Z}_2^\times$ with $f(x') = 0$. Then $Q(x', 0, z, w) = 0$.

(iii) If $t = 6$, then let $y \in \mathbf{Z}_2$ with $v(y) = 1$, and construct $x' \in \mathbf{Z}_2^\times$ with $Q(x', y, z, w) = 0$ as in (ii).

(iv) If $t = 3$, then let $y, z, w \in \mathbf{Z}_2^\times$, and construct $x' \in \mathbf{Z}_2^\times$ with $Q(x', y, z, w) = 0$ as in (ii).

Thus Q is soluble over $\mathbf{Q}_2$, completing the claim.

Since Q is soluble over $\mathbf{Q}$, there exists $0 \neq (x, y, z, w) \in \mathbf{Q}^4$ with $x^2 + y^2 + z^2 = Nw^2$. Moreover $w = 0$, else $x = y = z = 0$, so

$$N = (x/w)^2 + (y/w)^2 + (z/w)^2$$

is a sum of three rational squares. But $R(X, Y, Z) = X^2 + Y^2 + Z^2$ satisfies hypothesis (6.1), since as in the proof of Theorem 6.21, if $(x, y, z) \in \mathbf{Q}^3$ and (x', y', z') is the nearest integer point, then

$$R(x - x', y - y', z - z') \leqslant (1/2)^2 + (1/2)^2 + (1/2)^2 \leqslant 3/4 < 1.$$

Thus N is a sum of three integer squares by Lemma 6.19. □

Theorem 6.24. *Every positive integer N is a sum of four integer squares.*

Proof. We need only treat the case $N = 4^r(8s + 7)$, since otherwise N is a sum of three integer squares (hence 4 integer squares, simply adding 0). As in the proof of Theorem 6.22, we may reduce to the case $r = 0$. Since $N - 1 \equiv 6 \pmod 8$, we have $N - 1 = a^2 + b^2 + c^2$ is a sum of three squares by Theorem 6.22, so

$$N = a^2 + b^2 + c^2 + 1^2$$

is a sum of 4 squares, as required. □

6.7. The Hasse principle $(*)$

Section 6.7 is non-examinable.

Let $f(X_1, ..., X_n) \in \mathbf{Q}[X_1, ..., X_n]$. We say f *fails the Hasse principle* if $f(X_1, ..., X_n) = 0$ has non-trivial solutions over $\mathbf{R}$ and $\mathbf{Q}_p$ for all p, not no solution over $\mathbf{Q}$. Then the Hasse–Minkowski theorem says that if f is homogeneous quadratic, then it never fails the Hasse principle. There are, however, examples of equations that fail: e.g. Selmer showed that

$$f(X, Y, Z) = 3X^3 + 4Y^3 + 5Z^3 = 0$$

has solutions over $\mathbf{R}$ and $\mathbf{Q}_p$ for all p, but no rational solution.

This leads to a very active area of modern research: if a polynomial f fails the Hasse principle, *how badly* does it fail? The *Brauer group*, an object from Galois cohomology, measures this.

Part III

The Galois Theory of Local Fields

In Part III, we focus on extensions of local fields. We've already had a glimpse into the Galois theory of local fields, using Hensel's lemma to classify the quadratic extensions of $\mathbf{Q}_p$. These results already hint that the Galois theory of $\mathbf{Q}_p$ (where there are a small, finite number of quadratic extensions) is simpler than that of $\mathbf{Q}$ (where there are infinitely many quadratic extensions).

A recurring theme in this part is the following set-up:

> *Let $\mathcal{O}_K$ be a domain, and $K = \mathrm{Frac}(\mathcal{O}_K)$. Let L/K be a finite extension, and let $\mathcal{O}_L$ be the integral closure of $\mathcal{O}_K$ in L, i.e.*

$$\begin{tikzcd}
 & L \arrow[d, dash] & \\
K \arrow[ur, dash, "\text{finite}"] \arrow[d, dash] & \mathcal{O}_L & := \text{ integral closure of } \mathcal{O}_K \text{ in } L \\
\mathcal{O}_K \arrow[ur, dash] & &
\end{tikzcd}$$

The settings we most want to consider are where $\mathcal{O}_K$ is the ring of integers in a number field, or a DVR, or a complete DVR; and then we want to study the behaviour of L/K and $\mathcal{O}_L$.

In Chapter 7, we develop a uniform setting in which to study such questions: that of *Dedekind domains*. This is a particularly nice class of rings that share many properties with rings of integers in number fields, such as unique factorisation. Very importantly, Dedekind domains come with a class of non-archimedean absolute values indexed by prime ideals, generalising the p-adic absolute value on $\mathbf{Q}$. We show that when $\mathcal{O}_K$ is a Dedekind domain, then so is $\mathcal{O}_L$.

In Chapter 8, we refine our study to the case where $\mathcal{O}_K$ is a complete DVR. We show that in this case, $\mathcal{O}_L$ is also a complete DVR; and in particular, there is a *unique* extension of the absolute value $|\cdot|_K$ to $|\cdot|_L$, and finite extensions of complete fields are complete. We use this to complete the classification of local fields by showing that every non-archimedean local field of characteristic 0 is a finite extension of $\mathbf{Q}_p$.

In Chapter 9, we study ramification in extensions. The ramification and residue indices are integers attached to extensions, that are multiplicative in towers, and which allow a refined study of Galois extensions, describing how primes of the smaller field behave in the larger field. By studying ramification we are able to prove that any local field admits only finitely many extensions of degree n for any n, and show that the Galois group of a local field is always solvable (hence e.g. there is no S_5 extension of $\mathbf{Q}_p$).

In Chapter 10, we give an arithmetic application of our study to prove the *Kronecker–Weber theorem*, namely that any abelian extension of $\mathbf{Q}$ is a subfield of a cyclotomic field. This is one of the founding results of Class Field Theory, one of the crowning glories of 20th century mathematics.

Chapter 7

Dedekind domains

We now introduce *Dedekind domains*, a particularly nice class of rings that includes rings of integers in number fields. Crucially, the valuation ring in a complete discretely valued field is a Dedekind domain, so the language of Dedekind domains gives us a uniform way to study rings of integers in number fields *and* in finite extensions of $\mathbf{Q}_p$.

7.1. Definition and important properties

In this section, we'll define Dedekind domains, state (without proof!) two fundamental theorems about them, and give some examples. We'll also define valuations on (the fraction field of) Dedekind domains, and hence see why they're so important in the context of this course.

Definition 7.1. A *Dedekind domain* is a ring R such that:

(i) R is an integral domain (so in particular R is commutative with 1),

(ii) R is Noetherian,

(iii) R is integrally closed (in $\mathrm{Frac}(R)$),

(iv) every non-zero prime ideal of R is maximal.

($*$) **Remark.** The last property is the same as R having Krull dimension 1. So in the language of Grothendieck, Dedekind domains are rings of functions on well-behaved algebraic curves (algebro-geometric spaces of dimension 1).

Examples.
- $\mathbf{Z}$ is a Dedekind domain.
- The ring of integers in a number field is a Dedekind domain. (You proved all the relevant properties in MA3 Algebraic Number Theory).
- More generally: any PID is a Dedekind domain. We prove this in a lemma below; the proof is non-examinable!
- In particular, we deduce immediately that any DVR is a Dedekind domain (since DVRs are PIDs).
- ($*$, non-examinable) For those of a geometric mindset: the co-ordinate ring $k[C]$ of a smooth curve C over a field k is a Dedekind domain (e.g. the co-ordinate ring of an elliptic curve).

Lemma. *If R is a PID, it is a Dedekind domain.*

Proof. We must check:

R **is a domain** by definition.

R **is Noetherian,** since every ideal in R is principal, hence finitely generated.

R **is integrally closed.** Suppose $x = r/s$ in $\mathrm{Frac}(R)$ is a root of a monic polynomial

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in R[X].$$

We must show that $r/s \in R$, that is, that $s \in R^\times$. Since PIDs are UFDs, we may divide by common factors and assume $\mathrm{GCD}(r, s) = 1$. Now we have

$$\left(\tfrac{r}{s}\right)^n + a_{n-1}\left(\tfrac{r}{s}\right)^{n-1} + \cdots + a_0 = 0,$$

and clearing denominators

$$r^s = -(a_{n-1}r^{n-1}s + a_{n-2}r^{n-2}s^2 + \cdots + a_0 s^n).$$

Since s divides the right-hand side, it divides the left-hand side. But since $\mathrm{GCD}(r, s) = 1$, this is only possible if $s \in R^\times$, as required.

Every non-zero prime of R is maximal. Let $0 \neq \mathfrak{p} \subset R$ be prime, and write $\mathfrak{p} = \pi R$. There exists a maximal ideal $\mathfrak{m} = mR$ with $\mathfrak{p} \subset \mathfrak{m}$, that is, $\pi = mr$ for $r \in R$. Since $\mathfrak{p}$ is prime, either m is a unit or r is a unit; but m is not a unit! Hence r is a unit, and $\mathfrak{p} = \pi R = mR = \mathfrak{m}$ is maximal. ☐

Perhaps *the* most important justification for this definition is the following theorem:

Theorem 7.2. *Let R be a Dedekind domain. Then every non-zero ideal can be written uniquely as a product of prime ideals*

$$I = \mathfrak{p}_1^{r_1} \cdots \mathfrak{p}_m^{r_m}, \qquad \mathfrak{p}_i \subset R \textit{ prime}, \qquad r_i \in \mathbf{Z}.$$

Proof. We omit this proof. However, you might like to consider the following exercise:

(1) Go to your MA3 Algebraic Number Theory notes (or equivalent).

(2) Observe just how many times the name 'Dedekind' occurs: this is not a coincidence!

(3) Find the proof of unique factorisation of ideals in the ring of integers $\mathcal{O}_K$ of a number field K. Go through the proof, and identify precisely where we use that: (a) $\mathcal{O}_K$ is Noetherian; (b) $\mathcal{O}_K$ is integrally closed[1] in $K = \mathrm{Frac}(\mathcal{O}_K)$ (which only makes sense if $\mathcal{O}_K$ is a domain); and (c) where every non-zero prime ideal of $\mathcal{O}_K$ is maximal. Convince yourself that *exactly the same proof* works if you replaced $\mathcal{O}_K$ with a general Dedekind domain.

(4) Feel satisfied that you now understand where the definition of Dedekind domain comes from: essentially, the definition is set up so this theorem is true! ☐

Why is this theorem so important? If R is a Dedekind domain, it gives us a ready class of valuations on $K = \mathrm{Frac}(R)$!

Definition 7.3. If R is a Dedekind domain, and $\mathfrak{p} \subset R$ a prime ideal. Define a normalised discrete valuation $v_{\mathfrak{p}}$ on $K = \mathrm{Frac}(R)$ as follows: for $x \in K^\times$, use Theorem 7.2 to factorise the principal ideal

$$(x) = \mathfrak{p}^r \cdot \mathfrak{q}_1^{s_1} \cdots \mathfrak{q}_m^{s_m},$$

[1] Hint: in Samir's notes from 2018, this is strongly related to the 'integral stability lemma'.

where $\mathfrak{p} \neq \mathfrak{q}_i$. Then define

$$v_{\mathfrak{p}}(x) := r.$$

Thus for any $0 < \alpha < 1$, we get an associated non-archimedean absolute value $|\cdot|_{\mathfrak{p}}$ by Lemma 3.7, i.e. $|x|_{\mathfrak{p}} = \alpha^{v_{\mathfrak{p}}(x)}$. Note that

$$|x|_{\mathfrak{p}} < 1 \iff v_{\mathfrak{p}}(x) > 0 \iff \mathfrak{p}|(x) \iff x \in \mathfrak{p}. \tag{7.1}$$

Remark. This directly generalises the $\mathfrak{p}$-adic absolute value on a number field to the general Dedekind setting.

Recall $\mathbf{Z}$ is a Dedekind domain; and its localisation $\mathbf{Z}_{(p)}$ at any non-zero prime ideal is a DVR. In fact, this is true much more generally. The following further connects Dedekind domains to the objects we've been studying throughout the course.

Lemma 7.4. *Let R be a Dedekind domain with $K = \mathrm{Frac}(R)$, let and $\mathfrak{p} \subset R$ any non-zero prime ideal. Then we have*

$$\begin{aligned}\Big[\textit{the localisation } R_{\mathfrak{p}}\Big] &= \{x \in K : v_{\mathfrak{p}} \geqslant 0\} \\ &= \Big[\textit{valuation ring } \mathcal{O} \textit{ of } v_{\mathfrak{p}} \textit{ in } K\Big].\end{aligned}$$

Proof. First, we need a claim. Let $\pi \in \mathfrak{p}\backslash\mathfrak{p}^2$. Then:

Claim. *Let $0 \neq x \in K$ with $v_{\mathfrak{p}}(x) = n$. There exist $r, s \in R$ with $v_{\mathfrak{p}}(r) = v_{\mathfrak{p}}(s) = 0$ such that*

$$x = \pi^n \frac{r}{s}.$$

To see the claim: using unique factorisation, write

$$xR = \mathfrak{p}^n \cdot \mathfrak{q}_1^{r_1} \cdots q_m^{r_m}.$$

We assume the r_i's are all non-zero (else we just omit the prime $\mathfrak{q}_i$ from the factorisation). Letting

$$I = \prod_{i:r_i>0} \mathfrak{q}_i^{r_i}, \qquad J = \prod_{i:r_i<0} \mathfrak{q}_i^{-r_i},$$

we see that $I, J \subset R$ are ideals prime to $\mathfrak{p}$ and

$$xR = \mathfrak{p}^n \cdot IJ^{-1}.$$

Similarly, we can write

$$\pi^n R = \mathfrak{p}^n K,$$

where $K \subset R$ is an integral ideal prime to $\mathfrak{p}$. Then

$$\frac{x}{\pi^n} R = I \cdot (JK)^{-1}.$$

We want to remove the $(JK)^{-1}$. By the Chinese Remainder Theorem (see Algebra 2 or Algebraic Number Theory), there exists $s \in R$ such that

$$s \in JK, \qquad s \notin \mathfrak{p}.$$

Then

$$sR = JK \cdot M, \qquad M \text{ prime to } \mathfrak{p},$$

and

$$s\frac{x}{\pi^n}R = JKM \cdot I(JK)^{-1} = IM \subset R.$$

We deduce that

$$xR \subset \frac{\pi^n}{s}R \qquad \Rightarrow \qquad x = \pi^n \frac{r}{s}, \qquad \text{some } r \in R.$$

Considering valuations, $v_{\mathfrak{p}}(r) = v_{\mathfrak{p}}(x) - n + v_{\mathfrak{p}}(s) = n - n + 0 = 0$. This proves the claim

To complete the proof, note

$$x \in \mathcal{O} \iff n \geqslant 0 \iff \pi^n r \in R \iff x \in R_{\mathfrak{p}},$$

as claimed. □

Proposition 7.5. *Let R be a Dedekind domain, and $\mathfrak{p} \subset R$ any non-zero prime ideal. Then the localisation $R_{\mathfrak{p}}$ is a DVR.*

Proof. Attached to $\mathfrak{p}$ we have a normalised discrete valuation $v_{\mathfrak{p}}$ on $K = \mathrm{Frac}(R)$ by above. Then $R_{\mathfrak{p}} \subset K$ is the valuation ring. But this is a DVR by Lemma 3.19. □

Corollary 7.6. *Let R be a ring. Then*

$$R \text{ is a DVR} \iff R \text{ is a Dedekind domain with exactly one non-zero prime ideal.}$$

Proof. If R is a DVR, then:

- R is a PID, hence a Dedekind domain, so every prime is maximal;
- if there were two non-zero prime ideals, then both would be maximal;
- but DVRs are local, so there is only one maximal ideal, so only one non-zero prime.

Conversely, suppose R is a Dedekind domain with precisely one non-zero prime $\mathfrak{p}$. Then R is local, and hence $R = R_{\mathfrak{p}}$ is equal to its localisation. But the localisation is a DVR by Proposition 7.5. □

Remark 7.7. Let R be a Dedekind domain. We've seen that for any prime $\mathfrak{p} \subset R$, the localisation $R_{\mathfrak{p}}$ is a DVR with discrete valuation $v_{\mathfrak{p}}$. In general, this DVR will *not* be complete, but we can always pass to a completion.

We thus have a **(very important)** picture:

$$\begin{array}{ccccc} \boxed{\text{Dedekind domains}} & \supset & \boxed{\text{DVRs}} & \supset & \boxed{\text{Complete DVRs}} \\ R & \overset{\text{localisation}}{\dashrightarrow} & R_{\mathfrak{p}} & \overset{\text{completion}}{\dashrightarrow} & \widehat{R}_{\mathfrak{p}}. \end{array}$$

The familiar examples of these rings to always keep in mind are

$$\mathbf{Z} \xrightarrow{\text{localisation at } (p)} \mathbf{Z}_{(p)} \xrightarrow{\text{completion}} \mathbf{Z}_p.$$

(∗) For the future, it is also worth keeping in mind that localisation does not change the fraction field (e.g. both $\mathbf{Z}$ and $\mathbf{Z}_{(p)}$ have fraction field $\mathbf{Q}$), so when studying an extension L/K of fields, we are often free to replace a Dedekind $\mathcal{O}_K$ with its localisation at a prime and assume $\mathcal{O}_K$ is a DVR. However, completion *drastically* enlarges the fraction field (e.g. $\mathbf{Z}_p$ has fraction field $\mathbf{Q}_p$, which is uncountable). We shall see that this process rather simplifies extensions. We will expand on all of this in later sections.

7.1.1. $(*)$ **Non-examinable: alternative proof of unique factorisation.** Proposition 7.5 tells us that being a Dedekind domain is a local property: if R is a Dedekind domain, then all its localisations $R_{\mathfrak{p}}$ are Dedekind domains.

In fact, one can prove these results even without using the unique factorisation of ideals. Even further, one can use this to give a new proof of unique factorisation! In this (non-examinable) subsection, we sketch a proof of this. First, we give a new proof of Corollary 7.6, which said that:

> *Let R be a ring. Then R is a DVR if and only if R is a Dedekind domain with exactly one non-zero prime ideal.*

Our previous proof of '$\Rightarrow$' is still valid. We now give a new proof of '$\Leftarrow$' that does not require unique factorisation. We need some lemmas:

Lemma A. *Let R be a Noetherian ring, and $I \subset R$ a non-zero ideal. There exists a non-empty finite set of prime ideals $\{\mathfrak{p}_1, \dots, \mathfrak{p}_m\}$ with $\mathfrak{p}_1 \cdots \mathfrak{p}_m \subset I$. (Note the $\mathfrak{p}_i$ need not be distinct).*

Proof. Suppose not. Then there exists an ideal I that is maximal with the property that no product of prime ideals is contained in I. In particular, I itself is not prime, so there exist $x, y \notin I$ with $xy \in I$. We thus have $I \subsetneq I + (x)$ and $I \subsetneq I + (y)$, and by maximality of I there exist primes $\mathfrak{p}_i, \mathfrak{q}_j$ with

$$\mathfrak{p}_1 \cdots \mathfrak{p}_m \subset I + (x), \qquad \mathfrak{q}_1 \cdots \mathfrak{q}_n \subset I + (y).$$

But then $\mathfrak{p}_1 \cdots \mathfrak{p}_m \mathfrak{q}_1 \cdots \mathfrak{q}_n \subset (I + (x))(I + (y)) = I$, which is a contradiction. $\square$

Lemma B. *Let R be an integrally closed domain, and $I \subset R$ a non-zero finitely generated ideal. Let $K = \mathrm{Frac}(R)$, and $a \in K$. If $aI \subset I$, then $x \in R$.*

Proof. Let $I = (r_1, \dots, r_n)$ for some $r_i \in R$. Then $ar_j = \sum_{i=1}^n c_{ij} r_i$, for some $c_{ij} \in R$.

Consider the $n \times n$ matrix

$$B(X) = X \cdot \mathrm{Id}_n - A, \qquad A_{ij} = a_{ij},$$

and let $f(X) = \det(B(X))$, a monic polynomial in $R[X]$. By construction, we have $B(a) \cdot r = 0$ for $r = (r_1, \dots, r_n)^t$. In particular, multiplying by $\mathrm{ad}(B(a))$ shows that $\det(B(a))c = f(a)c = 0$, and since R is a domain, we must have $f(a) = 0$. But R is integrally closed, so $a \in R$. $\square$

Proof of Corollary 7.6, $\Leftarrow$: Let R be a Dedekind domain with exactly one non-zero prime; then R is local, with unique maximal ideal $\mathfrak{m} \neq 0$. Note that if $I \subset R$ is an ideal, then either $I \subset \mathfrak{m}$ or $I = R$.

Claim. $\mathfrak{m}$ *is principal.*

Proof of claim. Let $0 \leqslant x \in \mathfrak{m}$. By Lemma A, we know that there exists n with

$$\mathfrak{m}^n \subset (x).$$

Choose such n minimal, i.e. $\mathfrak{m}^n \subset (x)$ but $\mathfrak{m}^{n-1} \not\subset (x)$. Fix an element

$$y \in \mathfrak{m}^{n-1} \backslash (x).$$

Then

$$y\mathfrak{m} \subset \mathfrak{m}^n \subset (x),$$

so, letting $\pi = x/y$, we have

$$\pi^{-1}\mathfrak{m} = \frac{y}{x}\mathfrak{m} \subset R.$$

If $\pi^{-1}\mathfrak{m} \subset \mathfrak{m}$, then Lemma B implies $y/x \in R$, so that $y \in (x)$, contradiction; so $\pi^{-1}\mathfrak{m} = R$. In particular, $\mathfrak{m} = (\pi)$ is principal, and the claim is proved.

Now we show R is a PID. Let $I \subset R$ be a non-zero ideal. Consider the chain of ideals

$$I \subset \pi^{-1}I \subset \pi^{-2}I \subset \cdots .$$

All of these inclusions are strict: indeed, if $\pi^{-n}I = \pi^{-n-1}I$, then Lemma B implies $\pi^{-1} \in R$, whence $\pi \in R^\times$ is a unit, contradiction as it generates $\mathfrak{m}$. Now since R is Noetherian, there is a maximal r such that $\pi^{-n}I \subset R$.

Now if $\pi^{-n}I \subset \mathfrak{m} = (\pi)$, then $\pi^{-(n+1)}I \subset R$, contradicting maximality of n; so $\pi^{-n}I = R$, and $I = \pi^n R$ is principal. Hence R is a PID, and thus a DVR, as required. This completes the (new!) proof of Corollary 7.6. □

We now sketch how Corollary 7.6 implies Theorem 7.2. If $\mathfrak{p} \subset R$ is a prime, and I an ideal, then we have an analogous 'localised' ideal $I_\mathfrak{p} \subset R_\mathfrak{p}$ in the localisation of R at $\mathfrak{p}$. We use the following facts:

(1) If $I \not\subset \mathfrak{p}$, then $I_\mathfrak{p} = R_\mathfrak{p}$.

(2) Let $I, J \subset R$ be two ideals. Then $I = J$ if and only if $I_\mathfrak{p} = J_\mathfrak{p}$ for all primes $\mathfrak{p}$.

(3) If R is Dedekind, then $R_\mathfrak{p}$ is Dedekind.

Combining (3) with Corollary 7.6, we see that if R is Dedekind, then every local ring $R_\mathfrak{p}$ is a DVR.

Now let R be Dedekind, and $I \subset R$ a non-zero ideal. By Lemma A, there is a product of primes $\mathfrak{p}_1^{\beta_1} \cdots \mathfrak{p}_r^{\beta_r} \subset I$. One can check that for each prime $\mathfrak{q}$ o R, we have

$$I_\mathfrak{q} = R_\mathfrak{q} \text{ if } \mathfrak{q} \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\},$$

and

$$I_{\mathfrak{p}_i} = \mathfrak{p}^{s_i} \text{ for some } 0 \leqslant \alpha_i \leqslant \beta_i.$$

In particular, by (2), we have $I = \mathfrak{p}_1^{\alpha_1} \cdots \mathfrak{p}_r^{\alpha_r}$, showing existence.

=Now if $I = \prod \mathfrak{p}^{\alpha_\mathfrak{p}} = \prod \mathfrak{p}^{\beta_\mathfrak{p}}$, then localising at $\mathfrak{p}$ we see $I_\mathfrak{p} = \mathfrak{p}^{\alpha_\mathfrak{p}} = \mathfrak{p}^{\beta_\mathfrak{p}}$, i.e. $\alpha_\mathfrak{p} = \beta_\mathfrak{p}$. This is true for all primes, so the prime factorisation is unique.

This ends the non-examinable material.

7.2. Prime ideals in extensions

We now consider the situation described in the introduction to Part III, namely

$$\begin{tikzcd} & L \arrow[dd, no head] \\ K \arrow[ur, no head, "\text{finite}"] \arrow[dd, no head] & \\ & \mathcal{O}_L \quad := \text{ integral closure of } \mathcal{O}_K \text{ in } L \\ \mathcal{O}_K \arrow[ur, no head] & \end{tikzcd} \tag{7.2}$$

We have a very natural result: if the domain $\mathcal{O}_K$ we start with is Dedekind, then so is the extension $\mathcal{O}_L$.

Theorem 7.8. *Let $\mathcal{O}_K$ be a Dedekind domain, and $K = \mathrm{Frac}(\mathcal{O}_K)$. Let L/K be a finite extension. Let $\mathcal{O}_L$ be the integral closure of $\mathcal{O}_K$ in L. Then $\mathcal{O}_L$ is a Dedekind domain.*

Remark. The proof of this theorem (at least, for separable L/K) is not particularly hard, but it is quite long, so we omit it from the lectures/examinable material. We give the proof for separable L/K in a non-examinable appendix to this section.

Example. Note that $\mathbf{Z}$ is a Dedekind domain. We take $\mathcal{O}_K = \mathbf{Z}$ in the theorem. Then:

- $K = \mathbf{Q}$,
- L is a number field,
- and (by definition) $\mathcal{O}_L$ is the ring of integers in L.

So the theorem tells us that the ring of integers in a number field is a Dedekind domain.

Remark. One is led to a natural question: *If $\mathcal{O}_K$ is a DVR, is $\mathcal{O}_L$ a DVR?*

In general, the answer is *no*. For example, localising $\mathbf{Z}$ at $p = 5$ gives a DVR $\mathcal{O}_K = \mathbf{Z}_{(5)}$ with $K = \mathbf{Q}$. Let $L = \mathbf{Q}(i)$; then $\mathcal{O}_L = \mathbf{Z}_{(5)}[i]$, i.e. we have

$$\begin{tikzcd}
 & \mathbf{Q}(i) \arrow[dl, dash] \arrow[dd, dash] & & & \\
\mathbf{Q} \arrow[dd, dash] & & & & \\
 & \mathbf{Z}_{(5)}[i] \arrow[dl, dash] & \supset & (2+i) \arrow[d, dash] & (2-i) \arrow[dl, dash] \\
\mathbf{Z}_{(5)} & & \supset & (5) &
\end{tikzcd}$$

Then p splits into two primes in L, generated by $2+i$ and $2-i$, and these elements generate distinct maximal ideals in $\mathcal{O}_L$. Thus $\mathcal{O}_L$ is not local, so is not a DVR.

This counterexample suggests immediately, however, that understanding the behaviour of *prime ideals* in extensions is crucial to understanding extensions of DVRs.

For the rest of the chapter, assume $\mathcal{O}_K$ is a Dedekind domain. We have the following generalisation of a notion you will have seen in MA3 Algebraic Number Theory. Let $\mathfrak{p} \subset \mathcal{O}_K$ be a prime ideal. Then we have an ideal

$$\mathfrak{p}\mathcal{O}_L = \{xy : x \in \mathfrak{p} \subset \mathcal{O}_K \subset \mathcal{O}_L, y \in \mathcal{O}_L\} \subset \mathcal{O}_L,$$

and by Theorem 7.2, we know that this factorises uniquely as a product

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r},$$

where the $\mathfrak{P}_i$ are distinct prime ideals of $\mathcal{O}_L$. Understanding the properties of this factorisation is fundamental in algebraic number theory. We introduce the following notion:

Definition 7.9. (i) If $\mathfrak{P}$ is a prime of $\mathcal{O}_L$ appearing in the factorisation, we say $\mathfrak{P}$ *lies above* $\mathfrak{p}$, and denote this $\mathfrak{P}|\mathfrak{p}$.

(ii) We write
$$e(\mathfrak{P}_i/\mathfrak{p}) := e_i,$$
and call it the *ramification index of* $\mathfrak{P}/\mathfrak{p}$.

(iii) We let
$$f_i = f(\mathfrak{P}_i/\mathfrak{p}) := [\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}],$$
and call it the *residue index* of $\mathfrak{P}_i/\mathfrak{p}$.

These notions are intimately related by the following formula, which is closely related to the Dedekind–Kummer theorem from Algebraic Number Thory.

Theorem 7.10. *We have*
$$\sum_{i=1}^{r} e_i f_i = [L : K]. \tag{7.3}$$

Proof. The general result is not that difficult, but we do not have time to cover it in the lectures. The general proof is thus non-examinable and is contained in the next subsection. We remark:

- In the special case that $\mathcal{O}_K$ is a complete DVR, this will be proved in the next chapter.
- In the special case where $K = \mathbf{Q}$ and L is a number field, for all but finitely many primes p this is implied by the Dedekind–Kummer theorem, which we recall below.

□

Recall from Algebraic Number Theory that one may compute these quantities by using the *Dedekind–Kummmer theorem*, which we now recap.

Theorem 7.11 (Dedekind–Kummer)**.** *Let* $L = \mathbf{Q}[X]/(f) = \mathbf{Q}(\alpha)$ *be a number field, where* α *is a root of an irreducible polynomial* $f(X) \in \mathbf{Z}[X]$*. Let* $\mathcal{O}_L$ *be the ring of integers in* L*, and suppose*
$$p \nmid [\mathcal{O}_L : \mathbf{Z}[\alpha]]. \tag{7.4}$$
If f *factorises* $(\mathrm{mod}\, p)$ *into irreducibles as*
$$f(X) \equiv f_1^{e_1}(X) \cdots f_r^{e_r}(X) \, (\mathrm{mod}\, p),$$
then
$$p\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r},$$
where
$$\mathfrak{P}_i = (p, f_i(\alpha)), \qquad e(\mathfrak{P}_i/p) = e_i, \qquad \textit{and } f(\mathfrak{P}_i/p) = \deg(f_i).$$

Remark 7.12. In practice, condition (7.4) is implied by the (more easily computable) condition
$$p^2 \nmid \mathrm{disc}(f),$$
where $\mathrm{disc}(f)$ is the discriminant of f. This is defined as
$$\mathrm{disc}(f) = \det \begin{pmatrix} 1 & \alpha & \cdots & \alpha^{n-1} \\ 1 & \beta & \cdots & \beta^{n-1} \\ \vdots & & \ddots & \vdots \\ 1 & \gamma & \cdots & \gamma^{n-1} \end{pmatrix}^2,$$
where $n = [L : \mathbf{Q}]$ and $\{\alpha, \beta, \dots, \gamma\}$ are the roots of f.

Example. Let $f(X) = X^3 - 6X^2 - 3X + 3$. This is Eisenstein for the prime 3 (as f is monic, 3 exactly divides the constant term, and 3 divides all the other non-leading coefficients), so f is irreducible. Let α be a root of f, and $L = \mathbf{Q}(\alpha)$, a cubic field. One may compute that

$$\mathrm{disc}(f) = 3753 = 3^3 \times 139.$$

In particular, we may apply Dedekind–Kummer for any prime $p \neq 3$.

- For $p = 5$, one may factorise
$$f(X) \equiv (X-1)(X^2+2) \pmod 5$$
into irreducibles, so
$$5\mathcal{O}_L = \mathfrak{P}_1\mathfrak{P}_2$$
where
$$\mathfrak{P}_1 = (5, \alpha - 1), \qquad \mathfrak{P}_2 = (5, \alpha^2 + 2).$$
We have $e(\mathfrak{P}_1/5) = e(\mathfrak{P}_2/5) = 1$, and $f(\mathfrak{P}_1/5) = 1$ and $f(\mathfrak{P}_2/5) = 2$.
- For $p = 7$, we check $f \pmod 7$ is irreducible. Therefore
$$7\mathcal{O}_L = \mathfrak{p}$$
where $\mathfrak{p} = (7, f(\alpha))$, with $e(\mathfrak{p}/7) = 1, f(\mathfrak{p}/7) = 3$.
- If $p = 23$, then
$$f(X) \equiv (X+2)(X+3)(X+12) \pmod{23}.$$
Then
$$23\mathcal{O}_K = \mathfrak{q}_1\mathfrak{q}_2\mathfrak{q}_3, \qquad \text{with } e(\mathfrak{q}_i/p) = f(\mathfrak{q}_i/p) = 1$$
and for example $\mathfrak{q}_1 = (23, \alpha + 2)$.

Corollary. *There is at least one prime $\mathfrak{P} \subset \mathcal{O}_L$ above $\mathfrak{p}$.*

Proof. Suppose not. Then $r = 0$ and

$$\sum_{i=}^{r} e_i f_i = 0 < 1 \leqslant [L : K],$$

contradiction. □

7.2.1. (∗) **A (non-examinable) proof of Theorem 7.10.** In the proof, we will need the following generalisation of the Chinese Remainder Theorem:

Theorem (Chinese Remainder Theorem[2])**.** *Let R be a ring, and $I_1, \ldots, I_r$ ideals such that $I_i + I_j = R$ for any pair i, j. Let $I = I_1 \cdots I_r$. Then*

$$R/I \cong R/I_1 \times \cdots \times R/I_r.$$

[2] To get a feel for how this works, you might like to check that in the special case where $R = \mathbf{Z}$, this does recover the usual Chinese Remainder Theorem!

Proof of Theorem 7.10. We split the proof into several steps. Let $k := \mathcal{O}_K/\mathfrak{p}$.

Step 1: Reduce to the case $\mathcal{O}_K$ is a PID.

To do this, we localise at $\mathfrak{p} \subset \mathcal{O}_K$. Let $S_\mathfrak{p} = \mathcal{O}_K\backslash\mathfrak{p}$; then the localisation $\mathcal{O}_{K,\mathfrak{p}} = S_\mathfrak{p}^{-1}\mathcal{O}_K$ is Dedekind and has only one prime ideal, so is a DVR, hence a PID. A general property of localisation is that $S_\mathfrak{p}^{-1}\mathcal{O}_L$ is the integral closure of $\mathcal{O}_{K,\mathfrak{p}}$ in L, and that $S_\mathfrak{p}^{-1}\mathfrak{P}_i$ is prime in this ring; hence we remain in the set-up of the theorem. Since $\mathrm{Frac}(\mathcal{O}_{K,\mathfrak{p}}) = K$, we don't change K or L, hence don't change $[L : K]$. The f_i do not change, because the residue fields are exactly the same. The e_i do not change, since

$$\mathfrak{p}(S_\mathfrak{p}^{-1}\mathcal{O}_L) = (S_\mathfrak{p}^{-1}\mathfrak{P}_1)^{e_1}\cdots(S_\mathfrak{p}^{-1}\mathfrak{P}_r)^{e_r}.$$

So if we prove the formula (7.3) for the localisation $\mathcal{O}_{K,p}$, which is a PID, then we obtain it for $\mathcal{O}_K$. Thus without loss of generality we may assume henceforth that $\mathcal{O}_K$ is a PID.

Step 2: Show $[L : K] = \dim_k \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$.

By step 1, we may assume $\mathcal{O}_K$ is a PID; then the structure theorem for modules over PIDs implies that $\mathcal{O}_L$ is a free $\mathcal{O}_K$-module of rank $n = [L : K]$, that is $\mathcal{O}_L \cong \mathcal{O}_K^n$ as $\mathcal{O}_K$-modules. Then

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong (\mathcal{O}_K/\mathfrak{p})^n = k^n$$

has dimension $n = [L : K]$ over k.

Step 3: Show $\dim_k \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L = \sum e_i f_i$.

First note that if $i \neq j$, then $\mathfrak{P}_i^{e_i} + \mathfrak{P}_j^{e_j} = \mathcal{O}_L$; since if not, then this ideal is proper, hence is contained in some prime ideal $\mathfrak{P}$ of $\mathcal{O}_L$. Then $\mathfrak{P}_i^{e_i} \subset \mathfrak{P}$ implies $\mathfrak{P}_i = \mathfrak{P}$, but similarly $\mathfrak{P}_j = \mathfrak{P}$, contradiction. By the Chinese Remainder Theorem, we deduce

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}_L/\mathfrak{P}_1^{e_1} \times \cdots \times \mathcal{O}_L/\mathfrak{P}_r^{e_r}. \tag{7.5}$$

Note that for $a \in \mathbf{Z}_{\geqslant 0}$, if $x \in \mathfrak{P}_i^a\backslash\mathfrak{P}_i^{a+1}$, then $(x)+\mathfrak{P}_i^{a+1} = \mathfrak{P}_i^a$ (by considering prime factorisations), so

$$\begin{aligned}\dim_{\mathcal{O}_L/\mathfrak{P}_i}\Big(\mathfrak{P}_i^a/\mathfrak{P}_i^{a+1}\Big) = 1 \quad &\Rightarrow \quad \dim_k\Big(\mathfrak{P}_i^a/\mathfrak{P}_i^{a+1}\Big) = 1 \\ &\Rightarrow \quad \dim_k\Big(\mathcal{O}_L/\mathfrak{P}_i^{e_i}\Big) = e_i f_i.\end{aligned}$$

By (7.5), we deduce $\dim_k \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L = \sum_{i=1}^r e_i f_i$.

Combining steps 2 and 3, we conclude that

$$[L : K] = \dim_k \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L = \sum_{i=1}^r e_i f_i,$$

completing the proof of Theorem 7.10. □

This ends the non-examinable material.

7.3. Classification of absolute values on number fields

Recall Ostrowski's theorem (Theorem 2.11): up to equivalence, the absolute values on $\mathbf{Q}$ are precisely $|\cdot|_\infty$ and $|\cdot|_p$, for p prime. We now prove the natural analogue for number fields.

We already saw (in Remark 2.14) that the equivalence classes of archimedean absolute values on a number field K are in bijection with real embeddings and pairs of complex embeddings. This leaves the non-archimedean absolute values, which we now study.

We first introduce a crucial and recurring definition.

Definition 7.13. Let $(K, |\cdot|_K)$ be a valued field, and let L/K be a field extension. If $|\cdot|_L$ is an absolute value, we say $|\cdot|_L$ *extends* $|\cdot|_K$ if $|\alpha|_L = |\alpha|_K$ for all $\alpha \in K$ (that is, the restriction of $|\cdot|_L$ to K is equal to $|\cdot|_K$).

Example 7.14. The following is the most important example of extensions of absolute values in the whole course. Let $\mathcal{O}_K$ be Dedekind, $K = \mathrm{Frac}(K)$, L/K a finite extension, and $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L. Then we are again in the set-up of (7.2):

$$\begin{tikzcd}
 & & L \arrow[dll, dash, "\text{finite}"'] \arrow[d, dash] & \\
K \arrow[d, dash] & & \mathcal{O}_L & := \text{integral closure of } \mathcal{O}_K \text{ in } L \\
\text{Dedekind} = \mathcal{O}_K \arrow[urr, dash] & & &
\end{tikzcd}$$

Then $\mathcal{O}_L$ is Dedekind by Theorem 7.8.

Let $\mathfrak{p} \subset \mathcal{O}_K$ prime, and write $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$. Note $v_{\mathfrak{P}_i}(x) = e_i v_{\mathfrak{p}}(x)$. Fix $0 < \alpha < 1$, and recall:

– To $\mathfrak{p}$, we can associate a non-archimedean absolute value $|\cdot|_{\mathfrak{p}}$ on K by

$$|x|_{\mathfrak{p}} = \alpha^{v_{\mathfrak{p}}(x)}, \qquad x \in K^\times.$$

(see Definition 7.3).

– For any $\mathfrak{P}_i$, we have $0 < \alpha^{\frac{1}{e_i}} < 1$, and we get an absolute value on L given by

$$|y|_{\mathfrak{P}_i} = \left(\alpha^{\frac{1}{e_i}}\right)^{v_{\mathfrak{P}_i}(y)}$$

If $x \in K$, then ; so

$$|x|_{\mathfrak{p}} = \alpha^{v_{\mathfrak{p}}(x)} = \alpha^{\frac{v_{\mathfrak{P}_i}(x)}{e_i}} = (\alpha^{\frac{1}{e_i}})^{v_{\mathfrak{P}_i}(x)} = |x_{\mathfrak{P}_i}|.$$

Thus for these normalisations, $|\cdot|_{\mathfrak{P}_i}$ extends $|\cdot|_{\mathfrak{p}}$. (Note that different normalisations give equivalent absolute values).

Finally, note that if $\mathfrak{P}_i \neq \mathfrak{P}_j$, then the associated absolute values $|\cdot|_{\mathfrak{P}_i}$ and $|\cdot|_{\mathfrak{P}_j}$ are not equivalent. To see this, note there exists $x \in \mathcal{O}_L$ with $x \in \mathfrak{P}_i$ but $x \notin \mathfrak{P}_j$, since $\mathfrak{P}_i \not\subset \mathfrak{P}_j$. But then $|x|_{\mathfrak{P}_i} < 1$ and $|x|_{\mathfrak{P}_j} \geqslant 1$, breaking Lemma 2.6.

Actually, in the setting of Dedekind domains, this example accounts for *all* extensions of $|\cdot|_{\mathfrak{p}}$.

Theorem 7.15. *Let $\mathcal{O}_K$ be Dedekind, $K = \mathrm{Frac}(K)$, L/K a finite extension, and $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L. Let $\mathfrak{p} \subset \mathcal{O}_K$ be prime, and write $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$.*

Then every absolute value on L extending $|\cdot|_{\mathfrak{p}}$ on K is one of $|\cdot|_{\mathfrak{P}_1}, \ldots, |\cdot|_{\mathfrak{P}_r}$.

Proof. Let $|\cdot|$ be an absolute value on L extending $|\cdot|_{\mathfrak{p}}$. It is non-archimedean by Lemma 2.4.

Claim 1: If $x \in \mathcal{O}_L$, then $|x| \leqslant 1$.

To see this, note We have

$$\mathcal{O}_K \subset \{x \in K : |x|_{\mathfrak{p}} \leqslant 1\} \subset \{x \in L : |x| \leqslant 1\}.$$

Since $\{x \in L : |x| \leqslant 1\}$ is the valuation ring for $(L, |\cdot|)$, it is integrally closed in L by Lemma 3.12; thus since $\mathcal{O}_L$ is by definition the integral closure of $\mathcal{O}_K$ in L, we deduce

$$\mathcal{O}_L \subset \{x \in L : |x| \leqslant 1\}, \tag{7.6}$$

proving the claim. (More generally: if $A \subset C \subset L$, and C is integrally closed in L, then the integral closure of A in L must lie in C).

Let

$$\mathfrak{P} \coloneqq \{x \in \mathcal{O}_L : |x| < 1\}. \tag{7.7}$$

This is a prime ideal:

- if $x, y \in \mathfrak{P}$ then $|x + y| \leqslant \max(|x|, |y|) < 1$ (since $|\cdot|$ is non-archimedean), so $x + y \in \mathfrak{P}$.
- if $r \in \mathcal{O}_L$, $x \in \mathfrak{P}$, then $|rx| = |r| \cdot |x| \leqslant |x| < 1$ (using Claim 1), so $rx \in \mathfrak{P}$. (Combining these, $\mathfrak{P}$ is an ideal).
- if $x, y \in \mathcal{O}_L$ but $x, y \notin \mathfrak{P}$, then $|x| = |y| = 1$, and $|xy| = |x| \cdot |y| = 1$, so $xy \notin \mathfrak{P}$ (so $\mathfrak{P}$ is prime).

Note also that $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$. Indeed, this follows as

$$\begin{aligned} x \in \mathfrak{P} \cap \mathcal{O}_K &\iff x \in \mathcal{O}_K, |x| < 1 \\ &\iff x \in \mathcal{O}_K, |x|_\mathfrak{p} < 1 \qquad (\text{since } |\cdot| \text{ extends } |\cdot|_\mathfrak{p}) \\ &\iff x \in \mathfrak{p} \qquad (\text{by } (7.1)). \end{aligned}$$

In particular,

$$\mathfrak{p}\mathcal{O}_L \subset \mathfrak{P} \qquad \Rightarrow \qquad \mathfrak{P} | \mathfrak{p}.$$

Thus we have an absolute value $|\cdot|_\mathfrak{P}$ on L attached to $\mathfrak{P}$, and this extends $|\cdot|_\mathfrak{p}$ by Example 7.14.

Claim 2: $|\cdot|$ is equivalent to $|\cdot|_\mathfrak{P}$.
To see this, recall that for $x \in L$, we have

$$\begin{aligned} |x|_\mathfrak{P} < 1 &\iff v_\mathfrak{P}(x) > 0 \\ &\iff \mathfrak{P} | (x) \\ &\iff x \in \mathfrak{P} \\ &\iff |x| < 1, \end{aligned}$$

where the last equivalence is the definition of $\mathfrak{P}$. But then $|\cdot|$ is equivalent to $|\cdot|_\mathfrak{P}$ by Lemma 2.6.

Finally, the theorem now follows from:

Claim 3: We have $|\cdot| = |\cdot|_\mathfrak{P}$.
To see this, note that since they are equivalent absolute values, there exists some $c > 0$ such that $|\cdot| = |\cdot|_\mathfrak{P}^c$ (see Lemma 2.6). But both absolute values extend $|\cdot|_\mathfrak{p}$ on K. Thus if $0 \neq x \in K$, then

$$|x|_\mathfrak{P}^c = |x| = |x|_\mathfrak{P},$$

where the first equality is equivalence, and the second is the definition of extensions (Definition 7.13). Thus $c = 1$ and $|\cdot| = |\cdot|_\mathfrak{P}$, as required. $\square$

Corollary 7.16. *Every non-archimedean absolue value on a number field K is equivalent to $|\cdot|_\mathfrak{p}$ for a unique prime ideal $\mathfrak{p} \subset \mathcal{O}_K$.*

Proof. If $|\cdot|$ is a non-archimedean absolute value, then by Ostrowski's (first) theorem (Theorem 2.11), its restriction to $\mathbf{Q}$ must be equivalent to $|\cdot|_p$ for some prime. By Theorem 7.15, $|\cdot| \sim |\cdot|_{\mathfrak{p}}$ for some $\mathfrak{p}|p$.

This prime ideal is unique, since if $\mathfrak{q}$ is another prime, then $|\cdot|_{\mathfrak{p}}$ and $|\cdot|_{\mathfrak{q}}$ are not equivalent (by the last part of Example 7.14). $\square$

7.4. (∗) A (non-examinable) proof of Theorem 7.8

We now give a non-examinable proof of Theorem 7.8. The proof is not hard, but requires introducing and studying the *trace pairing.*

7.4.1. (∗) The trace pairing. Let K be a field, and R/K a finite commutative K-algebra of degree n. Note R is a K-vector space.

Definition 7.17. For every $x \in R$, we have a K-linear map

$$\begin{aligned}[x] : R &\longrightarrow R\\ y &\longmapsto xy.\end{aligned}$$

We define

$$\mathrm{Tr}_{R/K}(x) := \mathrm{Trace}([x]), \qquad N_{R/K}(x) := \mathrm{Norm}([x]).$$

Suppose L/K is a finite field extension, and let $\sigma_1, \dots, \sigma_m : L \hookrightarrow \overline{K}$ be the distinct field embeddings of L into the algebraic closure of K. Recall (from Galois theory) that L/K is *separable* if $m = n$, in which case we have the following result (also from Galois theory):

Lemma 7.18. *If L/K is a separable field extension, then*

$$\mathrm{Tr}_{L/K}(x) = \sum_{i=1}^{n} \sigma_i(x), \qquad N_{L/K}(x) = \prod_{i=1}^{n} \sigma_i(x).$$

Definition 7.19. The *trace form/pairing* attached to L/K is the pairing

$$\begin{aligned}\mathrm{T}_{L/K} : L \times L &\longrightarrow K,\\ (x,y) &\longmapsto \mathrm{Tr}_{L/K}(xy).\end{aligned}$$

Lemma 7.20. *If L/K is separable, then the trace pairing is non-degenerate (that is, if $\mathrm{T}_{L/K}(x,y) = 0$ for all $y \in L$, then $x = 0$).*

Proof. Suppose there exists $x \in L^\times$ such that $\mathrm{T}_{L/K}(x,y) = 0$ for all $y \in L$. In particular for $y = x^{-1}$ we have

$$0 = \mathrm{T}_{L/K}(x, x^{-1}) = \mathrm{Tr}_{L/K}(xx^{-1}) = \mathrm{Tr}_{L/K}(1) = [L:K]$$

by Lemma 7.18. But this is absurd. $\square$

7.4.2. (∗) Extensions of Dedekind domains are Dedekind. Now we prove Theorem 7.8, at least in the case L/K is separable. Recall this said:

> *Let $\mathcal{O}_K$ be a Dedekind domain, and $K = \mathrm{Frac}(\mathcal{O}_K)$. Let L/K be a finite extension. Let $\mathcal{O}_L$ be the integral closure of $\mathcal{O}_K$ in L. Then $\mathcal{O}_L$ is a Dedekind domain.*

Proof. For simplicity, we assume that L/K is separable. The result also holds for non-separable finite extensions, but we omit the proof (see [Jan96], §I.6).

We must check that:

(i) $\mathcal{O}_L$ is a domain, since it is a subring of a field.

(ii) $\mathcal{O}_L$ is Noetherian; this is a bit involved; see below.

(iii) $\mathcal{O}_L$ is integrally closed, since integral closures are integrally closed.

(iv) every non-zero prime in $\mathcal{O}_L$ is maximal.

Proof that extensions of Dedekind domains are Noetherian: Let $x_1, \dots, x_n$ be a K-basis for L. Rescaling denominators we may assume each $x_i \in \mathcal{O}_L$. Since we assume L/K is separable, the trace pairing $\mathrm{T}_{L/K}$ is non-degenerate by Lemma 7.20, and thus there exists a dual basis $y_1, \dots, y_n$ of L over K with respect to $\mathrm{T}_{L/K}$, that is, such that

$$\mathrm{T}_{L/K}(x_i, y_j) = \delta_{ij} = \begin{cases} 1 & : \ i = j \\ 0 & : \ i \neq j. \end{cases}$$

Let $z \in \mathcal{O}_L$, and write

$$z = \sum_{i=1}^{n} \lambda_i y_i, \qquad \lambda_i \in K.$$

Then

$$\lambda_j = \mathrm{T}_{L/K}(x_j, z) = \mathrm{Tr}(x_j z) = \sum_{i=1}^{n} \sigma_i(x_j z) \in \mathcal{O}_K,$$

since $x_j z \in \mathcal{O}_L$ implies $\sigma_i(x_j z) \in \mathcal{O}_L$. Therefore we deduce

$$\mathcal{O}_L \subset \mathcal{O}_K y_1 + \cdots + \mathcal{O}_K y_n. \tag{7.8}$$

Now we use two basic properties of Noetherian rings:

(a) if R is a Noetherian ring, and M is a finitely generated R-module, then every submodule of M is finitely generated;

(b) and an R-algebra S is finitely generated if and only if it is Noetherian.

By (7.8) and (a), $\mathcal{O}_L$ is a finitely generated $\mathcal{O}_K$-module, and hence by (b) it is Noetherian.

To see (iv): let $\mathfrak{P} \subset \mathcal{O}_L$ be a non-zero prime ideal. Then $\mathfrak{p} := \mathfrak{P} \cap \mathcal{O}_K$ is a prime ideal in $\mathcal{O}_K$. Since $\mathfrak{P} \neq 0$, there exists $0 \neq x \in \mathfrak{P}$, and $0 \neq N_{L/K}(x) \in \mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$. (It lies in $\mathfrak{P}$ because $\mathfrak{P}$ is an ideal, and in $\mathcal{O}_K$ by basic properties of norm). We deduce $\mathfrak{p} \neq 0$, so since $\mathcal{O}_K$ is Dedekind, $\mathfrak{p}$ is maximal. Thus $k := \mathcal{O}_K/\mathfrak{p}$ is a field.

Now inclusion $\mathcal{O}_K \hookrightarrow \mathcal{O}_L$ induces an embedding

$$k = \mathcal{O}_K/\mathfrak{p} \hookrightarrow \mathcal{O}_L/\mathfrak{P}.$$

By above, we know $\mathcal{O}_L/\mathfrak{P}$ is a finite dimensional k-algebra; and since $\mathfrak{P}$ is prime, it is an integral domain. Hence if $0 \neq y \in \mathcal{O}_L/\mathfrak{P}$, then multiplication by y is injective; so as a k-linear map, it has nullity 0. By the rank-nullity theorem, multiplication by y is then invertible. Hence $\mathcal{O}_L/\mathfrak{P}$ is a field, and $\mathfrak{P}$ is maximal. □

Chapter 8

Extensions of complete fields

In the previous chapter, we considered the picture:

$$\begin{tikzcd} & L \\ K \arrow[ur, dash, "\text{finite}"] & \\ & \mathcal{O}_L \arrow[uu, dash] \quad := \ \text{integral closure of } \mathcal{O}_K \text{ in } L \\ \mathcal{O}_K \arrow[uu, dash] \arrow[ur, dash] & \end{tikzcd} \tag{8.1}$$

We developed a hierarchy of rings

$$\boxed{\text{Dedekind domains}} \supset \boxed{\text{DVRs}} \supset \boxed{\text{Complete DVRs}},$$

and saw that:

- if $\mathcal{O}_K$ is Dedekind, then $\mathcal{O}_L$ is Dedekind (Theorem 7.8);
- if $\mathcal{O}_K$ is a DVR with maximal ideal $\mathfrak{p}$, then the absolute values on L extending $|\cdot|_K$ biject with the primes of $\mathcal{O}_L$ above $\mathfrak{p}$ (via Theorem 7.15).

In this chapter, a central result – shown in Theorem 8.6 – is:

If $\mathcal{O}_K$ is a complete DVR, then $\mathcal{O}_L$ is a complete DVR.

Equivalently, we show:

- there is only one prime of $\mathcal{O}_L$ above $\mathfrak{p}$;
- hence there is a *unique* extension of $|\cdot|_K$ to L;
- and that L is complete with respect to this extension.

By studying the properties of extensions of complete fields, in §8.3 we complete the classification of local fields (Theorem 4.3), by proving that any non-archimedean local field of characteristic 0 is a finite extension of $\mathbf{Q}_p$.

We finally apply these results to study completions of Galois extensions of number fields.

8.1. Norms on vector spaces

Let $(K, |\cdot|)$ be a valued field. We first study a more general setting: that of norms on K-vector spaces. Extensions of absolute values are simply a special case of a norm.

Definition 8.1. Let V be a K-vector space. A *norm* on V is a function

$$||\cdot|| : V \to \mathbf{R}$$

such that:

(i) $||v|| \geqslant 0$, with equality if and only if $v = 0$;

(ii) $||\lambda v|| = |\lambda| \cdot ||v||$ for all $\lambda \in K$ and $v \in V$;

(iii) $||v + w|| \leqslant ||v|| + ||w||$ for all $v, w \in V$.

Examples. (1) If L/K is an extension, and $|\cdot|_L$ is an extension of $|\cdot|$, then $|\cdot|_L$ is a norm on L (considered as a K-vector space). (This explains the relevance of the theory of norms to our current problems of interest).

(2) If $V \cong K$ is 1-dimensional, with basis e, then $||\lambda e|| := |\lambda|$ is a norm on V.

(3) More generally if $V \cong K^d$ with basis $e_1, ..., e_d$, then

$$||\lambda_1 e_1 + \cdots \lambda_d e_d||_{\sup} := \sup_i |\lambda_i|$$

is a norm (where $\lambda_i \in K$).

Lemma 8.2. *Suppose $(K, |\cdot|)$ is complete, and $d = \dim_K(V) < \infty$. Then V is complete with respect to $||\cdot||_{\sup}$.*

Proof. Let (v_n) be a sequence. Writing $v_n = (v_{n,1}, ..., v_{n,d}) \in K^d$, we see

$$v_n \text{ is Cauchy for } ||\cdot||_{\sup} \iff v_{n,i} \text{ is Cauchy in } K \text{ for all } i = 1, \ldots, d.$$

Since K is complete, in this case $v_{n,i} \to v_i \in K$, so $v_n \to v := (v_1, \ldots, v_d)$. ☐

Definition 8.3. We say two norms $||\cdot||_1$ and $||\cdot||_2$ on V are *equivalent* if there exist $a, b > 0$ such that

$$a||v||_1 \leqslant ||v||_2 \leqslant b||v||_1 \qquad \forall v \in V.$$

Theorem 8.4. *Suppose $(K, |\cdot|)$ is complete. If $d = \dim_K(V) < \infty$, then any two norms on V are equivalent, and V is complete (with respect to any of them).*

Remarks. (1) This is essentially a general version of the Heine–Borel theorem, which says that 'the unit sphere in any finite-dimensional real vector space is compact'.

(2) With this in mind, morally the proof goes as follows: let $||\cdot||$ be a norm on V. Then:

- Show that $||\cdot||$ is continuous with respect to the topology from $||\cdot||_{\sup}$.
- 'The unit sphere $S = \{x \in V : ||x||_{\sup} = 1\}$ is compact.'
- Since $||\cdot||$ is continuous and S is compact, we deduce there exist $a, b > 0$ such that $a < ||v|| < b$ for all $v \in S$.
- Deduce that $a||v||_{\sup} \leqslant ||v|| \leqslant b||v||_{\sup}$ for all $v \in V$, so $||\cdot||$ and $||\cdot||_{\sup}$ are equivalent.

(3) The relevance to our study is when we take L/K a finite extension of a complete field $(K, |\cdot|_K)$, whence all absolute values on L extending $|\cdot|_K$ are equivalent, from which we'll deduce such an extension is unique!

Proof. Without loss of generality, we may write $V = K^d$, with the obvious basis

$$e_1 = (1, 0, ..., 0), e_2 = (0, 1, 0, ..., 0), ..., e_d = (0, ..., 0, 1).$$

Let $||\cdot||$ be any norm on V. By induction on d, we will show it is equivalent to $||\cdot||_{\sup}$, and that V is complete with respect to $||\cdot||$.

When $d = 1$, we have $||\lambda e_1|| = |\lambda| ||e_1||$ is totally determined by $||e_1||$, which may take any value > 0. Hence all norms are scalar multiples of each other, and in particular are equivalent to each other. Since K is complete, V is complete.

For general d, let $v = v_1e_1 + \cdots + v_de_d \in V$. Then

$$\begin{aligned} ||v|| = ||v_1e_1 + \cdots + v_de_d|| &\leqslant \sum_{i=1}^{d} |v_i| \cdot ||e_d|| \\ &\leqslant \Big(\sum_{i=1}^{d} ||e_i||\Big) \cdot \max_i |v_i| = b||v||_{\sup}, \end{aligned} \tag{8.2}$$

where $b \coloneqq \sum_i ||e_i||$ is constant. This is the 'bounded above' inequality in the definition of equivalence.

Note (8.2) also implies that $||\cdot||$ is continuous with respect to the topology given by $||\cdot||_{\sup}$: for $\epsilon > 0$, let $\eta\epsilon/b$. Then if $||x - y||_{\sup} < \eta$, then

$$||x - y|| \leqslant b||x - y||_{\sup} < b\eta = \epsilon.$$

It remains to bound below. Let

$$S \coloneqq \{v \in V : ||v||_{\sup} = 1\}.$$

Claim. *There exists $a > 0$ such that $a \leqslant ||v||$ for all $v \in S$.*

Proof of claim: Suppose not; then there exists a sequence

$$x^{(n)} = x_1^{(n)}e_1 + \cdots x_d^{(n)}e_d \in S, \qquad ||x^{(n)}|| \to 0 \text{ as } n \to \infty.$$

By the pigeonhole principle, for at least one value of i, $1 \leqslant i \leqslant d$, we must have

$$||x^{(n)}||_{\sup} = |x_i^{(n)}|$$

for infinitely many n; without loss of generality, take $i = d$. Passing to this subsequence, we may assume $||x^{(n)}||_{\sup} = |x_d^{(n)}|$ for all n; and since $x^{(n)} \in S$, we have $|x_d^{(n)}| = 1$. Replacing $x^{(n)}$ with $x^{(n)}/x_d^{(n)}$, we may assume $x_d^{(n)} = 1$ for all n. (Note this is ok, since

$$||x^{(n)}/x_d^{(n)}|| = |1/x_d^{(n)}| \cdot ||x^{(n)}|| = ||x^{(n)}|| \to 0 \qquad \text{as } n \to \infty,$$

i.e. this modified sequence still tends to 0 for $||\cdot||$). Write $x^{(n)} = y^{(n)} + e_d$, for $y^{(n)} \in \langle e_1, ..., e_{d-1}\rangle$.

Now, since $||x^{(n)}|| \to 0$, the sequence $(x^{(n)})$ is Cauchy for $||\cdot||$. Thus $(y^{(n)})$ is also Cauchy in K^{d-1} for $||\cdot||_{K^{d-1}}$, where $||\cdot||_{K^{d-1}}$ is the restriction of $||\cdot||$ to $\langle e_1, ..., e_{d-1}\rangle$. By the induction hypothesis, K^{d-1} is complete with respect to $||\cdot||_{K^{d-1}}$, so $y^{(n)} \to y \in \langle e_1, \cdots, e_{d-1}\rangle$ converges to some limit. But

$$y^{(n)} = x^{(n)} - e_d \to -e_d \notin \langle e_1, ..., e_d\rangle.$$

We derive a contradiction, which proves the claim.

Now, given the claim, let $0 \neq v \in V$. Write $||v||_{\sup} = |v_i|$, some $1 \leqslant i \leqslant d$. Then $||v/v_i||_{\sup} = 1$, so $v/v_i \in S$. Thus by the claim, we have

$$a \leqslant ||v/v_i|| \Rightarrow a||v||_{\sup} = a|v_i| \leqslant ||v||.$$

Thus $||\cdot||$ is equivalent to $||\cdot||_{\sup}$.

To complete the induction, we observe that since K is complete, we have V complete with respect to $||\cdot||_{\sup}$ by Lemma 8.2. As equivalent norms induce the same topology, V is complete with respect to any norm. □

Corollary 8.5. *Let K be complete with respect to an absolute value $|\cdot|_K$. Let L/K be a finite extension. If $|\cdot|_1$ and $|\cdot|_2$ are two absolute values on L extending $|\cdot|_K$, then $|\cdot|_1 = |\cdot|_2$, and L is complete with respect to $|\cdot|_i$.*

Proof. Both $|\cdot|_1$ and $|\cdot|_2$ are norms on the finite-dimensional K-vector space L, so by Theorem 8.4 they are equivalent. Thus they induce the same topology on L, so $|\cdot|_1$ and $|\cdot|_2$ are equivalent absolute values (as in Lemma 2.6). Thus there exists $c > 0$ such that $|\cdot|_1 = |\cdot|_2^c$. Now if $x \in K$, then $|x|_1 = |x|_2^c = |x|_2$, so $c = 1$.

Finally, completeness follows from Theorem 8.4. □

8.2. Extending absolute values on complete fields

Let:

- K be complete with respect to a discrete valuation v_K,
- $\mathcal{O}_K$ be the valuation ring, which is a complete DVR by Lemmas 3.13 and 3.19,
- $\mathfrak{m}_K = \pi_K\mathcal{O}_K$ be the maximal ideal,
- and $|\cdot|_K$ be the attached absolute value.

Remark. Note that, provided we choose the same $0 < \alpha < 1$ in the definitions of both $|\cdot|_K$ and $|\cdot|_{\mathfrak{m}_K}$, we have

$$|\cdot|_K = |\cdot|_{\mathfrak{m}_K}$$

as absolute values on K, allowing us to use the frameworks of Chapter 7 to study $|\cdot|_K$. To see this, it suffices to show that the normalised discrete valuation v_K on K is the same as the one $v_{\mathfrak{m}_K}$ attached to $\mathfrak{m}_K$ in Definition 7.3, since then

$$|x|_K = \alpha^{v_K(x)} = \alpha^{v_{\mathfrak{m}_K}(x)} = |x|_{\mathfrak{m}_K}.$$

Recall by Proposition 3.18 that any $x \in K$ uniquely has form $u\pi_K^r$, for some $r \in \mathbf{Z}$. Then:

(1) In the proof of Lemma 3.19, v_K is shown to satisfy/defined via

$$v_K(x) = v_K(u\pi^r) = r.$$

(2) We have $x\mathcal{O}_K = u\pi_K^r\mathcal{O}_K = \mathfrak{m}_K^r$ as fractional ideals of K, so by Definition 7.3 we see

$$v_{\mathfrak{m}_K}(x) = v_{\mathfrak{m}_K}(u\pi^r) = r.$$

So $v_K = v_{\mathfrak{m}_K}$, and hence $|\cdot|_K = |\cdot|_{\mathfrak{m}_k}$.

Let L/K be a finite extension, and let $\mathcal{O}_L$ be the integral closure of $\mathcal{O}_K$ in L. In other words, we're in the usual set-up with picture:

$$\begin{tikzcd} & & L \arrow[dd, dash] \\ \text{complete discretely valued field} = & K \arrow[ur, dash, "\text{finite}"] \arrow[dd, dash] & \\ & & \mathcal{O}_L \quad := \text{ integral closure of } \mathcal{O}_K \text{ in } L \\ \text{complete DVR} = & \mathcal{O}_K \arrow[ur, dash] & \end{tikzcd}$$

The main result of this section is the following.

Theorem 8.6. *(i) There is a unique extension $|\cdot|_L$ of $|\cdot|_K$ to L.*

(ii) L is complete with respect to $|\cdot|_L$.

(iii) There is a unique prime $\mathfrak{m}_L \subset \mathcal{O}_L$ above $\mathfrak{m}_K$.

(iv) $\mathcal{O}_L$ is a DVR. It is the valuation ring in L.

In other words, if $\mathcal{O}_K$ is a complete DVR in (8.1), then so is $\mathcal{O}_L$.

Proof. (i) By Lemma 3.19, we know $\mathcal{O}_K$ is a DVR, hence Dedekind, and $|\cdot|_K = |\cdot|_{\mathfrak{m}_K}$. By Theorem 7.10, there exists a prime $\mathfrak{m}_L \subset \mathcal{O}_L$ above $\mathfrak{m}_K$ (else the right-hand side of the theorem is 0, which cannot be true). By Example 7.14, $|\cdot|_{\mathfrak{m}_L}$ is an absolute value on L extending $|\cdot|_{\mathfrak{m}_K} = |\cdot|_K$. This gives *existence* of an extension. But extensions are unique by Corollary 8.5.

(ii) This is Corollary 8.5.

(iii) Using Theorem 7.2, write

$$\pi_K\mathcal{O}_L = \mathfrak{P}_1^{e_1}\cdots\mathfrak{P}_r^{e_r}$$

By Example 7.14, for each i we get an absolute value $|\cdot|_{\mathfrak{P}_i}$ on L extending $|\cdot|_K$. Note the $|\cdot|_{\mathfrak{P}_i}$ are all inequivalent: since $\mathfrak{P}_i + \mathfrak{P}_j = \mathcal{O}_L$, there exist $x \in \mathfrak{P}_i, y \in \mathfrak{P}_j$ with $x + y = 1$, then $|x|_{\mathfrak{P}_i} < 1$ and $|x|_{\mathfrak{P}_j} = 1$ (cf. Lemma 2.6). This contradicts (i) unless $r = 1$ whence $\mathfrak{P}_1 = \mathfrak{m}_L$ is the unique prime ideal above $\mathfrak{m}_K$.

(iv) If $\mathfrak{P} \subset \mathcal{O}_L$ is any prime, then $\mathfrak{P} \cap \mathcal{O}_K$ is a non-zero prime of $\mathcal{O}_K$; since $\mathcal{O}_K$ is a DVR, we thus have $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{m}_K$. But the only prime ideal of $\mathcal{O}_L$ above $\mathfrak{m}_K$ is $\mathfrak{m}_L$, so $\mathcal{O}_L$ has a unique non-zero prime ideal. Thus $\mathcal{O}_L$ is a DVR by Corollary 7.6. By Lemma 3.19(ii), $\mathcal{O}_L$ is the valuation ring in its fraction field L. □

Corollary 8.7. *Let L/K a finite Galois extension, and let $|\cdot|$ be the unique absolute value on L extending $|\cdot|_K$. Then*

$$|\sigma(x)|_L = |x|_L \qquad \forall \sigma \in \mathrm{Gal}(L/K), x \in L.$$

Proof. For fixed σ, the function $x \mapsto |\sigma(x)|_L$ is an absolute value on L extending $|\cdot|_K$; so it must be equal to $x \mapsto |x|_L$. □

(∗) **Remark.** We deduce there is a unique valuation $|\cdot|$ on the algebraic closure $\overline{\mathbf{Q}}_p$ extending $|\cdot|_p$.

To see this, note that if $x \in \overline{\mathbf{Q}}_p$, then $x \in K$ for some finite extension $L/\mathbf{Q}_p$. Let $|\cdot|_L$ be the unique absolute value on L extending $|\cdot|_p$, given by Theorem 8.6. We define

$$|x| := |x|_L.$$

If $M/\mathbf{Q}_p$ is another finite extension with $x \in M$, then $x \in ML$, and $|x|_L = |x|_{ML} = |x|_M$ by uniqueness of extensions; thus $|x|$ is independent of the choice of field L. One can check the axioms, e.g. $|x+y| \leqslant |x| + |y|$, by computing in the finite field extension $\mathbf{Q}_p(x,y)$.

(∗) **Remark.** An extension L/K is *algebraic* if $K(\alpha)/K$ is finite for every $\alpha \in L$ (that is, every $\alpha \in L$ is a root of a polynomial over K). We've seen that if $L/\mathbf{Q}_p$ is an algebraic extension, there is a unique absolute value on L extending $|\cdot|_p$, and that if $L/\mathbf{Q}_p$ is *finite*, then L is complete. If $L/\mathbf{Q}_p$ is *infinite*, however, then L is *never* complete! There is a non-credit question about this on the final example sheet.

By above, $\overline{\mathbf{Q}}_p$ is algebraically closed, but not complete. We let $\mathbf{C}_p$ denote the completion of $\overline{\mathbf{Q}}_p$. Fortunately, this remains algebraically closed. The upshot is that it is $\mathbf{C}_p$, not $\overline{\mathbf{Q}}_p$, that is the 'p-adic analogue' of $\mathbf{C}$ (the algebraic closure of $\mathbf{R}$).

8.3. The invariants $e(L/K)$ and $f(L/K)$

If K is complete non-archimedean and L/K finite, we've shown L is complete and there is only one prime of L. We now consider L/K possibly infinite, but assume L is complete. With that in mind:

– Let $K \subset L$ both be complete with respect to normalised discrete valuations v_K and v_L. (We allow L/K infinite).

– We assume that v_L 'extends' v_K, in the following sense[1]: we demand that the restriction of v_L to K is a discrete valuation equivalent to v_K, that is, there exists a positive real number $e > 0$ such that
$$v_L(x) = e v_K(x) \qquad \forall x \in K^\times. \tag{8.3}$$

– Let $\mathcal{O}_K$ and $\mathcal{O}_L$ be the valuation rings, which are complete DVRs by Lemma 3.19. By (8.3) we have $\mathcal{O}_K \subset \mathcal{O}_L$.

– let $\mathfrak{m}_K = \pi_K\mathcal{O}_K$ and $\mathfrak{m}_L = \pi_L\mathcal{O}_L$ be the maximal ideals. Note $\mathfrak{m}_K \subset \mathfrak{m}_L$ by (8.3). As $\mathfrak{m}_K$ and $\mathfrak{m}_L$ are the *unique* non-zero prime ideals in $\mathcal{O}_K$ and $\mathcal{O}_L$ respectively, we must have $\mathfrak{m}_L | \mathfrak{m}_K$.

– Let $k = \mathcal{O}_K/\mathfrak{m}_K$ and $k_L = \mathcal{O}_L/\mathfrak{m}_L$ be the residue fields; as $\mathfrak{m}_L|\mathfrak{m}_K$, we have k_L/k is a field extension.

Definition 8.8. (i) The *ramification index* of L/K is

$$e(L/K) = v_L(\pi_K) \in \mathbf{Z}.$$

(ii) The *residue index* of L/K is

$$f(L/K) = [k_L : k] \in \mathbf{Z} \cup \{\infty\}.$$

[1] In particular, this means it is possible to normalise the associated absolute values so that $|\cdot|_L$ extends $|\cdot|_K$ in the sense described in Definition 7.13. This process is exactly as described in Example 7.14.

This extension property is not guaranteed. For example, it is not satisfied if we take $K = \mathbf{Q}_p$ with valuation v_p, and $L = \mathbf{Q}_p((T))$ with valuation v_T. The extension $\mathbf{Q}_p((T))/\mathbf{Q}_p$ is not covered by the results of this section.

By definition, we have

$$f(L/K) = f(\mathfrak{m}_L/\mathfrak{m}_K),$$

so when L/K is finite, the residue index agrees with the notion from before (in Definition 7.9). This is also true for the ramification index:

Lemma 8.9. *We have*

$$\mathfrak{m}_K\mathcal{O}_L = \mathfrak{m}_L^{e(L/K)}.$$

In particular:

(1) We have $|\pi_K|_L = |\pi_L^{e(L/K)}|_L$,

(2) When L/K is finite, we have

$$e(L/K) = e(\mathfrak{m}_L/\mathfrak{m}_K).$$

Proof. Let $e = e(L/K) = v_L(\pi_K)$. Since $v_L(\pi_L) = 1$, we have

$$v_L(\pi_L^e) = v_L(\pi_K) = e. \tag{8.4}$$

Thus by Lemma 3.11 we have $\pi_K\mathcal{O}_L = \pi_L^e\mathcal{O}_L$, and hence

$$\mathfrak{m}_K\mathcal{O}_L = \pi_K\mathcal{O}_L = \pi_L^e\mathcal{O}_L = \mathfrak{m}_L^e\mathcal{O}_L.$$

Statement (1) follows immediately from (8.4), and (2) follows from the definition of $e(\mathfrak{m}_L/\mathfrak{m}_K)$ (Definition 7.9). □

In practical terms, there are easy ways to compute $e(L/K)$. The following is particularly useful. Let

$$|K^\times|_K = \{|x|_K : x \in K^\times\} \subset \mathbf{R}_{>0}.$$

Note that since $|K^\times|_K = |K^\times|_L$, we have $|K^\times|_K \subset |L^\times|_L$ as a multiplicative subgroup.

Lemma 8.10. *We have*

$$e(L/K) \textit{ is the index of the subgroup } |K^\times|_K \subset |L^\times|_L.$$

Proof. Recall every element of $K^\times$ has form $u\pi_K^r$ for $|u|_K = 1$ and $r \in \mathbf{Z}$ (by Lemma 3.18). Hence

$$|K^\times|_K = |\pi_K|_K^{\mathbf{Z}} = |\pi_K|_L^{\mathbf{Z}}.$$

Similarly

$$|L^\times|_L = |\pi_L|_L^{\mathbf{Z}}.$$

Writing $e = e(L/K)$, we know from Lemma 8.9(1) that

$$|\pi_K|_L = |\pi_L|_L^e,$$

hence

$$\begin{aligned} |K^\times|_K &= |\pi_K|_L^{\mathbf{Z}} \\ &= |\pi_L|_L^{e\mathbf{Z}} \text{ is contained with index } e \text{ in } |\pi_L|_L^{\mathbf{Z}} = |L^\times|_L, \end{aligned}$$

as required. □

Examples. (1) Consider $K = \mathbf{Q}_5$ and $L = \mathbf{Q}_5(\sqrt{2})$. Then L/K is quadratic. Note that

$$\sqrt{2} \,(\mathrm{mod}\, \mathfrak{m}_L) \in k_L, \qquad \text{but } \sqrt{2} \notin k = \mathbf{F}_5$$

(since 2 is not a quadratic residue modulo 5). In particular $k_L \neq k$, so $f(L/K) \geqslant 2$.

(2) Consider $K = \mathbf{Q}_5$ and $L = \mathbf{Q}(\sqrt{5})$. Note

$$\begin{aligned} |\sqrt{5}|_L^2 = |5|_L &= |5|_K \qquad \text{since } |\cdot|_L \text{ extends } |\cdot|_K \\ &= 1/5, \end{aligned}$$

so that

$$|\sqrt{5}|_L = \frac{1}{\sqrt{5}} \notin 5^{\mathbf{Z}} = |K^\times|_K.$$

Thus $|L^\times|_L \neq |K^\times|_L$, and $e(L/K) \geqslant 2$.

To complete these examples, we use:

Proposition 8.11. *If L/K is finite, then*[2]

$$[L : K] = e(L/K)f(L/K).$$

This allows us to finish the above examples:

- In example (1), we must have $2 = [\mathbf{Q}_5(\sqrt{2}) : \mathbf{Q}_5] = ef$, and $f \geqslant 2$. This forces

$$f(\mathbf{Q}_5(\sqrt{2})/\mathbf{Q}_5) = 2 \qquad \text{and} \qquad e(\mathbf{Q}_5(\sqrt{2})/\mathbf{Q}_5) = 1.$$

- Similarly in example (2) we see

$$e(\mathbf{Q}_5(\sqrt{5})/\mathbf{Q}_5) = 2 \qquad \text{and} \qquad f(\mathbf{Q}_5(\sqrt{5})/\mathbf{Q}_5) = 1.$$

Proof. (Proposition 8.11). I made hard work of this proof during lectures, so the entire proof of Proposition 8.11 is ***non-examinable****.*

Let $n = [L : K]$, $e = e(L/K)$ and $f = f(L/K)$.

Step 1: Show $[L : K] = \dim_k \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L$.
Note that:

- Note $\mathcal{O}_L$ is a torsion-free $\mathcal{O}_K$-module (since it is a domain; so if $x \in \mathcal{O}_L$ and $r \in \mathcal{O}_K$ with $rx = 0$, then either $r = 0$ or $x = 0$).
- $\mathcal{O}_K$ is a PID, since it is a DVR.
- $\mathcal{O}_L$ is finitely generated as an $\mathcal{O}_K$-module, since L/K is finite.

Then the structure theorem for modules over PIDs implies that $\mathcal{O}_L$ is a free $\mathcal{O}_K$-module of some rank $m \in \mathbf{N}$, that is,

$$\mathcal{O}_L \cong \mathcal{O}_K^m$$

[2] This is a special case of Theorem 7.10; however, the proof of that theorem was non-examinable, and this is very important, so we give a complete proof of this special case.

as $\mathcal{O}_K$-modules. We can check that $m = n = [L : K]$, since

$$\begin{aligned} K^n \cong L = \mathcal{O}_L\left[\tfrac{1}{\pi_K}\right] & \qquad \text{by Lemma 3.11(2)} \\ \cong \left(\mathcal{O}_K\left[\tfrac{1}{\pi_K}\right]\right)^m & \qquad \text{by above} \\ = K^m & \qquad \text{by Lemma 3.11(2) again} \end{aligned}$$

(where the isomorphisms are as $\mathcal{O}_K$-modules or K-vector spaces). Thus $m = n$ and $\mathcal{O}_L \cong \mathcal{O}_K^n$ as $\mathcal{O}_K$-modules, so

$$\mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L \cong (\mathcal{O}_K/\mathfrak{m}_K)^n = k^n$$

has dimension $n = [L : K]$ over k.

Step 2: Show $\dim_k \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L = ef$.
We know that

$$\mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L = \mathcal{O}_L/\mathfrak{m}_L^e.$$

Now we consider π_L-adic expansions. Let $B \subset \mathcal{O}_L$ be a set of coset representatives for $k_L = \mathcal{O}_L/\mathfrak{m}_L$. We may write any element $x \in \mathcal{O}_L/\mathfrak{m}_L^e$ uniquely in the form

$$x = a_0 + a_1\pi_L + \cdots + a_{e-1}\pi_L^{e-1} \pmod{\pi_L^e}, \qquad a_i \in B.$$

In particular, we get an identification

$$\begin{aligned} \frac{\mathcal{O}_L}{\pi_L^e\mathcal{O}_L} \cong \quad \frac{\mathcal{O}_L}{\pi_L} \quad \oplus \quad \frac{\pi_L\mathcal{O}_L}{\pi_L^2\mathcal{O}_L} \quad \oplus \ \cdots \ \oplus \quad \frac{\pi_L^{e-1}\mathcal{O}_L}{\pi_L^e\mathcal{O}_L}, \\ x \mapsto (a_0 \pmod{\pi_L}, a_1\pi_L \pmod{\pi_L^2}, ..., a_{e-1}\pi_L^{e-1} \pmod{\pi_L^e}). \end{aligned}$$

as $\mathcal{O}_L/\pi_L = k_L$ vector spaces. Moreover we have isomorphisms

$$\frac{\pi_L^i\mathcal{O}_L}{\pi_L^{i+1}\mathcal{O}_L} \xrightarrow{\sim} \frac{\mathcal{O}_L}{\pi_L\mathcal{O}_L}, \qquad \pi_L^i x \pmod{\pi_L^{i+1}} \mapsto x \pmod{\pi_L}$$

of k_L-vector spaces, so each summand is a 1-dimensional k_L-vector space. As there are e such summands, we see

$$\mathcal{O}_L/\mathfrak{m}_L^e = \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L \text{ is a } k_L\text{-vector space of dimension } e.$$

Since $\dim_k k_L = f$, we deduce

$$\dim_k \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L = (\dim_{k_L} \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L) \times f = ef,$$

proving step 2.

Combining steps 1 and 2, we conclude that

$$[L : K] = \dim_k \mathcal{O}_L/\mathfrak{m}_K\mathcal{O}_L = ef,$$

as required. □

Proposition 8.12. *If $f(L/K)$ is finite, then L/K is finite.*

Proof. Note e is always finite, so by assumption ef is finite.

In Step 1 of the proof of Proposition 8.11, we showed that:

$$\textit{if } \mathcal{O}_L \textit{ is a finitely generated } \mathcal{O}_K\textit{-module, then } \dim_k\big(\mathcal{O}_L/\pi_K\mathcal{O}_L\big) = [L:K]. \tag{8.5}$$

In Step 2 of the same proof, we showed (unconditionally) that if f is finite, then

$$\dim_k\Big(\mathcal{O}_L/\pi_K\mathcal{O}_L\Big) = ef =: n < \infty. \tag{8.6}$$

(We implicitly use Lemma 8.9 here). Combining (8.5) and (8.6), we see that:

If $\mathcal{O}_K$ is a finitely generated $\mathcal{O}_K$-module, then L/K is finite.

It remains to show $\mathcal{O}_L$ is finitely generated. Let $\overline{x}_1, ..., \overline{x}_n$ be a k-basis of $\mathcal{O}_L/\pi_K\mathcal{O}_L$, and lift to elements $x_1, ..., x_n$ of $\mathcal{O}_L$. If $\overline{b} \in \mathcal{O}_L/\pi_K\mathcal{O}_L$, then we may write

$$\overline{b} = \overline{a}_1\overline{x}_1 + \cdots + \overline{a}_n\overline{x}_n, \qquad \overline{a}_i \in k.$$

If a_i is any lift of $\overline{a}_i$ to $\mathcal{O}_K$, we see that

$$a_1x_1 + \cdots + a_nx_n \equiv \overline{b} \pmod{\pi_K\mathcal{O}_L}$$

is a coset representative of $\overline{b}$ in $\mathcal{O}_Kx_1 + \cdots + \mathcal{O}_Kx_n$, so there exists a set

$$B \subset \mathcal{O}_Kx_1 + \cdots + \mathcal{O}_Kx_n \subset \mathcal{O}_L$$

of coset representatives of $\mathcal{O}_L/\pi_K\mathcal{O}_L$.

Note $\pi_K \in \mathfrak{m}_L$, so if $y \in \mathcal{O}_L$, then by Proposition 3.21(i) we can write

$$\begin{aligned} y &= \sum_{i=0}^{\infty} b_i\pi_K^i, & b_i \in B \\ &= \sum_{i=0}^{\infty}\Big(\sum_{j=1}^{n} a_{ij}x_j\Big)\pi_K^i, & a_{ij} \in \mathcal{O}_K \\ &= \sum_{j=1}^{n}\Big(\sum_{i=0}^{\infty} a_{ij}\pi_K^i\Big)x_j. \end{aligned}$$

Note $\sum_{i=0}^{\infty} a_{ij}\pi_K^i \in \mathcal{O}_K$ since K (hence $\mathcal{O}_K$) is complete (cf. Proposition 3.21(ii)). Hence $\mathcal{O}_L$ is generated over $\mathcal{O}_L$ by the x_i, are we are done. ☐

We can finally complete the classification of local fields.

Theorem 8.13. *Let K be a non-archimedean local field of characteristic 0. Then K is a finite extension of $\mathbf{Q}_p$ for some prime p.*

Proof. As K has characteristic 0, it contains $\mathbf{Q}$. By Ostrowski's theorem, the restriction of its absolute value $|\cdot|_K$ to $\mathbf{Q}$ must be equivalent to $|\cdot|_p$ for some prime p. As K is complete, it must contain $\mathbf{Q}_p$. Hence the residue field k of K is an extension of $\mathbf{F}_p$.

Since K is a local field, k is finite, so $[k:\mathbf{F}_p] = f(K/\mathbf{Q}_p)$ is finite. But by Proposition 8.12, this forces $K/\mathbf{Q}_p$ to be finite, as required. ☐

Definition. A *p-adic field* is a finite extension of $\mathbf{Q}_p$.

8.4. Extensions of number fields under completion

Let L/K be an extension of number fields.

If v is a place (i.e. equivalence class of absolute values) of K, and w a place of L, we say w lies above v, written $w|v$, if $|\cdot|_w$ extends $|\cdot|_v$.

Example. By Theorem 7.15, if $\mathfrak{p}$ and $\mathfrak{P}$ are primes of K and L respectively, then $\mathfrak{P}|\mathfrak{p}$ (as primes) if and only if $w_{\mathfrak{P}}|v_{\mathfrak{p}}$ (as the associated places).

Let L/K be an extension of number fields, $\mathfrak{p}$ a prime of K, and $\mathfrak{P}|\mathfrak{p}$ a prime of L. Let $K_{\mathfrak{p}} :=$ completion of K with respect to $|\cdot|_{\mathfrak{p}}$, and $L_{\mathfrak{P}} :=$ completion of L for $|\cdot|_{\mathfrak{P}}$. Then $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is an extension, i.e. we have

$$\begin{tikzcd}
& L_{\mathfrak{P}} \arrow[d, dash] \\
K_{\mathfrak{p}} \arrow[ur, dash] \arrow[d, dash] & L \\
K \arrow[ur, dash, "\text{finite}"'] &
\end{tikzcd}$$

Remark. Recall *tensor products* are an algebraic construction. For us, their relevance is as follows: if L/K is a field extension, then tensor product defines a functor

$$\begin{aligned} \{K - \text{vector spaces}\} &\longrightarrow \{L - \text{vector spaces}\} \\ V &\longmapsto V \otimes_K L, \end{aligned}$$

and if $\{x_i\}$ is a basis of V over K, then $\{x_i \otimes 1\}$ is a basis of $V \otimes_K L$ over L.

Of particular relevance for us is the fact that when $V = K[X]$ is the ring of polynomials over K, then $K[X] \otimes_K L \cong L[X]$ (as rings, not just as vector spaces!).

Lemma 8.14. *(i) The natural map $\phi : L \otimes_K K_{\mathfrak{p}} \to L_{\mathfrak{P}}$, $\ell \otimes k \mapsto \ell k$, is surjective.*

(ii) We have $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] \leqslant [L : K]$. In particular, $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is finite.

Proof. Let $x_1, ..., x_n$ be a basis of L as a K-vector space, and let

$$M = \mathrm{Im}(\phi) = LK_{\mathfrak{p}} \subset L_{\mathfrak{P}}.$$

Note

$$LK_{\mathfrak{p}} = (Kx_1 + \cdots + Kx_n)K_{\mathfrak{p}} = x_1K_{\mathfrak{p}} + \cdots + x_nK_{\mathfrak{p}},$$

so the x_i span M as a $K_{\mathfrak{p}}$-vector space; thus some subset of the x_i is a basis, and we deduce that

$$[M : K_{\mathfrak{p}}] \leqslant [L : K]$$

is finite. Thus by Theorem 8.6, M is complete with respect to a unique absolute value $|\cdot|_M$ extending $|\cdot|_{K_{\mathfrak{p}}}$.

Note $L \subset M \subset L_{\mathfrak{P}}$. Moreover:

- the absolute value $|\cdot|_{L_{\mathfrak{P}}}$ extends $|\cdot|_{K_{\mathfrak{p}}}$ by construction (since $|\cdot|_{\mathfrak{P}}$ on L extends $|\cdot|_{\mathfrak{p}}$ on K),
- so its restriction to M must be $|\cdot|_M$ (as this is the *unique* absolute value on M extending $|\cdot|_{K_{\mathfrak{p}}}$).

– We deduce finally that if $x \in L \subset M$, then

$$|x|_M = |x|_{L_{\mathfrak{P}}} = |x|_{\mathfrak{P}},$$

where the first equality is because $|\cdot|_{L_{\mathfrak{P}}}$ extends $|\cdot|_M$, and the second because $|\cdot|_{L_{\mathfrak{P}}}$ extends $|\cdot|_{\mathfrak{P}}$. Thus $|\cdot|_M$ on M extends $|\cdot|_{\mathfrak{P}}$ on L.

Now $L_{\mathfrak{P}}$ is the smallest complete field containing $(L, |\cdot|_{\mathfrak{P}})$ (by properties of completion). Since $M \subset L_{\mathfrak{P}}$, and $(L, |\cdot|_{\mathfrak{P}}) \subset M$, we deduce $M = L_{\mathfrak{P}}$. Thus $\mathrm{Im}(\phi) = M = L_{\mathfrak{P}}$, so ϕ is surjective, giving (i); and $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [M : K_{\mathfrak{p}}] \leqslant [L : K]$, giving (ii). $\square$

Remark. By considering $K = \mathbf{Q}$, as a special case we see that if L is any number field, and $\mathfrak{P}$ any prime of L above p, then the completion $L_{\mathfrak{P}}$ is a finite extension of $\mathbf{Q}_p$ (and in particular is a local field).

Lemma 8.15. *The ramification and residue indices do not change under completion, that is,*

(a) $e(\mathfrak{P}/\mathfrak{p}) = e(L_{\mathfrak{P}}/K_{\mathfrak{p}})$,

(b) $f(\mathfrak{P}/\mathfrak{p}) = f(L_{\mathfrak{P}}/K_{\mathfrak{p}})$.

Proof. (a) Pick $x \in K$ with $v_{\mathfrak{p}}(x) = 1$. Then:

– By definition $x\mathcal{O}_K = \mathfrak{p} \cdot I$, with I prime to $\mathfrak{p}$. So

$$x\mathcal{O}_L = \mathfrak{P}^{e(\mathfrak{P}/\mathfrak{p})} \cdot I', \qquad I' \text{ prime to } \mathfrak{P}.$$

Thus

$$v_{\mathfrak{P}}(x) = e(\mathfrak{P}/\mathfrak{p}). \tag{8.7}$$

– x is a uniformiser in $\mathcal{O}_{K_{\mathfrak{p}}}$, so $x\mathcal{O}_{K_{\mathfrak{p}}} = \mathfrak{p}\mathcal{O}_{K_{\mathfrak{p}}}$. Thus

$$x\mathcal{O}_{L_{\mathfrak{P}}} = \mathfrak{p}\mathcal{O}_{L_{\mathfrak{P}}} = \mathfrak{P}^{e(L_{\mathfrak{P}}/K_{\mathfrak{p}})}\mathcal{O}_{L_{\mathfrak{P}}},$$

so

$$v_{\mathfrak{P}}(x) = e(L_{\mathfrak{P}}/K_{\mathfrak{p}}). \tag{8.8}$$

Combining (8.7) and (8.8) gives

$$e(\mathfrak{P}/\mathfrak{p}) = v_{\mathfrak{P}}(x) = e(L_{\mathfrak{P}}/K_{\mathfrak{p}}).$$

(b) Note that since $\mathcal{O}_K \cap \mathfrak{p}\mathcal{O}_{K_{\mathfrak{p}}} = \mathfrak{p}$, the inclusion $\mathcal{O}_K \hookrightarrow \mathcal{O}_{K_{\mathfrak{p}}}$ induces an injective homomorphism

$$\phi : \mathcal{O}_K/\mathfrak{p} \hookrightarrow \mathcal{O}_{K_{\mathfrak{p}}}/\mathfrak{p}\mathcal{O}_{K_{\mathfrak{p}}}.$$

This map is actually an isomorphism. To show this, it suffices to prove it is surjective. Write $\mathfrak{p}\mathcal{O}_{K_{\mathfrak{p}}} = \pi\mathcal{O}_{K_{\mathfrak{p}}}$ (i.e. fix a uniformiser π in $K_{\mathfrak{p}}$). Let

$$\overline{y} \in \mathcal{O}_{K_{\mathfrak{p}}}/\pi\mathcal{O}_{K_{\mathfrak{p}}},$$

and lift to some $y \in \mathcal{O}_{K_{\mathfrak{p}}}$. Then $y = \lim_{n\to\infty} y_n$ is a limit of a Cauchy sequence, with $y_n \in \mathcal{O}_K$. In particular,

$$y \equiv y_n \pmod{\mathfrak{p}\mathcal{O}_{K_{\mathfrak{p}}}} \tag{8.9}$$

for sufficiently large n. Let

$$\overline{y}_n := y_n \pmod{\mathfrak{p}};$$

then $\phi(\overline{y}_n) = \overline{y}$ by (8.9), and hence ϕ is surjective, and

$$\mathcal{O}_K/\mathfrak{p} \cong \mathcal{O}_{K_\mathfrak{p}}/\mathfrak{p}\mathcal{O}_{K_\mathfrak{p}}.$$

The same argument shows that $\mathcal{O}_L/\mathfrak{P} \cong \mathcal{O}_{L_\mathfrak{P}}/\mathfrak{P}\mathcal{O}_{L_\mathfrak{P}}$. Thus the residue fields are unchanged by completion, and $f(\mathfrak{P}/\mathfrak{p}) = f(L_\mathfrak{P}/K_\mathfrak{p})$. □

Theorem 8.16. *The natural map*

$$\begin{aligned} L \otimes_K K_\mathfrak{p} &\longrightarrow \bigoplus_{\mathfrak{P}|\mathfrak{p}} L_\mathfrak{P} \\ \ell \otimes k &\longmapsto (\ell k, \dots, \ell k) \end{aligned}$$

is an isomorphism.

Proof. Write $L = K(\alpha)$, and let $f(X)$ be the minimal polynomial of α over K. Note $f(X)$ may not be irreducible any more over $K_\mathfrak{p}$; factorise

$$f(X) = f_1(X)\cdots f_r(X), \qquad f_i \in K_\mathfrak{p}[X] \text{ distinct irreducibles.}$$

(Note all the f_i are distinct as f is separable, as we are in characteristic 0). Note $L = K[X]/(f)$. We have

$$\begin{aligned} L \otimes_K K_\mathfrak{p} &= K[X]/(f) \otimes_K K_\mathfrak{p} \\ &= K_\mathfrak{p}[X]/(f) \cong K_\mathfrak{p}[X]/(f_1) \oplus \cdots \oplus K_\mathfrak{p}[X]/(f_r), \end{aligned}$$

where the last isomorphism is the Chinese Remainder Theorem. These isomorphisms are as $K_\mathfrak{p}$-algebras, hence continuous with respect to any norm.

Let $L_i := K_\mathfrak{p}[X]/(f_i)$. This contains $K_\mathfrak{p}$ trivially.

Claim. *L_i contains L as a dense subfield.*

Proof of claim: We have a natural map $K[X] \hookrightarrow K_\mathfrak{p}[X]$. We have a prime ideal $(f_i) \subset K_\mathfrak{p}[X]$. Then $(f_i) \cap K[X]$ is a prime ideal of $K[X]$, and $f \in (f_i) \cap K[X]$. Since $(f) \subset K[X]$ is maximal, we deduce $(f_i) \cap K[X] = (f) \subset K[X]$.

We deduce that (f) is the kernel of the composition

$$K[X] \hookrightarrow K_\mathfrak{p}[X] \twoheadrightarrow K_\mathfrak{p}[X]/(f_i),$$

that is, this induces an injective map

$$L = K[X]/(f) \hookrightarrow K_\mathfrak{p}[X]/(f_i).$$

Moreover L is dense in L_i since K is dense in $K_\mathfrak{p}$.

Claim. *Each L_i is isomorphic to $L_\mathfrak{P}$ for some $\mathfrak{P}|\mathfrak{p}$.*

Proof of claim: We know $[L_i : K_\mathfrak{p}] < \infty$, so by Theorem 8.6 there exists a unique absolute value $|\cdot|_i$ on L_i extending $|\cdot|_\mathfrak{p}$ on $K_\mathfrak{p}$, and L_i is complete with respect to $|\cdot|_i$. By Theorem 7.15, the restriction of $|\cdot|_i$ to L is equivalent to $|\cdot|_\mathfrak{P}$ for some $\mathfrak{P}|\mathfrak{p}$. Since L_i is complete and $L \subset L_i$ is dense, we then have $L_i \cong L_\mathfrak{P}$, proving the claim.

Claim. *Each $\mathfrak{P}|\mathfrak{p}$ appears at least once.*

Proof of claim: By Lemma 8.14, $L_{\mathfrak{P}} = LK_{\mathfrak{p}} = K_{\mathfrak{p}}(\alpha)$. Let $g(X)$ be the minimal polynomial of $\alpha \in L_{\mathfrak{P}}$ over $K_{\mathfrak{p}}$. Then $g(X)$ is an irreducible factor of $f(X) \in K_{\mathfrak{p}}[X]$, so $g(X) = f_i(X)$ for some i, and $L_{\mathfrak{P}} \cong L_i$.

We complete the proof. By the second and third claims, we know

$$\bigoplus_{\mathfrak{P}|\mathfrak{p}} L_{\mathfrak{P}} \hookrightarrow L \otimes_K K_{\mathfrak{p}}$$

as $K_{\mathfrak{p}}$-vector spaces. We compute their dimensions:

$$\begin{aligned}
\dim_{K_{\mathfrak{p}}} \Big(\bigoplus_{\mathfrak{P}|\mathfrak{p}} L_{\mathfrak{P}}\Big) &= \sum_{\mathfrak{P}|\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] \\
&= \sum_{\mathfrak{P}|\mathfrak{p}} e(L_{\mathfrak{P}}/K_{\mathfrak{p}}) \cdot f(L_{\mathfrak{P}}/K_{\mathfrak{p}}) \qquad \text{by Proposition 8.11} \\
&= \sum_{\mathfrak{P}|\mathfrak{p}} e(\mathfrak{P}/\mathfrak{p}) \cdot f(\mathfrak{P}/\mathfrak{p}) \qquad \text{by Lemma 8.15} \\
&= [L : K] \qquad \text{by Theorem 7.10} \\
&= \dim_{K_{\mathfrak{p}}} \Big(L \otimes_K K_{\mathfrak{p}}\Big).
\end{aligned}$$

Thus we have an injective map of vector spaces of the same dimension, which must be an isomorphism, giving the theorem. □

Example. Let

$$f(X) = X^3 - 6X^2 - 3X + 3 \in \mathbf{Q}[X],$$

which is Eisenstein for 3 and hence irreducible, and let $L = \mathbf{Q}[X]/(f)$. We will show:

$$L \otimes_{\mathbf{Q}} \mathbf{Q}_5 \cong \mathbf{Q}_5 \oplus \mathbf{Q}_5(\sqrt{2}).$$

Step 1: compute the primes above 5. We have $5^2 \nmid \mathrm{disc}(f) = 3^3 \times 139$, so to compute the primes above 5, we use Dedekind–Kummer.

Note $f(1) = -5 \equiv 0 \pmod 5$, so f has a factor $(X-1)$ modulo 5. Writing

$$\begin{aligned}
f(X) \equiv& (X-1)(X^2 + aX + b) \pmod 5 \\
&= X^3 + (a-1)X^2 + (b-a)X - b \equiv X^3 - X^2 + 2X - 2 \pmod 5,
\end{aligned}$$

and solving, we find $a = 0$ and $b = 2$, whence f factorises as

$$f(X) \equiv (X-1)(X^2+2) \pmod 5.$$

Thus Dedekind–Kummer (Theorem 7.11 says

$$5\mathcal{O}_L = \mathfrak{p}\mathfrak{q}, \qquad \mathfrak{p} = (5, \alpha - 1), \qquad \mathfrak{q} = (5, \alpha^2 + 2),$$

where α is a root of f, and that

$$e(\mathfrak{p}/5) = 1, \qquad f(\mathfrak{p}/5) = \deg(X-1) = 1,$$

and

$$e(\mathfrak{q}/5) = 1, \qquad f(\mathfrak{q}/5) = \deg(X^2+2) = 2.$$

Step 2: Apply the theorem. By Theorem 8.16,

$$L \otimes_{\mathbf{Q}} \mathbf{Q}_5 \cong L_{\mathfrak{p}} \oplus L_{\mathfrak{q}}.$$

Step 3: Compute the fields $L_{\mathfrak{p}}$ and $L_{\mathfrak{q}}$. What are the fields $L_{\mathfrak{p}}$ and $L_{\mathfrak{q}}$? By Proposition 8.12 and Lemma 8.15, we know that

$$[L_{\mathfrak{p}} : \mathbf{Q}_5] = e(\mathfrak{p}/5) \cdot f(\mathfrak{p}/5) = 1,$$

so $L_{\mathfrak{p}} = \mathbf{Q}_5$. Similarly we know

$$[L_{\mathfrak{q}} : \mathbf{Q}_5] = e(\mathfrak{q}/5) \cdot f(\mathfrak{q}/5) = 2,$$

so $L_{\mathfrak{q}}$ is a quadratic extension of $\mathbf{Q}_5$. By Proposition 5.5(3), we have

$$L_{\mathfrak{q}} \in \{\mathbf{Q}_5(\sqrt{2}), \mathbf{Q}_5(\sqrt{5}), \mathbf{Q}_5(\sqrt{10})\}.$$

One may compute (e.g. by the methods of the examples before and after Proposition 8.12) that these extensions of $\mathbf{Q}_5$ have $(e,f) = (1,2), (2,1), (2,1)$ respectively; thus $L_{\mathfrak{q}} = \mathbf{Q}_5(\sqrt{2})$.

Summarising, as claimed we have

$$L \otimes_{\mathbf{Q}} \mathbf{Q}_5 \cong \mathbf{Q}_5 \oplus \mathbf{Q}_5(\sqrt{2}).$$

Remark. Combining this example with the proof of Theorem 8.16, this is showing that f factorises in $\mathbf{Q}_5$ as the product

$$f(X) = (X - \alpha_1)(X^2 + aX + b), \qquad \alpha_1, a, b \in \mathbf{Q}_5,$$

where

$$X - \alpha_1 \equiv X - 1 \pmod 5, \qquad X^2 + aX + b \equiv X^2 + 2 \pmod 5.$$

One can prove this directly by proving an analogue of Hensel's lemma for polynomials!

Example. Let $K = \mathbf{Q}$, $L = \mathbf{Q}(i)$.

- If $p \equiv 1 \pmod 4$, then p splits as $\mathfrak{P}_1\mathfrak{P}_2$ in L. Since $2 = [L : K] = efr$ and $r = 2$, we have $e = f = 1$, so $[L_{\mathfrak{P}_i} : \mathbf{Q}_p] = ef = 1$. Thus

  $$\mathbf{Q}(i) \otimes_{\mathbf{Q}} \mathbf{Q}_p \cong \mathbf{Q}_p \oplus \mathbf{Q}_p.$$

  Note this is consistent with Hensel's lemma, which says $X^2 + 1$ is solvable in $\mathbf{Q}_p$.
- If $p \equiv 3 \pmod 4$, then $p\mathcal{O}_L = \mathfrak{P}$ is inert. Then $e = r = 1$, so $f = 2$, and

  $$\mathbf{Q}(i) \otimes_{\mathbf{Q}} \mathbf{Q}_p \cong L_{\mathfrak{P}}$$

  is a quadratic extension of $\mathbf{Q}_p$.

8.5. Completions of Galois extensions

Let L/K now be a *Galois* extension of number fields, and $\mathfrak{p}$ a prime of K. We now study how the Galois group acts on the primes above $\mathfrak{p}$ (and the attached invariants).

Proposition 8.17. *If L/K is Galois, then* $\mathrm{Gal}(L/K)$ *acts transitively on the primes above* $\mathfrak{p}$.

Proof. Suppose not; then there exist primes $\mathfrak{P}_1, \mathfrak{P}_2|\mathfrak{p}$ such that $\sigma(\mathfrak{P}_1) \neq \mathfrak{P}_2$ for all $\sigma \in G := \mathrm{Gal}(L/K)$. By the Chinese Remainder Theorem, there exists $x \in \mathcal{O}_L$ such that

$$x \equiv 0 \pmod{\mathfrak{P}_2}, \qquad x \equiv 1 \pmod{\sigma(\mathfrak{P}_1)} \ \ \forall \sigma \in G.$$

Define

$$N_{L/K}(x) := \prod_{\sigma \in G} \sigma(x).$$

Since $x \in \mathfrak{P}_2$, by the property of being an ideal, we have $N_{L/K}(x) \in \mathfrak{P}_2$. But also this product is invariant under every element of the Galois group, as e.g. if $\tau \in G$, we have

$$\tau\left(N_{L/K}(x)\right) = \prod_{\sigma \in G} \tau\sigma(x) = \prod_{\sigma' \in G} \sigma'(x) = N_{L/K}(x),$$

where $\sigma' = \tau\sigma$. As $N_{L/K}(x)$ is invariant under G, it lies in K, and

$$N_{L/K}(x) \in \mathfrak{P}_2 \cap K = \mathfrak{p} \subset \mathfrak{P}_1.$$

Since $\mathfrak{P}_1$ is prime, there exists $\tau \in G$ such that $\tau(x) \in \mathfrak{P}_1$. But then $\tau(x) \equiv 0 \pmod{\mathfrak{P}_1}$, contradicting the definition of x. ☐

Corollary. *If L/K is Galois, then*

$$e_1 = \cdots = e_r =: e, \qquad f_1 = \cdots = f_r =: f.$$

In particular,

$$[L : K] = efr.$$

Proof. The statement is empty if $r = 1$. If $r \geqslant 2$, without loss of generality, it suffices to prove that $e_1 = e_2$ and $f_1 = f_2$. Let $\sigma \in \mathrm{Gal}(L/K)$ with $\sigma(\mathfrak{P}_1) = \mathfrak{P}_2$. Then

$$\begin{aligned}\mathfrak{P}_1^{e_1}\mathfrak{P}_2^{e_2}\cdots\mathfrak{P}_r^{e_r} = \mathfrak{p}\mathcal{O}_L = \sigma(\mathfrak{p}\mathcal{O}_L) &= \sigma(\mathfrak{P}_1)^{e_1}\sigma(\mathfrak{P}_2)^{e_2}\cdots\sigma(\mathfrak{P}_r)^{e_r} \\ &= \mathfrak{P}_2^{e_1}\sigma(\mathfrak{P}_2)^{e_2}\cdots\sigma(\mathfrak{P}_r)^{e_r}.\end{aligned}$$

By unique factorisation, the power of $\mathfrak{P}_2$ appearing in each side is equal; thus $e_1 = e_2$.

Note that σ induces an isomorphism $\mathcal{O}_L/\mathfrak{P}_1 \xrightarrow{\sim} \mathcal{O}_L/\mathfrak{P}_2$. We deduce immediately that $f_1 = f_2$, as required. ☐

Definition 8.18. Let $\mathcal{O}_K$ be Dedekind, $K = \mathrm{Frac}(\mathcal{O}_K)$, L/K a Galois extension, and $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L. Let $\mathfrak{P}|\mathfrak{p}$ primes of L and K. The *decomposition group* of $\mathfrak{P}|\mathfrak{p}$ is

$$D_{\mathfrak{P}} := \{\sigma \in \mathrm{Gal}(L/K) : \sigma(\mathfrak{P}) = \mathfrak{P}\}.$$

Lemma 8.19. $\#D_{\mathfrak{P}} = e(\mathfrak{P}/\mathfrak{p}) \cdot f(\mathfrak{P}/\mathfrak{p})$.

Proof. We know $\mathrm{Gal}(L/K)$ has size efr. It acts transitively on the set $\{\mathfrak{P}|\mathfrak{p}\}$, which has size r, and $D_{\mathfrak{P}}$ is the stabiliser of $\mathfrak{P}$; so by the orbit-stabiliser theorem $D_{\mathfrak{P}}$ has size ef. ☐

Proposition 8.20. *Let L/K be a Galois extension of number fields, and $\mathfrak{P}|\mathfrak{p}$ primes of L and K. Then:*

(i) If $\mathfrak{P}'$ is another prime above $\mathfrak{p}$, then $L_{\mathfrak{P}} \cong L_{\mathfrak{P}'}$ over $K_{\mathfrak{p}}$.

(ii) $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is Galois.

Proof. (i) By Proposition 8.17, there exists $\sigma \in \mathrm{Gal}(L/K)$ with $\sigma(\mathfrak{P}) = \mathfrak{P}'$. But then σ induces a field isomorphism $L_{\mathfrak{P}} \xrightarrow{\sim} L_{\mathfrak{P}'}$. Since σ fixes K, it must also fix its completion $K_{\mathfrak{p}}$ (since σ acts continuously, so preserves limits of Cauchy sequences); hence the field isomorphism fixes $K_{\mathfrak{p}}$.

(ii) Recall that in characteristic 0:

a field extension E/F is Galois $\iff$ E is the splitting field[3] of a polynomial in $F[X]$.

Thus as L/K is Galois, L is the splitting field of some $f \in K[X]$. Since $L \subset L_{\mathfrak{P}}$, we also know f splits[4] over $L_{\mathfrak{P}}$; and since $L_{\mathfrak{P}} = K_{\mathfrak{p}}(\alpha)$ for some root α of f, we know that any intermediate extension $K_{\mathfrak{p}} \subset M \subset L_{\mathfrak{P}}$ does not contain α, so f cannot split over such an M. Thus $L_{\mathfrak{P}}$ is the splitting field of f over $K_{\mathfrak{p}}$, and $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is Galois. $\square$

We now relate the Galois groups of $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ and L/K.

Since L/K is Galois, it is a *normal* extension, that is, if $x \in L$, then all the conjugates of x (roots of its minimal polynomial over K) also lie in L. Let $\sigma \in \mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}})$. Since $\sigma(x)$ is a conjugate of x, we have $\sigma(x) \in L$. In particular,

$$\sigma(L) \subset L,$$

and the restriction $\sigma|_L$ is a well-defined element of $\mathrm{Gal}(L/K)$. We get a map

$$\begin{aligned} \mathrm{res} : \mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}}) &\longrightarrow \mathrm{Gal}(L/K) \\ \sigma &\longmapsto \sigma|_L. \end{aligned}$$

Proposition 8.21. *The restriction map $\mathrm{res} : \mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}}) \to \mathrm{Gal}(L/K)$ is injective. It defines an isomorphism*

$$\mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}}) \xrightarrow{\sim} D_{\mathfrak{P}} \subset \mathrm{Gal}(L/K)$$

Proof. For $\sigma \in \mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}})$, we know $\mathrm{res}(\sigma) = 1$ if and only if σ fixes every element of L. Note σ fixes $K_{\mathfrak{p}}$ by definition, so if σ fixes L, then it also fixes $LK_{\mathfrak{p}}$; but by Lemma 8.14, we know $LK_{\mathfrak{p}} = L_{\mathfrak{P}}$, whence $\sigma = 1$ and restriction is injective.

To conclude, it suffices to prove $\mathrm{Im(res)} = D_{\mathfrak{P}}$. Suppose $\sigma \in \mathrm{Im(res)}$. We first claim:

Claim. $\sigma \in D_{\mathfrak{P}}$.

To see this, let $\mathfrak{P}' = \sigma^{-1}(\mathfrak{P})$; the claim is that $\mathfrak{P}' = \mathfrak{P}$. For all $x \in L$, we have

$$\sigma(x)\mathcal{O}_L = \mathfrak{P}^s \cdot I \qquad \Rightarrow \qquad x\mathcal{O}_L = \sigma^{-1}(\mathfrak{P})^s \cdot \sigma^{-1}(I) = (\mathfrak{P}')^s \cdot \sigma^{-1}(I),$$

so

$$v_{\sigma^{-1}(\mathfrak{P})}(x) = v_{\mathfrak{P}}(\sigma(x)),$$

and

$$|x|_{\mathfrak{P}'} = |x|_{\sigma^{-1}(\mathfrak{P})} = |\sigma(x)|_{\mathfrak{P}} = |x|_{\mathfrak{P}} \tag{8.10}$$

[3] Recall that the *splitting field* of a polynomial $f \in F[X]$ is the smallest extension E/F such that f splits into linear factors over E; i.e. the smallest extension containing all the roots of f.

[4] Thanks to Kenji Terao, who pointed this slick proof out after I gave a much uglier proof in the lectures. Clearly I'm losing the plot late in the term!

where the last equality is Corollary 8.7. By the Chinese Remainder Theorem, if $\mathfrak{P}' \neq \mathfrak{P}$, we may take $x \in \mathfrak{P}$ and $x \notin \mathfrak{P}'$; but then $|x|_{\mathfrak{P}} < 1$ and $|x|_{\mathfrak{P}'} = 1$, contradicting (8.10). This means $\sigma(\mathfrak{P}) = \mathfrak{P}$, so $\sigma \in D_{\mathfrak{P}}$, as claimed.

Thus $\mathrm{Im}(\mathrm{res}) \subset D_{\mathfrak{P}}$. As res is injective, we conclude since

$$\begin{aligned}\#\mathrm{Im}(\mathrm{res}) &= \#\mathrm{Gal}(L_{\mathfrak{P}}/K_{\mathfrak{p}}) \\ &= e(\mathfrak{P}/\mathfrak{p})f(\mathfrak{P}/\mathfrak{p}) = \#D_{\mathfrak{P}}.\end{aligned}$$

□

8.6. (∗) The exponential and logarithm maps

The exponential map gives us a nice description of the ring $\mathcal{O}^\times$ of units in a non-archimedean local field. We now describe this (though this section is non-examinable).

Let $K/\mathbf{Q}_p$ be finite, with valuation ring $\mathcal{O}_K$, normalised absolute value v_K, uniformiser π and residue field $k = \mathcal{O}_K/\pi$. Let

$$e := e(K/\mathbf{Q}_p) = v_K(p).$$

Definition 8.22. (i) Define a map

$$\begin{aligned}(1+\pi\mathcal{O}_K, \times) &\longrightarrow (\pi\mathcal{O}_K, +) \\ 1+x &\longmapsto \log(1+x) := \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .\end{aligned}$$

(ii) If $r > \frac{e}{p-1}$, define a map

$$\begin{aligned}(\pi^r\mathcal{O}_K, +) &\longrightarrow (1+\pi^r\mathcal{O}_K, \times) \\ x &\longmapsto \exp(x) := 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .\end{aligned}$$

Proposition 8.23. *Both* log *and* exp *are well-defined group homomorphisms, and are inverse to each other. If $r > \frac{e}{p-1}$, they thus give an isomorphism of groups*

$$(\pi^r\mathcal{O}_K, +) \xrightarrow{\sim} (1+\pi^r\mathcal{O}_K, \times).$$

Proof. As formal power series in $\mathbf{Q}[\![X,Y]\!]$ (i.e. ignoring convergence), we have the identities

$$\exp(X+Y) = \exp(X)\exp(Y), \qquad \log((1+X)(1+Y)) = \log(1+X) + \log(1+Y)$$

and

$$\log(\exp(X)) = X, \qquad \exp(\log(1+X)) = 1+X.$$

So providing these series converge, the maps are well-defined and inverse to each other.

To see $\log(1+x)$ converges, we simply note that since $x \in \pi\mathcal{O}$, we have

$$v_p\left(\frac{x^n}{n}\right) = v_p(x^n) - v_p(n) \geqslant n - v_p(n) \to \infty \qquad \text{as } n \to \infty.$$

Now we check $\exp(x)$ converges if $x \in \pi^r\mathcal{O}_K$. Note

$$v_K(n!) = e \cdot v_p(n!) \leqslant e \cdot \frac{n-1}{p-1},$$

where the last inequality was proved on the example sheets. So

$$\begin{aligned} v_K\left(\frac{x^n}{n!}\right) = ev_p(x^n) - v_p(n!) &\geqslant nr - e\frac{n-1}{p-1} \\ &= r + (n-1)\left(r - \frac{e}{p-1}\right). \end{aligned}$$

Since $r > e/(p-1)$, this tends to ∞, so $\exp(x)$ converges. This completes the proof. □

Corollary 8.24. *$\mathcal{O}_K^\times$ has a subgroup of finite index isomorphic to $(\mathcal{O}_K, +)$.*

Proof. For any $r \geqslant 1$, the map

$$\begin{aligned} \mathcal{O}_K^\times &\longrightarrow (\mathcal{O}_K/\pi^r)^\times \\ a &\longmapsto a \pmod{\pi^r} \end{aligned}$$

is surjective with kernel $1 + \pi^r\mathcal{O}_K$, which is hence a subgroup of finite index.

Choosing $r \geqslant e/(p-1)$, the proposition shows $1 + \pi^r\mathcal{O}_K \cong (\pi^r\mathcal{O}_K, +) \cong (\mathcal{O}_K, +)$. □

Example. Let $K = \mathbf{Q}_p$, so $e = 1$. If $p > 2$, then $r = 1 > e/(p-1) = 1/(p-1)$, so we see

$$\begin{aligned} \mathbf{Z}_p^\times &\xrightarrow{\sim} (\mathbf{Z}/p)^\times \times 1 + p\mathbf{Z}_p, \\ x &\mapsto (\overline{x}, x/\overline{x}), \end{aligned}$$

where $\overline{x} = x \pmod p$.

If $p = 2$, then $r = 2 > 1 = e/(p-1)$, and

$$\begin{aligned} \mathbf{Z}_p^\times &\xrightarrow{\sim} (\mathbf{Z}/4)^\times \times 1 + 4\mathbf{Z}_p, \\ x &\mapsto (\overline{x}, x/\overline{x}), \end{aligned}$$

where $\overline{x} = x \pmod 4$.

Remark 8.25. This gives another proof that

$$\mathbf{Z}_p^\times/(\mathbf{Z}_p^\times)^2 = \begin{cases} \mathbf{Z}/2\mathbf{Z} & : p > 2 \\ (\mathbf{Z}/2\mathbf{Z})^2 & : p = 2. \end{cases}$$

Chapter 9

Ramification theory

If L/K is an extension of complete fields, we earlier introduced the *ramification index* $e(L/K)$ and *residue index* $f(L/K)$ of the extension. We now study these in more detail, as they play an important role in classifying Galois extensions of local fields.

One of the most important results of this section is that for a Galois extension L/K of local fields, the Galois group is *solvable*, that is, it admits a chain of normal subgroups with cyclic quotients. This puts a very strong restriction on the possible Galois groups that can appear: for example, S_5 is not solvable, so is never the Galois group of an extension of local fields.

We also use ramification theory to prove that there are only finitely many extensions of a local field of any given degree.

9.1. Ramified and unramified extensions

Now let L/K be finite extensions of $\mathbf{Q}_p$, with valuation rings $\mathcal{O}_L$ and $\mathcal{O}_K$, normalised discrete valuations v_L and v_K, uniformisers π_L and π_K, and residue fields k_L and k. Recall the invariants $e(L/K)$ and $f(L/K)$ from Definition 8.8, and that $[L:K] = e(L/K)f(L/K)$ by Proposition 8.11.

Definition 9.1. We say L/K is:

- *unramified* if $e(L/K) = 1$ ($\iff$ $f(L/K) = [L:K]$),
- *ramified* if $e(L/K) > 1$,
- *totally ramified* if $e(L/K) = [L:K]$ ($\iff$ $f(L/K) = 1$).

Lemma 9.2. *If $K \hookrightarrow L \hookrightarrow M$ are extensions of such fields, then*

$$e(M/K) = e(M/L)\cdot e(L/K), \qquad f(M/K) = f(M/L)\cdot f(L/K).$$

In particular, if L/K and M/L are both unramified (resp. totally ramified), then M/L is unramified (resp. totally ramified).

Proof. For f, this follows since $k_M/k_L/k$ are finite field extensions, and in general we have

$$[k_M : k] = [k_M : k_L]\cdot[k_L : k]$$

(i.e. degrees of field extensions are multiplicative in towers).

For e, we argue in the same manner, using that $[L:K] = e(L/K)f(L/K)$, and that $[L:K]$ and $f(L/K)$ are multiplicative in towers[1]. □

[1]Thanks to Ruth Pugh for pointing out this argument, which is nicer than the one from lectures.

Theorem 9.3. *Let L/K be a finite extension, with residue fields $\mathbf{F}_{q^n}$ and $\mathbf{F}_q$. Let $m = q^n - 1$, and ζ_m a primitive mth root of 1. Then $K \subset K(\zeta_m) \subset L$, with*

$$K(\zeta_m)/K \text{ unramified}, \qquad L/K(\zeta_m) \text{ totally ramified.}$$

Proof. Recall the Teichmüller map gives an embedding

$$[-] : \mathbf{F}_{q^n}^\times \hookrightarrow L,$$

and $\mathbf{F}_{q^n}^\times \cong C_{q^n-1} = C_m$. Let $\overline{\zeta}_m$ be a (multiplicative) generator of $\mathbf{F}_{q^n}^\times$, and let $\zeta_m = [\overline{\zeta}_m] \in L^\times$ be its image. Thus $K(\zeta_m) \subset L$. Note that there is an equality of residue fields

$$k_{K(\zeta_m)} = \mathbf{F}_q(\overline{\zeta}_m) = \mathbf{F}_{q^n} = k_L,$$

so $f(L/K(\zeta_m)) = 1$ and hence $L/K(\zeta_m)$ is totally ramified.

It remains to show $K(\zeta_m)/K$ is unramified. Let π be a uniformiser in $K(\zeta_m)$.

Claim. *The natural map*

$$\begin{aligned} \phi : \mathrm{Gal}(K(\zeta_m)/K) &\longrightarrow \mathrm{Gal}(\mathbf{F}_{q^n}/\mathbf{F}_q) \\ \sigma &\longmapsto \sigma \pmod{\pi} \end{aligned}$$

is injective.

Proof of claim: if $\sigma \in \mathrm{Gal}(K(\zeta_m)/K)$, then

$$\begin{aligned} \phi(\sigma) = 1 &\iff \sigma(\zeta_m) = \zeta_m \equiv \overline{\zeta}_m \pmod{\pi} \\ &\iff \sigma(\zeta_m) = \zeta_m \qquad \text{(by uniqueness of Teichmüller lift)} \\ &\iff \sigma = 1. \end{aligned}$$

Thus $\ker(\phi) = \{1\}$, and ϕ is injective, as claimed.

The claim then implies

$$e(K(\zeta_m)/K)f(K(\zeta_m)/K) = [K(\zeta_m) : K] \leqslant [\mathbf{F}_{q^n} : \mathbf{F}_q] = f(K(\zeta_m)/K),$$

which forces

$$e(K(\zeta_m)/K) = 1, \qquad f(K(\zeta_m)/K) = n.$$

In particular, $K(\zeta_m)/K$ is unramified. ∎

This is already enough to completely classify the Galois theory of unramified extensions.

Corollary 9.4. *(i) For each $n \geqslant 1$, the extension*

$$L_n := K(\zeta_{q^n-1})$$

is the unique unramified extension of K of degree n.

(ii) L_n/K is Galois and the natural map $\mathrm{Gal}(L_n/K) \to \mathrm{Gal}(k_{L_n}/k)$ is an isomorphism.

(iii) $\mathrm{Gal}(L_n/K) = \langle \mathrm{Frob}_{L_n/K} \rangle$ is cyclic, where

$$\mathrm{Frob}_{L_n/K}(x) = x^q \pmod{\pi} \qquad \forall x \in \mathcal{O}_L.$$

Proof. (i) In Theorem 9.3, we showed that L_n/K is unramified of degree n. The theorem also shows that if L/K is unramified of degree n, then $L_n \subset L$. Since they have the same degree, we deduce $L_n = L$ is unique.

(ii) In the proof of Theorem 9.3, we showed the natural map on Galois groups is injective, hence surjective by counting sizes.

(iii) This follows from (ii) and the Galois theory of finite fields. $\square$

Corollary. *If L/K is Galois, then the natural map* $\mathrm{Gal}(L/K) \to \mathrm{Gal}(k_L/k)$ *is surjective.*

Proof. Restriction defines a surjective map $\mathrm{Gal}(L/K) \twoheadrightarrow \mathrm{Gal}(K(\zeta_m)/K)$, which is isomorphic to $\mathrm{Gal}(k_L/k)$. $\square$

It is natural to make the following definition now, but we will study it properly later.

Definition. Let L/K Galois. The *inertia group* is

$$I_{L/K} := \ker\Big[\mathrm{Gal}(L/K) \to \mathrm{Gal}(k_L/k)\Big].$$

Note since $[L:K] = e(L/K)f(L/K)$, and $\#\,\mathrm{Gal}(k_L/k) = f(L/K)$, we have $\#I_{L/K} = e(L/K)$ by the first isomorphism theorem of group theory.

We now turn to existence of totally ramified extensions.

Definition 9.5. A polynomial $g(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathcal{O}_K[X]$ is *Eisenstein* if $v_K(a_i) \geqslant 1$ for all i, and $v_K(a_0) = 1$.

Proposition 9.6. *If $g(X) \in \mathcal{O}_K[X]$ is Eisenstein, it is irreducible.*

Proof. This is the same as the proof that Eisenstein polynomials over $\mathbf{Z}$ are irreducible (cf. MA2 Algebra II). $\square$

Remark 9.7. In the next proof, we'll repeatedly use the following fact: if v is a valuation on L, and $\alpha, \beta \in L$ with $v(\alpha) < v(\beta)$, then $v(\alpha+\beta) = v(\alpha)$. This is a valuation analogue of Lemma 2.1(i), and can be deduced by combining that result with Lemma 3.7. Another way of thinking about this is that

$$v(\alpha+\beta) \geqslant \min(v(\alpha), v(\beta)), \qquad \textbf{with equality if the minimum is unique.}$$

Perhaps this is easiest to grasp by example. If $m < n$, then $p^m + p^n = p^m(1+p^{n-m})$. The first term has valuation m, and the second is $\equiv 1 \pmod p$, so has valuation 0; so

$$v_p(p^m + p^n) = m = v_p(p^m).$$

Theorem 9.8. *(i) If $g \in \mathcal{O}_K[X]$ is Eisenstein, α is a root of g, and $L = K(\alpha)$, then L/K is totally ramified and α is a uniformiser for L.*

(ii) Conversely, if L/K is totally ramified, then the minimal polynomial of π_L over $\mathcal{O}_K$ is Eisenstein, and

$$\mathcal{O}_L = \mathcal{O}_K[\pi_L].$$

Proof. (i) Write $g(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$. Let $e = e(L/K) \leqslant n = [L:K]$; then $v_L(a_i) \geqslant e$ and $v_L(a_0) = e$.

We are trying to prove:

(a) L/K is totally ramified, i.e. $e = n$;

(b) α is a uniformiser, i.e. $v_L(\alpha) = 1$.

We first prove a less lofty goal:

Claim. *If α is a root of g, then $v_L(\alpha) > 0$.*

Note first that $v_L(\alpha) \geqslant 0$: this is because α is the root of an integral monic polynomial in $\mathcal{O}_K[X]$, hence lies in the integral closure $\mathcal{O}_L$ of $\mathcal{O}_K$ in L. (We implicitly use Theorem 8.6 here, which says that this integral closure $\mathcal{O}_L$ is also the valuation ring in L for v_L).

To see that moreover $v_L(\alpha) > 0$, note that since $g(\alpha) = 0$, we have

$$\alpha^n = -\sum_{i=0}^{n-1} a_i\alpha^i.$$

Thus

$$\begin{aligned} nv_L(\alpha) = v_L(\alpha^n) = v_L\left(\sum_{i=0}^{n-1} a_i\alpha^i\right) \\ &\geqslant \min(v_L(a_i\alpha^i)) \\ &\geqslant \min(v_L(a_i)) \\ &\geqslant e, \end{aligned}$$

so $v_L(\alpha) \geqslant e/n > 0$, giving the claim.

Now note that $v_L(a_0) = e$, and

$$v_L(a_{n-1}\alpha^{n-1}), \cdots, v_L(a_1\alpha) > e,$$

since if $i > 0$ then $v_L(a_i\alpha^i) = v_L(a_i) + iv_L(\alpha) > v_L(a_i) \geqslant e$. In particular, $a_i\alpha^i$ achieves its (strict) minimal valuation at $i = 0$; and by Remark 9.7, we deduce

$$nv_L(\alpha) = v_L(\alpha^n) = v_L\left(-\sum_{i=0}^{n-1} a_i\alpha^i\right) = v_L(a_0) = e.$$

As $n \geqslant e$ and $v_L(\alpha) \geqslant 1$, this forces $e = n$ and $v_L(\alpha) = 1$. Thus L/K is totally ramified and α is a uniformiser.

(ii) *I ran out of time to cover this in lectures, so the proof of part (ii) is **non-examinable**.*

Since L/K is totally ramified, $[L : K] = e$. Let

$$g(X) = X^m + a_{m-1}X^{m-1} + \cdots + a_0 \in \mathcal{O}_K[X]$$

be the minimal polynomial for π_L (so $m \leqslant e$). Then

$$\pi_L^m = -\sum_{i=0}^{m-1} a_i\pi_L^i.$$

Note

$$v_L(a_i\pi_L^i) = v_L(a_i) + v_L(\pi_L^i) = e \cdot v_K(a_i) + i \equiv i \pmod{e}.$$

Hence all the terms on the right have distinct valuations modulo e, so have distinct valuations. In particular, as in Lemma 2.1(i), we deduce

$$m = v_L(\pi_L^m) = \min_{0 \leqslant i \leqslant m-1} (i + ev_K(a_i))$$

(i.e. we have equality in the ultrametric law, not just inequality). Since $m > i$, we see $v_K(a_i) \geqslant 1$ for all i. But for $i > 0$, we then have $i + ev_K(a_i) > e \geqslant m$, so this minimum must be obtained at $i = 0$, whence $v_K(a_i) = 1$ and $m = e$. Thus g is Eisenstein and $L = K(\pi_L)$.

We show $\mathcal{O}_L = \mathcal{O}_K[\pi_L]$. Let $y \in L$; then since we've shown $L = K(\pi_L)$, write

$$y = \sum_{i=0}^{e-1} b_i \pi_L^i,$$

with $b_i \in K$. Exactly as before, we have

$$v_L(y) = \min_{0 \leqslant i \leqslant e-1} (i + ev_K(b_i)).$$

Thus:

$$\begin{aligned} y \in \mathcal{O}_L &\iff v_K(b_i) \geqslant 0 \text{ for all } i \\ &\iff y \in \mathcal{O}_K[\pi_L]. \end{aligned}$$

$\square$

Example. Consider the polynomial $f(X) = X^3 + 6X + 3 \in \mathbf{Q}[X]$, and let $L = \mathbf{Q}[X]/(f)$. This is Eisenstein at $p = 3$, so f is irreducible and $L/\mathbf{Q}$ is cubic. We also deduce that $f(X)$ remains irreducible in $\mathbf{Q}_3[X]$, and that

$$\begin{aligned} L \otimes_{\mathbf{Q}} \mathbf{Q}_3 &\cong \mathbf{Q}_3[X]/(f) \\ &= \mathbf{Q}_3(\alpha) =: L_{\mathfrak{p}} \end{aligned}$$

is a field, a cubic totally ramified extension of $\mathbf{Q}_3$. By Theorem 8.16, there is a unique prime $\mathfrak{p}|3$, and $e(L_{\mathfrak{p}}/\mathbf{Q}_3) = e(\mathfrak{p}/3) = 3$. In particular, we deduce that 3 splits in L as

$$3\mathcal{O}_L = \mathfrak{p}^3.$$

Note $\mathrm{disc}(f) = -4 \cdot 6^3 - 27 \cdot 3^2 = -1107 = -3^3 \cdot 41$, which is divisible by 3^2; so we *couldn't* deduce this from the naive version of Kummer–Dedekind[2].

Theorem 9.8 shows that totally ramified extensions correspond to Eisenstein polynomials. However, this correspondence is far from bijective. We will show in the next subsection that in fact, there are many – even infinitely many! – Eisenstein polynomials corresponding to each individual totally ramified extension.

9.2. Krasner's lemma

Recall that we showed, in §5.3, that $\mathbf{Q}_p$ has only 3 (resp. 7) quadratic extensions if p is odd (resp. $p = 2$). The main result of this section is a huge generalisation: that for any n, there are only finitely many extensions of $\mathbf{Q}_p$ (or more generally any p-adic field $K/\mathbf{Q}_p$) of degree n.

Let us sketch the idea of the proof:

[2] In fact we have $\mathcal{O}_L = \mathbf{Z}[\alpha]$, so we *are* allowed to apply Kummer–Dedekind, but I only know this because SageMath told me so!

(1) By Theorem 9.3, it suffices to consider $K/\mathbf{Q}_p$ either unramified or totally ramified, since every extension is built from an unramified part and a totally ramified part.

(2) For each n, we already showed there is a *unique* unramified extension of degree n. So we are reduced to totally ramified extensions.

(3) We showed that totally ramified extensions are generated by the roots of Eisenstein polynomials. Show that if two Eisenstein polynomials are 'nearby', then they give the same totally ramified extension. Deduce that if Z is the set of all Eisenstein polynomials of degree n, then the subset Z_L corresponding to any fixed totally ramified L is *open* in Z.

(4) Show Z is compact, hence there are only finitely many totally ramified extensions.

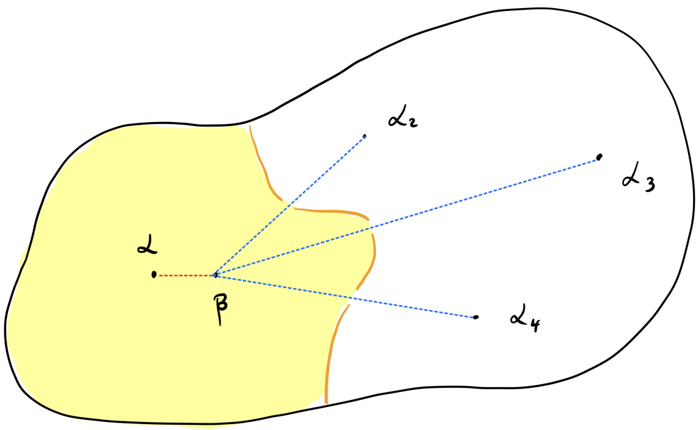
The difficult step is (3). For this, we need:

Lemma 9.9 (Krasner). *Let $K/\mathbf{Q}_p$ be a finite extension. Let $f \in \mathcal{O}_K[X]$ monic irreducible, with roots $\alpha = \alpha_1, ..., \alpha_n \in \overline{K}$. Suppose $\beta \in \overline{K}$ with*

$$|\beta - \alpha| < |\beta - \alpha_i| \qquad i = 2, ..., n.$$

Then $\alpha \in K(\beta)$.

Remark. This is perhaps best understood graphically. In the following picture, $\alpha, \alpha_2, \alpha_3$ and α_4 are the roots of an irreducible quartic polynomial over K, and the shape represents $\overline{K}$:



If β is in the yellow shaded region, i.e. if β is strictly closer to α than any of the other α_i, Krasner's lemma says $\alpha \in K(\beta)$. In particular, α lies in "many" extensions. (Of course, this picture is a bit misleading, as for it to be accurate we'd need to draw it on p-adic paper. This is, however, an intuitive approximation).

Proof. Let $L = K(\beta)$ and $L' = L(\alpha_1, ..., \alpha_n)$. Then L' is the splitting field of f over L, hence is Galois. If $\sigma \in \mathrm{Gal}(L'/L)$, then

$$|\beta - \sigma(\alpha)| = |\sigma(\beta - \alpha)| = |\beta - \alpha|.$$

If $\sigma(\alpha) = \alpha_i$ for $i \geqslant 2$, then $|\beta - \sigma(\alpha)| < |\beta - \alpha|$, contradiction; so $\sigma(\alpha) = \alpha$ for all σ, and hence $\alpha \in L = K(\beta)$. $\square$

The following theorem says that 'nearby polynomials define the same extensions'.

Theorem 9.10. *Let*

$$f(X) = X^n + a_{n-1}X^n + \cdots + a_0 \in \mathcal{O}_K[X]$$

be irreducible. There exists $\varepsilon > 0$ such that if

$$g(X) = X^n + b_{n-1}X^n + \cdots + b_0 \in \mathcal{O}_K[X]$$

with $|b_i - a_i| < \varepsilon$ for all i, then g is irreducible, and

$$K[X]/(f) = K[X]/(g).$$

Proof. (Non-examinable). Write the (distinct) roots of f as $\alpha = \alpha_1, ..., \alpha_n \in \mathcal{O}_K$, so:

- $f(\alpha) = 0$ and $f'(\alpha) \neq 0$,
- $K(\alpha) = K[X]/(f)$ has degree n.

Let β be a root of g, so that $K(\beta)/K$ is an extension of degree $\leqslant n$. It suffices to prove that

$$\alpha \in K(\beta), \tag{9.1}$$

as then:

- $K(\alpha) \subset K(\beta)$,
- $K(\alpha)/K$ has degree n and $K(\beta)/K$ has degree $\leqslant n$,
- so both have degree n over K and $K(\alpha) = K(\beta)$.

Morevoer $K(\beta)/K$ has degree n if and only if g is irreducible.

We'll prove $\alpha \in K(\beta)$ using Krasner's lemma[3].

We'll pick a good choice of β using Hensel's lemma, and the following claims:

Claim. *For sufficiently small ε, we have*

$$|g(\alpha)| < |f'(\alpha)|^2 \tag{9.2}$$

$$\text{and} \qquad |g'(\alpha) - f'(\alpha)| < |f'(\alpha)|. \tag{9.3}$$

To see this is possible, note

$$\begin{aligned} |g(\alpha)| = |g(\alpha) - f(\alpha)| &= \left|\sum_{i=0}^{n-1}(a_i - b_i)\cdot\alpha^i\right| \\ &\leqslant \max_{i=0}^{n-1}(|a_i - b_i|\cdot|\alpha|^i) \\ &\leqslant \varepsilon\cdot\max(1, |\alpha|^{n-1}). \end{aligned}$$

Similarly we have

$$\begin{aligned} |g'(\alpha) - f'(\alpha)| &= \left|\sum_{i=1}^{n-1} i\cdot(a_i - b_i)\cdot\alpha^{i-1}\right| \\ &\leqslant \max_{i=1}^{n-1}(|i|\cdot|a_i - b_i|\cdot|\alpha^{i-1}|) \\ &\leqslant 1\cdot\varepsilon\cdot\max(1, |\alpha|^{n-2}). \end{aligned}$$

In particular, we may take ε to be small enough that (9.2) and (9.3) hold. For the rest of the proof, suppose they hold.

[3] Morally, the argument is as follows: "as $g \to f$, the roots of g tend to the roots of f". In particular, if ε is sufficiently small, then g is so close to f that one of the roots β of g must be much closer to the distinguished root α of f than any of the other roots of f. Then Krasner tells us that α is contained in $K(\beta)$.

Claim. *$|g'(\alpha)| = |f'(\alpha)|$, i.e. $|g(\alpha)| < |g'(\alpha)|^2$ (by (9.2)).*

If this were not true, then by Lemma 2.1 we have *equality* in the ultrametric law

$$|g'(\alpha) - f'(\alpha)| \leqslant \max(|f'(\alpha)|, |g'(\alpha)|),$$

i.e.

$$|g'(\alpha) - f'(\alpha)| = \max(|f'(\alpha)|, |g'(\alpha)|) \geqslant |f'(\alpha)|.$$

But this contradicts (9.3), and

$$|g'(\alpha)| = |f'(\alpha)|,$$

as claimed.

By Hensel's lemma (Theorem 5.2), g has a root $\beta \in K(\alpha)$ with $|\beta - \alpha| < |g'(\alpha)|$.

Claim. *We have $|\beta - \alpha| < |\beta - \alpha_j| \qquad \forall j = 2, ..., n$.*

Given the claim, Krasner's lemma says $\alpha \in K(\beta)$, and we are done by the arguments above.

To see the claim, note

$$|g'(\alpha)| = |f'(\alpha)| = \Big|\prod_{i=2}^{n}(\alpha - \alpha_i)\Big| \leqslant |\alpha - \alpha_j|, \qquad 2 \leqslant j \leqslant n,$$

since each $|\alpha - \alpha_i| \leqslant 1$. But then

$$|\beta - \alpha| < |g'(\alpha) \leqslant |\alpha - \alpha_j|.$$

Again using Lemma 2.1(i), we deduce

$$|\beta - \alpha_j| = |(\beta - \alpha) + (\alpha - \alpha_j)| = \max(|\beta - \alpha|, |\alpha - \alpha_j|) = |\alpha - \alpha_j|,$$

so

$$|\beta - \alpha| < |\beta - \alpha_j| \qquad \forall j = 2, ..., n,$$

as claimed. □

Example. Our primary application is to show that there are only finitely many extensions of any degree n. Of course, we already saw this for quadratic extensions, and we can make Theorem 9.10 explicit in this case. Suppose p is odd, and let $f(X) = X^2 - u$, with u prime to p a quadratic non-residue modulo p. Suppose $g(X) = X^2 - v$; then, taking $\varepsilon = 1$, we have

$$\begin{aligned} |f - g| < 1 &\iff |u - v|_p < 1 \\ &\iff u \equiv v \pmod p \\ &\iff \mathbf{Q}_p(\sqrt{u}) = \mathbf{Q}_p(\sqrt{v}), \end{aligned}$$

where in the last equivalence we use Proposition 5.5 (which said that u and v define the same class in $\mathbf{Q}_p^\times/(\mathbf{Q}_p^\times)^2$ if $u \equiv v \pmod p$). This implies f and g define the same extension, since $\mathbf{Q}_p(\sqrt{u}) = \mathbf{Q}_p[X]/(f)$ and $\mathbf{Q}_p(\sqrt{v}) = \mathbf{Q}_p[X]/(g)$.

This was not quite the full force of Theorem 9.10. More generally, we must consider $g(X) = X^2 + bX + v$, so we have

$$\beta = \frac{-b \pm \sqrt{b^2 - 4v}}{2},$$

and hence $\mathbf{Q}_p[X]/(g) = \mathbf{Q}_p(\sqrt{b^2 - 4v})$.

I claim we can take $\varepsilon = 1$, as:

$$\begin{aligned} |f - g| < 1 &\iff |b - 0|_p < 1, |v - u| < 1 \\ &\iff b \equiv 0 \pmod{p}, v \equiv u \pmod{p} \\ &\iff b^2 - 4v \equiv 4u \pmod{p}. \end{aligned}$$

In particular, by Proposition 5.5, we have

$$b^2 - 4v \equiv 4u \equiv u \pmod{(\mathbf{Q}_p^\times)^2},$$

so if $|f - g| < 1$, we have

$$\mathbf{Q}_p[X]/(f) = \mathbf{Q}_p(\sqrt{u}) = \mathbf{Q}_p(\sqrt{b^2 - 4v}) = \mathbf{Q}_p[X]/(g).$$

Corollary. *Let $K/\mathbf{Q}_p$ be a finite extension. Then K is the completion of some number field.*

Proof. Write $K = \mathbf{Q}_p(\alpha)$ for some α, with minimal polynomial $f(X) \in \mathbf{Z}_p[X]$. Theorem 9.10 says there exists $g(X) \in \mathbf{Z}[X]$ irreducible such that $K \cong \mathbf{Q}_p[X]/(g) = \mathbf{Q}_p(\beta)$, for $\beta \in \overline{\mathbf{Q}}$ a root of g. Thus $K = \mathbf{Q}_p(\beta)$ is a completion of the number field $\mathbf{Q}(\beta)$. ☐

We finally arrive at the main result of this section.

Theorem 9.11. *Let $K/\mathbf{Q}_p$ be finite, $n \geqslant 1$. There are finitely many extensions of K of degree n.*

Proof. By Theorem 9.3 and Corollary 9.4, it suffices to prove there are only finitely many *totally ramified* extensions. By Theorem 9.8, the defining polynomial of a totally ramified extension is Eisenstein.

Let

$$Z = \{(a_0, ..., a_{n-1}) \in \mathcal{O}_K^n : v_K(a_i) \geqslant 1, v_K(a_0) = 1\} \subset \mathcal{O}_K^n.$$

Note Z is compact. This is because Z is closed in $\mathcal{O}_K^n$; and $\mathcal{O}_K^n$ is compact because $\mathcal{O}_K$ is (see Corollary 4.12).

We have a function

$$\begin{array}{cccccc} \phi : & Z & \xrightarrow{1:1} & \Big\{\text{Eisenstein polys of deg } n\Big\} & \twoheadrightarrow & \Big\{\text{tot. ram. ext'ns of deg } n\Big\}, \\ & \Psi & & \Psi & & \Psi \\ & (a_0, ..., a_{n-1}) & \longmapsto & \Big[f(X) := X^n + \sum_{i=0}^{n-1} a_i X^i\Big] & \longmapsto & K[X]/(f) \end{array}$$

which is surjective by Theorem 9.8. If L is totally ramified, let $Z_L := \phi^{-1}(L) \subset Z$; then

$$Z = \bigsqcup_{L \text{ tot. ram.}} Z_L.$$

By Theorem 9.10, each Z_L is open. Since Z is compact, there is a finite subcover; and hence only finitely many totally ramified extensions, as required. ☐

$(*)$ **Remark.** Krasner actually gave a formula for the number of extensions by making these arguments effective. The formula is pretty complicated, however!

$(*)$ **Remark.** A non-examinable exercise: why do we need to use Theorem 9.3, Corollary 9.4 and Theorem 9.8 in this proof? In other words, *what's wrong with the following sketch argument?*

– If f is *any* polynomial over $\mathcal{O}_K$ of degree n, then $K[X]/(f)$ is an extension of K of degree at most n.

– Any extension of degree $\leqslant n$ extension (with arbitrary ramification) is of the form $K(\alpha)$, i.e. is $K[X]/(f)$ for some polynomial f of degree n. (e.g. if L/K has degree $m \leqslant n$, and $L = K[X]/(g)$ for some irreducible g of degree m, then take $f = g(X) \cdot X^{n-m}$).

– We get a surjective map

$$\mathcal{O}_K^n \twoheadrightarrow \big\{\text{extensions of degree} \leqslant n\big\}.$$

– The inverse image of any L in the right hand side is open in $\mathcal{O}_K^n$ by Theorem 9.10, and $\mathcal{O}_K^n$ is compact; so as above, we deduce there are only finitely many extensions of degree $\leqslant n$, and hence finitely many of degree exactly n.

9.3. (∗) Higher ramification groups

We now prove that if L/K is a Galois extension of p-adic fields, then there is a very strong restriction on $\mathrm{Gal}(L/K)$: it is *soluble* group. Recall:

Definition 9.12. A finite group G is *soluble* if there is a chain $G \supset G_0 \supset G_1 \supset \cdots \supset G_r = \{1\}$ of normal subgroups such that each quotient G/G_0 and G_n/G_{n+1} for $n \geqslant 0$ is abelian.

Example. Any finite abelian group is soluble. S_4 is soluble via

$$S_4 \supset A_4 \supset C_2 \times C_2 \supset \{1\}.$$

A_5 and S_5 are not soluble; nor is S_n for any $n \geqslant 5$.

Let L/K be a finite Galois extension of p-adic fields, and $G = \mathrm{Gal}(L/K)$. We now show that G is soluble. Recall restriction defines a surjective map $\mathrm{res} : G \twoheadrightarrow \mathrm{Gal}(k_L/k)$.

Definition 9.13. The *inertia group* is

$$I_{L/K} = \ker(\mathrm{res}) = \{\sigma \in G : \sigma(x) \equiv x \pmod{\pi_L} \forall x \in \mathcal{O}_L\}.$$

Note that $I_{L/K}$ is a normal subgroup, as the kernel of a homomorphism; and that $\#I_{L/K} = e(L/K)$ (since $\#G = ef$ and $\#\,\mathrm{Gal}(k_L/k) = f$).

To define our chain of normal subgroups, we use a higher analogue.

Definition 9.14. The *nth ramification group* is

$$G_n := \{\sigma \in G : \sigma(x) \equiv x \pmod{\pi_L^{n+1}} \forall x \in \mathcal{O}_L\}.$$

Then $I_{L/K} = G_0$, and $G_n = \ker(G \to \mathrm{Aut}(\mathcal{O}_L/\pi_L^n\mathcal{O}_L)$ is normal in G.

Theorem 9.15. *1.* $G_n = \{\sigma \in I : v_L(\sigma(\pi_L) - \pi_L) \geqslant n\}$.

2. $\cap_{n \geqslant 0} G_n = \{1\}$.

3. We have injective homomorphisms

$$G_0/G_1 \hookrightarrow k_L^\times$$

and

$$G_n/G_{n+1} \hookrightarrow (k_L, +) \qquad \forall n \geqslant 1.$$

Proof. By Theorem 9.3, up to replacing K with an unramified extension inside L, we may assume L/K is totally ramified.

(i) As L/K totally ramified, $\mathcal{O}_L = \mathcal{O}_K[\pi_L]$. Thus if $\sigma(\pi_L) \equiv \pi_L \pmod{\pi_L^{n+1}}$, then $\sigma(x) =\equiv x \pmod{\pi_L^{n+1}}$ for all $x \in \mathcal{O}_L$.

(ii) If $\sigma \neq 1$, then $\sigma(\pi_L) \neq \pi_L$ (since $L = K(\pi_L)$). Thus $v(\sigma(\pi_L) - \pi_L) < \infty$, so $\sigma \notin G_n$ for n sufficiently large.

(iii) Let $\sigma, \tau \in G_n$. Write $\tau(\pi_L) = u\pi_L$ for some $u \in \mathcal{O}_L^\times$. We first note that

$$\begin{aligned} \frac{\sigma \circ \tau(\pi_L)}{\pi_L} &= \frac{\sigma(\tau(\pi_L))}{\tau(\pi_L)} \cdot \frac{\tau(\pi_L)}{\pi_L} \\ &= \frac{\sigma(u\pi_L)}{u\pi_L} \cdot \frac{\tau(\pi_L)}{\pi_L} \\ &\equiv \frac{\sigma(\pi_L)}{\pi_L} \cdot \frac{\tau(\pi_L)}{\pi_L} \pmod{\pi_L^{n+1}}, \end{aligned} \tag{9.4}$$

since $\sigma \in G_n$.

Suppose $n = 0$. Write $\sigma(\pi_L) = a_\sigma \pi_L$ with $a_\sigma \in \mathcal{O}_L^\times$. Then (9.4) implies

$$a_{\sigma\tau} \equiv a_\sigma a_\tau \pmod{\pi_L},$$

so

$$\begin{aligned} G_0 &\longrightarrow k_L^\times, \\ \sigma &\longmapsto a_\sigma \pmod{\pi_L} \end{aligned}$$

is a group homomorphism with kernel G_1.

Suppose $n \geqslant 1$. Write $\sigma(\pi_L) = \pi_L + a_\sigma \pi_L^{n+1}$ with $a_\sigma \in \mathcal{O}_L$. Then (9.4) implies

$$(1 + a_{\sigma\tau}\pi_L^n) \equiv (1 + a_\sigma \pi_L^n)(1 + a_\tau \pi_L^n) \pmod{\pi_L^{n+1}},$$

so $a_{\sigma\tau} = a_\sigma + a_\tau \pmod{\pi_L}$. Thus

$$\begin{aligned} G_n &\longrightarrow k_L, \\ \sigma &\longmapsto a_\sigma \pmod{\pi_L} \end{aligned}$$

is a group homomorphism with kernel G_{n+1}. □

Corollary 9.16. *If L/K is a finite Galois extension of p-adic fields, then $G = \mathrm{Gal}(L/K)$ is soluble.*

Proof. Each of the higher ramification groups G_i is normal in G. We know $G/G_0 \cong \mathrm{Gal}(k_L/k)$ is cyclic, hence abelian. We know G_0/G_1 is a subgroup of $k_L^\times$, hence is abelian; and finally, G_n/G_{n+1} is a subgroup of k_L, hence is abelian. □

Remark 9.17. Another way to view this is as follows: let $L_n := L^{G_n}$ be the fixed field of G_n, a sub extension of L/K. Then we have a tower of Galois extensions

$$\begin{tikzcd}
L \arrow[r, equal] \arrow[ddd, dash] & L_r \arrow[d, dash, "\text{totally ramified of } p\text{-power degree}"] \\
& \vdots \arrow[d, dash] \\
& L_1 \arrow[d, dash, "\text{totally ramified degree prime to } p"] \\
K(\zeta_m) \arrow[r, equal] \arrow[d, dash] & L_0 \arrow[d, dash, "\text{unramified}"] \\
K \arrow[r, equal] & K.
\end{tikzcd}$$

The left-hand tower $L/K(\zeta_m)/K$ is that of Theorem 9.3, and the right-hand tower is a refinement. Here every subextension L_{n+1}/L_n and L_0/K is abelian; only the bottom extension L_0/K is unramified, all the other extensions are totally ramified. Moreover L_1/L_0 has degree prime to p (since the Galois group injects into a group of order $p^d - 1$, for some d) and for $n \geqslant 1$, the extension L_{n+1}/L_n has p-power degree (since the Galois group injects into a group of order p^d).

Exercise. Show that G_1 is the unique p-Sylow subgroup of G.

We give one immediate corollary of this study. Recall the Inverse Galois Problem asks: *is every finite group the Galois group of a Galois extension of* $\mathbf{Q}$*?* The analogue over $\mathbf{Q}_p$ is *false*; since S_5 is not soluble, we have:

Corollary 9.18. *There is no Galois extension* $L/\mathbf{Q}_p$ *with* $\mathrm{Gal}(L/\mathbf{Q}_p) \cong S_5$.

9.4. Cyclotomic extensions of $\mathbf{Q}_p$

Definition 9.19. Let $\zeta_n \in \overline{\mathbf{Q}}$ be a choice of primitive nth root of unity. A *cyclotomic extension of* $\mathbf{Q}_p$ is a field of the form $\mathbf{Q}_p(\zeta_n)$.

Lemma 9.20. *If m, n are coprime, then* $\mathbf{Q}_p(\zeta_{mn}) = \mathbf{Q}_p(\zeta_m)\mathbf{Q}_p(\zeta_n)$.

Proof. Since m, n coprime, $\zeta_m\zeta_n$ is a primitive mnth root of unity, so $\mathbf{Q}_p(\zeta_{mn}) \subset \mathbf{Q}_p(\zeta_m)\mathbf{Q}_p(\zeta_n)$. But ζ_{mn}^m is a primitive nth root of unity, and ζ_{mn}^n is a primitive mth root of unity, so $\mathbf{Q}_p(\zeta_{mn}) \supset \mathbf{Q}_p(\zeta_m)\mathbf{Q}_p(\zeta_n)$. □

Thus we can reduces the study of cyclotomic extensions to those of prime power order. For those prime to p, we've essentially already classified these.

Lemma 9.21. *Suppose $p \nmid n$. Then $\mathbf{Q}_p(\zeta_n)/\mathbf{Q}_p$ is unramified and cyclic of degree dividing the order of p in* $(\mathbf{Z}/n\mathbf{Z})^\times$.

Proof. Let $r :=$ order of p in $(\mathbf{Z}/n\mathbf{Z})^\times$. Then $p^r - 1 = Mn$ for some M. By Corollary 9.4, $\mathbf{Q}_p(\zeta_{Mn})$ is unramified of degree r. The result follows since $\mathbf{Q}_p(\zeta_n)$ is a subfield. □

We now study the case where we adjoin ζ_{p^m}.

Lemma 9.22. *Let $p \neq 2$ and $m \geqslant 1$.*

- $\mathbf{Q}_p(\zeta_{p^m})/\mathbf{Q}_p$ *is totally ramified of degree* $p^{m-1}(p-1)$*, with uniformiser* $\zeta_{p^m}-1$.
- $\mathbf{Q}_p(\zeta_{p^m})/\mathbf{Q}_p$ *is cyclic.*

Proof. (i) For $m=1$, we have

$$X^p-1=(X-1)(X^{p-1}+X^{p-2}+\cdots+1).$$

First we treat $m=1$. Let $f(X)=X^{p-1}+\cdots+X+1$, an irreducible factor of X^p-1. Then let

$$g(X):=f(X+1)=X^{p-1}+\binom{p}{1}X^{p-2}+\cdots+\binom{p}{2}X+\binom{p}{1}.$$

(To see the second equality, consider $(X+1)^p-1$). Note g is Eisenstein, so it is irreducible, and $\mathbf{Q}_p[X]/(g)$ is totally ramified of degree $p-1$. Since ζ_p-1 is a root of g, we have $\mathbf{Q}_p(\zeta_p)=\mathbf{Q}_p(\zeta_p-1)=\mathbf{Q}_p[X]/(g)$; and also ζ_p-1 is a uniformiser by Theorem 9.8, giving the result for $m=1$.

We proceed by induction. Suppose the result is known for $m-1$. Then without loss of generality $\zeta_{p^{m+1}}$ is a root of $f_m(X)=X^p-\zeta_{p^{m-1}}$; so $\zeta_{p^{m+1}}-1$ is a root of

$$g_m(X):=f_m(X+1)=X^p+\binom{p}{1}X^{p-1}+\cdots+\binom{p}{1}X-(\zeta_{p^m}-1).$$

By the induction hypothesis, $\zeta_{p^m}-1$ is a uniformiser, so g_m is Eisenstein. As above, we deduce that $\mathbf{Q}_p(\zeta_{p^{m+1}})/\mathbf{Q}_p(\zeta_{p^m})$ is totally ramified of degree p, with uniformiser $\zeta_{p^{m+1}}-1$.

(ii) We know $\mathbf{Q}(\zeta_{p^m})/\mathbf{Q}\cong(\mathbf{Z}/p^m\mathbf{Z})^\times$ is cyclic of degree $p^{m-1}(p-1)$. The restriction map $\mathrm{Gal}(\mathbf{Q}_p(\zeta_{p^m})/Qp)\to\mathrm{Gal}(\mathbf{Q}(\zeta_{p^m})/Q)$ is injective by Proposition 8.20(iii). But both groups have the same size, so this map is an isomorphism. □

Lemma 9.23. *For* $p=2$ *and* $m\geqslant 2$*, the extension* $\mathbf{Q}_2(\zeta_{2^m})/\mathbf{Q}_2$ *is totally ramified with Galois group* $\mathbf{Z}/2\mathbf{Z}\times\mathbf{Z}/2^{m-2}\mathbf{Z}$.

Proof. Exercise. □

9.5. (∗) Appendix: Ramification in Dedekind domains

(Provisionally this is ***non-examinable****, depending on how much time I have left when the lectures reach this point).*

We now study ramification for general Dedekind domains (e.g. the ring of integers in a number field). We return to the setting: $\mathcal{O}_K$ is a Dedekind domain, $K=\mathrm{Frac}(\mathcal{O}_K)$, L/K an extension of degree n, $\mathcal{O}_L$ the integral closure of $\mathcal{O}_K$ in L, and $k=\mathcal{O}_K/\mathfrak{p}$.

Definition 9.24. Let $\mathfrak{p}\subset\mathcal{O}_K$. We say $\mathfrak{p}$ *ramifies in* L if there exists $\mathfrak{P}|\mathfrak{p}$ with $e(\mathfrak{P}/\mathfrak{p})>1$, that is, if when we write $\mathfrak{p}\mathcal{O}_L=\mathfrak{P}_1^{e_1}\cdots\mathfrak{P}_r^{e_r}$, then some $e_i>1$.

Definition 9.25. (i) Let R/K be a finite commutative K-algebra. If $x_1,\dots,x_n\in R$, we define

$$\Delta_{R/K}(x_1,\dots,x_n)=\det(\mathrm{Tr}_{R/K}(x_ix_j)).$$

(ii) The *discriminant* of L/K is the ideal $d_{L/K}\subset\mathcal{O}_K$ generated by $\Delta_{L/K}(x_1,\dots,x_n)$ as $x_1,\dots,x_n$ range over all possible choices in $\mathcal{O}_L$.

Lemma 9.26. *(i)* $\Delta_{R/K}(x_1,\dots,x_n)=0$ *unless* $x_1,\dots,x_n$ *is a* K*-basis of* R.

(ii) If $y_1,\dots,y_n\in R$ *with* $y_i=\sum_{j=1}^n a_{ij}x_j$ *for* $a_{ij}\in K$*, then*

$$\Delta_{R/K}(y_1,\dots,y_n)=\det(A)^2\cdot\Delta(x_1,\dots,x_n),$$

where $A=(a_{ij})\in M_n(K)$.

Proof. Exercise. □

Example. Suppose $L=K$. Clearly $\Delta_{K/K}(x)=x$ for all $x\in\mathcal{O}_L=\mathcal{O}_K$, so $d_{K/K}=\mathcal{O}_K$.

Let $K=\mathbf{Q}$ and $L=\mathbf{Q}(\sqrt{D})$ for $D\in\mathbf{Z}$ square-free. We may compute that

$$\begin{aligned}\Delta_{L/K}(1,\sqrt{D})&=\det\begin{pmatrix}\mathrm{Tr}(1)&\mathrm{Tr}(\sqrt{D})\\ \mathrm{Tr}(\sqrt{D})&\mathrm{Tr}(D)\end{pmatrix}\\ &=\det\begin{pmatrix}2&0\\0&2D\end{pmatrix}=4D.\end{aligned}$$

We deduce immediately that $4D\in d_{\mathbf{Q}(\sqrt{D})/\mathbf{Q}}$.

On the exercise sheets, we'll prove that

$$d_{\mathbf{Q}(\sqrt{D})/\mathbf{Q}}=\begin{cases}(D) & : \mathcal{O}_{\mathbf{Q}(\sqrt{D})}=\mathbf{Z}\left[\frac{1+\sqrt{D}}{2}\right]\\ (4D) & : \mathcal{O}_{\mathbf{Q}(\sqrt{D})}=\mathbf{Z}[\sqrt{D}].\end{cases}$$

Theorem 9.27. *Let* $\mathfrak{p}\subset\mathcal{O}_K$ *be a prime, and assume* $k=\mathcal{O}_K/\mathfrak{p}$ *is finite. Then:*

(i) If $\mathfrak{p}$ *ramifies in* L*, then*

$$\mathfrak{p}|\Delta_{L/K}(x_1,\dots,x_n)\mathcal{O}_K$$

for all choices of $x_1,\dots,x_n\in\mathcal{O}_K$.

(ii) If $\mathfrak{p}$ *does not ramify in* L*, then there exist* $x_1,\dots,x_n\in\mathcal{O}_K$ *such that*

$$\mathfrak{p}\nmid\Delta_{L/K}(x_1,\dots,x_n)\mathcal{O}_K.$$

(iii) A prime $\mathfrak{p}\subset\mathcal{O}_K$ *ramifies in* L *if and only if* $\mathfrak{p}$ *divides* $d_{L/K}$.

Note part (iii) is an immediate corollary of parts (i) and (ii). To prove (i) and (ii), we need some lemmas.

Lemma 9.28. *There is a commutative diagram*

$$\begin{tikzcd}\mathcal{O}_L \arrow[r] \arrow[d, "\mathrm{Tr}_{L/K}"'] & \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \arrow[d, "\mathrm{Tr}_{R/k}"] & =: R\\ \mathcal{O}_K \arrow[r] & \mathcal{O}_K/\mathfrak{p} & =: k.\end{tikzcd}$$

Proof. After localising at $\mathfrak{p}$, we may assume $\mathcal{O}_K$ is a PID. (Explicitly: letting $S_{\mathfrak{p}}=\mathcal{O}_K\backslash\mathfrak{p}$, we may replace $\mathcal{O}_L$ with $S_{\mathfrak{p}}^{-1}\mathcal{O}_L$ and $\mathcal{O}_K$ with $S_{\mathfrak{p}}^{-1}\mathcal{O}_K=\mathcal{O}_{K,\mathfrak{p}}$. Since $\mathcal{O}_L$ and $\mathcal{O}_K$ are subsets of their localised cousins, and the trace maps and quotient rings are the same, showing the diagram commutes on the localisations implies it commutes on the non-localisations).

If $\mathcal{O}_K$ is a PID, then $\mathcal{O}_L$ is a free $\mathcal{O}_K$-module by the structure theorem for modules over PIDs. Let $x_1,\dots,x_n$ be an $\mathcal{O}_K$-basis of $\mathcal{O}_L$. Then $x_1,\dots,x_n$ is also a K-basis for L. Letting $\overline{x}:=x\,(\mathrm{mod}\,\mathfrak{p})$, we also have $\overline{x}_1,\dots\overline{x}_n$ is a k-basis of R.

For any $z \in \mathcal{O}_L$, write $zx_i = \sum_{i=1}^n a_{ij}x_j$, for some $a_{ij} \in \mathcal{O}_K$. Write $A = (a_{ij})$. Then in $\mathcal{O}_K/\mathfrak{p}$, we have

$$\overline{\mathrm{Tr}_{L/K}(z)} = \overline{\mathrm{Tr}(A)} = \overline{\sum_{i=1}^n a_{ii}}$$
$$= \sum_{i=1}^n \overline{a_{ii}} = \mathrm{Tr}(\overline{A}) = \mathrm{Tr}_{R/k}(\overline{z}),$$

as required. ☐

Proof of Theorem 9.27. By the Chinese remainder theorem, we have an isomorphism

$$R := \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}_L/\mathfrak{P}_1^{e_1} \times \cdots \mathcal{O}_L/\mathfrak{P}_r^{e_r}$$

of k-algebras.

(i) Suppose $\mathfrak{p}$ ramifies. Then without loss of generality $e_1 > 1$, and thus $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$ contains nilpotent elements (e.g. if $0 \neq x \in \mathfrak{P}_1\backslash\mathfrak{P}_1^2$, then $x \neq 0$ in $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$, but $x^{e_1} \equiv 0 \pmod{\mathfrak{p}\mathcal{O}_L}$). We claim this forces

$$0 = \Delta_{R/k}(\overline{x}_1, \ldots, \overline{x}_n) \in k$$

for any $\overline{x}_1, ..., \overline{x}_n \in R$. We use Lemma 9.26: by (i) this is always true unless the $\overline{x}_i$ form a basis, and by (ii) $\Delta_{R/k}$ is zero for one basis if and only if it is zero for all bases. Let $\overline{x}_1 \in R$ be nilpotent, and extend to a basis $\overline{x}_1, \overline{x}_2, ..., \overline{x}_n$. Then $\overline{x}_1\overline{x}_j$ is nilpotent for all j. Nilpotent elements have trace 0, since all their eigenvalues must be 0; so $\mathrm{Tr}_{R/k}(\overline{x}_1\overline{x}_j)$ is 0 for all j. Thus the entire top row of $\mathrm{Tr}_{R/k}(\overline{x}_i\overline{x}_j)$ vanishes, and this matrix must have determinant zero.

Now we compute that for any $x_1, ..., x_n \in \mathcal{O}_L$, we have

$$\Delta_{L/K}(x_1, \ldots, x_n) \equiv \Delta_{R/k}(\overline{x}_1, \ldots, \overline{x}_n) = 0 \pmod{\mathfrak{p}\mathcal{O}_L},$$

where the first congruence is Lemma 9.28. Thus $\Delta_{L/K}(x_1, \ldots, x_n) \in \mathfrak{p}\mathcal{O}_L$ for all $x_1, \ldots, x_n$, as required.

(ii) If $\mathfrak{p}$ is unramified in L, then $R = k_1 \times \cdots \times k_r$ is a product of finite field extensions. Since k is finite each k_j/k is separable, of degree s_j, say. By Lemma 7.20, for each j there exists a basis $\overline{x}_{j,1}, \ldots, \overline{x}_{j,s_j}$ of k_j/k such that $\Delta_{k_j/k}(\overline{x}_{j,1}, \ldots, \overline{x}_{j,s_j}) \neq 0$. Note that $\overline{x}_{j,m}\overline{x}_{k,n} = 0$ unless $j = k$. Lift each $\overline{x}_{j,i}$ to an element $x_{j,i} \in L$. Then

$$0 \neq \prod_{j=1}^r \Delta_{k_j/k}(\overline{x}_{j,1}, \ldots, \overline{x}_{j,s_j}) = \Delta_{R/k}(\overline{x}_{1,1}, \ldots, \overline{x}_{r,s_r})$$
$$\equiv \Delta_{L/K}(x_{1,1}, \ldots, x_{r,s_r}) \pmod{\mathfrak{p}\mathcal{O}_L}.$$

This completes (ii).

(iii) Recall $d_{L/K}$ is the ideal of $\mathcal{O}_K$ generated by $\Delta_{L/K}(x_1, \ldots, x_n)$ as x_i range over $\mathcal{O}_L$. From (i), we know that if $\mathfrak{p}$ ramifies, then $\mathfrak{p}|\Delta_{L/K}(x_1, \ldots, x_n)$ for all x_i, and hence $\mathfrak{p}|d_{L/K}$. If $\mathfrak{p}$ does not ramify, by (ii) there exist x_i with $\mathfrak{p} \nmid \Delta_{L/K}(x_1, ..., x_n)$, and thus $\mathfrak{p} \nmid d_{L/K}$. This proves Theorem 9.27. ☐

Corollary 9.29. *Only finitely many primes of K ramify in L.*

Proof. The discriminant $d_{L/K}$ has only finitely many prime divisors. ☐

Chapter 10

$(*)$ The Kronecker–Weber theorem

This entire chapter is non-examinable. There was not time to cover this in the lectures, but it serves as a beautiful application of local fields to studying the arithmetic of $\mathbf{Q}$*, and if you've followed all the examinable material – and, also, the non-examinable section §9.3 on higher ramification groups – then you should be able to read this chapter.*

Recall an extension L/K of fields is *abelian* if it is Galois and its Galois group is an abelian group. In this chapter, we prove:

Theorem 10.1 (Kronecker–Weber)**.** *Every abelian extension of* $\mathbf{Q}$ *is a subfield of a cyclotomic field* $\mathbf{Q}(\zeta_n)$*, for some* n*.*

Sometimes we will refer to this as the 'global' Kronecker–Weber theorem. Our proof will follow the account given by Washington in [Was82, Chapter 14].

Note $\mathrm{Gal}(\mathbf{Q}(\zeta_n)/\mathbf{Q}) \cong (\mathbf{Z}/n\mathbf{Z})^\times$, a particularly simple group. We completely understand the subgroups of $(\mathbf{Z}/n\mathbf{Z})^\times$, and thus – via the fundamental theorem of Galois theory – we completely understand the subfields of $\mathbf{Q}(\zeta_n)$; so via this theorem, we completely understand all abelian extensions of $\mathbf{Q}$.

Remark. We sketch the proof strategy:

- Let $K/\mathbf{Q}$ be an abelian extension.
- For all primes p, and $\mathfrak{p}|p$ a prime of K, completing at $\mathfrak{p}|p$ gives an abelian extension $K_\mathfrak{p}/\mathbf{Q}_p$ of $\mathbf{Q}_p$.
- Finite abelian groups are products of cyclic groups, so $K_\mathfrak{p}/\mathbf{Q}_p$ is a compositum of cyclic extensions of degree ℓ^s for ℓ prime.
- The Galois theory of $\mathbf{Q}_p$ is simple enough to show the every cyclic extension of $\mathbf{Q}_p$ lies in a cyclotomic field.
- We deduce the *Local Kronecker–Weber theorem:* every abelian extension of $\mathbf{Q}_p$ is a subfield of a cyclotomic extension.
- We deduce the global Kronecker–Weber theory from the local Kronecker–Weber theorem.

In the proof, we'll use some basic facts about abelian extensions (see e.g. Galois theory):

Lemma 10.2. *(i) If* $K \subset L \subset M$ *and* M/K *is abelian, then* M/L *and* L/K *are abelian.*

(ii) If L_1/K *and* L_2/K *are abelian, then* L_1L_2/K *is abelian.*

(iii) If $L_1 \cap L_2 = K$*, then* $\mathrm{Gal}(L_1L_2/K) \cong \mathrm{Gal}(L_1/K) \times \mathrm{Gal}(L_2/K)$*.*

10.1. Prime power extensions of $\mathbf{Q}_p$

We will start our study in the special case of abelian extensions of $\mathbf{Q}_p$ of prime power degree. We first treat the 'generic' case where $\ell \neq p$.

Proposition 10.3. *Let $K/\mathbf{Q}_p$ be an extension of degree ℓ^s, where $\ell \neq p$ is a prime. Then K is contained in a cyclotomic extension.*

Proof. We have a filtration

$$G = \mathrm{Gal}(K/\mathbf{Q}_p) \supset G_0 = I_{K/\mathbf{Q}_p} \supset G_1 \supset G_2 \supset \cdots \supset G_t = \{1\}$$

of G into its ramification groups (from §9.3), and $e(K/\mathbf{Q}_p) = \#I(K/\mathbf{Q}_p)$. Let

$$F \coloneqq K^{G_0} = \mathbf{Q}_p(\zeta_m) \text{ be the maximal unramified subextension (from Theorem 9.3).}$$

Then $f(F/\mathbf{Q}_p) = f(K/\mathbf{Q}_p)$.

Note that G_i/G_{i+1} has p-power order for $i \geqslant 1$, so as $p \nmid \ell^s$, we have $G_1 = \{1\}$. Thus

$$G_0 = G_0/G_1 \hookrightarrow \mathbf{F}_p^\times$$

by Theorem 9.15, so $e(K/\mathbf{Q}_p) | p-1$. Since $e(K/\mathbf{Q}_p)|\ell^s$, we know it is a power of ℓ. Let ℓ^r be the exact power of ℓ dividing $p-1$; then we've shown

$$e(K/\mathbf{Q}_p) \leqslant \ell^r,$$

no matter how large s is. Now if $p=2$, then $\ell^r = e(K/\mathbf{Q}_p) = 1$, so $K/\mathbf{Q}_p$ is unramified and $K = \mathbf{Q}_p(\zeta_m)$ is cyclotomic, as required. This reduces us to the case p odd.

Recall $\mathbf{Q}_p(\zeta_p)$ is totally ramified and cyclic of degree $p-1$ (Lemma 9.22). Let

$$E \coloneqq \text{ unique subextension of } \mathbf{Q}_p(\zeta_p)/\mathbf{Q}_p \text{ of order } \ell^r$$

(which exists by Galois theory). Replacing K with EK, we may assume $E \subset K$; since if $EK \subset \mathbf{Q}(\zeta_n)$ for some n, then $K \subset EK \subset \mathbf{Q}(\zeta_n)$. Then

$$\ell^r = e(E/\mathbf{Q}_p) \leqslant e(K/\mathbf{Q}_p) \leqslant \ell^r \qquad \Rightarrow \qquad e(E/\mathbf{Q}_p) = e(K/\mathbf{Q}_p) = \ell^r.$$

Now $EF \subset K$ by assumption, so $[EF:\mathbf{Q}_p] \leqslant [K:\mathbf{Q}_p]$. Since

$$e(EF/\mathbf{Q}_p) \geqslant e(E/\mathbf{Q}_p) = e(K/\mathbf{Q}_p) \qquad \text{and} \qquad f(EF/\mathbf{Q}_p) \geqslant f(F/\mathbf{Q}_p) = f(K/\mathbf{Q}_p),$$

we have

$$[EF:\mathbf{Q}_p] = e(EF/\mathbf{Q}_p)\cdot f(EF/\mathbf{Q}_p) \geqslant e(K/\mathbf{Q}_p)\cdot f(K/\mathbf{Q}_p) = [K:\mathbf{Q}_p].$$

But $EF \subset K$, so this forces equality and

$$K = EF \subset \mathbf{Q}(\zeta_p)\mathbf{Q}(\zeta_m) = \mathbf{Q}(\zeta_{mp}),$$

as required. □

This leaves the case of degree p^s extensions of $\mathbf{Q}_p$. The proof of this case uses Kummer theory. It is not hard, but it was too long to fit into even the intended lectures. A proof is deferred to an appendix (see §10.4.3).

Using this, we deduce:

Proposition 10.4. *Let $K/\mathbf{Q}_p$ be an cyclic extension of exponent p^s, for $s \geqslant 1$. Then K is contained in a cyclotomic extension.*

When $p = 2$, we defer the proof to a (non-examinable) appendix (see §10.4.2).

Let p be odd. In the proof, we require the following result:

Proposition 10.5. *If p is odd, there is no Galois extension of $\mathbf{Q}_p$ with Galois group $(\mathbf{Z}/p\mathbf{Z})^3$.*

Proof of Proposition 10.4: Let $F = \mathbf{Q}_p(\zeta_m)$ be the unique unramified extension of $\mathbf{Q}_p$ of degree p^s, and let $E \subset \mathbf{Q}_p(\zeta_{p^{s+1}})$ be the (totally ramified) unique subfield of degree p^s over $\mathbf{Q}_p$. Note EF is abelian with Galois group $\mathbf{Z}/p^s\mathbf{Z} \times \mathbf{Z}/p^s\mathbf{Z}$, and EFK is an abelian extension with Galois group of form

$$\mathrm{Gal}(EFK/\mathbf{Q}_p) = \mathbf{Z}/p^s\mathbf{Z} \times \mathbf{Z}/p^s\mathbf{Z} \times \mathbf{Z}/p^t\mathbf{Z},$$

with $0 \leqslant t \leqslant s$.

We claim $K \subset EF \subset \mathbf{Q}_p(\zeta_{mp^{s+1}})$. Suppose not; then $s \geqslant 1$, and thus $EFK/\mathbf{Q}_p$ has a subextension with Galois group $(\mathbf{Z}/p\mathbf{Z})^3$. But no such extension exists by Proposition 10.5, contradiction. □

10.2. The local Kronecker–Weber theorem

We'll deduce the (global) Kronecker–Weber theorem from its local analogue:

Theorem 10.6 (Local Kronecker–Weber)**.** *Every abelian extension of $\mathbf{Q}_p$ is a subfield of a cyclotomic field $\mathbf{Q}_p(\zeta_n)$.*

Proof. Suppose $K/\mathbf{Q}$ is abelian. Then by the structure theorem for finite abelian groups, we have

$$\mathrm{Gal}(K/\mathbf{Q}) \cong \mathbf{Z}/p_1^{s_1}\mathbf{Z} \times \mathbf{Z}/p_2^{s_2}\mathbf{Z} \times \cdots \times \mathbf{Z}/p_r^{s_r}\mathbf{Z}$$

for (not necessarily distinct) primes p_i and $s_i \in \mathbf{Z}_{\geqslant 1}$. By Galois theory, for each i, there is a cyclic subextension $K_i/\mathbf{Q}$ of degree $p_i^{s_i}$ such that K is the compositum of all the K_i's. By Propositions 10.3 and 10.4, there exist cyclotomic fields $\mathbf{Q}(\zeta_{n_i})$ such that $K_i \subset \mathbf{Q}(\zeta_{n_i})$. Then

$$K = K_1 \cdots K_r \subset \mathbf{Q}(\zeta_n), \qquad n = n_1 \cdots n_r.$$

□

10.3. Proof of the Kronecker–Weber theorem

We now prove the Kronecker–Weber theorem (Theorem 10.1). We'll need some standard auxiliary results.

Definition 10.7. Let $K/\mathbf{Q}$ be a finite Galois extension, and let p be a prime. If $\mathfrak{p}|p$ is a prime of K, the *inertia group* of $\mathfrak{p}|p$ is

$$I_{\mathfrak{p}/p} : \{\sigma \in \mathrm{Gal}(K/\mathbf{Q}) : \sigma(x) \equiv x \pmod{\mathfrak{p}} \ \forall x \in \mathcal{O}_K\}.$$

Note $\#I_{\mathfrak{p}/p} = e(\mathfrak{p}/p)$ (as can be seen by considering the completion $K_{\mathfrak{p}}/\mathbf{Q}_p$).

Lemma 10.8. *The natural map $\mathrm{Gal}(K_{\mathfrak{p}}/\mathbf{Q}_p) \hookrightarrow \mathrm{Gal}(K/\mathbf{Q})$ induces an isomorphism $I_{K_{\mathfrak{p}}/\mathbf{Q}_p} \cong I_{\mathfrak{p}/p}$. Thus $\#I_{\mathfrak{p}/p} = e(\mathfrak{p}/p)$, and $I_{\mathfrak{p}/p}$ is non-trivial if and only if p is unramified in K.*

Proof. This is easy by combining the definitions with Proposition 8.20 and Lemma 8.15. □

Lemma 10.9. *If $K/\mathbf{Q}$ is abelian, then $I_{\mathfrak{p}/p}$ is the same for all $\mathfrak{p}|p$.*

Proof. Let $\mathfrak{q}$ be a different choice of prime; since the Galois action on primes is transitive (Proposition 8.17), there exists $\tau \in \mathrm{Gal}(K/\mathbf{Q})$ with $\mathfrak{q} = \tau(\mathfrak{p})$. Then

$$I_{\mathfrak{q}/p} = \tau I_{\mathfrak{p}/p}\tau^{-1} \subset \mathrm{Gal}(K/\mathbf{Q}).$$

But if $K/\mathbf{Q}$ is abelian, then $\tau I_{\mathfrak{p}/p}\tau^{-1} = I_{\mathfrak{p}/p}$. □

Definition 10.10. If $K/\mathbf{Q}$ is abelian, and p prime, we write I_p for $I_{\mathfrak{p}/p}$ (for one, hence any, choice of $\mathfrak{p}|p$).

Theorem 10.11. *A prime p ramifies in K if and only if it divides the discriminant of $K/\mathbf{Q}$. In particular, at most finitely many primes ramify.*

Proof. This was Theorem 9.27. □

Theorem 10.12 (Minkowski)**.** *Every finite extension $K/\mathbf{Q}$ is ramified at at least one prime p.*

Proof. Minkowski proved that every number field has discriminant > 1, so always has a prime divisor. □

Corollary 10.13. *Let $K/\mathbf{Q}$ be a finite abelian extension. Then* $\mathrm{Gal}(K/\mathbf{Q})$ *is generated by the subgroups I_p for primes p.*

Proof. Let H be the subgroup of $\mathrm{Gal}(K/\mathbf{Q})$ generated by I_p for all p. The fixed field K^H for H is a subextension, and every prime is unramified in K^H (because all the inertia groups are now trivial). This forces $K^H = \mathbf{Q}$ by Minkowski's theorem, so $H = \mathrm{Gal}(K/\mathbf{Q})$. □

Proof of the Kronecker–Weber Theorem: Let $K/\mathbf{Q}$ be an abelian extension. Let p be prime, and choose some $\mathfrak{p}|p$. Then $K_{\mathfrak{p}}/\mathbf{Q}_p$ is abelian by Proposition 8.20, and by local Kronecker–Weber we know $K_{\mathfrak{p}}$ lies in a cyclotomic extension. We may break this into unramified and ramified parts

$$K_{\mathfrak{p}} \subset \mathbf{Q}_p(\zeta_m)\cdot\mathbf{Q}_p(\zeta_{p^{e_p}}), \qquad p \nmid m, e_p \geqslant 0.$$

Note $e_p = 0$ if and only if p is unramified in K (by Lemmas 9.22 and 9.23).

Let $n = \prod_p p^{e_p}$, and $E \coloneqq \mathbf{Q}(\zeta_n) = \prod_p \mathbf{Q}(\zeta_{p^{e_p}}),$. This is well-defined since $e_p \neq 0$ for only finitely many p. Note E is a cyclotomic, hence abelian, extension of degree

$$[E:\mathbf{Q}] = \prod_p \phi(p^{e_p}),$$

where ϕ is Euler's totient function. Replacing K with EK, we may assume $E \subset K$, so that

$$[K:\mathbf{Q}] \geqslant [E:\mathbf{Q}] = \prod_p \phi(p^{e_p}).$$

Since $K_{\mathfrak{p}} \subset \mathbf{Q}_p\left(\zeta_m, \zeta_{p^{e_p}}\right)$, we have

$$\#I_p = \#I_{K_{\mathfrak{p}}/\mathbf{Q}_p} \leqslant e\Big(\mathbf{Q}_p\left(\zeta_m, \zeta_{p^{e_p}}\right)/\mathbf{Q}_p\Big) = \phi(p^{e_p}).$$

Since $\mathrm{Gal}(K/\mathbf{Q})$ is generated by the I_p, we have

$$[K:\mathbf{Q}] = \#\,\mathrm{Gal}(K/\mathbf{Q}) \leqslant \prod_p \#I_p \leqslant \prod_p \phi(p^{e_p}) = [E:\mathbf{Q}].$$

We deduce that $[E:\mathbf{Q}] = [K:\mathbf{Q}]$, hence $K = E$ is cyclotomic, as required. □

10.4. Appendix: filling the gaps

In this (even more) **non-examinable** section, we provide brief historical context for the Kronecker–Weber theory, and prove some results that we left out earlier in this chapter. These were always going to be omitted from the course for space and time purposes, and we include them here for completeness.

10.4.1. Class Field Theory and Hilbert's 12th problem. The Kronecker–Weber theorem is one of the first results in (and inspired the study of) *Class Field Theory.* We give some brief context.

In a strong sense, every question of interest in the field of Algebraic Number Theory is related to the arithmetic of finite extensions of $\mathbf{Q}$. On a basic level, understanding number fields (and their Galois theory) means we understand solutions to polynomial equations. On a deeper level, much of modern research in the field is taken with the *Langlands program*, a conjectural universal theory of mathematics linking number theory, arithmetic geometry and analysis; a heavily brutalised/loose version is:

> *"Objects of interest in number theory – e.g. number fields, Dirichlet characters, elliptic curves, modular forms, etc. – are equivalent to representations of the absolute Galois group* $\mathrm{Gal}(\overline{K}/K)$ *of a number field* K*."*

As such, understanding the extensions, and by extension the absolute Galois group, is of paramount importance in number theory.

The Langlands program generalises Class Field Theory, itself a crowning glory of early-to-mid 20th century mathematics. Class Field Theory gives a complete theoretical description of the *abelian* extensions of number fields. These results are abstract, in the sense that given an abelian extension, it completely describes its properties (e.g. its Galois group and the splitting behaviour of all of the primes); but they are not explicit, in the sense that you can write down generating polynomials/primitive elements.

Note that the Kronecker–Weber theorem goes further still: it gives an *explicit* description of all of the abelian extensions of $\mathbf{Q}$! Such explicit descriptions are still quite poorly understood in general; for a general number field K, this is Hilbert's 12th problem (from his famous list of 23 open problems, compiled in 1900; they shaped huge areas of 20th century mathematics). For K imaginary quadratic, a complete explicit description is given by the beautiful theory of elliptic curves with complex multiplication (see, for example, Chapter II of Silverman's book *Advanced topics in the arithmetic of elliptic curves* [Sil94]). For totally real fields, there is recent exciting work of Darmon–Vonk [DV21] and Dasgupta–Kakde [DK] in this direction.

10.4.2. Extensions of degree 2^s over $\mathbf{Q}_2$. We now prove that abelian extensions over $\mathbf{Q}_2$ of degree a power of 2 are contained in a cyclotomic field (the $p = 2$ case of Proposition 10.4).

Lemma 10.14. *There is no extension $L/\mathbf{Q}_2$ that both:*

- *is cyclic of degree 4,*
- *contains $i = \sqrt{-1}$.*

Proof. We first prove the following claim:

Claim 10.15. *Let $L/\mathbf{Q}_2$ be cyclic of degree 4 containing i. Then there exist $a, b \in \mathbf{Q}_2$ with*

$$a^2 + b^2 = -1.$$

Proof of claim: By Kummer theory, we may write $L = \mathbf{Q}_2(i, \alpha)$, with $\alpha^2 \in \mathbf{Q}_2(i)$. Let σ be a generator of $\mathrm{Gal}(L/\mathbf{Q})$. Then σ^2 generates $\mathrm{Gal}(L/\mathbf{Q}_2(i))$, so

$$\sigma(i) = -i, \sigma^2(\alpha) = -\alpha.$$

Also note that

$$(\sigma(\alpha))^2 = \sigma(\alpha^2) \in \mathbf{Q}_2(i),$$

so $\sigma^3(\alpha) = \sigma^2(\sigma(\alpha)) = -\sigma(\alpha)$. We deduce

$$\frac{\sigma(\alpha)}{\alpha} \text{ is fixed by } \sigma^2 \Rightarrow \frac{\sigma(\alpha)}{\alpha} = a + bi \in \mathbf{Q}_2(i), \qquad a, b \in \mathbf{Q}_2.$$

Also

$$\frac{\sigma^2(\alpha)}{\sigma(\alpha)} = \sigma\left(\frac{\sigma(\alpha)}{\alpha}\right) = a - bi,$$

and

$$-1 = \frac{\sigma^2(\alpha)}{\alpha} = \frac{\sigma^2(\alpha)}{\sigma(\alpha)} \cdot \frac{\sigma(\alpha)}{\alpha} = (a - bi)(a + bi) = a^2 + b^2,$$

proving the claim.

Now note that $\mathbf{Q}_2(X, Y, Z) = X^2 + Y^2 + Z^2$ is insoluble over $\mathbf{Q}_2$ by Proposition 6.13. Thus there cannot exist a, b with $a^2 + b^2 = -1$, so by the claim, there cannot exist $L/\mathbf{Q}$ as in the lemma. $\square$

If G is a finite abelian group, we say G has *exponent* m if $g^m = 0$ for all $g \in G$.

Proposition 10.16. *Let $K/\mathbf{Q}_2$ be an abelian extension of exponent 2^s, for $s \geqslant 1$. Then K is contained in a cyclotomic extension.*

Proof. If $s = 1$, then K is quadratic, whence K is contained in $\mathbf{Q}_2(\sqrt{-1}, \sqrt{2}, \sqrt{-3}) \subset \mathbf{Q}_2(\zeta_{24})$. (To see this inclusion, note $\zeta_4 = \sqrt{-1}, \zeta_8 = \frac{1+\sqrt{-2}}{2}$ and $\zeta_3 = \frac{1+\sqrt{-3}}{2}$). So assume $s \geqslant 2$.

Let $F = \mathbf{Q}_2(\zeta_m)$ be the unramified extension of degree 2^s. By Lemma 9.23, we may write $\mathbf{Q}_2(\zeta_{2^{s+2}}) = E(i)$ where E is totally ramified of degree 2^s and $i = \sqrt{-1}$. By replacing K with $EFK(i)$, we do not increase the exponent 2^s of $\mathrm{Gal}(K/\mathbf{Q})$ (though we may increase the degree!) so we may assume $EF(i) \subset K$.

Now note $\mathrm{Gal}(EF/\mathbf{Q}_2) \cong \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2^s\mathbf{Z}$, and $\mathrm{Gal}(EF(i)/\mathbf{Q}_2) \cong \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$.

Claim 10.17. *If $\mathrm{Gal}(K/\mathbf{Q}_2)$ has exponent 2^s and $EF(i) \subset K$, then $K = EF(i)$.*

Given the claim, the result follows, since $K = EF(i) = \mathbf{Q}(\zeta_{m2^{s+2}})$ is cyclotomic.

Proof of claim: Suppose not, so $EF(i) \subsetneq K$. If any such field exists, then (passing to a subfield if necessary) there exists K of degree 2 over $EF(i)$. Since the exponent is 2^s, we have

$$\mathrm{Gal}(K/\mathbf{Q}) \cong \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2^s\mathbf{Z} \times H,$$

where H has size 4. There are then two cases:

- $H \cong \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$, i.e.

$$\mathrm{Gal}(K/\mathbf{Q}) \cong \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}.$$

  By Galois theory K has a subfield with Galois group $(\mathbf{Z}/2\mathbf{Z})^4$ over $\mathbf{Q}_2$. By Galois theory this has 15 distinct quadratic subextensions of $\mathbf{Q}_2$; but $\mathbf{Q}_2$ has only 7 quadratic extensions, contradiction.

- $H = \mathbf{Z}/4\mathbf{Z}$. Then the fixed field of $\mathbf{Z}/2^s\mathbf{Z} \times \mathbf{Z}/2^s\mathbf{Z} \times \{1\} \subset \mathrm{Gal}(K/\mathbf{Q})$ is EF, and we deduce $K = EFL$, where $L/\mathbf{Q}_2$ is cyclic of degree 4 and contains i. But this contradicts Lemma 10.14.

Hence no such K exists, and the claim – hence Proposition 10.16 – is proved. □

10.4.3. Results in Kummer theory. We now prove Proposition 10.5, which was used to study degree p^s extensions of $\mathbf{Q}_p$ when p is odd. We first develop the required Kummer theory.

Let $n = p^r$ be an odd prime power, and let K be a field with $\mathrm{char}(K) \neq p$. Let $L = K(\zeta_n)$, and $M = L(\sqrt[n]{a})$ for $a \in L^\times$. Fix an isomorphism

$$\omega : \mathrm{Gal}(L/K) \longrightarrow (\mathbf{Z}/n\mathbf{Z})^\times$$

such that $g(\zeta_n) = \zeta_n^{w(g)}$.

Lemma 10.18. *In the set-up above, if M/K is abelian, then*

$$\frac{g(a)}{a^{w(g)}} \in (L^\times)^n \qquad \forall g \in \mathrm{Gal}(L/K).$$

Proof. This is [Was82], Lemma 14.7. □

Proposition. *If p is odd, there is no Galois extension of $\mathbf{Q}_p$ with Galois group $(\mathbf{Z}/p\mathbf{Z})^3$.*

Proof. See [Was82], Lemma 14.8.

□

Chapter 11

(∗) Appendix: What was the point of Part III?

Students that got to the end of the course might now be asking themselves:

> *What was the point*[1] *of all that extensions stuff?...*
>
> *...Why do I care about anything that we did in Part III?!*

This course may be over, but at this point the story of p-adic numbers and their place in mathematics is only just beginning. Part III was really the introduction to this story.

The aim of this (super-non-examinable) appendix is to hint where all of this is going, looking at how – on a deeper level – the topics we've discussed fit with some of your other algebraic pure courses, and into the framework of modern mathematics research.

Disclaimer: I wrote all of this as a stream of consciousness in week 10 of term, as the course was ending. It's very low on rigour and detail, and I've frequently sacrificed mathematical accuracy to allow me to simplify the picture. Students that find any of this interesting are:

(a) strongly encouraged to come and chat to me about it, as it would be a lot easier to discuss this in person;

(b) even more strongly encouraged to pursue a PhD in Algebraic Number Theory!

11.1. Infinite Galois theory

In your MA3 Galois theory course, you spent a lot of time studying *finite* Galois extensions, and applications therein to ancient problems like insolubility of the quintic and trisecting the angle. You learnt that if $L/\mathbf{Q}$ is a finite Galois extension, then the Galois group $\mathrm{Gal}(L/\mathbf{Q})$ tracks all the symmetries of amn irreducible polynomial f (where L is the splitting field of f).

In this study, we're looking at one polynomial at a time. The modern approach is to ask:

(†) *Why study one f at a time – is it not better to study them all at once?*

This leads you to the notion of *infinite Galois theory.*

Let $\mathbf{Q} \subsetneq L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq \cdots$ be an infinite tower of finite field extensions, with each $L_n/\mathbf{Q}$ Galois.

[1] Beyond the potentially frustrating answer 'Galois theory is beautiful, p-adic fields are beautiful, so what's not to love about the Galois theory of p-adic fields?'

Example 11.1. A good, and important, example of this is to take p to be prime, and consider

$$L_n = \mathbf{Q}(\zeta_{p^n}), \qquad \zeta_{p^n} := \text{ primitive } p^n\text{th root of unity.}$$

Then $L_n/\mathbf{Q}$ is Galois, and $\mathrm{Gal}(L_n/\mathbf{Q}) \cong (\mathbf{Z}/p^n\mathbf{Z})^\times$.

Let $L = \bigcup_n L_n$ be the union of these extensions. Since $[L_n : \mathbf{Q}]$ must tend to infinity, and L contains every L_n, this is an infinite extension of $\mathbf{Q}$. Our usual notion of (finite) Galois theory no longer works here; L cannot be the splitting field of a polynomial, and it doesn't make sense to say that the number of field automorphisms of $L/\mathbf{Q}$ is equal to the degree!

Note that for each n, restriction from L_n to L_{n-1} induces a map $\mathrm{Gal}(L_n/\mathbf{Q}) \to \mathrm{Gal}(L_{n-1}/\mathbf{Q})$: this is just a consequence of the Galois correspondence of fields. In particular, we get a tower

$$\cdots \to \mathrm{Gal}(L_n/\mathbf{Q}) \to \mathrm{Gal}(L_{n-1}/\mathbf{Q}) \to \cdots \to \mathrm{Gal}(L_2/\mathbf{Q}) \to \mathrm{Gal}(L_1/\mathbf{Q}).$$

This should look familiar: we studied such towers in Chapter 4.2! In particular, we introduced inverse limits.

Definition 11.2. If L arises in the manner above, we say $L/\mathbf{Q}$ is Galois, and define

$$\mathrm{Gal}(L/\mathbf{Q}) := \varprojlim_n \mathrm{Gal}(L_n/\mathbf{Q}).$$

This has surjective projection maps to all the $\mathrm{Gal}(L_n/\mathbf{Q})$, so encapsulates all the symmetries of all the $L_n/\mathbf{Q}$.

As we saw in Chapter 4.2, $\mathrm{Gal}(L/\mathbf{Q})$ is a profinite group, and hence it is a compact topological space when equipped with the inverse limit topology. This is very important! There is still a version of the fundamental theorem of Galois theory, here:

Theorem 11.3 (Fundamental theorem of Galois theory)**.** *There is a bijective correspondence*

$$\{\textit{subextensions } \mathbf{Q} \subset K \subset L\} \longleftrightarrow \{\text{closed } \textit{subgroups of } \mathrm{Gal}(L/\mathbf{Q})\}.$$

Under this correspondence, the *finite* subextensions correspond to subgroups of finite index; and similarly to the study we did of profinite groups in Chapter 4.2, these correspond to the *open* subgroups of $\mathrm{Gal}(L/\mathbf{Q})$.

Note this generalises the correspondence for finite extensions, since a finite Galois group carries the discrete topology, hence all subgroups are closed.

11.2. The most interesting object in mathematics

In our question (†), we said we wanted to study all polynomials at once. The above only lets us study the polynomials f_n attached to $L_n/\mathbf{Q}$. To go further, we need the more general notion of inverse limit, introduced in the non-examinable section §4.5.

Definition 11.4. Let $\overline{\mathbf{Q}}$ be the algebraic closure of $\mathbf{Q}$ (that is, the field obtained by adjoining the roots of all irreducible polynomials over $\mathbf{Q}$). The *absolute Galois group of* $\mathbf{Q}$ is the Galois group of $\overline{\mathbf{Q}}/\mathbf{Q}$, defined as the (generalised) inverse limit

$$G_{\mathbf{Q}} := \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) := \varprojlim_{L/\mathbf{Q} \text{ finite Galois}} \mathrm{Gal}(L/\mathbf{Q}).$$

This has a reasonable claim[2] to being *the most interesting object in pure mathematics*©.

[2] After I wrote this, I tried to find something to back it up. First thing I found: $G_{\mathbf{Q}}$ is the top voted answer on the MathOverflow question 'What's the most intricate and beautiful structure in mathematics?'.

On a basic level, $G_{\mathbf{Q}}$ is an huge group with deeply horrible structure. It has, for example, infinitely many distinct index 2 subgroups, which already tells you the group is *massive.* Beyond this it's about as complicated as can possibly be: indeed, the *inverse Galois problem* asks if literally *every* finite group appears as the Galois group of an extension of $\mathbf{Q}$, which would mean that somehow $G_{\mathbf{Q}}$ encompasses all of finite group theory at once.

As per our motivation, $G_{\mathbf{Q}}$ encompasses all possible symmetries of all possible solutions to all possible rational polynomials. In itself, this makes it a very deep object. However, from an algebraic number theory perspective, it is the *representations* of $G_{\mathbf{Q}}$, rather than the group itself, that hold the most interest, as we shall later sketch.

11.3. Galois representations attached to elliptic curves

One of the most beautiful, and relevant, examples comes from elliptic curves; for this, it will have been helpful to have taken **MA4 Elliptic Curves**. This is all rather nicely explained in Chapter 9 of [DS05].

Let $E : y^2 = x^3 + ax + b$ be an elliptic curve over the rationals.

Definition 11.5. For an integer N, define

$$E[N] \coloneqq \{P \in E(\overline{\mathbf{Q}}) : NP = 0\}$$

to be the subgroup of points of order N.

By identifying $E(\overline{\mathbf{Q}}) \subset E(\mathbf{C})$ with a complex torus, you see that:

Proposition 11.6. *$E[N] \cong (\mathbf{Z}/N\mathbf{Z})^2$ as groups.*

Since these points are defined over $\overline{\mathbf{Q}}$, there exists some number field L_N such that $E[N] \subset E(L_N)$. Then $\mathrm{Gal}(L_N/\mathbf{Q})$ acts on $E[N]$: for example, $\sigma \in \mathrm{Gal}(L_N/\mathbf{Q})$ sends a point $P = (x, y) \mapsto \sigma(P) = (\sigma(x), \sigma(y))$. Moreover:

Lemma 11.7. *The action of $\mathrm{Gal}(L_N/\mathbf{Q})$ on $E[N]$ is linear, i.e.*

$$\sigma(P + Q) = \sigma(P) + \sigma(Q),$$

and the resulting linear transformation of $(\mathbf{Z}/N\mathbf{Z})^2$ is invertible.

Proof. (Sketch). The action is linear as the addition law on E is defined by polynomials with rational coefficients; so the coefficients are fixed by any σ. This linear transformation is invertible as the action σ^{-1} gives an inverse. $\square$

Choose a basis P_1, P_2 of $E[N] \cong (\mathbf{Z}/N\mathbf{Z})^2$; then every element of $E[N]$ can be written uniquely as $P = a_1P_1 + a_2P_2$, with $a_i \in \{0, ..., N-1\}$. This choice identifies the action of σ on $(\mathbf{Z}/N\mathbf{Z})^2$ with an linear invertible transformation of $(\mathbf{Z}/N\mathbf{Z})^2$, that is, with a matrix $A_\sigma \in \mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z})$.

Definition 11.8. Let ρ_N denote the 'mod N Galois representation'

$$\begin{aligned} \rho_N : \mathrm{Gal}(L_N/\mathbf{Q}) &\longrightarrow \mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z}), \\ \sigma &\longmapsto A_\sigma. \end{aligned}$$

Via the projection map in the theory of inverse limits, we may consider this as a map

$$\rho_N : G_{\mathbf{Q}} \twoheadrightarrow \mathrm{Gal}(L_N/\mathbf{Q}) \longrightarrow \mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z}).$$

(Note that different choices of bases P_1, P_2 give equivalent representations).

Now fix a prime ℓ. (We will later talk about the Galois group of $\mathbf{Q}_p$, so we need to consider two different primes; and it is customary to use a prime ℓ for the coefficients). One can consider the maps ρ_N as $N = \ell^n$ with n varying. This brings us into the realm of ℓ-adic numbers! It's not so hard to see that for all $\sigma \in G_{\mathbf{Q}}$, we have

$$\rho_{\ell^{n+1}}(\sigma) \equiv \rho_{\ell^n}(\sigma) \pmod{\ell^n}.$$

In particular, we can consider

$$\begin{aligned}\rho_E := \varprojlim_n \rho_{\ell^n} : G_{\mathbf{Q}} \longrightarrow & \varprojlim_n \mathrm{GL}_2(\mathbf{Z}/\ell^n\mathbf{Z}) \\ & \cong \mathrm{GL}_2(\mathbf{Z}_\ell).\end{aligned}$$

(Another way of thinking about this is taking

$$T_\ell E := \varprojlim_n E[\ell^n] \cong \varprojlim_n (\mathbf{Z}/\ell^n\mathbf{Z})^2 \cong \mathbf{Z}_\ell^2,$$

which has a natural action of $G_{\mathbf{Q}}$ from all the above, which induces ρ_E. The module $T_\ell E$ is called the *Tate module* of E.)

This is a 2-dimensional continuous *p-adic Galois representation* attached to E! It is an *extremely* rich object. In fact, it was the central object in Wiles' proof of Fermat's last theorem.

Remark 11.9. It's worth noting that we needed to use ℓ-adic numbers here. In fact, if you tried to consider continuous Galois representations valued in $\mathrm{GL}_2(\mathbf{C})$, you find that they are rather limited: they all have to factor through a finite extension of $\mathbf{Q}$ (for topological reasons). This is a major advantage of the p-adic coefficients: it unlocks this whole realm of 'more interesting' Galois representations.

Ok, so we want to study this new and exciting representation. However... As previously mentioned, the Galois group $G_{\mathbf{Q}}$ is very rich, and this makes studying ρ_E much harder. This is where Part III of the course comes into its own.

11.4. The absolute Galois group of $\mathbf{Q}_p$

First, though: a theme of Part III of the course has been 'if you can do something for $\mathbf{Q}$, then you should also try it for $\mathbf{Q}_p$, and there it might be simpler'. This is *especially* true here.

Definition 11.10. The *absolute Galois group of* $\mathbf{Q}_p$ is

$$G_{\mathbf{Q}_p} := \mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p) := \varprojlim_{L/\mathbf{Q}_p \text{ finite Galois}} \mathrm{Gal}(L/\mathbf{Q}).$$

Again, this encapsulates all symmetries of all polynomials over $\mathbf{Q}_p$. Now, though, this 'local' object is *significantly* simpler than its 'global' analogue $G_{\mathbf{Q}}$:

- For one thing, it has only finitely many index 2 subgroups, corresponding to the quadratic extensions of $\mathbf{Q}_p$.
- More generally, Theorem 9.11 tells us it has only finitely many index n subgroups for any n.
- Moreover, in the non-examinable section §9.3, we proved that any Galois extension of $\mathbf{Q}_p$ has *soluble* Galois group, so certainly $G_{\mathbf{Q}_p}$ does not encapsulate all of finite group theory, as $G_{\mathbf{Q}}$ (conjecturally) does: for example, it does not admit the symmetric group Sym_5 as a quotient, since this would correspond to an extension $K/\mathbf{Q}_p$ with Galois group Sym_5 under the Galois correspondence.

Even better, we've seen that extensions of $\mathbf{Q}_p$ can be broken into an unramified and a totally ramified part. On the level of Galois groups, if $L/\mathbf{Q}_p$ is a finite Galois extension, suppose $\mathbf{Q}_p(\zeta_m)$ is the intermediate field with

$$L/\mathbf{Q}_p(\zeta_m) \text{ totally ramified,} \qquad \mathbf{Q}_p(\zeta_m)/\mathbf{Q}_p \text{ unramified.}$$

Then $\mathrm{Gal}(\mathbf{Q}_p(\zeta_m)/\mathbf{Q}_p)$ is a finite cyclic group $\mathbf{Z}/n\mathbf{Z}$ for some n, and (as studied in the non-examinable section §9.3), we have

$$\begin{aligned}\mathrm{Gal}(L/\mathbf{Q}_p(\zeta_m)) &\cong I_{L/\mathbf{Q}_p} \\ &= \ker[\mathrm{Gal}(L/\mathbf{Q}_p) \twoheadrightarrow \mathrm{Gal}(k_L/\mathbf{F}_p)] \\ &= \{\sigma \in \mathrm{Gal}(L/\mathbf{Q}_p) : \sigma(x) \equiv x \pmod{\pi_L}\ \forall x \in \mathcal{O}_L\},\end{aligned}$$

the *inertia subgroup* of L/K (which has size $e(L/K)$). Whilst it's not quite true that $\mathrm{Gal}(L/\mathbf{Q}_p) \cong \mathbf{Z}/n\mathbf{Z} \times I_{L/K}$, this is *close* to being true: we have

$$\mathrm{Gal}(L/\mathbf{Q}_p) \cong \mathbf{Z}/n\mathbf{Z} \rtimes I_{L/K}$$

is a *semi-direct product.* Morally, one should imagine this result as saying

$$\mathrm{Gal}(L/\mathbf{Q}_p) \cong [\text{unramified part}] + [\text{ramified part}],$$

where the unramified part is cyclic, generated by the Frobenius element

$$\mathrm{Frob}_{L/\mathbf{Q}_p}(x) \equiv x^p \pmod{\pi_L},$$

and the ramified part is some soluble group with (via the higher ramifications groups of §9.3) a prime-to-p part of size at most $p-1$, and the rest is p-power order.

Passing to the inverse limit, we get

$$\begin{aligned}G_{\mathbf{Q}_p} &\cong \Big[\varprojlim_n \mathbf{Z}/n\mathbf{Z}\Big] \rtimes \Big[\varprojlim_{L/\mathbf{Q}_p} I_{L/\mathbf{Q}_p}\Big] \\ &= \Big[\prod_{q \text{ prime}} \mathbf{Z}_q\Big] \rtimes I_{\mathbf{Q}_p} \\ &= \text{“[unramified part] + [ramified part]”},\end{aligned}$$

where

$$I_{\mathbf{Q}_p} := \varprojlim_{L/\mathbf{Q}_p \text{ finite Galois}} I_{L/\mathbf{Q}_p}$$

is the *inertia subgroup* of $G_{\mathbf{Q}_p}$. Here, the fact that

$$\varprojlim_n \mathbf{Z}/n\mathbf{Z} \cong \prod_{q \text{ prime}} \mathbf{Z}_q$$

was a non-credit question on the exercise sheets.

Even better, one can check that the Frobenius elements $\mathrm{Frob}_{L/\mathbf{Q}_p}$ fit together into an element $\mathrm{Frob}_p \in G_{\mathbf{Q}_p}$ in the inverse limit. This generates a 'diagonal' subgroup $\mathbf{Z} \subset \prod_{q \text{ prime}} \mathbf{Z}_q$, where

$$\begin{aligned}\mathbf{Z} = \langle \mathrm{Frob}_p \rangle &\longrightarrow \prod_q \mathbf{Z}_q \\ \mathrm{Frob}_p^n &\longmapsto (n, n, n, ...) \in \mathbf{Z}_2 \times \mathbf{Z}_3 \times \mathbf{Z}_5 \times \cdots .\end{aligned}$$

Additionally, in the profinite topology, the completion of this subgroup $\mathbf{Z} = \langle \mathrm{Frob}_p \rangle$ is simply the product of the completions of $\mathbf{Z}$ in the $\mathbf{Z}_q$, i.e. the product $\prod \mathbf{Z}_q$ of the $\mathbf{Z}_q$. We thus call Frob_p a *topological generator* of $\prod \mathbf{Z}_q$; it is, 'up to topology', a cyclic group, 'generated' by Frob_p.

Now, this is all particularly important when we come to study *representations* of $G_{\mathbf{Q}_p}$. If I have a (continuous) representation

$$\rho_p : G_{\mathbf{Q}_p} \longrightarrow \mathrm{GL}(V),$$

for some vector space V, then it is natural to consider the restrictions to:

- the ramified part $I_{\mathbf{Q}_p}$, and
- the unramified part $\prod_q \mathbf{Z}_q$.

By continuity, and topological generation, the restriction $\rho_p|_{\prod \mathbf{Z}_q}$ is completely determined by the value of $\rho_p(\mathrm{Frob}_p)$; so the unramified part of ρ_p boils down to a single element of $\mathrm{GL}(V)$.

The ramified part is much more involved, as $I_{\mathbf{Q}_p}$ is a more complicated group. However, frequently in nature we find that ρ_p has *trivial* restriction to $I_{\mathbf{Q}_p}$! Mimicking terminology we saw in the course, we call such ρ_p an *unramified* representation of $G_{\mathbf{Q}_p}$.

11.5. Restriction from $\mathbf{Q}_p$ to $\mathbf{Q}$

One of the key insights in studying

$$\rho_E : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \longrightarrow \mathrm{GL}_2(\mathbf{Z}_\ell)$$

is that we can use the results of Part III to simplify things. Recall we showed that if $L/\mathbf{Q}$ is a Galois extension, and $\mathfrak{P}|p$ are primes, then $L_{\mathfrak{P}}/\mathbf{Q}_p$ is finite Galois, and restriction from $L_{\mathfrak{P}}$ to L defines an injective homomorphism

$$\mathrm{Gal}(L_{\mathfrak{P}}/\mathbf{Q}_p) \hookrightarrow \mathrm{Gal}(L/\mathbf{Q}).$$

Taking this over inverse limits, we find there is an injective homomorphism

$$G_{\mathbf{Q}_p} = \mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p) \hookrightarrow \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) = G_{\mathbf{Q}}.$$

Thus... *We can restrict ρ_E to $G_{\mathbf{Q}_p}$!!*

(Technical note: in this process, in the background we have chosen a prime $\mathfrak{P}$ above p in every finite extension of $\mathbf{Q}_p$. This is tracked by fixing an embedding of $\overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_p$, which then lets us make sense of restriction from $\overline{\mathbf{Q}}_p$ to $\overline{\mathbf{Q}}$).

Definition 11.11. For p a prime, let

$$\rho_{E,p} : G_{\mathbf{Q}_p} \longrightarrow \mathrm{GL}_2(\mathbf{Z}_\ell)$$

be the restriction of ρ_E to $G_{\mathbf{Q}_p}$.

This is a continuous representation of $G_{\mathbf{Q}_p}$, and we've already highlighted that such an object is simpler to study. Importantly, it still carries lots of interesting information, particularly about the behaviour of the elliptic curve at the prime p. In particular, we have the following wonderful theorems. First, the consider the ramified part:

Theorem 11.12 (Néron–Ogg–Shafarevich)**.** *Let $p \neq \ell$ be a prime[3]. Then the restriction to inertia*

$$\rho_{E,p} : I_{\mathbf{Q}_p} \subset G_{\mathbf{Q}_p} \longrightarrow \mathrm{GL}_2(\mathbf{Z}_\ell)$$

is trivial if and only if $E \pmod p$ is an elliptic curve over $\mathbf{F}_p$ (that is, E has good reduction *at p).*

This occurs if and only if p does not divide the conductor N_E of E.

Definition 11.13. If $\rho_{E,p}\Big|_{I_{\mathbf{Q}_p}}$ is trivial, we say ρ_E is *unramified at p.*

Thus the restriction of $\rho_{E,p}$ to the ramified part is telling us about the behaviour of E modulo p. We can glean even more information from restriction to the unramified part. Recall that this restriction is completely determined by the value of $\rho_{E,p}$ on the Frobenius element Frob_p.

Theorem 11.14. *If E is unramified at p, then*

$$\mathrm{Trace}(\rho_{E,p}(\mathrm{Frob}_p)) = a_p(E) = p + 1 - \#E(\mathbf{F}_p).$$

In particular, the restriction to the unramified part is giving us data on the number of points modulo p, a crucial invariant of the elliptic curve, which has all sorts of interesting consequences (for example, being vital to define the L-function of E, the central object of Birch and Swinnerton-Dyer conjecture – one of the millennium problems!).

This pattern is repeated in *much* more general settings. A huge variety of arithmetic objects of interest (number fields, Dirichlet characters, more general versions of elliptic curves, and so on) can be given an attached 'global' Galois representation of $G_{\mathbf{Q}}$ (or G_K, for K a number field); and by restricting to primes, we get a family of 'local' Galois representations of $G_{\mathbf{Q}_p}$ (or $G_{K_\mathfrak{p}}$), for $K_\mathfrak{p}$ a finite extension of $\mathbf{Q}_p$), which contain data about the arithmetic at p.

11.6. The Langlands program

What follows is now a (potentially misguided) attempt to sketch how p-adic numbers play a role in the Langlands program, a universal theory of arithmetic that occupies a huge amount of current research in number theory and arithmetic geometry. I can't possibly do this justice here without omitting essentially *every* detail, but hopefully this gives a hint at a bigger picture. As motivation, let me say out of the gate that the modularity theorem, as proved in many cases by Wiles, is a part of this program, and led to his proof of Fermat's Last Theorem.

If you don't understand any of what follows, that's my fault for trying to summarise a very technical and demanding subject in a few pages at MA4 level!

11.6.1. The Local Langlands program: arithmetic to analysis, I. Let me poorly summarise the (2-dimensional) Local Langlands conjectures as saying:

Conjecture 11.15. *"There is a meaningful bijective correspondence between*

$$\boxed{\textit{Nice representations } G_{\mathbf{Q}_p} \to \mathrm{GL}_2(\mathbf{Q}_\ell)} \longleftrightarrow \boxed{\textit{`Analytic' representations of } \mathrm{GL}_2(\mathbf{Q}_p)},$$

$$\rho_p \longleftrightarrow \Pi_p \qquad \textit{(say).} \qquad "$$

[3]The cases $p = \ell$ and $p \neq \ell$ often need to be treated very differently: the case $p = \ell$ is the realm of *p-adic Hodge theory*, which seeks to find analogues of this theorem in this case.

I've not given you any meanining or definition for Π_p. Consider it just as the 'object coming out of the correspondence'. For the purposes of the MA4 Local Fields course, you can probably draw most intuition by thinking:

> *'The objects on the left are arithmetic, symmetries, and an example comes from elliptic curves. The objects on the right, though, are analytic. Actually they are* very *closely connected with modular forms. This is a collision between two very different worlds and it is therefore beautiful and surprising'.*

In particular, to anchor this back to the course: we saw that the Galois theory of $\mathbf{Q}_p$ is relatively simple (at least compared to that of $\mathbf{Q}$). Moreover Local Langlands is saying that representations of $G_{\mathbf{Q}_p}$ have this surprising connection to analytic representation theory.

11.6.2. The Langlands program: arithmetic to analysis, II. Next we consider this on an even deeper level. We've 'seen' that to an elliptic curve E, one can attach:

– A Galois representation
$$\rho_E : G_{\mathbf{Q}} \to \mathrm{GL}_2(\mathbf{Z}_\ell),$$

– For each prime q, a restricted Galois representation
$$\rho_{E,p} : G_{\mathbf{Q}_p} \to \mathrm{GL}_2(\mathbf{Z}_|ell)$$
that sees the 'arithmetic of E at p',

– and that to each $\rho_{E,p}$, there is an associated 'analytic representation' Π_p of $\mathrm{GL}_2(\mathbf{Q}_p)$.

The next step: why don't we consider all the primes p at once? In particular, can we reconstruct ρ_E (which is very complicated) if we know $\rho_{E,p}$ (which is simpler) for all primes p?

Studying all the $\rho_{E,p}$ at once is equivalent, once you have Local Langlands, to studying all the Π_p at once. If we're going to do that, then why not consider all the Π_p simultaneously, for example as a representation
$$\prod_{p \text{ prime}} \Pi_p \qquad \text{of} \qquad \prod_{p \text{ prime}} \mathrm{GL}_2(\mathbf{Q}_p).$$
...well, we don't do this because this group $\prod_p \mathrm{GL}_2(\mathbf{Q}_p)$ is pretty nasty. But there *is* a way of fixing it: we can introduce the so-called *adeles*, which look much closer to $\prod_p \mathrm{GL}_2(\mathbf{Z}_p)$, which is isomorphic to $\mathrm{GL}_2(\widehat{\mathbf{Z}})$ where $\widehat{\mathbf{Z}} = \prod_p \mathbf{Z}_p$, and which behaves *much* better. Without getting into details, there is a ring $\mathbf{A}$ – the ring of adeles of $\mathbf{Q}$ – which is somewhere between $\prod_q \mathbf{Z}_q$ and $\mathbf{R} \times \prod_p \mathbf{Q}_p$. In particular, $\mathbf{R}$ and $\mathbf{Q}_p \subset \mathbf{A}$, and hence $\mathbf{A}$ contains all possible completions of $\mathbf{Q}$ by Ostrowski's theorem. Attached to all the Π_p, we get an analytic representation Π of $\mathrm{GL}_2(\mathbf{A})$, which 'sees all the primes at once'. Under Local Langlands, this Π 'sees the arithmetic of E at all the primes p'.

The *global* Langlands program says this is enough to reconstruct the data coming from our original elliptic curve E!

Conjecture 11.16 (Global Langlands program over $\mathbf{Q}$)**.** *"There is a bijective and meaningful correspondence*

$$\boxed{\textit{Nice representations of } G_{\mathbf{Q}}} \longleftrightarrow \boxed{\textit{Analytic representations of } \mathrm{GL}_2(\mathbf{A})}."$$

Under this correspondence, $\Pi = \prod \Pi_p$ is mapped back to ρ_E again! So we recover the object on $G_{\mathbf{Q}}$ from its restrictions.

So we've obtained the picture:

$$\begin{tikzcd}
E \arrow[r] & \rho_E \arrow[r, "\text{restriction}"] \arrow[d, leftrightarrow, "\text{Global Langlands}"'] & \{\rho_{E,p} : p\} \arrow[d, leftrightarrow, "\text{Local Langlands}"] \\
& \Pi = \prod_p \Pi_p \text{ on } \mathrm{GL}_2(\mathbf{A}) & \{\Pi_p \text{ on } \mathrm{GL}_2(\mathbf{Q}_p) : p\} \arrow[l]
\end{tikzcd}$$

11.6.3. Modular forms!. Our whistle-stop tour is now, presumably, miles away from any of your comfort zones. Let's finally display the deep links to another course in the Warwick curriculum: to **MA4 Modular Forms**.

It turns out, when you look carefully at the definition of Π, it yields up a particularly nice function $\Phi : \mathrm{GL}_2(\mathbf{A}) \to \mathbf{C}$. Ok, you may say: I don't know what $\mathrm{GL}_2(\mathbf{A})$ really is. But I did tell you that $\mathbf{R} \subset \mathbf{A}$, so we have

$$\mathrm{GL}_2(\mathbf{R}) \subset \mathrm{GL}_2(\mathbf{A}).$$

Let

$$F : \mathrm{GL}_2(\mathbf{R}) \to \mathbf{C}$$

be the restriction of Φ to this subset.

Now, in MA4 Modular Forms you saw the upper half-plane

$$\mathcal{H} = \{z \in \mathbf{C} : \mathrm{Im}(z) > 0\} = \{x + iy : x \in \mathbf{R}, y \in \mathbf{R}_{>0}\}.$$

Actually, this is a *quotient* of $\mathrm{GL}_2(\mathbf{R})$ (or at least, the subgroup $\mathrm{GL}_2^+(\mathbf{R})$ of elements with positive determinant): we have

$$\mathrm{GL}_2^+(\mathbf{R})/\mathbf{R}_{>0} \cdot \mathrm{SO}_2(\mathbf{R}) \cong \mathcal{H}, \qquad \begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix} \mapsto x + iy.$$

Moreover, the function F is *invariant* under translation by $\mathbf{R}_{>0} \cdot \mathrm{SO}_2(\mathbf{R})$; so under this identification, we get a function

$$f : \mathcal{H} \to \mathbf{C}.$$

Here's the magic part. Recall we started with an elliptic curve E of conductor N_E. We attached to this ρ_E, then $\{\rho_{E,q}\}$, then $\{\Pi_q\}$, then Π, then f. And...

Theorem 11.17 (Modularity)**.** *The function f is a cuspidal modular form (even, a newform) of weight 2 and level $\Gamma_0(N_E)$. It is an eigenform for all the Hecke operators. Writing its q-expansion as*

$$f(q) = \sum_{n \geqslant 1} a_n q^n,$$

for all primes p we have

$$a_p = a_p(E) = p + 1 - \#E(\mathbf{F}_p).$$

Summarising, this says that to any elliptic curve, there is an attached weight 2 cuspidal modular eigenform!! This is a beautiful and extremely deep result: to the arithmetic/algebraic object E, we pull out a very analytic object f. There are questions about E that are easier to study by passing them over to f, and studying them analytically; and thus we obtain a wonderful, powerful new tool for studying the arithmetic of elliptic curves.

Wiles proved this theorem for a large class of elliptic curves in 1994, and this was *the* key ingredient in his proof of Fermat's Last Theorem! The proof was finished off for all elliptic curves by Breuil–Conrad–Diamond–Taylor in 2000.

More general versions of modularity (that is, the general Langlands conjectures) – and the use of 'local' methods, i.e. over $\mathbf{Q}_p$ rather than $\mathbf{Q}$ – are among the most intensely researched areas of mathematics to this day.

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